

Hermes automata compendium

This compendium records the finite transition-table corpus as executed domain scenes, radial transition diagrams, five-tuples, and tables. A table can witness its recorded graph; it cannot by itself witness a stack or establish a richer computational class. The rows come from knowledge/strategies/transition_tables/, with authored class attestations kept separate. Regenerate it with `python3 scripts/research/build_machine_typology.py && python3 scripts/research/render_automaton_svg.py && python3 scripts/research/render_automaton_context_svg.py && python3 scripts/research/build_automata_compendium.py`.

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Legend

Each entry writes $M = (Q, \Sigma, \delta, q_0, F)$: states, action alphabet, transition relation, start state, and accepting states. ASCII atoms remain beside the indexed state notation. In transition diagrams, **—** dark edges are conserving actions, **—** rust edges are deforming actions, and **—** muted edges are neutral actions. Dashed edges are observed-only rows. The gold arrow and state ring mark q_0 ; accepting states have two rings. Domain scenes use rust for an executed deformation and gold or dark marks for the admitted representation roles.

linear_trace no recorded branch or loop; the table records one accepting run.

- branching** one or more recorded states have multiple outgoing transitions; no loop edge is recorded.
- looping** one or more recorded edges return to an earlier state; no branch state is recorded.
- branching_looping** the recorded graph contains both a branch state and a loop edge.

Computed structure reports only the table graph. An authored class appears on a separate line only when module prose names it and supplies a source location.

Computed typology

These classes summarize the graphs recorded by all 222 transition tables. They do not add claims about unrecorded runs or memory.

Structural class	Machines	Recorded shape
linear_trace	214	no recorded branch or loop; the table records one accepting run
branching	7	one or more recorded states have multiple outgoing transitions; no loop edge is recorded
looping	0	one or more recorded edges return to an earlier state; no branch state is recorded
branching_looping	1	the recorded graph contains both a branch state and a loop edge

Authored class attestations

These class names come from module prose and remain separate from the computed table-graph typology.

Machine	Authored class	Source
addition/base_ones_chunking	finite_state_machine	knowledge/strategies/math/sar_add_chunking.pl:4
addition/make_base_transfer	finite_state_machine	knowledge/strategies/math/sar_add_rmb.pl:4
counting/recursive_place_value_inscription	pushdown_counter	knowledge/strategies/math/counting2.pl:1

Machine	Authored class	Source
fraction/unit_fraction_partition	finite_state_automaton	knowledge/strategies/math/fraction_partitioning.pl:3

Coincidence atlas

A deformation pair (f, g) coincides on the equalizer of the pair, the inputs where $f(x) = g(x)$. These profiles measure samples of those sets on finite grids; they are not proofs over every input.

A recognizer may charge a misconception only on inputs where the deformation separates. A problem set teaches against a misconception only on inputs where it separates.

Kind	Validity label	Measured coincidence	Sample coinciding input	Sample separating input	Separation
addition/append_column_sum_without_carrying	contextually_correct with condition(raw_column_concatenation_preserves_place_value)	45 / 405 (11%)	1,9	1,19	It separates where raw column concatenation does not preserve place value.
addition/count_all_instead_of_known_fact	correct_but_inefficient	190 / 190 (100%)	1,1	none recorded	Nothing: answer-agreement is total; only the trace separates it.
addition/count_all_when_count_on_available	correct_but_inefficient	900 / 900 (100%)	1,1	none recorded	Nothing: answer-agreement is total; only the trace separates it.
addition/drop_carry_to_next_column	incorrect	0 / 405 (0%)	none recorded	1,9	It separates on every input that ran in this measured grid.
addition/dropped_ones_chunk	contextually_correct with condition(decomposed_addend_has_no_ones_chunk)	90 / 900 (10%)	1,10	1,1	It separates where the decomposed addend has a ones chunk.
addition/make_ten_drop_leftover	contextually_correct with condition(no_leftover_after_making_base)	45 / 603 (7%)	1,9	2,9	It separates where a leftover remains after making the base.
addition/rotate_derived_fact_rule_misfire	incorrect	0 / 110 (0%)	none recorded	1,3	It separates on every input that ran in this measured grid.

Kind	Validity label	Measured coincidence	Sample coinciding input	Sample separating input	Separation
addition/round_without_adjusting	contextually_correct with condition(selected_addend_requires_no_rounding_adjustment)	9 / 900 (1%)	10,10	1,1	It separates where the selected addend requires a rounding adjustment.
addition/unbalanced_make_base_compensation	contextually_correct with condition(selected_addend_already_at_target_base)	117 / 720 (16%)	1,10	1,9	It separates where the selected addend is not already at the target base.
addition/wrong_carry_amount_to_next_column	incorrect	0 / 360 (0%)	none recorded	1,19	It separates on every input that ran in this measured grid.
counting/spatial_extent_as_cardinality	incorrect	0 / 756 (0%)	none recorded	counts(1,1), extents(2,1)	It separates on every input that ran in this measured grid.
decimal/decimal_add_unaligned_numerals	accidentally_correct	3200 / 6400 (50%)	decimal_pair(1,10,1,10), ignored	decimal_pair(1,10,1,100), ignored	The outcome records no condition atom; separation is the complement of the measured coincidence set.
decimal/decimal_numeral_comparison_without_scale_alignment	accidentally_correct	4880 / 6400 (76%)	decimal_pair(1,10,1,10), ignored	decimal_pair(1,10,1,100), ignored	The outcome records no condition atom; separation is the complement of the measured coincidence set.
decimal/decimal_scale_loss_comparison	accidentally_correct	4880 / 6400 (76%)	decimal_pair(1,10,1,10), ignored	decimal_pair(1,10,1,100), ignored	The outcome records no condition atom; separation is the complement of the measured coincidence set.
decimal/decimal_subtract_unaligned_numerals	accidentally_correct	1640 / 2524 (64%)	decimal_pair(1,10,1,10), ignored	decimal_pair(1,10,1,100), ignored	The outcome records no condition atom; separation is the complement of the measured coincidence set.
division/divide_larger_by_smaller	contextually_correct with condition(given_dividend_at_least_given_divisor)	654 / 720 (90%)	1,1	1,2	It separates where the given dividend is less than the given divisor.
division/name_group_count_as_share_size	contextually_correct with condition(group_count_equals_share_size)	7 / 184 (3%)	1,1	2,1	It separates where group count does not equal share size.

Kind	Validity label	Measured coincidence	Sample coinciding input	Sample separating input	Separation
division/name_reached_to tal_as_quotient	contextually_correct with condition(reached_to tal_equals_missing_factor)	60 / 184 (32%)	1,1	2,2	It separates where reached total does not equal missing factor.
division/share_into_divisor_groups	incorrect	0 / 536 (0%)	none recorded	1,2	It separates on every input that ran in this measured grid.
division/stop_after_first_partial_quotient	contextually_correct with condition(first_partial_quotient_completes_quotient_and_remainder)	192 / 612 (31%)	3,2	3,1	It separates where first partial quotient does not complete quotient and remainder.
division/stop_after_one_known_fact	contextually_correct with condition(first_known_fact_completes_quotient_and_remainder)	192 / 612 (31%)	3,2	3,1	It separates where first known fact does not complete quotient and remainder.
division/stop_at_nearby_product_in_search	incorrect	0 / 172 (0%)	none recorded	2,1	It separates on every input that ran in this measured grid.
fraction/add_numerator_denominator_comparison	contextually_correct with condition(numerator_denominator_sum_order_agrees_with_common_unit_order)	828 / 1764 (46%)	fraction_pair(1,2,1,2),unit(whole)	fraction_pair(1,2,1,3),unit(whole)	It separates where numerator denominator sum order does not agree with common unit order.
fraction/add_numerator_denominator_sum	incorrect	0 / 1764 (0%)	none recorded	fraction_addend_pair(frac(1,2),frac(1,2)),unit(whole)	It separates on every input that ran in this measured grid.
fraction/are_a_model_unequal_partition_pie_count	contextually_correct with condition(pie_count_order_agrees_with_fraction_magnitude)	1248 / 1764 (70%)	fraction_pair(1,2,1,2),unit(whole)	fraction_pair(1,2,1,3),unit(whole)	It separates where piece count order does not agree with fraction magnitude.
fraction/cross_multiplication_rule_without_ground	correct	1764 / 1764 (100%)	fraction_pair(1,2,1,2),unit(whole)	none recorded	Nothing: answer-agreement is total; only the trace separates it.

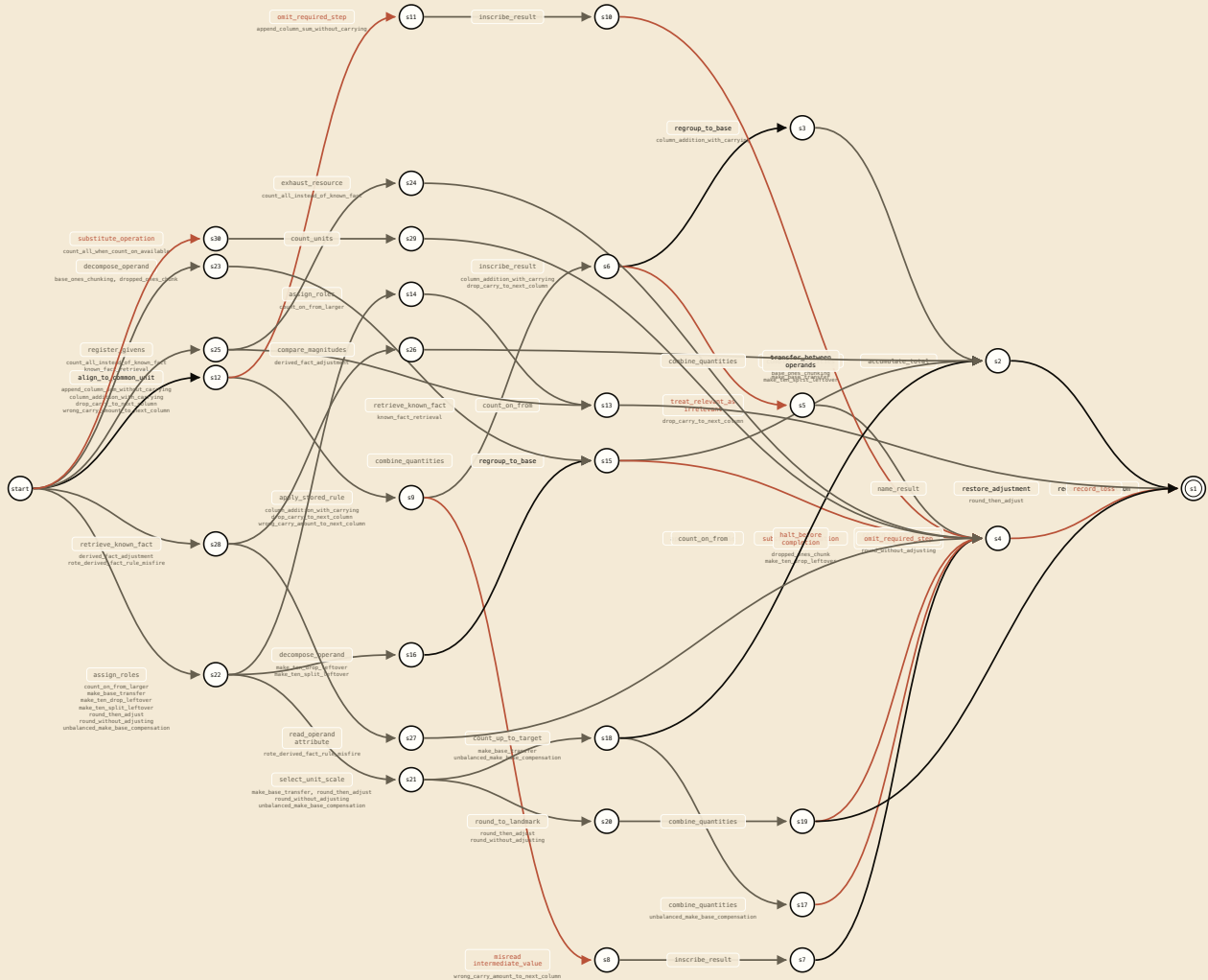
Kind	Validity label	Measured coincidence	Sample coinciding input	Sample separating input	Separation
fraction/gap_thinking_fraction_comparison	contextually_correct with condition(gap_order_coincides_with_fraction_order)	1164 / 1764 (65%)	fraction_pair(1,2,1,2),unit(whole)	fraction_pair(1,2,2,3),unit(whole)	It separates where gap order does not coincide with fraction order.
fraction/number_line_count_marks_not_intervals	contextually_correct with condition(mark_count_order_agrees_with_interval_measure)	1640 / 1764 (92%)	fraction_pair(1,2,1,2),unit(whole)	fraction_pair(1,2,2,3),unit(whole)	It separates where mark count order does not agree with interval measure.
fraction/set_model_subset_size_focus	contextually_correct with condition(subset_size_order_agrees_with_fraction_magnitude)	1248 / 1764 (70%)	fraction_pair(1,2,1,2),unit(whole)	fraction_pair(1,2,1,3),unit(whole)	It separates where subset size order does not agree with fraction magnitude.
multiplication/add_counts_without_composite_unit	accidentally_correct	1 / 400 (0%)	2,2	1,1	The outcome records no condition atom; separation is the complement of the measured coincidence set.
multiplication/add_instead_of_multiply	accidentally_correct	1 / 400 (0%)	2,2	1,1	The outcome records no condition atom; separation is the complement of the measured coincidence set.
multiplication/drop_regrouping_remainder	incorrect	0 / 292 (0%)	none recorded	1,1	It separates on every input that ran in this measured grid.
multiplication/drop_second_partial_product	incorrect	0 / 380 (0%)	none recorded	1,2	It separates on every input that ran in this measured grid.
multiplication/known_product_without_adjustment	incorrect	0 / 380 (0%)	none recorded	2,1	It separates on every input that ran in this measured grid.
multiplication/repeat_group_size_by_itself	contextually_correct with condition(group_count_equals_group_size)	20 / 400 (5%)	1,1	1,2	It separates where group count does not equal group size.
subtraction/add_instead_of_subtract_column	incorrect	0 / 465 (0%)	none recorded	1,1	It separates on every input that ran in this measured grid.

Kind	Validity label	Measured coincidence	Sample coinciding input	Sample separating input	Separation
subtraction/ answer_as_endpoint_count_up	incorrect	0 / 465 (0%)	none recorded	1, 1	It separates on every input that ran in this measured grid.
subtraction/ borrow_without_reducing_bases	incorrect	0 / 162 (0%)	none recorded	10, 1	It separates on every input that ran in this measured grid.
subtraction/ compare_returns_larger_count	incorrect	0 / 870 (0%)	none recorded	1, 2	It separates on every input that ran in this measured grid.
subtraction/ drop_ones_after_base_takeway	incorrect	0 / 198 (0%)	none recorded	11, 11	It separates on every input that ran in this measured grid.
subtraction/ slide_subtend_only	incorrect	0 / 297 (0%)	none recorded	10, 1	It separates on every input that ran in this measured grid.
subtraction/ smaller_from_larger_in_column	incorrect	0 / 162 (0%)	none recorded	10, 1	It separates on every input that ran in this measured grid.

Limits

The structural classes summarize generated table rows, including observed-only states named by those rows. They do not establish determinism, language coverage, or memory bounds beyond the recorded graph. Coincidence rates describe the finite grids named by their generator and do not generalize beyond those grids without further proof.

addition



Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

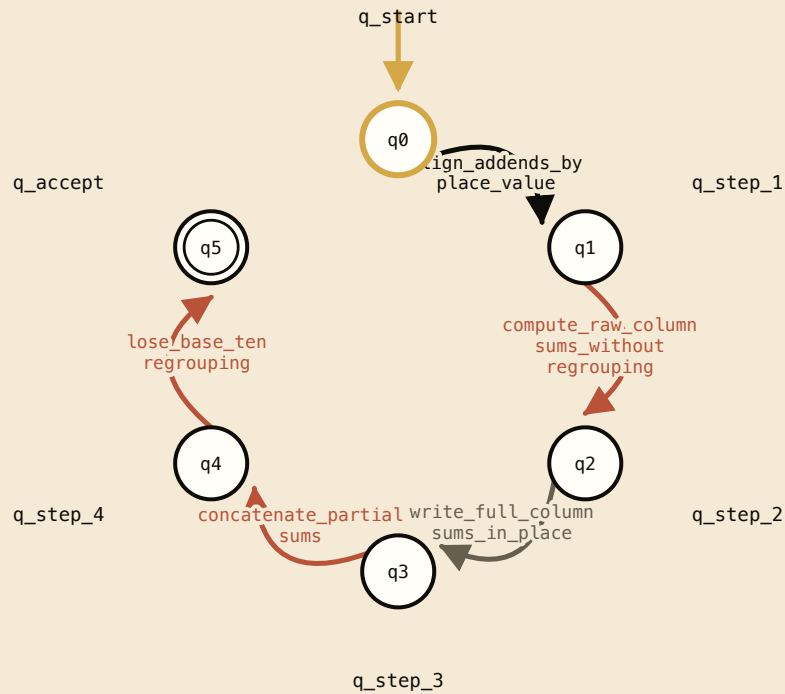
1. append_column_sum_without_carrying

append_column_sum_without_carrying

	overflow	tens	ones
47		4	7
28		2	8
615	6	1	5

The run produced 615; the recorded expected result is 75.

The run produced 615; the recorded expected result is 75. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{align_addends_by_place_value, compute_raw_column_sums_without_regrouping, concatenate_partial_sums, lose_base_ten_regrouping, write_full_column_sums_in_place\} \text{ (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

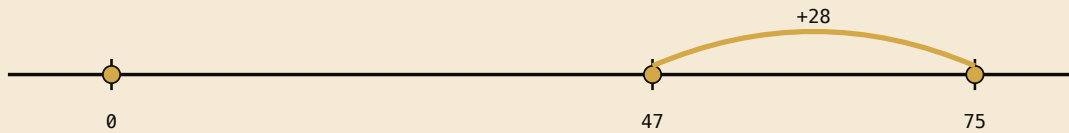
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	align_addends_by_place_value	q_step_1	sar_add_action_pairs.pl:385 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	compute_raw_column_sums_without_regrouping	q_step_2	sar_add_action_pairs.pl:385 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	write_full_column_sums_in_place	q_step_3	sar_add_action_pairs.pl:385 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	concatenate_partial_sums	q_step_4	sar_add_action_pairs.pl:385 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	lose_base_ten_regrouping	q_accept	sar_add_action_pairs.pl:385 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

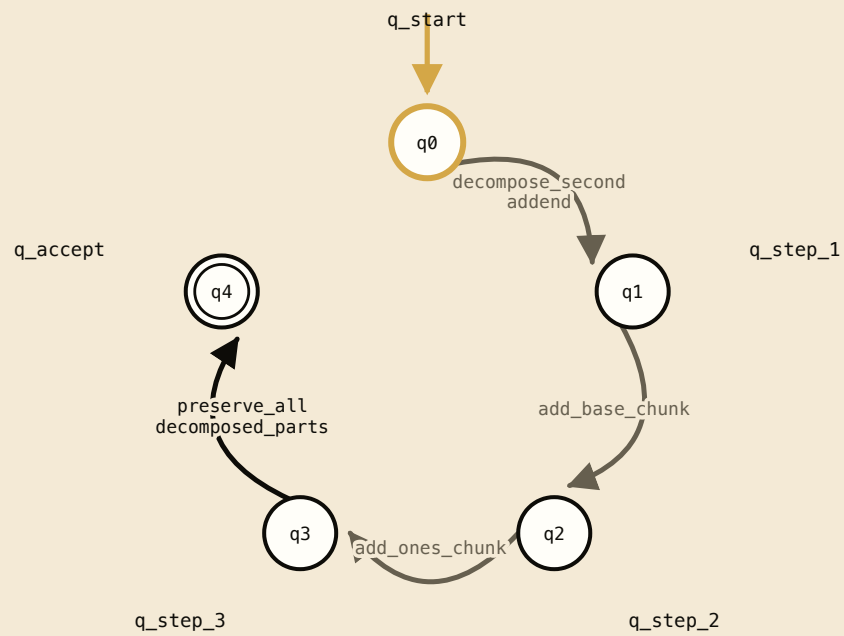
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

2. base_ones_chunking



The run produced 75.

The run produced 75. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{add_base_chunk}, \text{add_ones_chunk}, \text{decompose_second_addend}, \text{preserve_all_decomposed_parts}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_second_addend	q_step_1	sar_add_action_pairs.pl:229 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	add_base_chunk	q_step_2	sar_add_action_pairs.pl:229 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	add_ones_chunk	q_step_3	sar_add_action_pairs.pl:229 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	preserve_all_decomposed_parts	q_accept	sar_add_action_pairs.pl:229 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

Authored class: finite_state_machine, attested at
knowledge/strategies/math/sar_add_chunking.pl:4. This claim comes from module prose,

not from the table graph.

3. column_addition_with_carrying

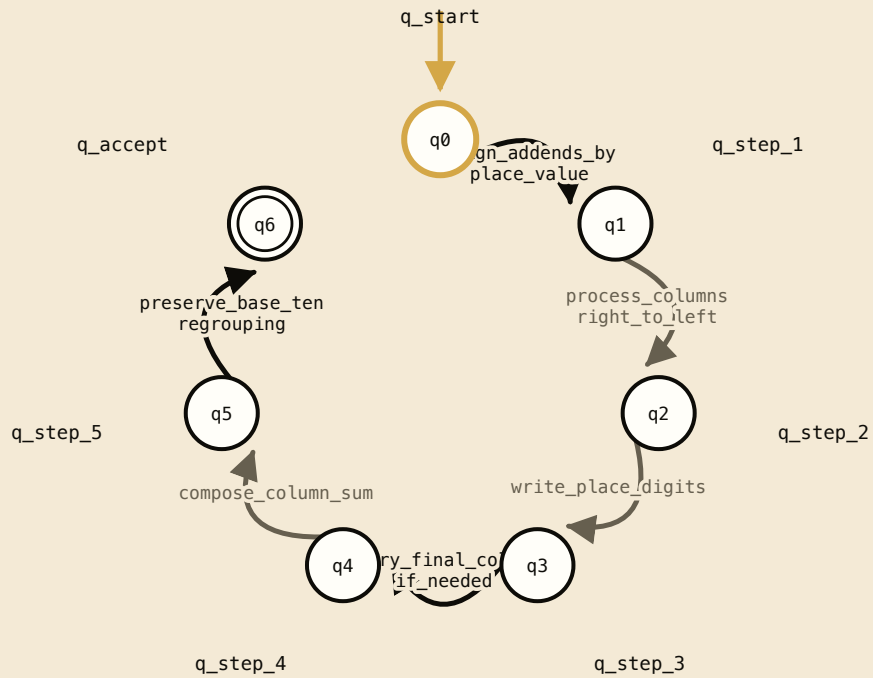
column_addition_with_carrying

+1 ten

	tens	ones
47	4	7
28	2	8
75	7	5

The run produced 75.

The run produced 75. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ align_addends_by_place_value, carry_final_column_if_needed, compose_column_sum, preserve_base_ten_regrouping, process_columns_right_to_left, write_place_digits (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

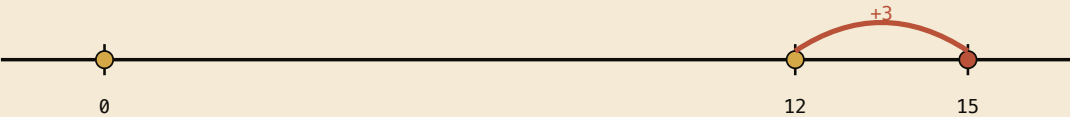
δ : recorded transitions

From	Action	To	Provenance
q_start	align_addends_by_place_value	q_step_1	sar_add_action_pairs.pl:336 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	process_columns_right_to_left	q_step_2	sar_add_action_pairs.pl:336 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	write_place_digits	q_step_3	sar_add_action_pairs.pl:336 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	carry_final_column_if_needed	q_step_4	sar_add_action_pairs.pl:336 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	compose_column_sum	q_step_5	sar_add_action_pairs.pl:336 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_5	preserve_base_ten_regrouping	q_accept	sar_add_action_pairs.pl:336 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

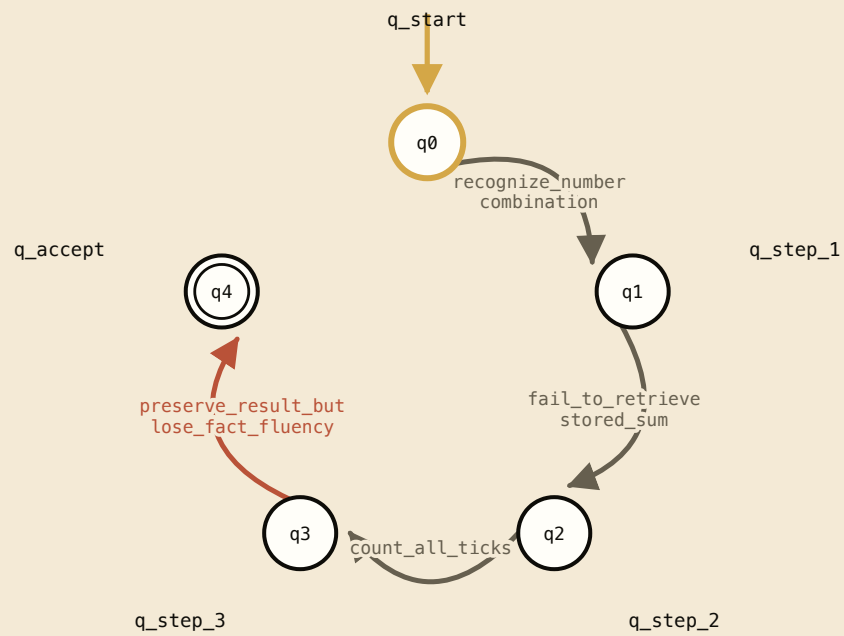
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

4. count_all_instead_of_known_fact



The run produced 15; the recorded expected result is 15.

The run produced 15; the recorded expected result is 15. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ count_all_ticks, fail_to_retrieve_stored_sum,
preserve_result_but_lose_fact_fluency, recognize_number_combination (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

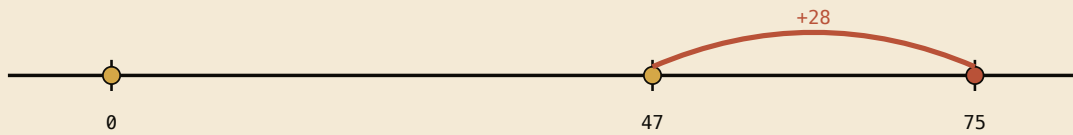
Input contract: {"a": "integer", "b": "integer"}; example {"a": 12, "b": 3};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	recognize_number_combination	q_step_1	sar_add_action_pairs.pl:461 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)
q_step_1	fail_to_retrieve_stored_sum	q_step_2	sar_add_action_pairs.pl:461 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)
q_step_2	count_all_ticks	q_step_3	sar_add_action_pairs.pl:461 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)
q_step_3	preserve_result_but_lose_fact_fluency	q_accept	sar_add_action_pairs.pl:461 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)

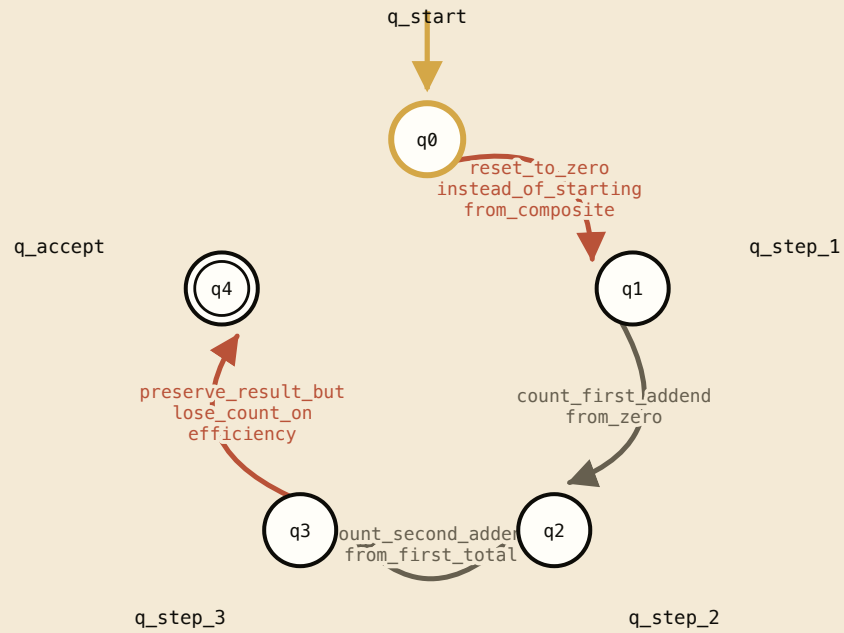
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

5. count_all_when_count_on_available



The run produced 75; the recorded expected result is 75.

The run produced 75; the recorded expected result is 75. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ `count_first_addend_from_zero`, `count_second_addend_from_first_total`,
`preserve_result_but_lose_count_on_efficiency`,
`reset_to_zero_instead_of_starting_from_composite` (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

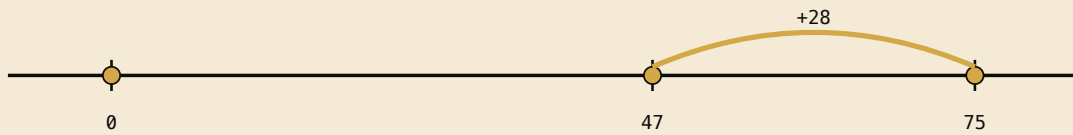
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	reset_to_zero_instead_of_starting_from_composite	q_step_1	sar_add_action_pairs.pl:90 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	count_first_addend_from_zero	q_step_2	sar_add_action_pairs.pl:90 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	count_second_addend_from_first_total	q_step_3	sar_add_action_pairs.pl:90 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	preserve_result_but_lose_count_on_efficiency	q_accept	sar_add_action_pairs.pl:90 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

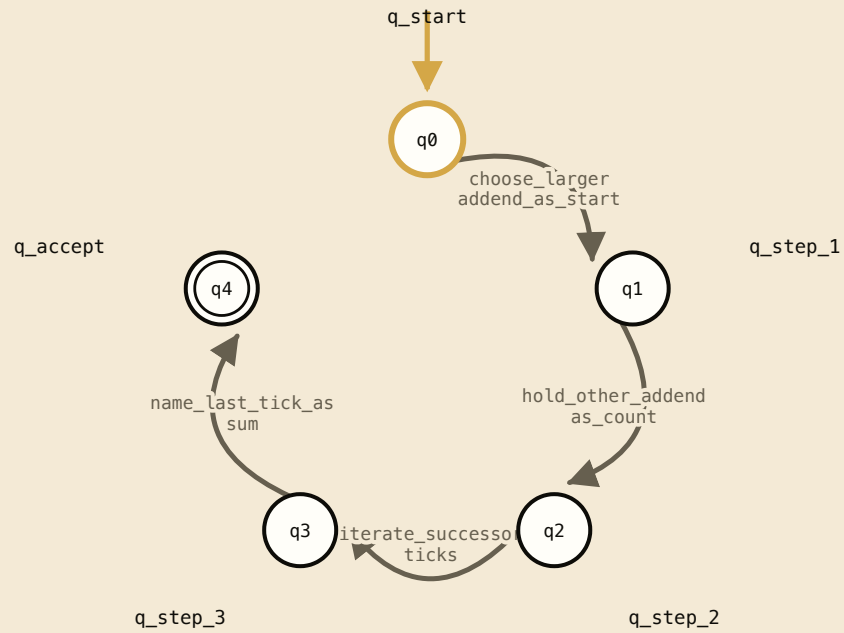
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

6. count_on_from_larger



The run produced 75.

The run produced 75. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{choose_larger_addend_as_start, hold_other_addend_as_count, iterate_successor_ticks, name_last_tick_as_sum (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

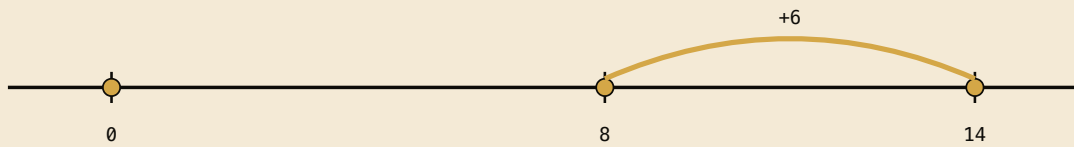
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	choose_larger_addend_as_start	q_step_1	sar_add_action_pairs.pl:65 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	hold_other_addend_as_count	q_step_2	sar_add_action_pairs.pl:65 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	iterate_successor_ticks	q_step_3	sar_add_action_pairs.pl:65 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	name_last_tick_as_sum	q_accept	sar_add_action_pairs.pl:65 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

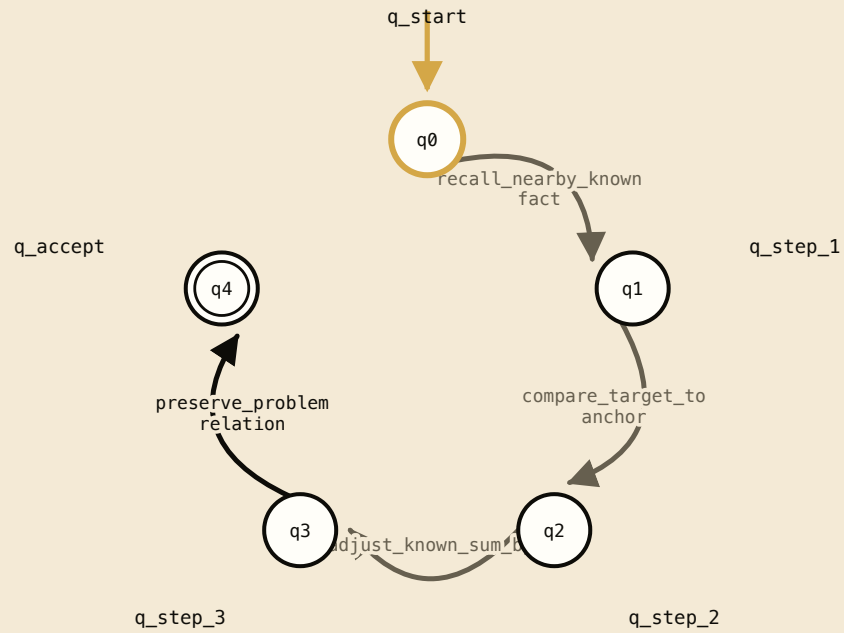
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

7. derived_fact_adjustment



The run produced 14.

The run produced 14. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{adjust_known_sum_by, compare_target_to_anchor, preserve_problem_relation, recall_nearby_known_fact (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 8, "b": 6};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	recall_nearby_known_fact	q_step_1	sar_add_action_pairs.pl:486 observed on {"a": 8, "b": 6} observed on {"a": 9, "b": 6} (derived template)
q_step_1	compare_target_to_anchor	q_step_2	sar_add_action_pairs.pl:486 observed on {"a": 8, "b": 6} observed on {"a": 9, "b": 6} (derived template)
q_step_2	adjust_known_sum_by	q_step_3	sar_add_action_pairs.pl:486 observed on {"a": 8, "b": 6} observed on {"a": 9, "b": 6} (derived template)
q_step_3	preserve_problem_relation	q_accept	sar_add_action_pairs.pl:486 observed on {"a": 8, "b": 6} observed on {"a": 9, "b": 6} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

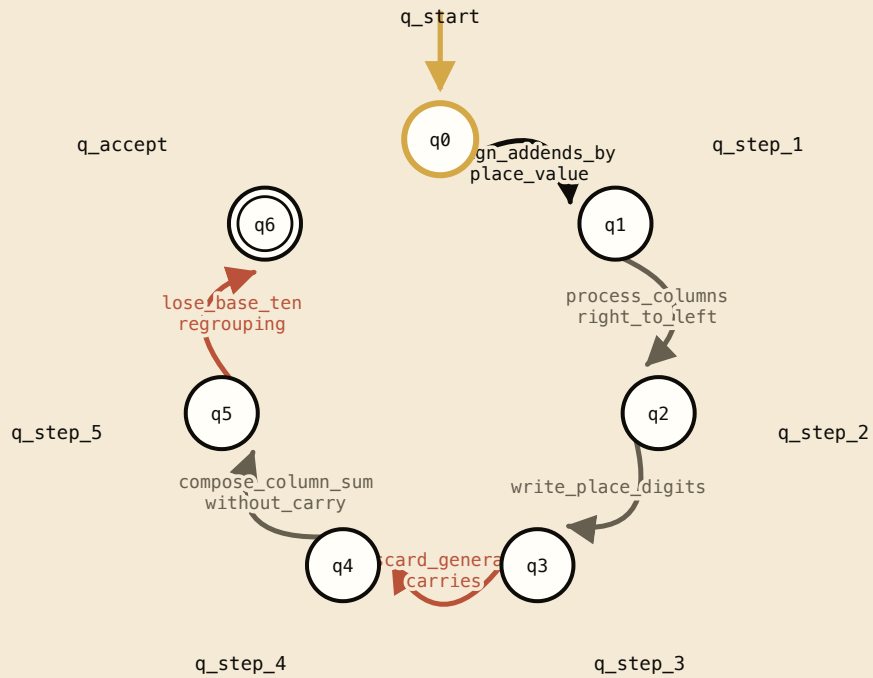
8. drop_carry_to_next_column

drop_carry_to_next_column

	tens	ones
47	4	7
28	2	8
65	6	5

The run produced 65; the recorded expected result is 75.

The run produced 65; the recorded expected result is 75. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ align_addends_by_place_value, compose_column_sum_without_carry,
discard_generated_carries, lose_base_ten_regrouping,
process_columns_right_to_left, write_place_digits (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

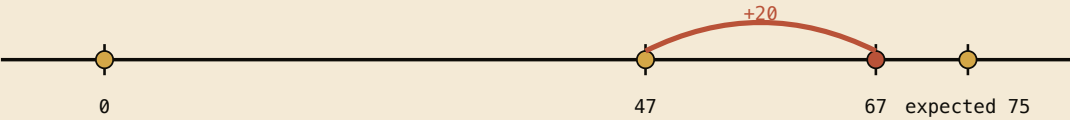
δ : recorded transitions

From	Action	To	Provenance
q_start	align_addends_by_place_value	q_step_1	sar_add_action_pairs.pl:359 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	process_columns_right_to_left	q_step_2	sar_add_action_pairs.pl:359 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	write_place_digits	q_step_3	sar_add_action_pairs.pl:359 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	discard_generated_carries	q_step_4	sar_add_action_pairs.pl:359 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	compose_column_sum_without_carry	q_step_5	sar_add_action_pairs.pl:359 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_5	lose_base_ten_regrouping	q_accept	sar_add_action_pairs.pl:359 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

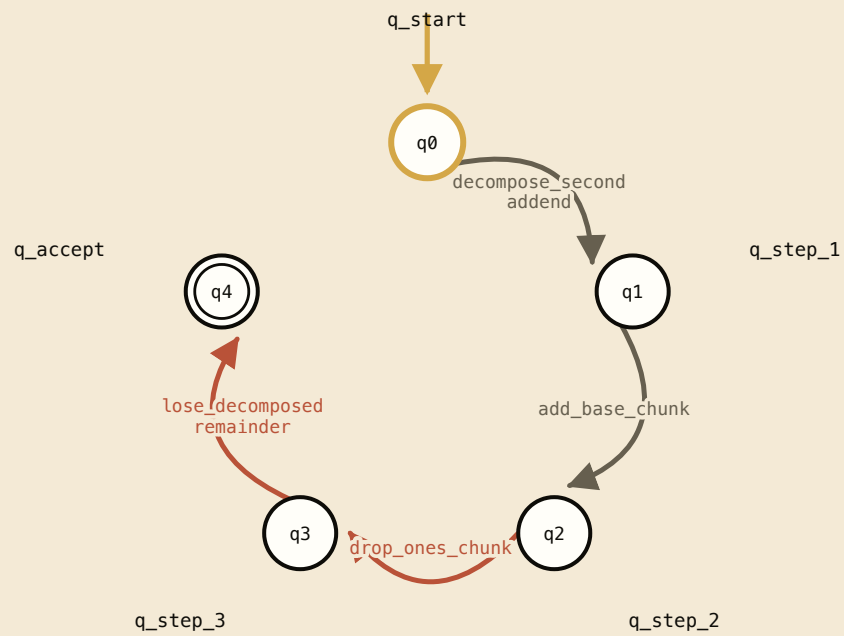
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

9. dropped_ones_chunk



The run produced 67; the recorded expected result is 75.

The run produced 67; the recorded expected result is 75. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{add_base_chunk, decompose_second_addend, drop_ones_chunk, lose_decomposed_remainder} \\ (4 \text{ distinct})$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

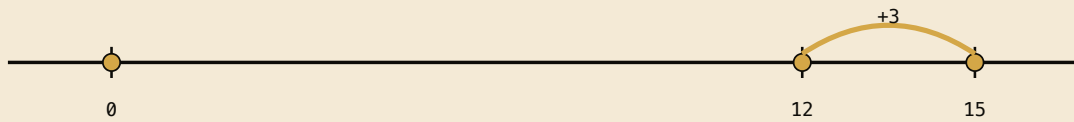
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_second_addend	q_step_1	sar_add_action_pairs.pl:251 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	add_base_chunk	q_step_2	sar_add_action_pairs.pl:251 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	drop_ones_chunk	q_step_3	sar_add_action_pairs.pl:251 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	lose_decomposed_remainder	q_accept	sar_add_action_pairs.pl:251 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

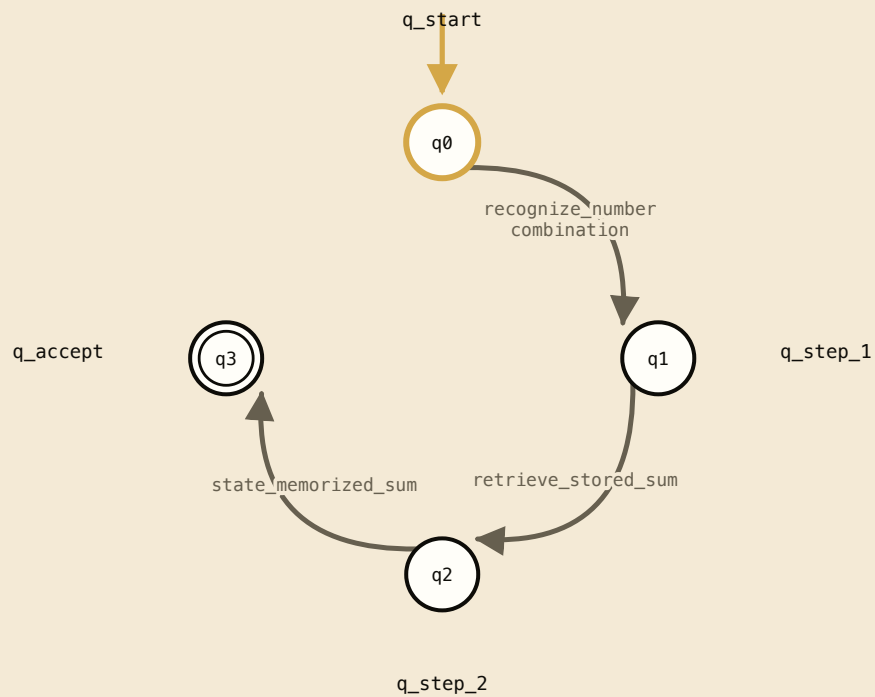
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

10. known_fact_retrieval



The run produced 15.

The run produced 15. The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \text{recognize_number_combination, retrieve_stored_sum, state_memorized_sum (3 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_3\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

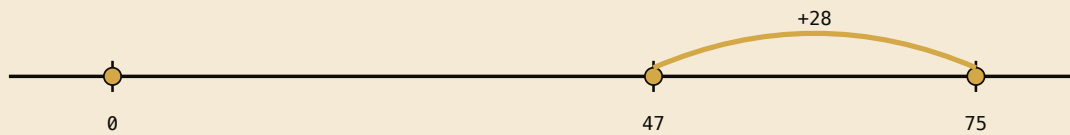
Input contract: {"a": "integer", "b": "integer"}; example {"a": 12, "b": 3};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	recognize_number_combination	q_step_1	sar_add_action_pairs.pl:442 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)
q_step_1	retrieve_stored_sum	q_step_2	sar_add_action_pairs.pl:442 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)
q_step_2	state_memorized_sum	q_accept	sar_add_action_pairs.pl:442 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)

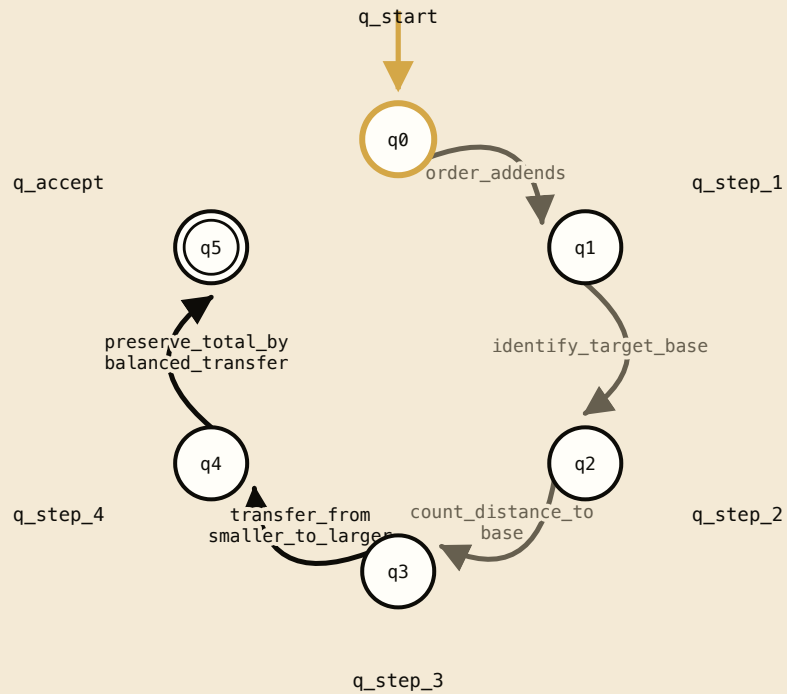
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 6 observed rows.

11. make_base_transfer



The run produced 75.

The run produced 75. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ count_distance_to_base, identify_target_base, order_addends,
preserve_total_by_balanced_transfer, transfer_from_smaller_to_larger (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

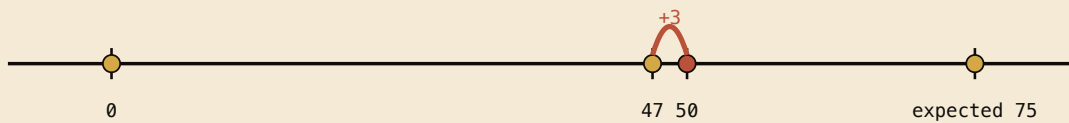
δ : recorded transitions

From	Action	To	Provenance
q_start	order_addends	q_step_1	sar_add_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	identify_target_base	q_step_2	sar_add_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	count_distance_to_base	q_step_3	sar_add_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	transfer_from_smaller_to_large r	q_step_4	sar_add_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	preserve_total_by_balanced_transfer	q_accept	sar_add_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

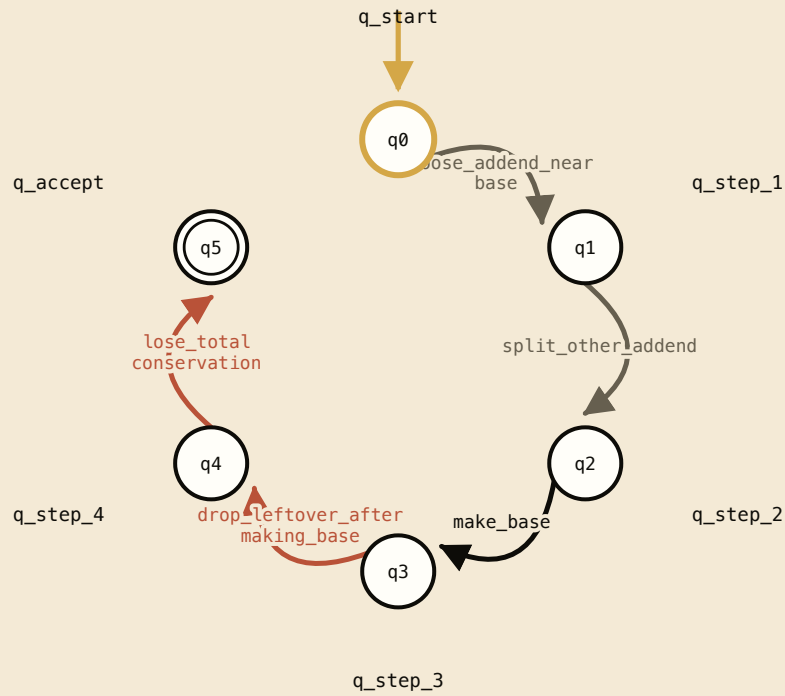
Authored class: finite_state_machine, attested at knowledge/strategies/math/sar_add_rmb.pl:4. This claim comes from module prose, not from the table graph.

12. make_ten_drop_leftover



The run produced 50; the recorded expected result is 75.

The run produced 50; the recorded expected result is 75. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{\text{choose_addend_near_base}, \text{drop_leftover_after_making_base}, \text{lose_total_conservation}, \text{make_base}, \text{split_other_addend}\} \text{ (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

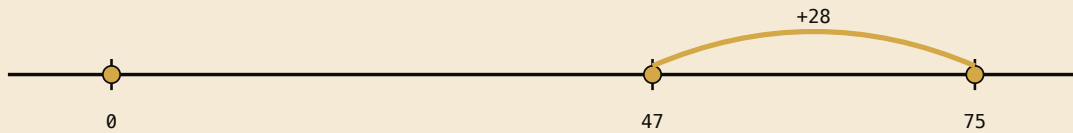
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	choose_addend_near_base	q_step_1	sar_add_action_pairs.pl:140 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	split_other_addend	q_step_2	sar_add_action_pairs.pl:140 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	make_base	q_step_3	sar_add_action_pairs.pl:140 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	drop_leftover_after_making_base	q_step_4	sar_add_action_pairs.pl:140 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	lose_total_conservation	q_accept	sar_add_action_pairs.pl:140 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

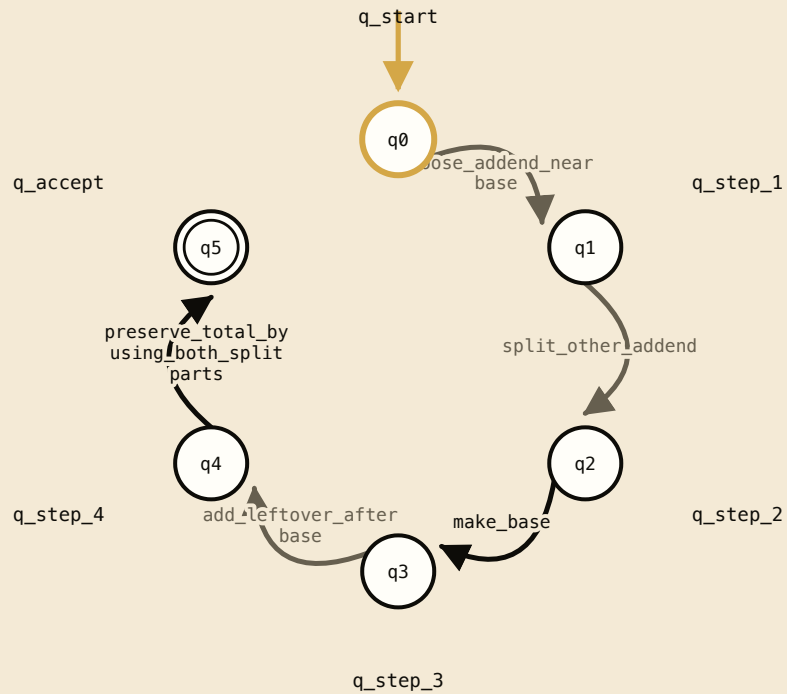
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

13. make_ten_split_leftover



The run produced 75.

The run produced 75. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ add_leftover_after_base, choose_addend_near_base, make_base,
preserve_total_by_using_both_split_parts, split_other_addend (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

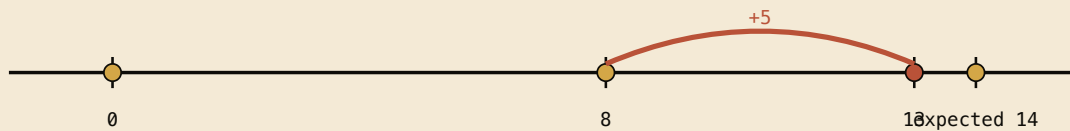
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	choose_addend_near_base	q_step_1	sar_add_action_pairs.pl:116 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	split_other_addend	q_step_2	sar_add_action_pairs.pl:116 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	make_base	q_step_3	sar_add_action_pairs.pl:116 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	add_leftover_after_base	q_step_4	sar_add_action_pairs.pl:116 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	preserve_total_by_using_both_split_parts	q_accept	sar_add_action_pairs.pl:116 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

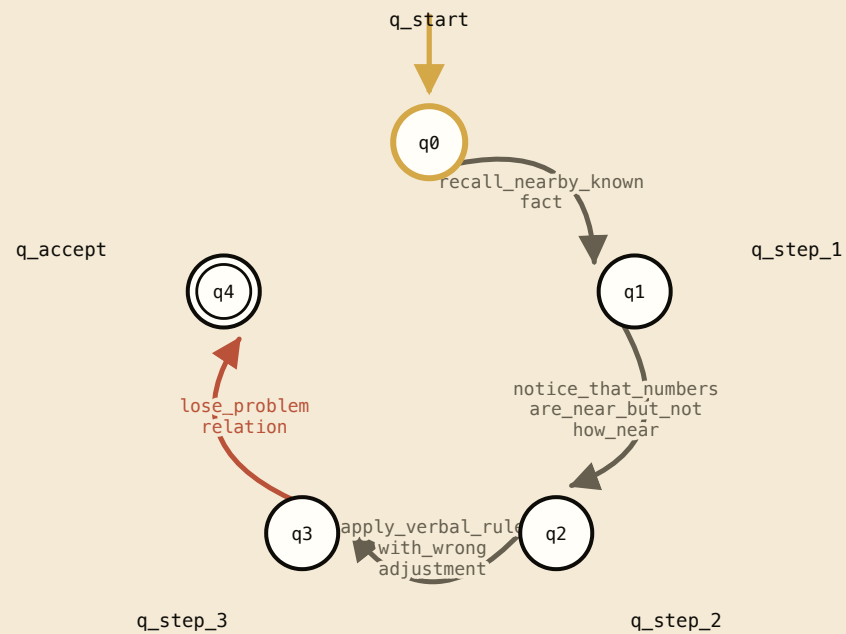
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

14. rote_derived_fact_rule_misfire



The run produced 13; the recorded expected result is 14.

The run produced 13; the recorded expected result is 14. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{apply_verbal_rule_with_wrong_adjustment}, \text{lose_problem_relation}, \text{notice_that_numbers_are_near_but_not_how_near}, \text{recall_nearby_known_fact}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

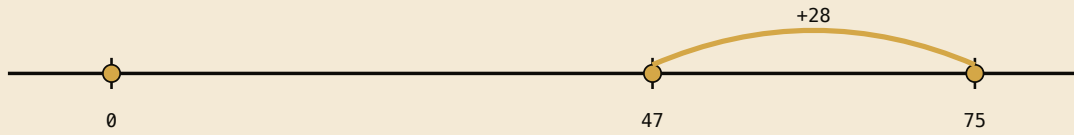
Input contract: {"a": "integer", "b": "integer"}; example {"a": 8, "b": 6};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	recall_nearby_known_fact	q_step_1	sar_add_action_pairs.pl:506 observed on {"a": 8, "b": 6} observed on {"a": 9, "b": 6} (derived template)
q_step_1	notice_that_numbers_are_near_but_not_how_near	q_step_2	sar_add_action_pairs.pl:506 observed on {"a": 8, "b": 6} observed on {"a": 9, "b": 6} (derived template)
q_step_2	apply_verbal_rule_with_wrong_adjustment	q_step_3	sar_add_action_pairs.pl:506 observed on {"a": 8, "b": 6} observed on {"a": 9, "b": 6} (derived template)
q_step_3	lose_problem_relation	q_accept	sar_add_action_pairs.pl:506 observed on {"a": 8, "b": 6} observed on {"a": 9, "b": 6} (derived template)

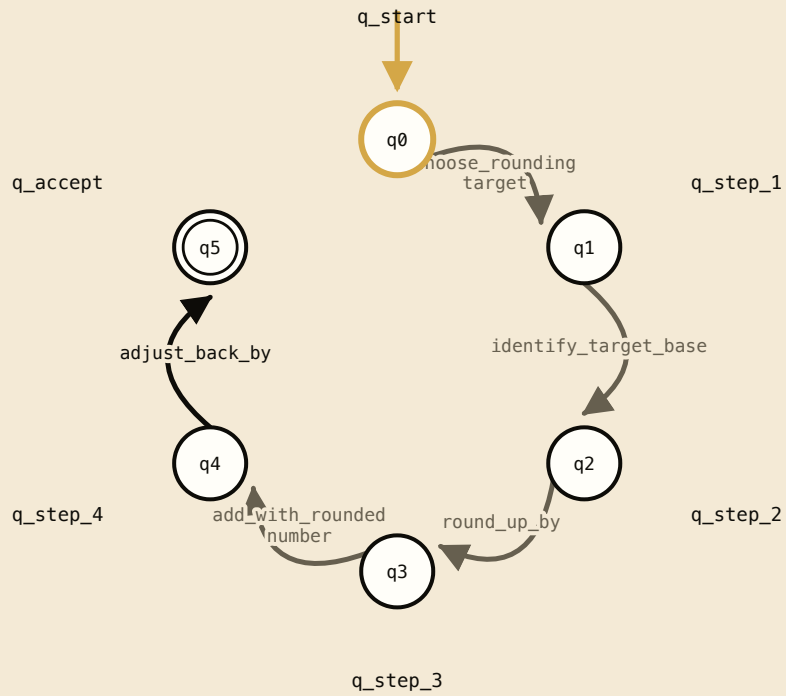
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

15. round_then_adjust



The run produced 75.

The run produced 75. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ add_with_rounded_number, adjust_back_by, choose_rounding_target, identify_target_base, round_up_by (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

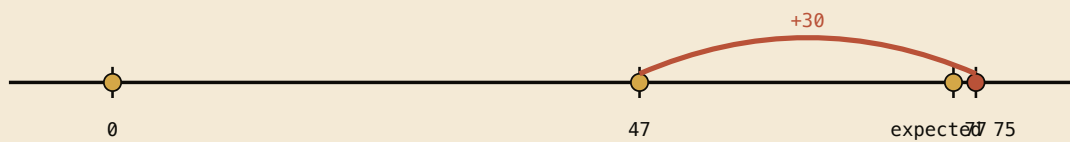
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	choose_rounding_target	q_step_1	sar_add_action_pairs.pl:280 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	identify_target_base	q_step_2	sar_add_action_pairs.pl:280 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	round_up_by	q_step_3	sar_add_action_pairs.pl:280 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	add_with_rounded_number	q_step_4	sar_add_action_pairs.pl:280 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	adjust_back_by	q_accept	sar_add_action_pairs.pl:280 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

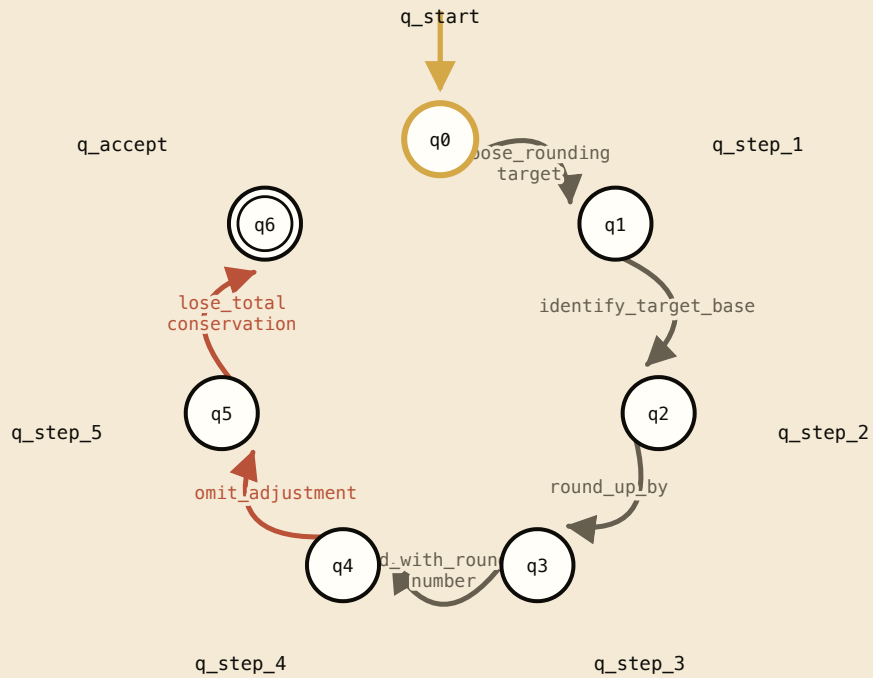
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

16. round_without_adjusting



The run produced 77; the recorded expected result is 75.

The run produced 77; the recorded expected result is 75. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \text{add_with_rounded_number, choose_rounding_target, identify_target_base, lose_total_conservation, omit_adjustment, round_up_by (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

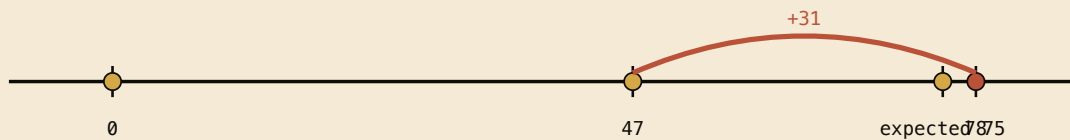
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	choose_rounding_target	q_step_1	sar_add_action_pairs.pl:305 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	identify_target_base	q_step_2	sar_add_action_pairs.pl:305 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	round_up_by	q_step_3	sar_add_action_pairs.pl:305 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	add_with_rounded_number	q_step_4	sar_add_action_pairs.pl:305 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	omit_adjustment	q_step_5	sar_add_action_pairs.pl:305 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_5	lose_total_conservation	q_accept	sar_add_action_pairs.pl:305 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

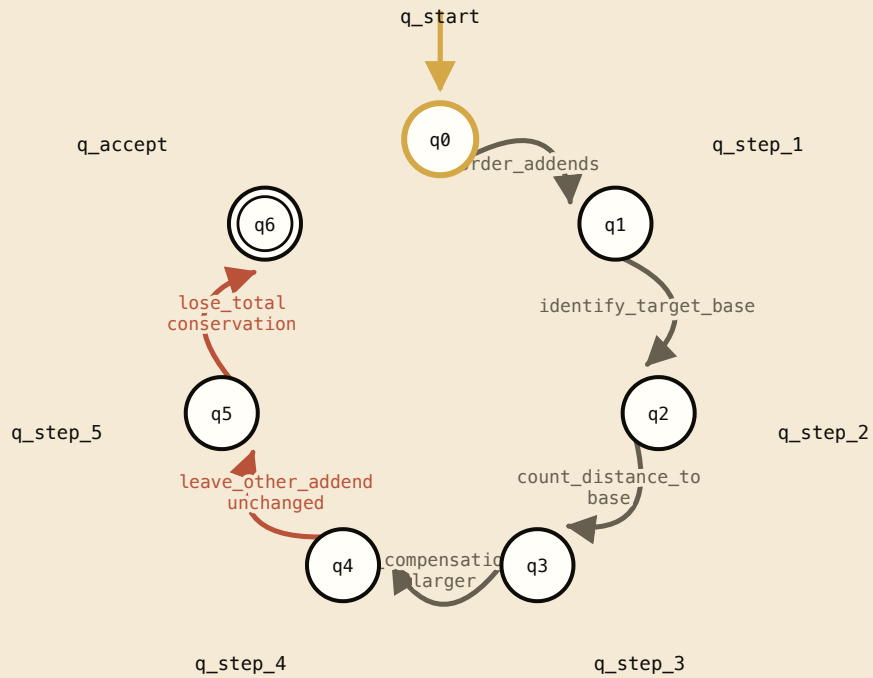
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

17. unbalanced_make_base_compensation



The run produced 78; the recorded expected result is 75.

The run produced 78; the recorded expected result is 75. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ add_compensation_to_larger, count_distance_to_base, identify_target_base, leave_other_addend_unchanged, lose_total_conservation, order_addends (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	order_addends	q_step_1	sar_add_action_pairs.pl:198 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	identify_target_base	q_step_2	sar_add_action_pairs.pl:198 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	count_distance_to_base	q_step_3	sar_add_action_pairs.pl:198 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	add_compensation_to_larger	q_step_4	sar_add_action_pairs.pl:198 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	leave_other_addend_unchanged	q_step_5	sar_add_action_pairs.pl:198 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_5	lose_total_conservation	q_accept	sar_add_action_pairs.pl:198 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

18. wrong_carry_amount_to_next_column

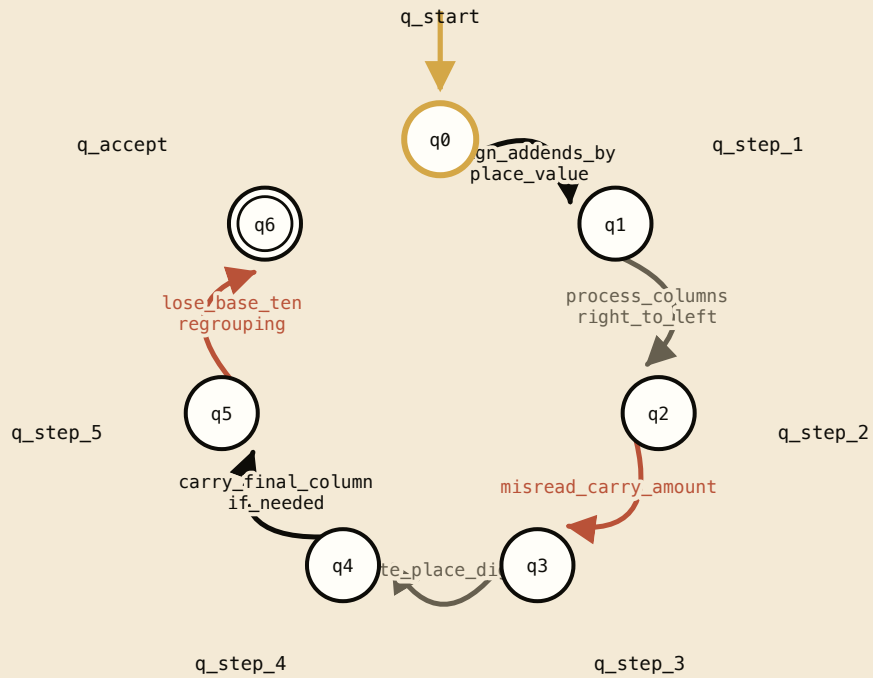
wrong_carry_amount_to_next_column

+1 ten

	tens	ones
47	4	7
28	2	8
85	8	5

The run produced 85; the recorded expected result is 75.

The run produced 85; the recorded expected result is 75. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ align_addends_by_place_value, carry_final_column_if_needed,
lose_base_ten_regrouping, misread_carry_amount, process_columns_right_to_left,
write_place_digits (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

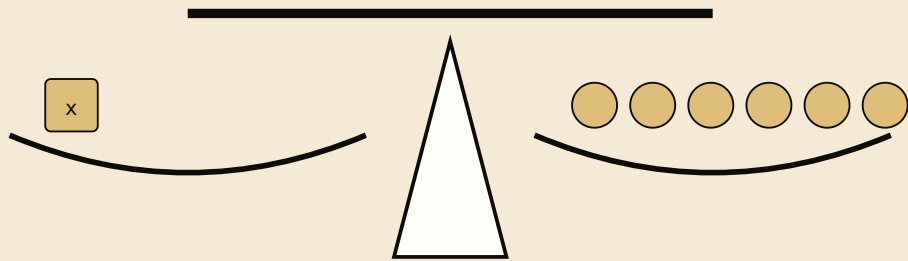
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	align_addends_by_place_value	q_step_1	sar_add_action_pairs.pl:416 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	process_columns_right_to_left	q_step_2	sar_add_action_pairs.pl:416 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	misread_carry_amount	q_step_3	sar_add_action_pairs.pl:416 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	write_place_digits	q_step_4	sar_add_action_pairs.pl:416 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	carry_final_column_if_needed	q_step_5	sar_add_action_pairs.pl:416 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

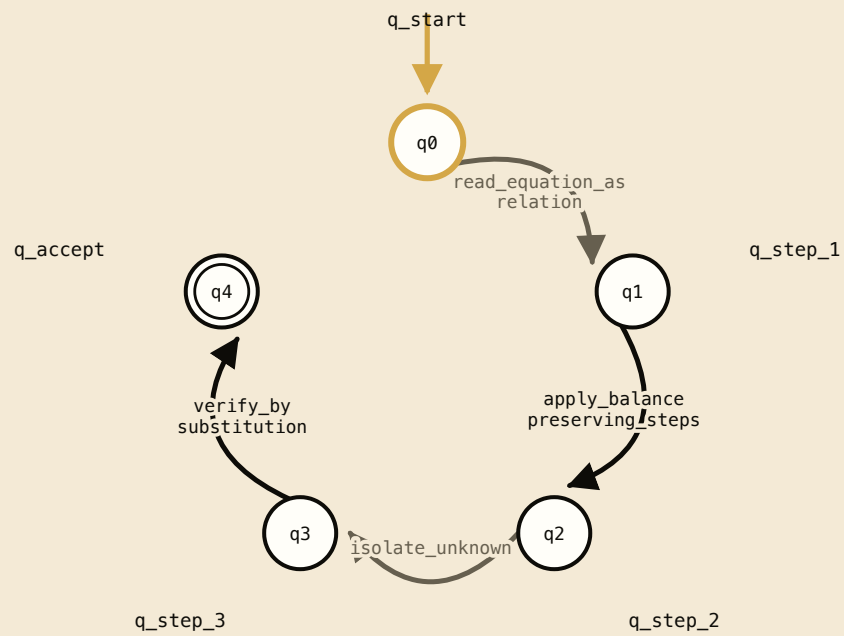
From	Action	To	Provenance
q_step_5	lose_base_ten_regrouping	q_accept	sar_add_action_pairs.pl:416 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.



The run produced value(6).

The run produced value(6). The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{apply_balance_preserving_steps}, \text{isolate_unknown}, \text{read_equation_as_relation}, \text{verify_by_substitution}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

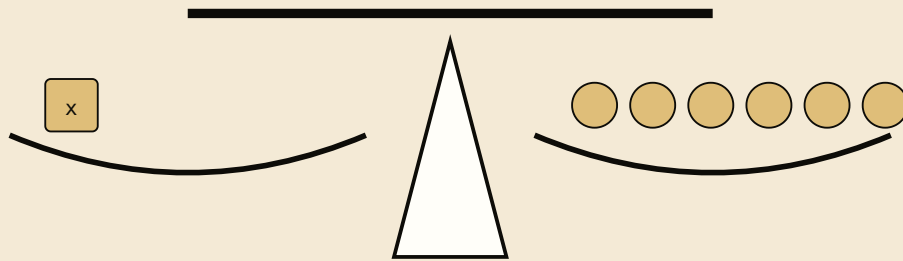
Input contract: {"a": "integer", "b": "integer", "c": "integer", "kind": "linear_equation"}; example {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	read_equation_as_relation	q_step_1	algebraic_action_pairs.pl:241 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}
q_step_1	apply_balance_preserving_steps	q_step_2	algebraic_action_pairs.pl:241 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}
q_step_2	isolate_unknown	q_step_3	algebraic_action_pairs.pl:241 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}
q_step_3	verify_by_substitution	q_accept	algebraic_action_pairs.pl:241 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}

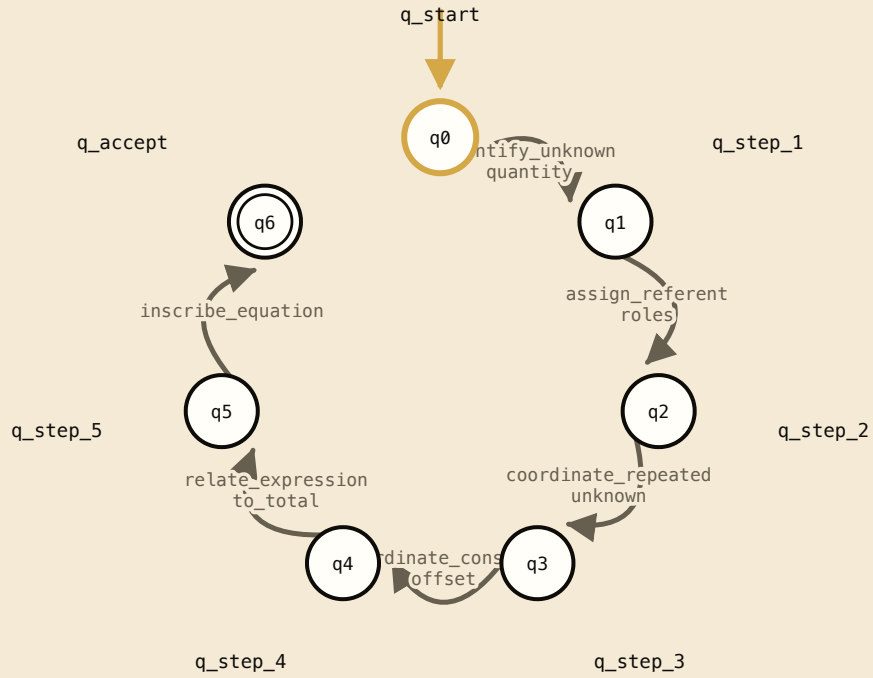
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

2. contextual_linear_equation_construction



The run produced `equation(add(mult(number(3),var(x)),number(2)),equals,number(20))`.

The run produced `equation(add(mult(number(3),var(x)),number(2)),equals,number(20))`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \text{assign_referent_roles, coordinate_constant_offset, coordinate_repeated_unknown, identify_unknown_quantity, inscribe_equation, relate_expression_to_total (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"coefficient": "number", "kind": "linear_context", "offset": "number", "referent_roles": ["atom"], "total": "number", "unknown": "atom"};
example {"coefficient": 3, "kind": "linear_context", "offset": 2, "referent_roles": ["groups", "loose_items", "total"], "total": 20, "unknown": "x"}; verified(strategy_trace_ok).

δ : recorded transitions

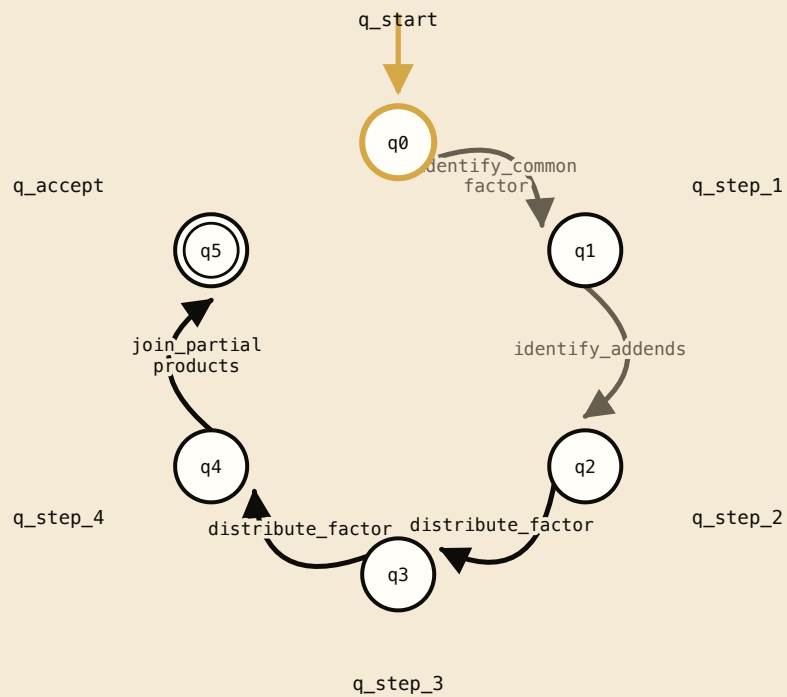
From	Action	To	Provenance
q_start	identify_unknown_quantity	q_step_1	algebraic_action_pairs.pl:140 observed on {"coefficient": 3, "kind": "linear_context", "offset": 2, "referent_roles": ["groups", "loose_items", "total"], "total": 20, "unknown": "x"}
q_step_1	assign_referent_roles	q_step_2	algebraic_action_pairs.pl:140 observed on {"coefficient": 3, "kind": "linear_context", "offset": 2, "referent_roles": ["groups", "loose_items", "total"], "total": 20, "unknown": "x"}
q_step_2	coordinate_repeated_unknown	q_step_3	algebraic_action_pairs.pl:140 observed on {"coefficient": 3, "kind": "linear_context", "offset": 2, "referent_roles": ["groups", "loose_items", "total"], "total": 20, "unknown": "x"}
q_step_3	coordinate_constant_offset	q_step_4	algebraic_action_pairs.pl:140 observed on {"coefficient": 3, "kind": "linear_context", "offset": 2, "referent_roles": ["groups", "loose_items", "total"], "total": 20, "unknown": "x"}

From	Action	To	Provenance
q_step_4	relate_expression_to_total	q_step_5	algebraic_action_pairs.pl:140 observed on {"coefficient": 3, "kind": "linear_context", "offset": 2, "referent_roles": ["groups", "loose_items", "total"], "total": 20, "unknown": "x"}
q_step_5	inscribe_equation	q_accept	algebraic_action_pairs.pl:140 observed on {"coefficient": 3, "kind": "linear_context", "offset": 2, "referent_roles": ["groups", "loose_items", "total"], "total": 20, "unknown": "x"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 6 observed rows.

3. distributive_expression_rewrite

No domain scene: the static scene emitter has no algebraic form for the expression_rewrite contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{\text{distribute_factor}, \text{identify_addends}, \text{identify_common_factor}, \text{join_partial_products}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"direction": "atom", "expression": {"left": {"node": "int", "value": "integer"}, "node": "mult", "right": {"left": {"name": "atom", "node": "var"}, "node": "add", "right": {"node": "int", "value": "integer"}}}, "kind": "expression_rewrite"}; **example** {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}; **verified(strategy_trace_ok).**

δ : recorded transitions

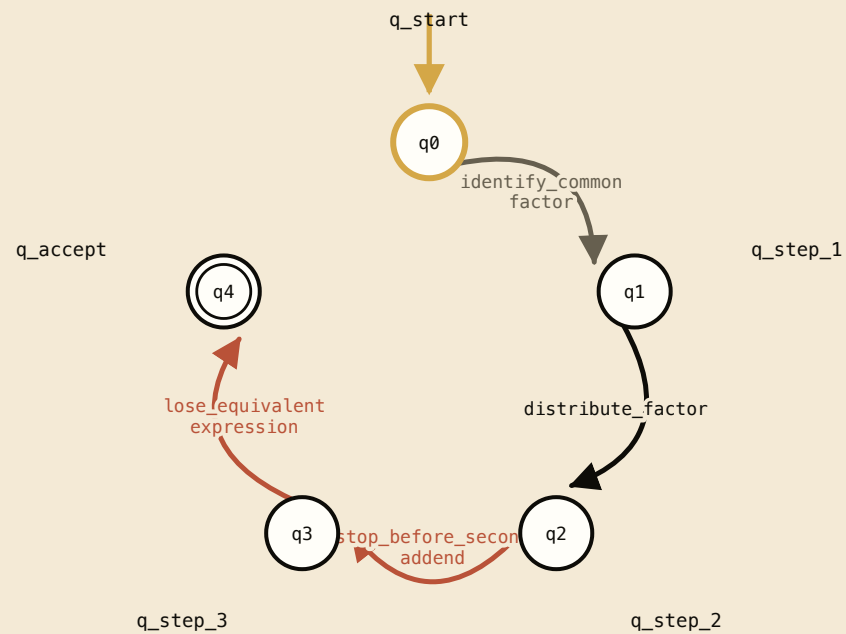
From	Action	To	Provenance
q_start	identify_common_factor	q_step_1	algebraic_action_pairs.pl:339 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}
q_step_1	identify_addends	q_step_2	algebraic_action_pairs.pl:339 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}
q_step_2	distribute_factor	q_step_3	algebraic_action_pairs.pl:339 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}

From	Action	To	Provenance
q_ste p_3	distribut e_factor	q_s tep _4	algebraic_action_pairs.pl:339 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}
q_ste p_4	join_part ial_produ cts	q_a cce pt	algebraic_action_pairs.pl:339 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 4 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

4. drop_distributed_term

No domain scene: the static scene emitter has no algebraic form for the expression_rewrite contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{distribute_factor}, \text{identify_common_factor}, \text{lose_equivalent_expression}, \text{stop_before_second_addend}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"direction": "atom", "expression": {"left": {"node": "int", "value": "integer"}, "node": "mult", "right": {"left": {"name": "atom", "node": "var"}, "node": "add", "right": {"node": "int", "value": "integer"}}}, "kind": "expression_rewrite"}; example {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}; verified(strategy_trace_ok).

δ : recorded transitions

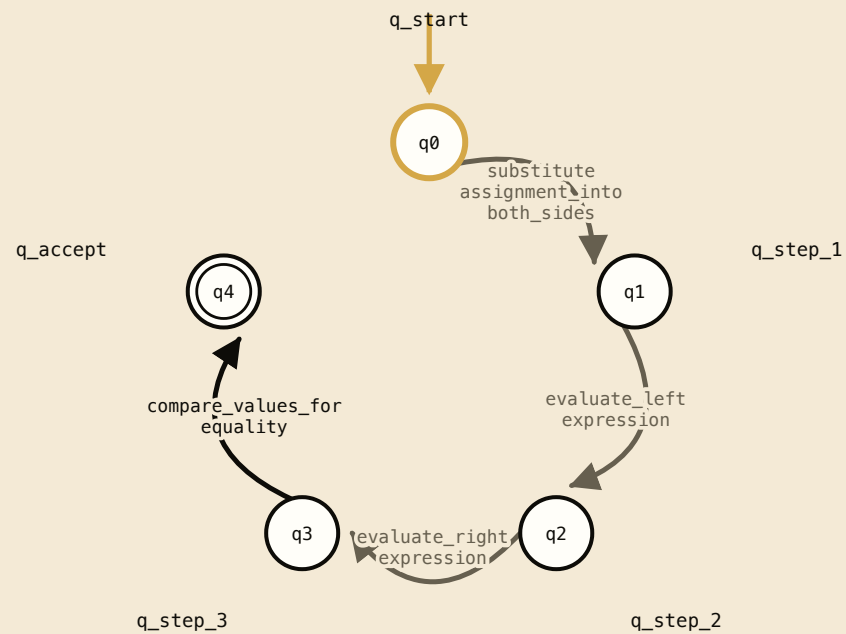
From	Action	To	Provenance
q_start	identify_common_factor	q_step_1	algebraic_action_pairs.pl:368 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}
q_step_1	distribute_factor	q_step_2	algebraic_action_pairs.pl:368 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}
q_step_2	stop_before_second_addend	q_step_3	algebraic_action_pairs.pl:368 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}

From	Action	To	Provenance
q_ste p_3	lose_equiv alent_expr ession	q_a cce pt	algebraic_action_pairs.pl:368 observed on {"direction": "expand", "expression": {"left": {"node": "int", "value": 3}, "node": "mult", "right": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 4}}}, "kind": "expression_rewrite"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

5. equation_truth_by_substitution

No domain scene: the static scene emitter has no algebraic form for the equation_assignment contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ compare_values_for_equality, evaluate_left_expression, evaluate_right_expression, substitute_assignment_into_both_sides (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"assignments": [{"value": "integer", "variable": "atom"}], "kind": "equation_assignment", "left": {"left": {"name": "atom", "node": "var"}, "node": "add", "right": {"node": "int", "value": "integer"}}, "right": {"node": "int", "value": "integer"}}; example {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}; verified(strategy_trace_ok).

δ : recorded transitions

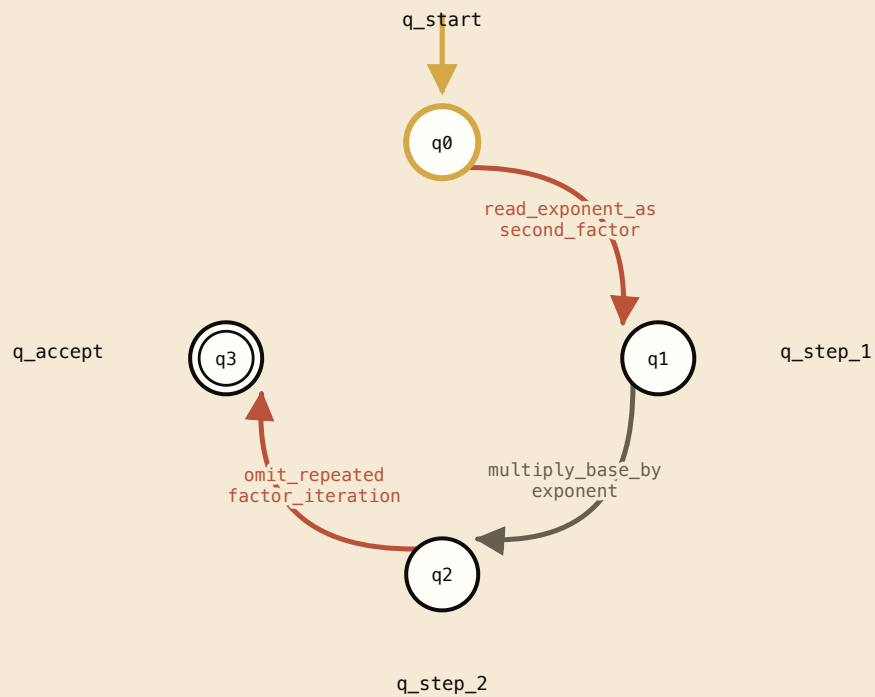
From	Action	To	Provenance
q_start	substitute_assignment_into_both_sides	q_step_1	algebraic_action_pairs.pl:174 observed on {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}
q_step_1	evaluate_left_expression	q_step_2	algebraic_action_pairs.pl:174 observed on {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}
q_step_2	evaluate_right_expression	q_step_3	algebraic_action_pairs.pl:174 observed on {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}

From	Action	To	Provenance
q_ste p_3	compare_value s_for_equalit y	q_a cce pt	algebraic_action_pairs.pl:174 observed on {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

6. exponent_as_multiplier

No domain scene: the static scene emitter has no algebraic form for the power_notation contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad multiply_base_by_exponent, omit_repeated_factor_iteration, read_exponent_as_second_factor \text{ (3 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_3\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

Input contract: {"base": {"node": "int", "value": "integer"}, "exponent": "integer", "kind": "power_notation"}; example {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}; verified(strategy_trace_ok).

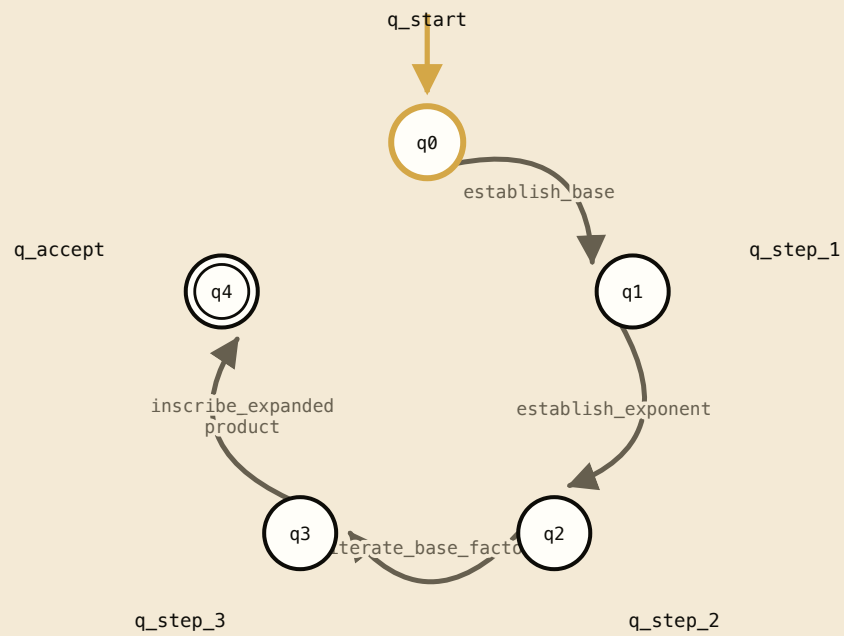
δ : recorded transitions

From	Action	To	Provenance
q_start	read_exponent_as_se cond_factor	q_step_1	algebraic_action_pairs.pl:419 observed on {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}
q_step_1	multiply_base_by_ex ponent	q_step_2	algebraic_action_pairs.pl:419 observed on {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}
q_step_2	omit_repeated_facto r_iteration	q_accept	algebraic_action_pairs.pl:419 observed on {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

7. exponent_as_repeated_factor

No domain scene: the static scene emitter has no algebraic form for the power_notation contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{establish_base, establish_exponent, inscribe_expanded_product, iterate_base_factor}\}$$

(4 distinct)

$$q_0 = q_{\text{start}}$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"base": {"node": "int", "value": "integer"}, "exponent": "integer", "kind": "power_notation"}; example {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}; verified(strategy_trace_ok).

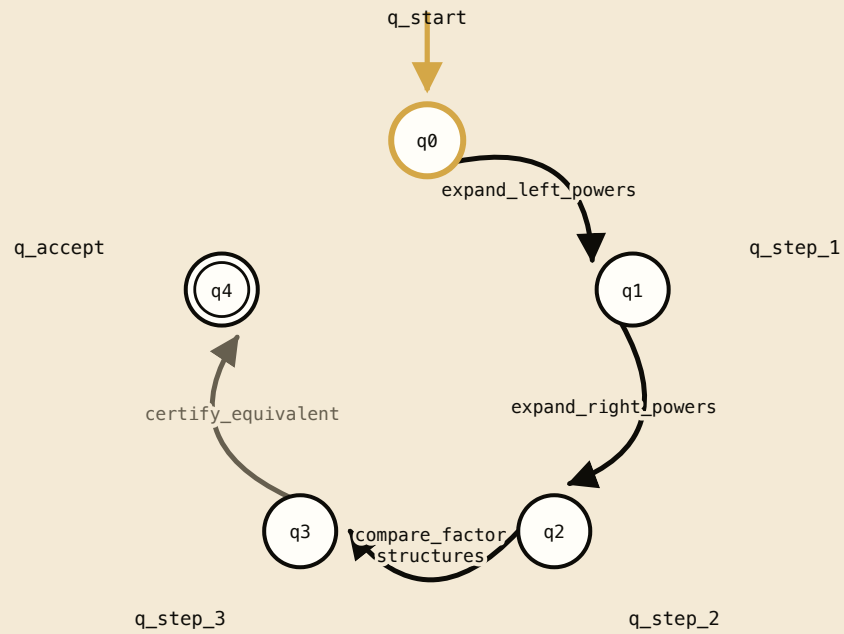
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_base	q_step_1	algebraic_action_pairs.pl:394 observed on {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}
q_step_1	establish_exponent	q_step_2	algebraic_action_pairs.pl:394 observed on {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}
q_step_2	iterate_base_factor	q_step_3	algebraic_action_pairs.pl:394 observed on {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}
q_step_3	inscribe_expanded_product	q_accept	algebraic_action_pairs.pl:394 observed on {"base": {"node": "int", "value": 3}, "exponent": 4, "kind": "power_notation"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

8. exponential_equivalence_by_expansion

No domain scene: the static scene emitter has no algebraic form for the expression_pair contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{certify_equivalent, compare_factor_structures, expand_left_powers, expand_right_powers (4 distinct)}$$

$$q_0 = q_start$$

$F \quad \{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"kind": "expression_pair", "left": {"base": {"node": "int", "value": "integer"}, "exponent": "integer", "node": "power"}, "right": {"left": {"node": "int", "value": "integer"}, "node": "mult", "right": {"left": {"node": "int", "value": "integer"}, "node": "mult", "right": {"node": "int", "value": "integer"}}}}}; example {"kind": "expression_pair", "left": {"base": {"node": "int", "value": 2}, "exponent": 3, "node": "power"}, "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"node": "int", "value": 2}}}}};
verified(strategy_trace_ok).

δ : recorded transitions

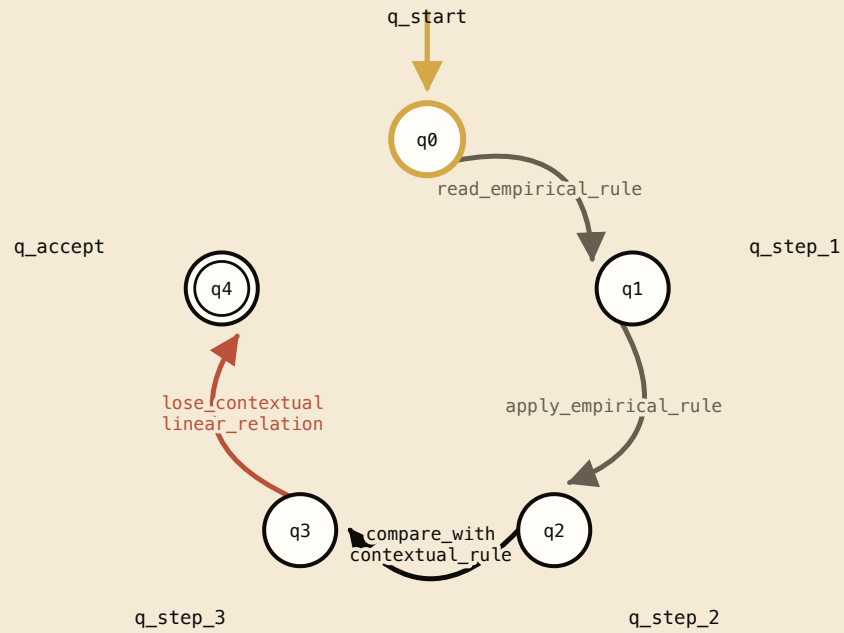
From	Action	To	Provenance
q_start	expand_left_powers	q_step_1	algebraic_action_pairs.pl:445 observed on {"kind": "expression_pair", "left": {"base": {"node": "int", "value": 2}, "exponent": 3, "node": "power"}, "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"node": "int", "value": 2}}}}
q_step_1	expand_right_powers	q_step_2	algebraic_action_pairs.pl:445 observed on {"kind": "expression_pair", "left": {"base": {"node": "int", "value": 2}, "exponent": 3, "node": "power"}, "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"node": "int", "value": 2}}}}

From	Action	To	Provenance
q_ste p_2	compare_factor_structures	q_step _3	algebraic_action_pairs.pl:445 observed on {"kind": "expression_pair", "left": {"base": {"node": "int", "value": 2}, "exponent": 3, "node": "power"}, "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"node": "int", "value": 2}}}}
q_ste p_3	certify_equivalent	q_accept pt	algebraic_action_pairs.pl:445 observed on {"kind": "expression_pair", "left": {"base": {"node": "int", "value": 2}, "exponent": 3, "node": "power"}, "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"left": {"node": "int", "value": 2}, "node": "mult", "right": {"node": "int", "value": 2}}}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

9. guess_and_check_rule

No domain scene: the static scene emitter has no algebraic form for the linear_pattern_empirical_rule contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ apply_empirical_rule, compare_with_contextual_rule,
lose_contextual_linear_relation, read_empirical_rule (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"change": "integer", "checked_rows": ["integer"], "constant": "integer", "first": "integer", "kind": "linear_pattern_empirical_rule", "multiplier": "integer", "row": "positive_integer"}; example {"change": 3, "checked_rows": [1, 2, 3], "constant": 1, "first": 4, "kind": "linear_pattern_empirical_rule", "multiplier": 2, "row": 6}; verified(strategy_trace_ok).

δ : recorded transitions

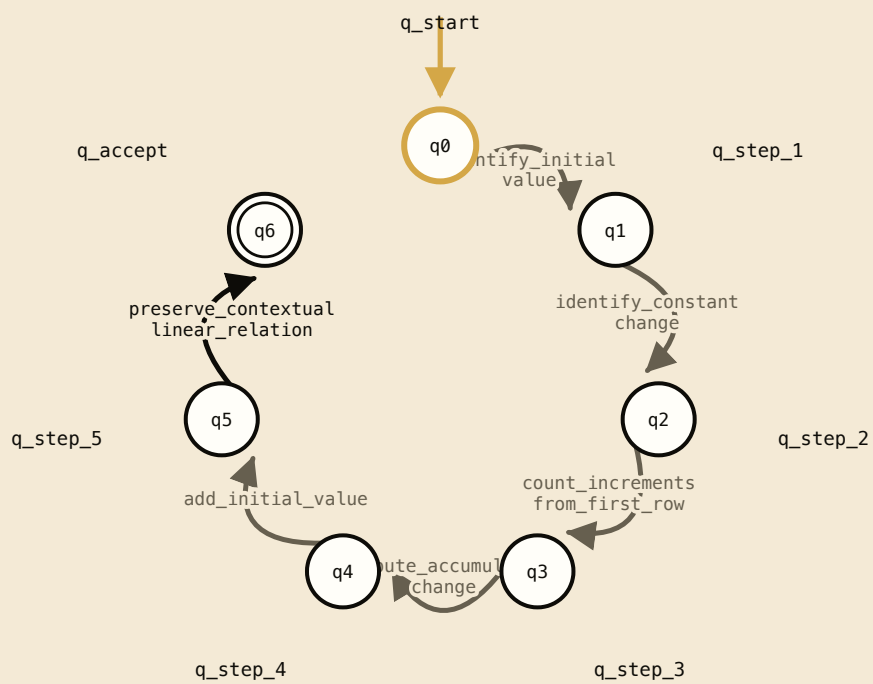
From	Action	To	Provenance
q_start	read_empirical_rule	q_step_1	algebraic_action_pairs.pl:503 observed on {"change": 3, "checked_rows": [1, 2, 3], "constant": 1, "first": 4, "kind": "linear_pattern_empirical_rule", "multiplier": 2, "row": 6}
q_step_1	apply_empirical_rule	q_step_2	algebraic_action_pairs.pl:503 observed on {"change": 3, "checked_rows": [1, 2, 3], "constant": 1, "first": 4, "kind": "linear_pattern_empirical_rule", "multiplier": 2, "row": 6}
q_step_2	compare_with_contextual_rule	q_step_3	algebraic_action_pairs.pl:503 observed on {"change": 3, "checked_rows": [1, 2, 3], "constant": 1, "first": 4, "kind": "linear_pattern_empirical_rule", "multiplier": 2, "row": 6}
q_step_3	lose_contextual_linear_relation	q_accept	algebraic_action_pairs.pl:503 observed on {"change": 3, "checked_rows": [1, 2, 3], "constant": 1, "first": 4, "kind": "linear_pattern_empirical_rule", "multiplier": 2, "row": 6}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and

4 observed rows.

10. linear_pattern_contextual_rule

No domain scene: the static scene emitter has no algebraic form for the linear_pattern_context contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$M = (Q, \Sigma, \delta, q_0, F)$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ add_initial_value, compute_accumulated_change, count_increments_from_first_row, identify_constant_change, identify_initial_value, preserve_contextual_linear_relation (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"change": "integer", "context": "atom", "first": "integer", "kind": "linear_pattern_context", "row": "positive_integer"}; example {"change": 3, "context": "tile_pattern", "first": 4, "kind": "linear_pattern_context", "row": 6}; verified(strategy_trace_ok).

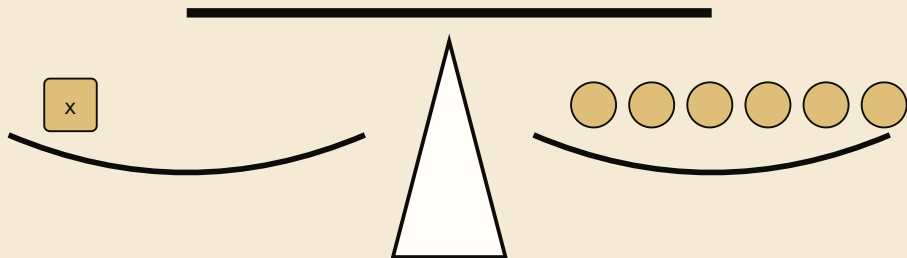
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_initial_value	q_step_1	algebraic_action_pairs.pl:473 observed on {"change": 3, "context": "tile_pattern", "first": 4, "kind": "linear_pattern_context", "row": 6}
q_step_1	identify_constant_change	q_step_2	algebraic_action_pairs.pl:473 observed on {"change": 3, "context": "tile_pattern", "first": 4, "kind": "linear_pattern_context", "row": 6}

From	Action	To	Provenance
q_step_2	count_increments_from_first_row	q_step_3	algebraic_action_pairs.pl:473 observed on {"change": 3, "context": "tile_pattern", "first": 4, "kind": "linear_pattern_context", "row": 6}
q_step_3	compute_accumulated_change	q_step_4	algebraic_action_pairs.pl:473 observed on {"change": 3, "context": "tile_pattern", "first": 4, "kind": "linear_pattern_context", "row": 6}
q_step_4	add_initial_value	q_step_5	algebraic_action_pairs.pl:473 observed on {"change": 3, "context": "tile_pattern", "first": 4, "kind": "linear_pattern_context", "row": 6}
q_step_5	preserve_contextual_linear_relation	q_accept	algebraic_action_pairs.pl:473 observed on {"change": 3, "context": "tile_pattern", "first": 4, "kind": "linear_pattern_context", "row": 6}

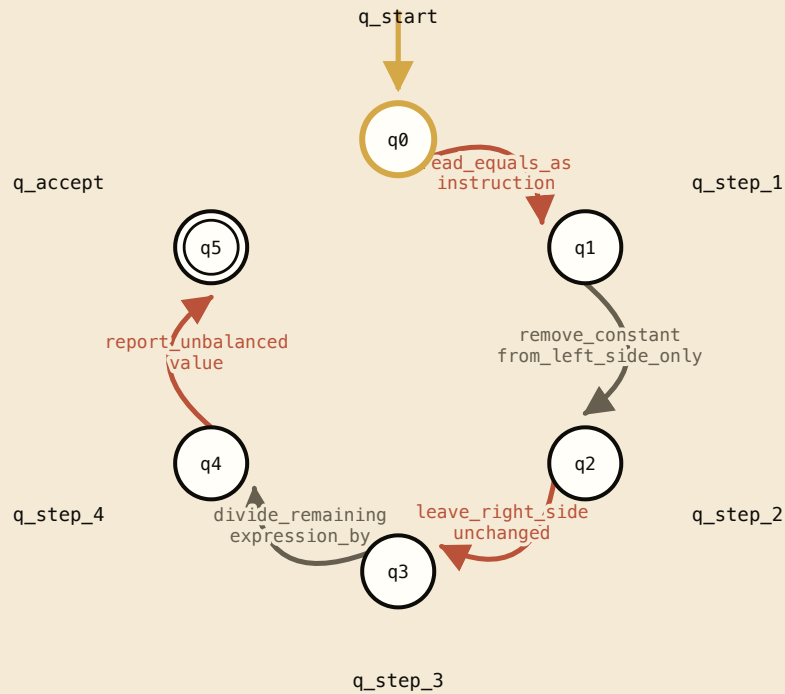
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 6 observed rows.

11. one_sided_equation_operation



The run produced value(20r3); the recorded expected result is value(6).

The run produced value(20r3); the recorded expected result is value(6). The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ divide_remaining_expression_by, leave_right_side_unchanged,
read_equals_as_instruction, remove_constant_from_left_side_only,
report_unbalanced_value (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer", "c": "integer", "kind": "linear_equation"}; example {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}; verified(strategy_trace_ok).

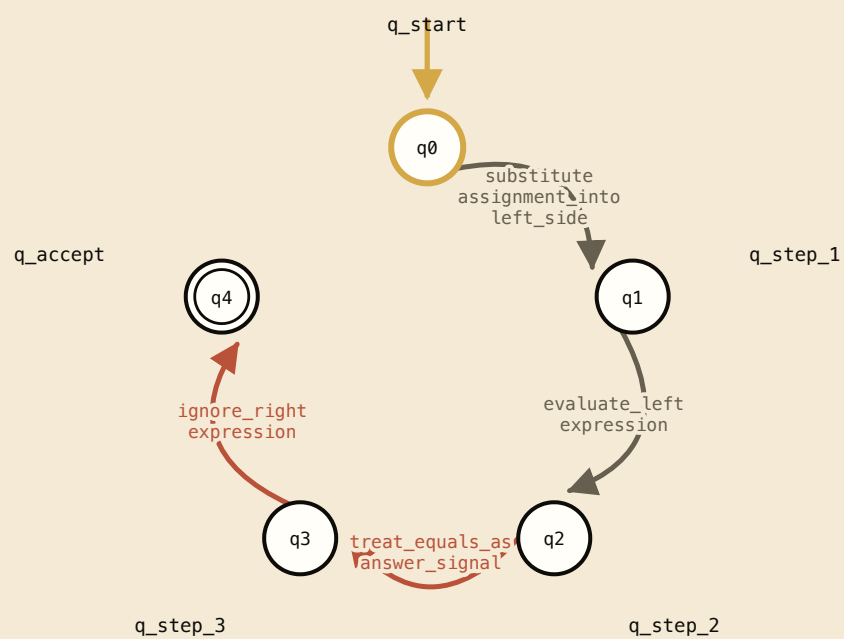
δ : recorded transitions

From	Action	To	Provenance
q_start	read_equals_as_instruction	q_step_1	algebraic_action_pairs.pl:271 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}
q_step_1	remove_constant_from_left_side_only	q_step_2	algebraic_action_pairs.pl:271 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}
q_step_2	leave_right_side_unchanged	q_step_3	algebraic_action_pairs.pl:271 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}
q_step_3	divide_remaining_expression_by	q_step_4	algebraic_action_pairs.pl:271 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}
q_step_4	report_unbalanced_value	q_accept	algebraic_action_pairs.pl:271 observed on {"a": 3, "b": 2, "c": 20, "kind": "linear_equation"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

12. operational_equals_left_value

No domain scene: the static scene emitter has no algebraic form for the equation_assignment contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{evaluate_left_expression, ignore_right_expression, substitute_assignment_into_left_side, treat_equals_as_answer_signal (4 distinct)}$$

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"assignments": [{"value": "integer", "variable": "atom"}], "kind": "equation_assignment", "left": {"left": {"name": "atom", "node": "var"}, "node": "add", "right": {"node": "int", "value": "integer"}}}, "right": {"node": "int", "value": "integer"}}}; example {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}; verified(strategy_trace_ok).

δ : recorded transitions

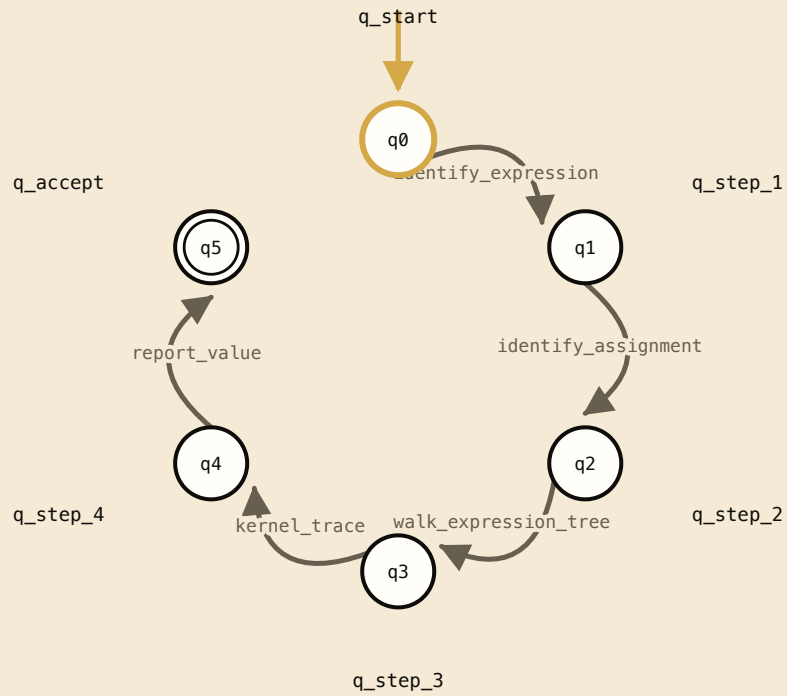
From	Action	To	Provenance
q_start	substitute_assignment_into_left_side	q_step_1	algebraic_action_pairs.pl:208 observed on {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}
q_step_1	evaluate_left_expression	q_step_2	algebraic_action_pairs.pl:208 observed on {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}
q_step_2	treat_equals_as_answer_signal	q_step_3	algebraic_action_pairs.pl:208 observed on {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}

From	Action	To	Provenance
q_ste p_3	ignore_right_ expression	q_a cce pt	algebraic_action_pairs.pl:208 observed on {"assignments": [{"value": 5, "variable": "x"}], "kind": "equation_assignment", "left": {"left": {"name": "x", "node": "var"}, "node": "add", "right": {"node": "int", "value": 2}}, "right": {"node": "int", "value": 7}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

13. programming_expression_evaluation

No domain scene: the static scene emitter has no algebraic form for the expression_assignment contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ identify_assignment, identify_expression, kernel_trace, report_value, walk_expression_tree (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"assignments": [{"value": "integer", "variable": "atom"}],
"expression": {"left": {"left": {"name": "atom", "node": "var"}, "node": "mult",
"right": {"node": "int", "value": "integer"}}, "node": "add", "right": {"node":
"int", "value": "integer"}}, "kind": "expression_assignment"}; example
{"assignments": [{"value": 4, "variable": "x"}], "expression": {"left": {"left":
{"name": "x", "node": "var"}, "node": "mult", "right": {"node": "int", "value":
3}}, "node": "add", "right": {"node": "int", "value": 2}}, "kind":
"expression_assignment"}; verified(strategy_trace_ok).

δ : recorded transitions

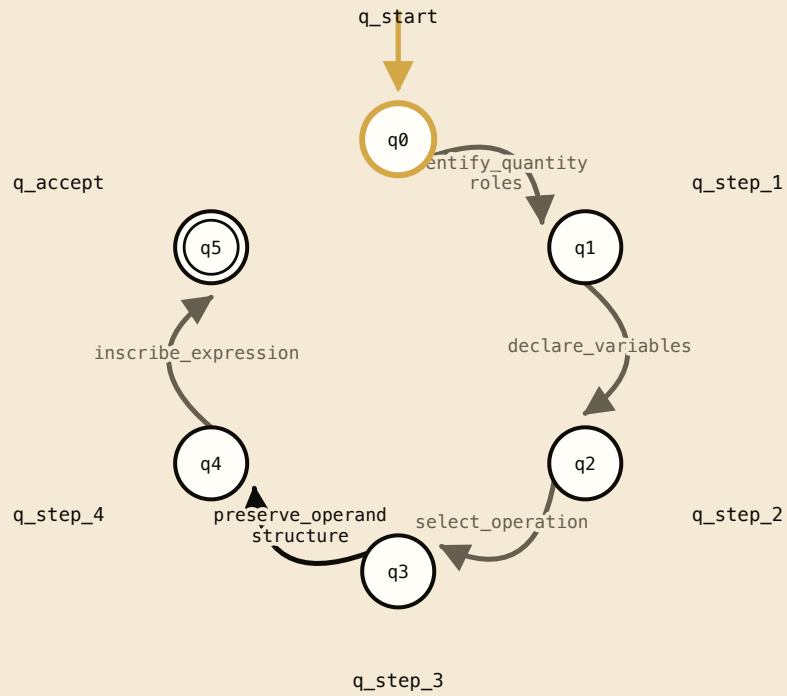
From	Action	To	Provenance
q_start	identify_expression	q_step_1	algebraic_action_pairs.pl:108 observed on {"assignments": [{"value": 4, "variable": "x"}], "expression": {"left": {"left": {"name": "x", "node": "var"}, "node": "mult", "right": {"node": "int", "value": 3}}, "node": "add", "right": {"node": "int", "value": 2}}, "kind": "expression_assignment"}
q_step_1	identify_assignment	q_step_2	algebraic_action_pairs.pl:108 observed on {"assignments": [{"value": 4, "variable": "x"}], "expression": {"left": {"left": {"name": "x", "node": "var"}, "node": "mult", "right": {"node": "int", "value": 3}}, "node": "add", "right": {"node": "int", "value": 2}}, "kind": "expression_assignment"}

From	Action	To	Provenance
q_step_2	walk_expression_tree	q_step_3	algebraic_action_pairs.pl:108 observed on {"assignments": [{"value": 4, "variable": "x"}], "expression": {"left": {"left": {"name": "x", "node": "var"}, "node": "mult", "right": {"node": "int", "value": 3}}, "node": "add", "right": {"node": "int", "value": 2}}, "kind": "expression_assignment"}
q_step_3	kernel_trace	q_step_4	algebraic_action_pairs.pl:108 observed on {"assignments": [{"value": 4, "variable": "x"}], "expression": {"left": {"left": {"name": "x", "node": "var"}, "node": "mult", "right": {"node": "int", "value": 3}}, "node": "add", "right": {"node": "int", "value": 2}}, "kind": "expression_assignment"}
q_step_4	report_value	q_accept	algebraic_action_pairs.pl:108 observed on {"assignments": [{"value": 4, "variable": "x"}], "expression": {"left": {"left": {"name": "x", "node": "var"}, "node": "mult", "right": {"node": "int", "value": 3}}, "node": "add", "right": {"node": "int", "value": 2}}, "kind": "expression_assignment"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

14. symbolic_expression_construction

No domain scene: the static scene emitter has no algebraic form for the quantity_relation contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ declare_variables, identify_quantity_roles, inscribe_expression, preserve_operand_structure, select_operation (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"declared_variables": ["atom"], "kind": "quantity_relation", "left": {"name": "atom", "node": "var"}, "operator": "atom", "referent_roles": ["atom"], "right": {"node": "int", "value": "integer"}}; example {"declared_variables": ["x"], "kind": "quantity_relation", "left": {"name": "x", "node": "var"}, "operator": "add", "referent_roles": ["unknown_quantity", "constant_offset"], "right": {"node": "int", "value": 4}}; verified(strategy_trace_ok).

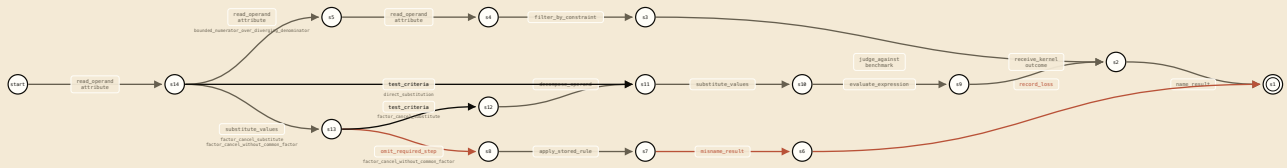
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_quantity_roles	q_step_1	algebraic_action_pairs.pl:303 observed on {"declared_variables": ["x"], "kind": "quantity_relation", "left": {"name": "x", "node": "var"}, "operator": "add", "referent_roles": ["unknown_quantity", "constant_offset"], "right": {"node": "int", "value": 4}}
q_step_1	declare_variables	q_step_2	algebraic_action_pairs.pl:303 observed on {"declared_variables": ["x"], "kind": "quantity_relation", "left": {"name": "x", "node": "var"}, "operator": "add", "referent_roles": ["unknown_quantity", "constant_offset"], "right": {"node": "int", "value": 4}}
q_step_2	select_operation	q_step_3	algebraic_action_pairs.pl:303 observed on {"declared_variables": ["x"], "kind": "quantity_relation", "left": {"name": "x", "node": "var"}, "operator": "add", "referent_roles": ["unknown_quantity", "constant_offset"], "right": {"node": "int", "value": 4}}

From	Action	To	Provenance
q_ste p_3	preserve_op erand_struc ture	q_s tep _4	algebraic_action_pairs.pl:303 observed on {"declared_variables": ["x"], "kind": "quantity_relation", "left": {"name": "x", "node": "var"}, "operator": "add", "referent_roles": ["unknown_quantity", "constant_offset"], "right": {"node": "int", "value": 4}}
q_ste p_4	inscribe_ex pression	q_a cce pt	algebraic_action_pairs.pl:303 observed on {"declared_variables": ["x"], "kind": "quantity_relation", "left": {"name": "x", "node": "var"}, "operator": "add", "referent_roles": ["unknown_quantity", "constant_offset"], "right": {"node": "int", "value": 4}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

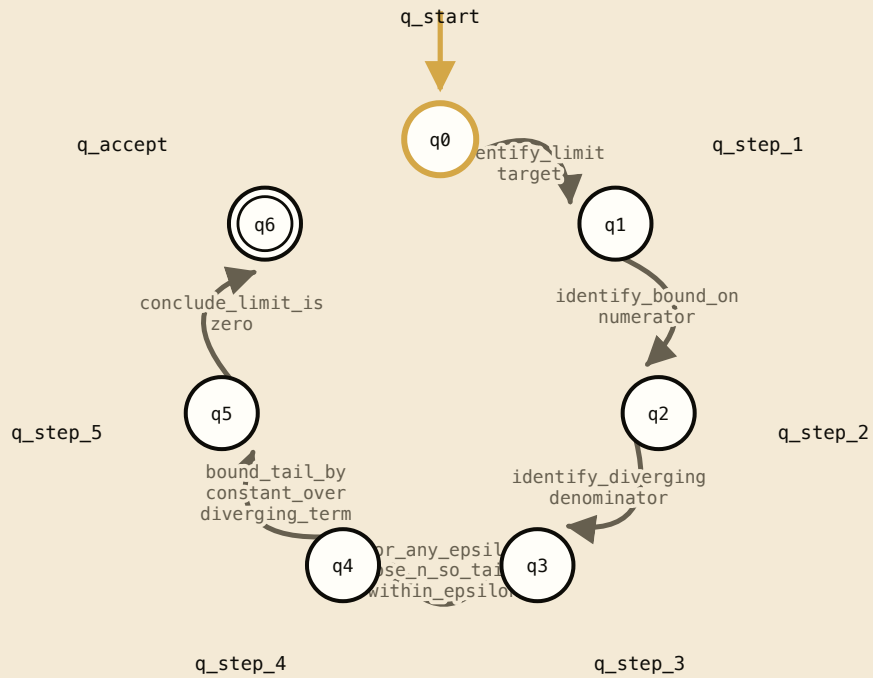
calculus



Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. bounded_numerator_over_diverging_denominator

No domain scene: the static scene emitter has no calculus form for the bounded_sequence_limit contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \{bound_tail_by_constant_over_diverging_term, conclude_limit_is_zero, for_any_epsilon_choose_n_so_tail_is_within_epsilon, identify_bound_on_numerator, identify_diverging_denominator, identify_limit_target\} \text{ (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"bound": "positive_integer", "denominator": "atom", "kind": "bounded_sequence_limit", "numerator": "atom"}; example {"bound": 1, "denominator": "linear_in_n", "kind": "bounded_sequence_limit", "numerator": "cosine_of_n_x"}; verified(strategy_trace_ok).

δ : recorded transitions

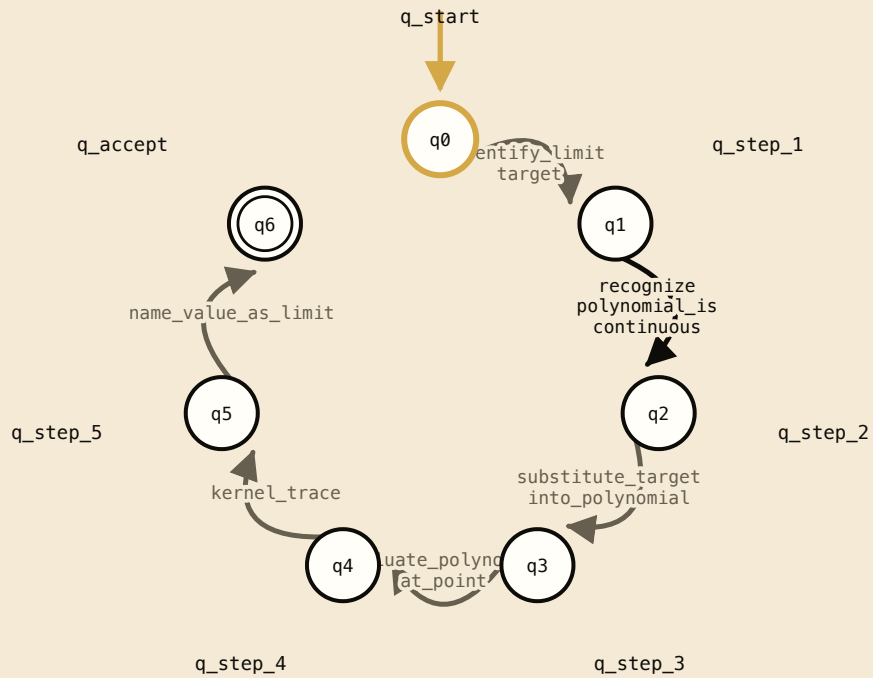
From	Action	To	Provenance
q_start	identify_limit_target	q_step_1	calculus_limits_action_pairs.pl:220 observed on {"bound": 1, "denominator": "linear_in_n", "kind": "bounded_sequence_limit", "numerator": "cosine_of_n_x"}
q_step_1	identify_bound_on_numerator	q_step_2	calculus_limits_action_pairs.pl:220 observed on {"bound": 1, "denominator": "linear_in_n", "kind": "bounded_sequence_limit", "numerator": "cosine_of_n_x"}
q_step_2	identify_diverging_denominator	q_step_3	calculus_limits_action_pairs.pl:220 observed on {"bound": 1, "denominator": "linear_in_n", "kind": "bounded_sequence_limit", "numerator": "cosine_of_n_x"}
q_step_3	for_any_epsilon_choose_n_so_tail_is_within_epsilon	q_step_4	calculus_limits_action_pairs.pl:220 observed on {"bound": 1, "denominator": "linear_in_n", "kind": "bounded_sequence_limit", "numerator": "cosine_of_n_x"}

From	Action	To	Provenance
q_step_4	bound_tail_by_constant_over_diverging_term	q_step_5	calculus_limits_action_pairs.pl:220 observed on {"bound": 1, "denominator": "linear_in_n", "kind": "bounded_sequence_limit", "numerator": "cosine_of_n_x"}
q_step_5	conclude_limit_is_zero	q_accepted	calculus_limits_action_pairs.pl:220 observed on {"bound": 1, "denominator": "linear_in_n", "kind": "bounded_sequence_limit", "numerator": "cosine_of_n_x"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 6 observed rows.

2. direct_substitution

No domain scene: the static scene emitter has no calculus form for the polynomial_limit contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ evaluate_polynomial_at_point, identify_limit_target, kernel_trace, name_value_as_limit, recognize_polynomial_is_continuous, substitute_target_into_polynomial (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"at": "integer", "coefficients": ["integer"], "kind": "polynomial_limit"}; example {"at": 2, "coefficients": [1, 2, 3], "kind": "polynomial_limit"}; verified(strategy_trace_ok).

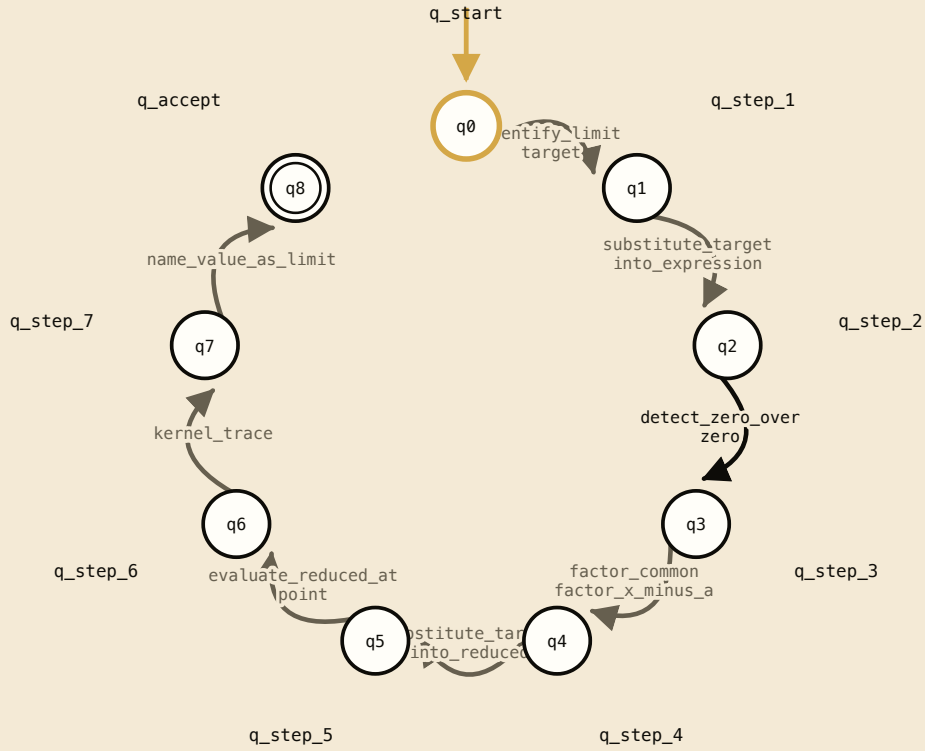
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_limit_target	q_step_1	calculus_limits_action_pairs.pl:96 observed on {"at": 2, "coefficients": [1, 2, 3], "kind": "polynomial_limit"}
q_step_1	recognize_polynomial_is_continuous	q_step_2	calculus_limits_action_pairs.pl:96 observed on {"at": 2, "coefficients": [1, 2, 3], "kind": "polynomial_limit"}
q_step_2	substitute_target_into_polynomial	q_step_3	calculus_limits_action_pairs.pl:96 observed on {"at": 2, "coefficients": [1, 2, 3], "kind": "polynomial_limit"}
q_step_3	evaluate_polynomial_at_point	q_step_4	calculus_limits_action_pairs.pl:96 observed on {"at": 2, "coefficients": [1, 2, 3], "kind": "polynomial_limit"}
q_step_4	kernel_trace	q_step_5	calculus_limits_action_pairs.pl:96 observed on {"at": 2, "coefficients": [1, 2, 3], "kind": "polynomial_limit"}
q_step_5	name_value_as_limit	q_accept	calculus_limits_action_pairs.pl:96 observed on {"at": 2, "coefficients": [1, 2, 3], "kind": "polynomial_limit"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 6 observed rows.

3. factor_cancel_substitute

No domain scene: the static scene emitter has no calculus form for the rational_limit contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ detect_zero_over_zero, evaluate_reduced_at_point, factor_common_factor_x_minus_a, identify_limit_target, kernel_trace, name_value_as_limit, substitute_target_into_expression, substitute_target_into_reduced (8 distinct)

q_0 $q_0 = q_{start}$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_step_6	none recorded
q_7	q_step_7	none recorded
q_8	q_accept	none recorded

Input contract: {"at": "integer", "denominator": {"coefficients": ["integer"]}, "kind": "rational_limit", "numerator": {"coefficients": ["integer"]}}; example {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}; verified(strategy_trace_ok).

δ : recorded transitions

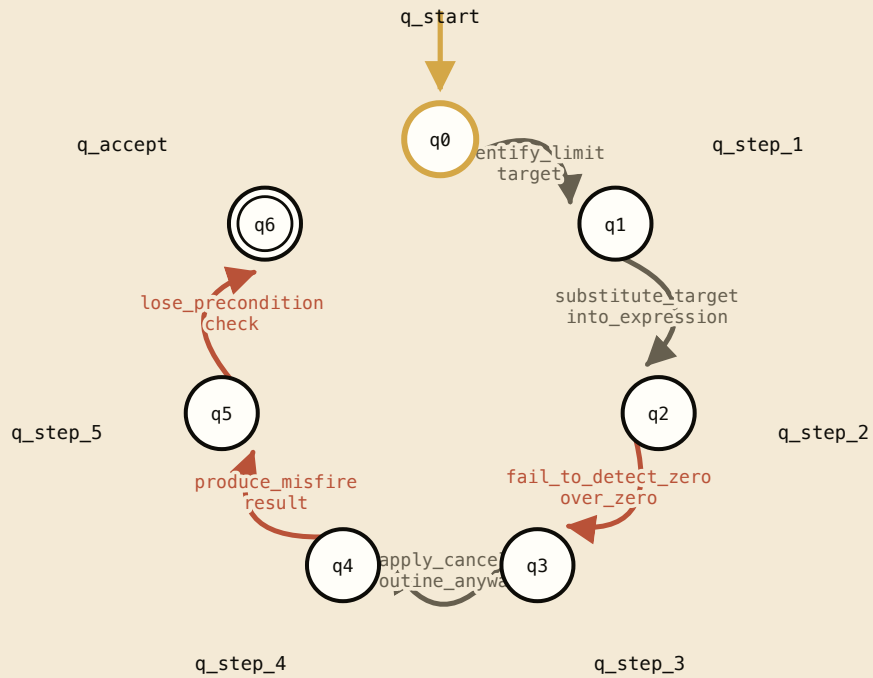
From	Action	To	Provenance
q_start	identify_limit_target	q_step_1	calculus_limits_action_pairs.pl:128 observed on {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}
q_step_1	substitute_target_into_expression	q_step_2	calculus_limits_action_pairs.pl:128 observed on {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}
q_step_2	detect_zero_over_zero	q_step_3	calculus_limits_action_pairs.pl:128 observed on {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}
q_step_3	factor_common_factor_x_minus_a	q_step_4	calculus_limits_action_pairs.pl:128 observed on {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}

From	Action	To	Provenance
q_step_4	substitute_target_into_reduced	q_step_5	calculus_limits_action_pairs.pl:128 observed on {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}
q_step_5	evaluate_reduced_at_point	q_step_6	calculus_limits_action_pairs.pl:128 observed on {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}
q_step_6	kernel_trace	q_step_7	calculus_limits_action_pairs.pl:128 observed on {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}
q_step_7	name_value_as_limit	q_accept	calculus_limits_action_pairs.pl:128 observed on {"at": 1, "denominator": {"coefficients": [-1, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [-2, 1, 1]}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 8 observed rows.

4. factor_cancel_without_common_factor

No domain scene: the static scene emitter has no calculus form for the rational_limit contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ $\text{apply_cancel_routine_anyway, fail_to_detect_zero_over_zero, identify_limit_target, lose_precondition_check, produce_misfire_result, substitute_target_into_expression}$
(6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"at": "integer", "denominator": {"coefficients": ["integer"]}, "kind": "rational_limit", "numerator": {"coefficients": ["integer"]}}; example {"at": 1, "denominator": {"coefficients": [2, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [1, 1]}}; verified(strategy_trace_ok).

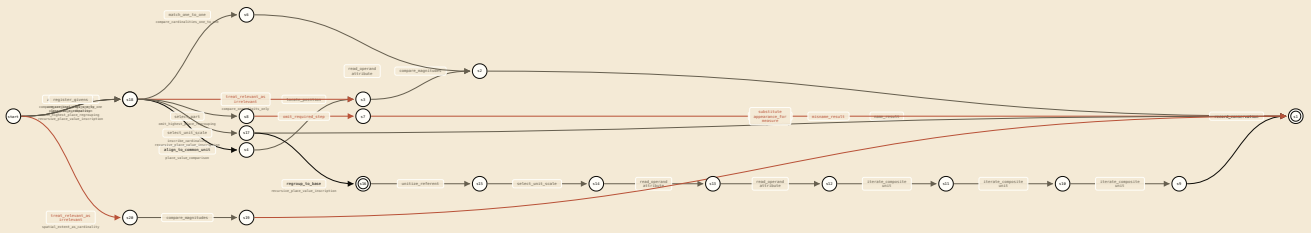
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_limit_target	q_step_1	calculus_limits_action_pairs.pl:179 observed on {"at": 1, "denominator": {"coefficients": [2, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [1, 1]}}
q_step_1	substitute_target_into_expression	q_step_2	calculus_limits_action_pairs.pl:179 observed on {"at": 1, "denominator": {"coefficients": [2, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [1, 1]}}
q_step_2	fail_to_detect_zero_over_zero	q_step_3	calculus_limits_action_pairs.pl:179 observed on {"at": 1, "denominator": {"coefficients": [2, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [1, 1]}}
q_step_3	apply_cancel_routine_anyway	q_step_4	calculus_limits_action_pairs.pl:179 observed on {"at": 1, "denominator": {"coefficients": [2, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [1, 1]}}

From	Action	To	Provenance
q_step_4	produce_misfire_result	q_step_5	calculus_limits_action_pairs.pl:179 observed on {"at": 1, "denominator": {"coefficients": [2, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [1, 1]}}
q_step_5	lose_precondition_check	q_accepted	calculus_limits_action_pairs.pl:179 observed on {"at": 1, "denominator": {"coefficients": [2, 1]}, "kind": "rational_limit", "numerator": {"coefficients": [1, 1]}}

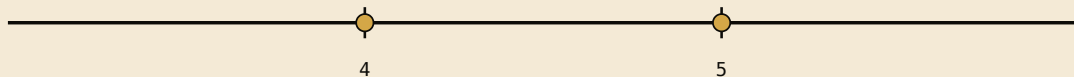
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 6 observed rows.

counting



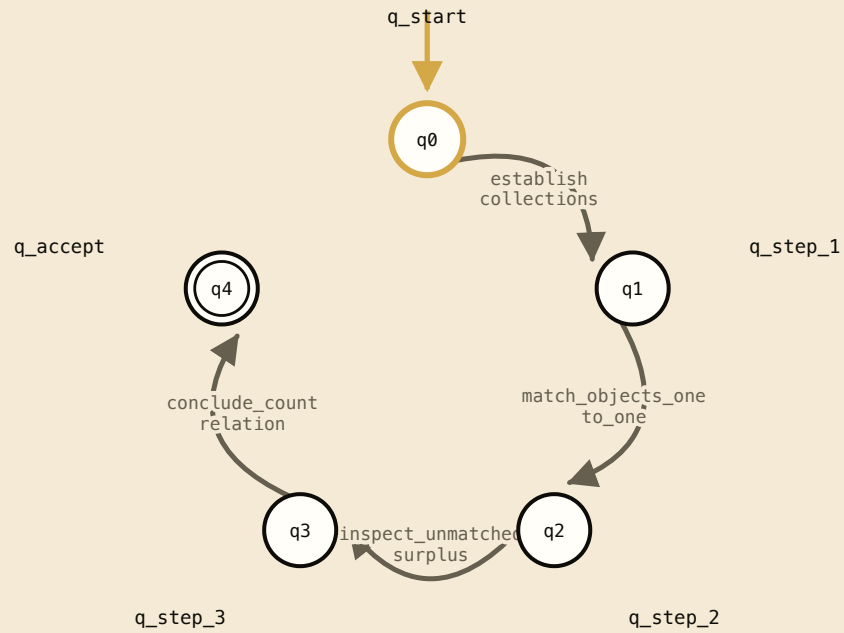
Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. compare_cardinalities_one_to_one



The run produced more.

The run produced more. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{conclude_count_relation, establish_collections, inspect_unmatched_surplus, match_objects_one_to_one (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"kind": "collection_pair", "left": "positive_integer", "left_extent": "positive_integer", "right": "positive_integer", "right_extent": "positive_integer"}; example {"kind": "collection_pair", "left": 5, "left_extent": 5, "right": 4, "right_extent": 4}; verified(strategy_trace_ok).

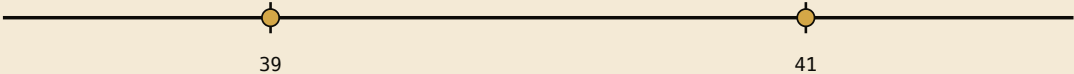
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_collections	q_step_1	counting_action_pairs.pl:177 observed on {"kind": "collection_pair", "left": 5, "left_extent": 5, "right": 4, "right_extent": 4} observed on {"kind": "collection_pair", "left": 6, "left_extent": 6, "right": 4, "right_extent": 4} (derived template)
q_step_1	match_objects_one_to_one	q_step_2	counting_action_pairs.pl:177 observed on {"kind": "collection_pair", "left": 5, "left_extent": 5, "right": 4, "right_extent": 4} observed on {"kind": "collection_pair", "left": 6, "left_extent": 6, "right": 4, "right_extent": 4} (derived template)
q_step_2	inspect_unmatched_surplus	q_step_3	counting_action_pairs.pl:177 observed on {"kind": "collection_pair", "left": 5, "left_extent": 5, "right": 4, "right_extent": 4} observed on {"kind": "collection_pair", "left": 6, "left_extent": 6, "right": 4, "right_extent": 4} (derived template)

From	Action	To	Provenance
q_step_3	conclude_count_relation	q_accept	counting_action_pairs.pl:177 observed on {"kind": "collection_pair", "left": 5, "left_extent": 5, "right": 4, "right_extent": 4} observed on {"kind": "collection_pair", "left": 6, "left_extent": 6, "right": 4, "right_extent": 4} (derived template)

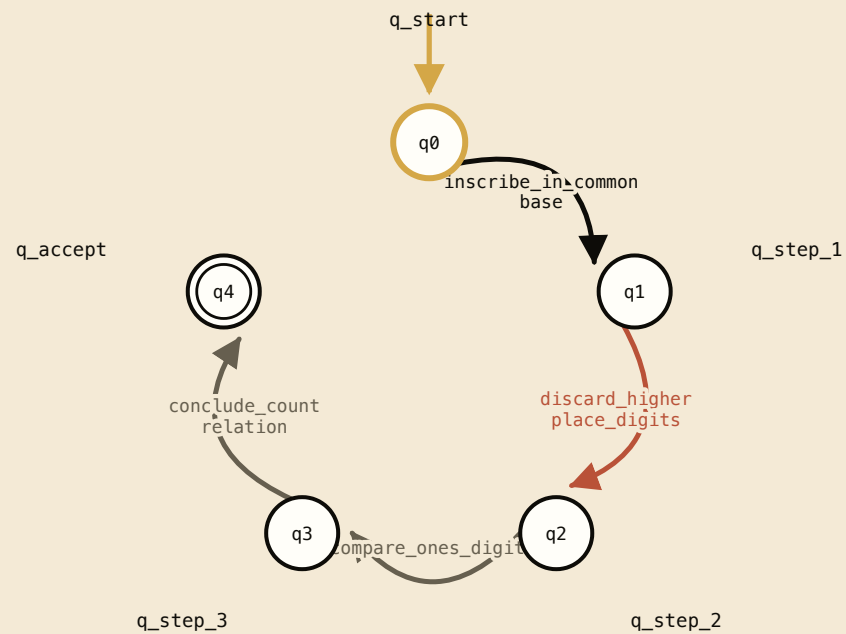
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

2. compare_ones_digits_only



The run produced more; the recorded expected result is fewer.

The run produced more; the recorded expected result is fewer. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{compare_ones_digits}, \text{conclude_count_relation}, \text{discard_higher_place_digits}, \text{inscribe_in_common_base}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

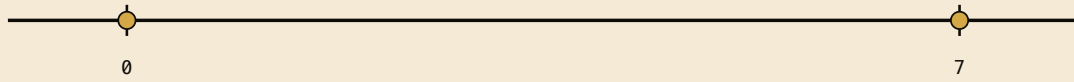
Input contract: {"base": "positive_integer", "kind": "count_pair", "left": "integer", "right": "integer"}; example {"base": 10, "kind": "count_pair", "left": 39, "right": 41}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	inscribe_in_common_base	q_step_1	counting_action_pairs.pl:152 observed on {"base": 10, "kind": "count_pair", "left": 39, "right": 41}
q_step_1	discard_higher_place_digits	q_step_2	counting_action_pairs.pl:152 observed on {"base": 10, "kind": "count_pair", "left": 39, "right": 41}
q_step_2	compare_ones_digits	q_step_3	counting_action_pairs.pl:152 observed on {"base": 10, "kind": "count_pair", "left": 39, "right": 41}
q_step_3	conclude_count_relation	q_accept	counting_action_pairs.pl:152 observed on {"base": 10, "kind": "count_pair", "left": 39, "right": 41}

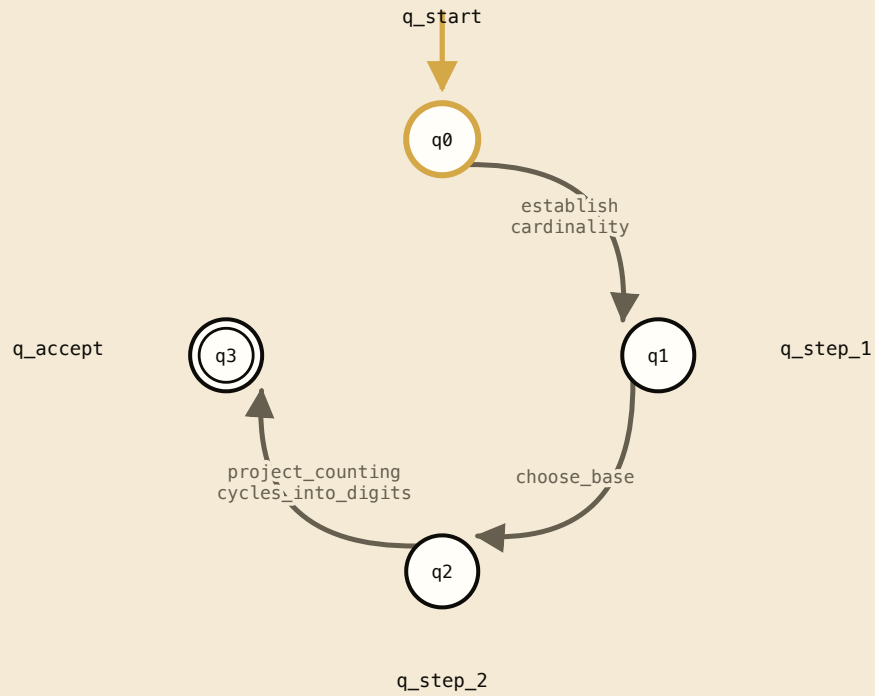
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

3. inscribe_cardinality



The run produced `numeral(10,positive,radix(1),[digit(7,"7")])`.

The run produced `numeral(10,positive,radix(1),[digit(7,"7")])`. The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad \text{choose_base, establish_cardinality, project_counting_cycles_into_digits (3 distinct)}$$

$$q_0 \quad q_0 = \text{q_start}$$

$$F \quad \{q_3\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

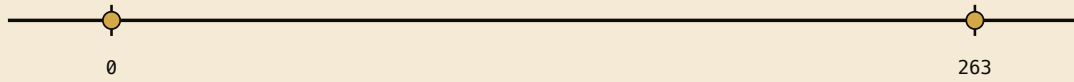
Input contract: {"base": "positive_integer", "count": "positive_integer", "kind": "cardinality"}; example {"base": 10, "count": 7, "kind": "cardinality"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_cardinality	q_step_1	counting_action_pairs.pl:52 observed on {"base": 10, "count": 7, "kind": "cardinality"} observed on {"base": 10, "count": 8, "kind": "cardinality"} (derived template)
q_step_1	choose_base	q_step_2	counting_action_pairs.pl:52 observed on {"base": 10, "count": 7, "kind": "cardinality"} observed on {"base": 10, "count": 8, "kind": "cardinality"} (derived template)
q_step_2	project_counting_cycles_into_digits	q_accept	counting_action_pairs.pl:52 observed on {"base": 10, "count": 7, "kind": "cardinality"} observed on {"base": 10, "count": 8, "kind": "cardinality"} (derived template)

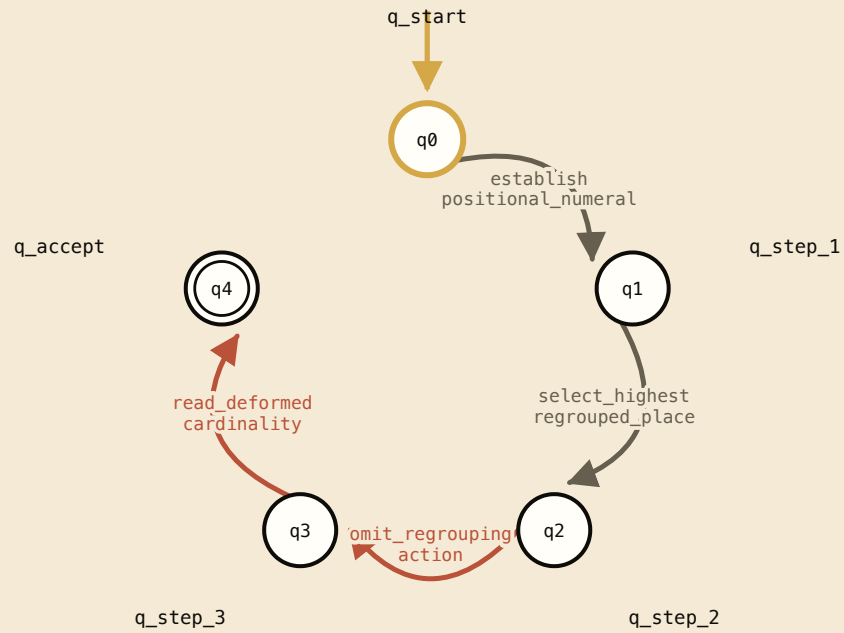
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 6 observed rows.

4. omit_highest_place_regrouping



The run produced cardinality(83); the recorded expected result is cardinality(263).

The run produced cardinality(83); the recorded expected result is cardinality(263). The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{establish_positional_numeral, omit_regrouping_action, read_deformed_cardinality, select_highest_regrouped_place (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"base": "positive_integer", "count": "integer", "kind": "cardinality"}; example {"base": 10, "count": 263, "kind": "cardinality"}; verified(strategy_trace_ok).

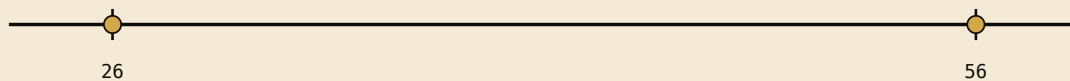
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_positional_numeral	q_step_1	counting_action_pairs.pl:96 observed on {"base": 10, "count": 263, "kind": "cardinality"} observed on {"base": 10, "count": 264, "kind": "cardinality"} (derived template)
q_step_1	select_highest_regrouped_place	q_step_2	counting_action_pairs.pl:96 observed on {"base": 10, "count": 263, "kind": "cardinality"} observed on {"base": 10, "count": 264, "kind": "cardinality"} (derived template)
q_step_2	omit_regrouping_action	q_step_3	counting_action_pairs.pl:96 observed on {"base": 10, "count": 263, "kind": "cardinality"} observed on {"base": 10, "count": 264, "kind": "cardinality"} (derived template)
q_step_3	read_deformed_cardinality	q_accept	counting_action_pairs.pl:96 observed on {"base": 10, "count": 263, "kind": "cardinality"} observed on {"base": 10, "count": 264, "kind": "cardinality"} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and

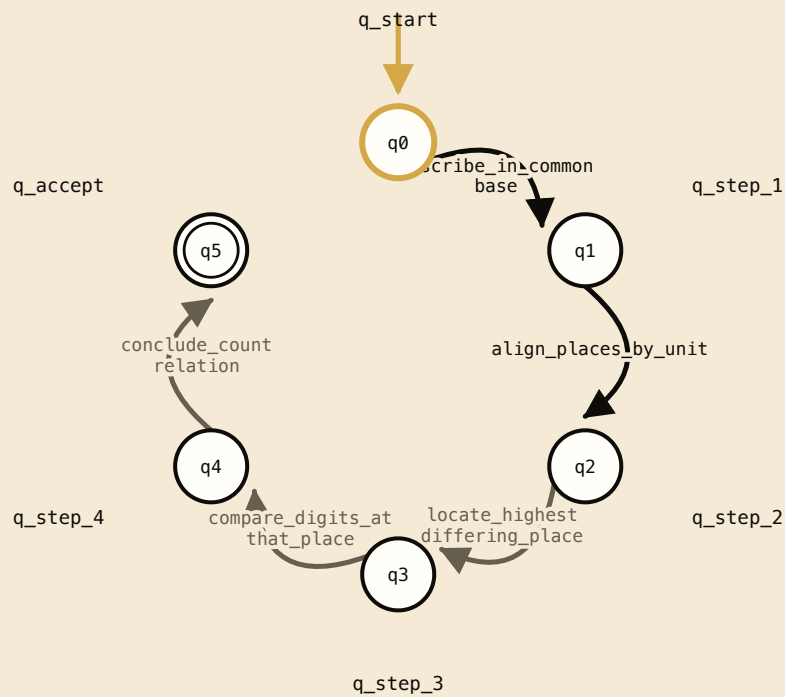
8 observed rows.

5. place_value_comparison



The run produced more.

The run produced more. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ align_places_by_unit, compare_digits_at_that_place, conclude_count_relation, inscribe_in_common_base, locate_highest_differing_place (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"base": "positive_integer", "kind": "count_pair", "left": "integer", "right": "integer"}; example {"base": 10, "kind": "count_pair", "left": 56, "right": 26}; verified(strategy_trace_ok).

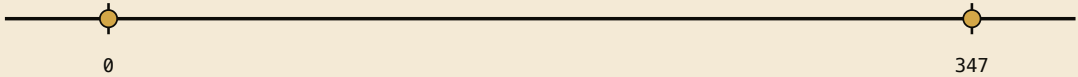
δ : recorded transitions

From	Action	To	Provenance
q_start	inscribe_in_common_base	q_step_1	counting_action_pairs.pl:127 observed on {"base": 10, "kind": "count_pair", "left": 56, "right": 26} observed on {"base": 10, "kind": "count_pair", "left": 57, "right": 26} (derived template)
q_step_1	align_places_by_unit	q_step_2	counting_action_pairs.pl:127 observed on {"base": 10, "kind": "count_pair", "left": 56, "right": 26} observed on {"base": 10, "kind": "count_pair", "left": 57, "right": 26} (derived template)
q_step_2	locate_highest_differing_place	q_step_3	counting_action_pairs.pl:127 observed on {"base": 10, "kind": "count_pair", "left": 56, "right": 26} observed on {"base": 10, "kind": "count_pair", "left": 57, "right": 26} (derived template)
q_step_3	compare_digits_at_that_place	q_step_4	counting_action_pairs.pl:127 observed on {"base": 10, "kind": "count_pair", "left": 56, "right": 26} observed on {"base": 10, "kind": "count_pair", "left": 57, "right": 26} (derived template)

From	Action	To	Provenance
q_step_4	conclude_count_relation	q_accept	counting_action_pairs.pl:127 observed on {"base": 10, "kind": "count_pair", "left": 56, "right": 26} observed on {"base": 10, "kind": "count_pair", "left": 57, "right": 26} (derived template)

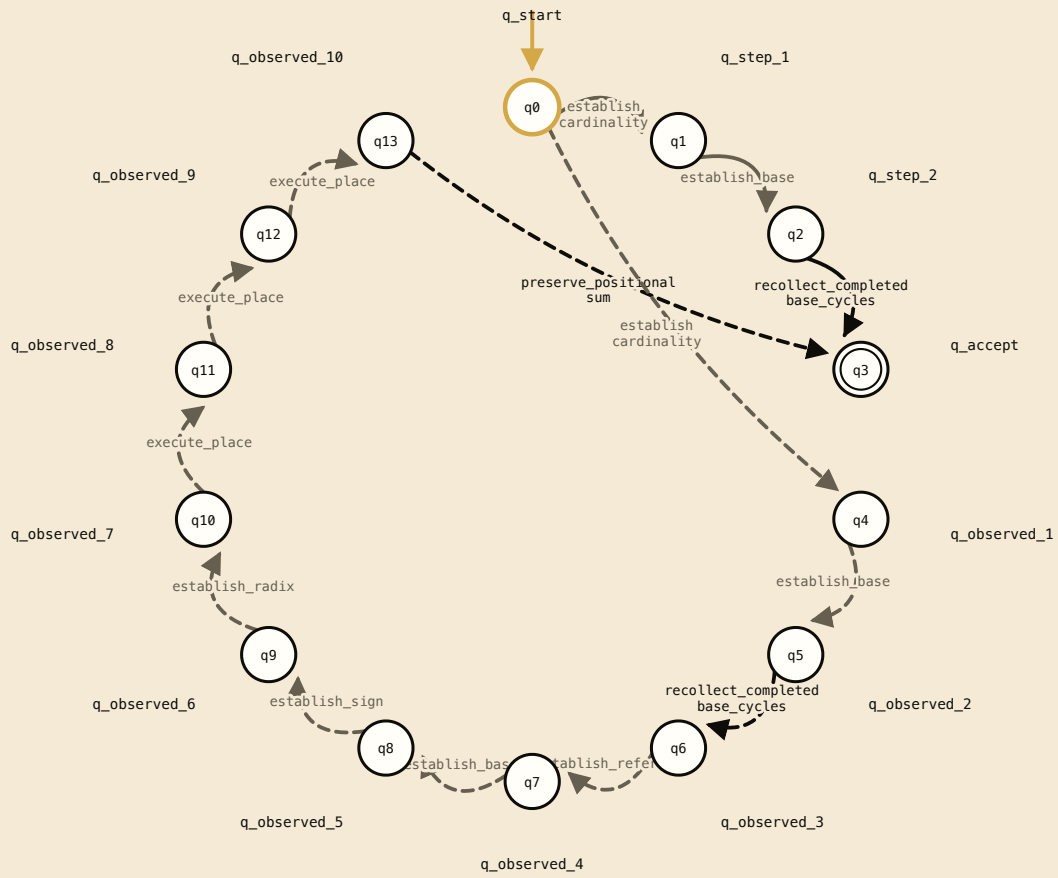
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

6. recursive_place_value_inscription



The run produced `numeral(10,positive,radix(3),[digit(3,"3"),digit(4,"4"),digit(7,"7")])`.

The run produced `numeral(10,positive,radix(3),[digit(3,"3"),digit(4,"4"),digit(7,"7")])`. The scene is derived from the contract run (11 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}\}$

Σ establish_base, establish_cardinality, establish_radix, establish_referent, establish_sign, execute_place, preserve_positional_sum, recollect_completed_base_cycles (8 distinct)

q_0 $q_0 = q_start$

F $\{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded
q_4	q_observed_1	none recorded
q_5	q_observed_2	none recorded
q_6	q_observed_3	none recorded
q_7	q_observed_4	none recorded
q_8	q_observed_5	none recorded
q_9	q_observed_6	none recorded
q_{10}	q_observed_7	none recorded
q_{11}	q_observed_8	none recorded
q_{12}	q_observed_9	none recorded
q_{13}	q_observed_10	none recorded

Input contract: {"base": "positive_integer", "count": "integer", "kind": "cardinality"}; example {"base": 10, "count": 347, "kind": "cardinality"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_cardinality	q_step_1	counting_action_pairs.pl:72
q_step_1	establish_base	q_step_2	counting_action_pairs.pl:72
q_step_2	recollect_completed_base_cycles	q_accept	counting_action_pairs.pl:72
q_start	establish_cardinality	q_observed_1	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)

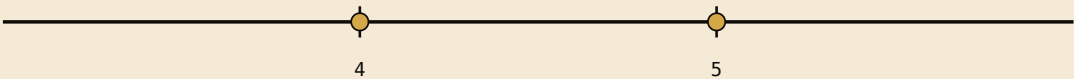
From	Action	To	Provenance
q_observe d_1	establish_base	q_observed_2	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)
q_observe d_2	recollect_completed_base_cycles	q_observed_3	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)
q_observe d_3	establish_referent	q_observed_4	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)
q_observe d_4	establish_base	q_observed_5	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)
q_observe d_5	establish_sign	q_observed_6	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)
q_observe d_6	establish_radix	q_observed_7	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)
q_observe d_7	execute_place	q_observed_8	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)
q_observe d_8	execute_place	q_observed_9	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)
q_observe d_9	execute_place	q_observed_10	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)

From	Action	To	Provenance
q_observe d_10	preserve_positional_ sum	q_accept	observed on {"base": 10, "count": 347, "kind": "cardinality"} observed on {"base": 10, "count": 348, "kind": "cardinality"} (derived template)

Computed structure: branching. one or more recorded states have multiple outgoing transitions; no loop edge is recorded. 14 states, 8 distinct actions, 1 branching states, 0 loop edges; 3 static rows and 22 observed rows.

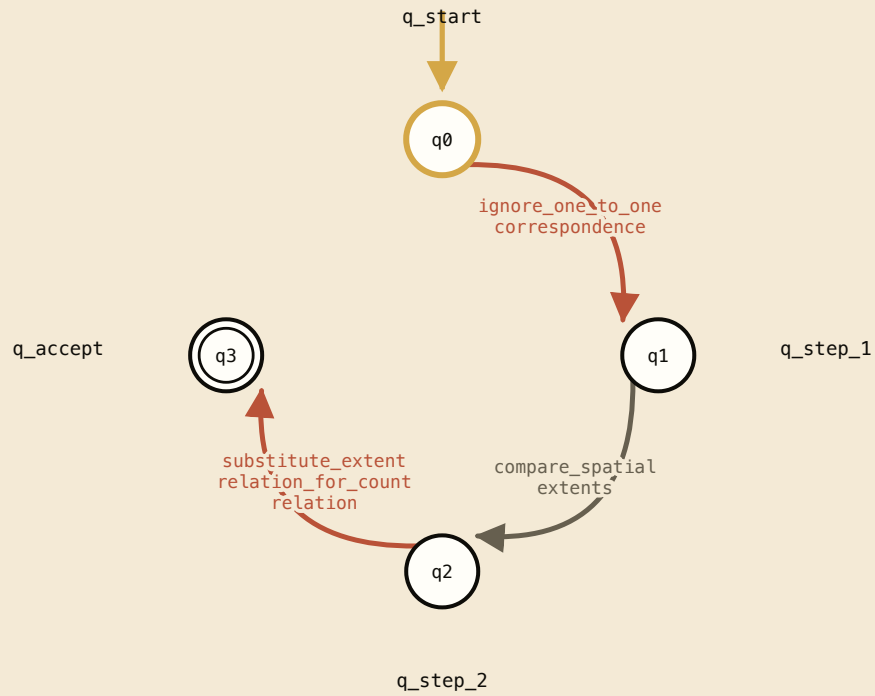
Authored class: pushdown_counter, attested at knowledge/strategies/math/counting2.pl:1. This claim comes from module prose, not from the table graph.

7. spatial_extent_as_cardinality



The run produced more; the recorded expected result is fewer.

The run produced more; the recorded expected result is fewer. The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad \text{compare_spatial_extents, ignore_one_to_one_correspondence,} \\ \text{substitute_extent_relation_for_count_relation (3 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_3\}$$

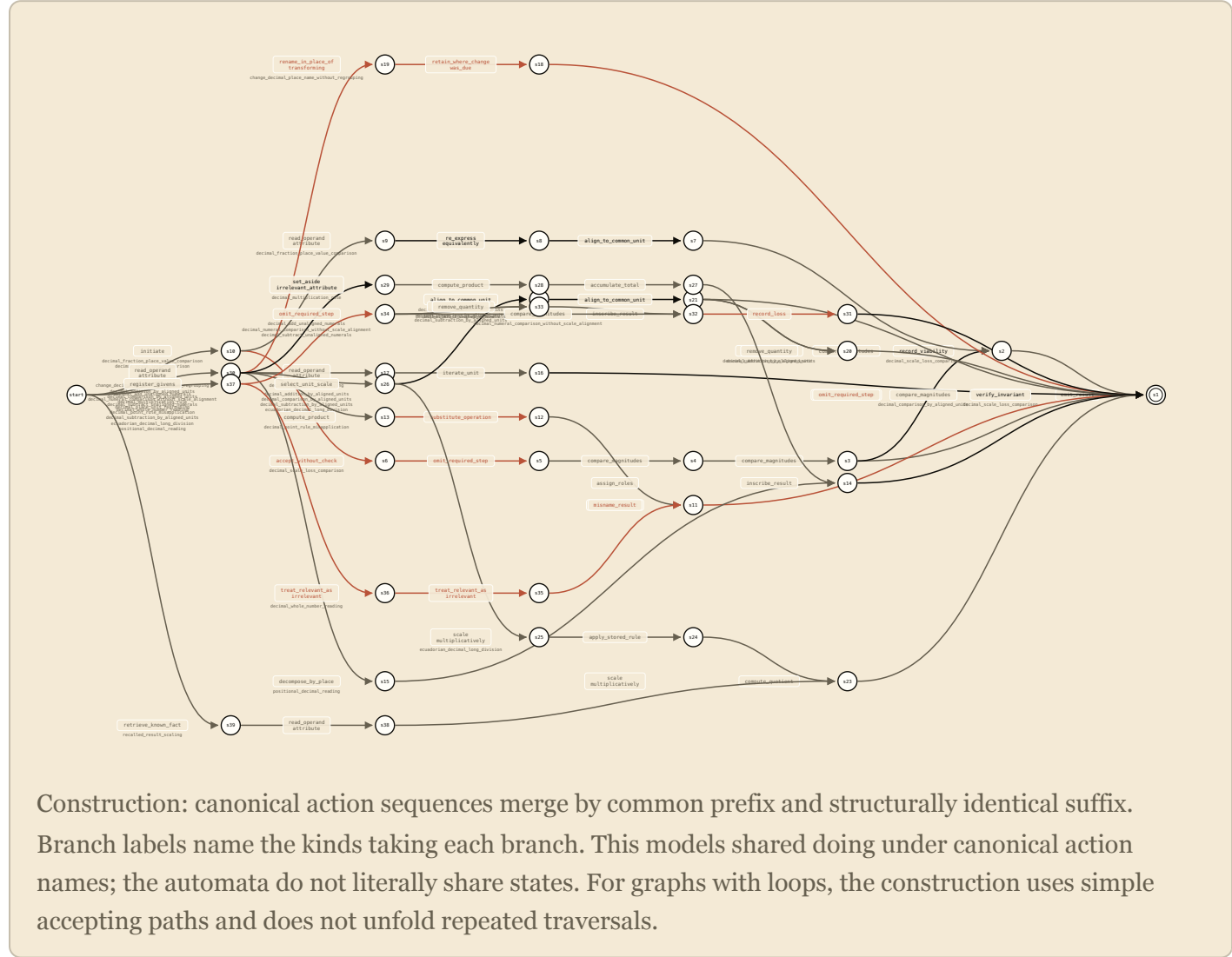
Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

Input contract: {"kind": "collection_pair", "left": "positive_integer", "left_extent": "positive_integer", "right": "positive_integer", "right_extent": "positive_integer"}; example {"kind": "collection_pair", "left": 4, "left_extent": 9, "right": 5, "right_extent": 3}; verified(strategy_trace_ok).

δ : recorded transitions

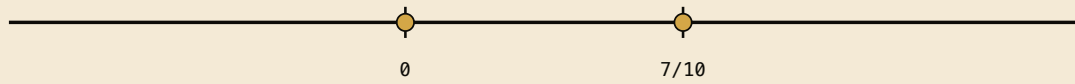
From	Action	To	Provenance
q_start	ignore_one_to_one_correspondence	q_step_1	counting_action_pairs.pl:200 observed on {"kind": "collection_pair", "left": 4, "left_extent": 9, "right": 5, "right_extent": 3} observed on {"kind": "collection_pair", "left": 5, "left_extent": 10, "right": 5, "right_extent": 3} (derived template)
q_step_1	compare_spatial_extents	q_step_2	counting_action_pairs.pl:200 observed on {"kind": "collection_pair", "left": 4, "left_extent": 9, "right": 5, "right_extent": 3} observed on {"kind": "collection_pair", "left": 5, "left_extent": 10, "right": 5, "right_extent": 3} (derived template)
q_step_2	substitute_extent_relation_for_count_relation	q_accept	counting_action_pairs.pl:200 observed on {"kind": "collection_pair", "left": 4, "left_extent": 9, "right": 5, "right_extent": 3} observed on {"kind": "collection_pair", "left": 5, "left_extent": 10, "right": 5, "right_extent": 3} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 6 observed rows.



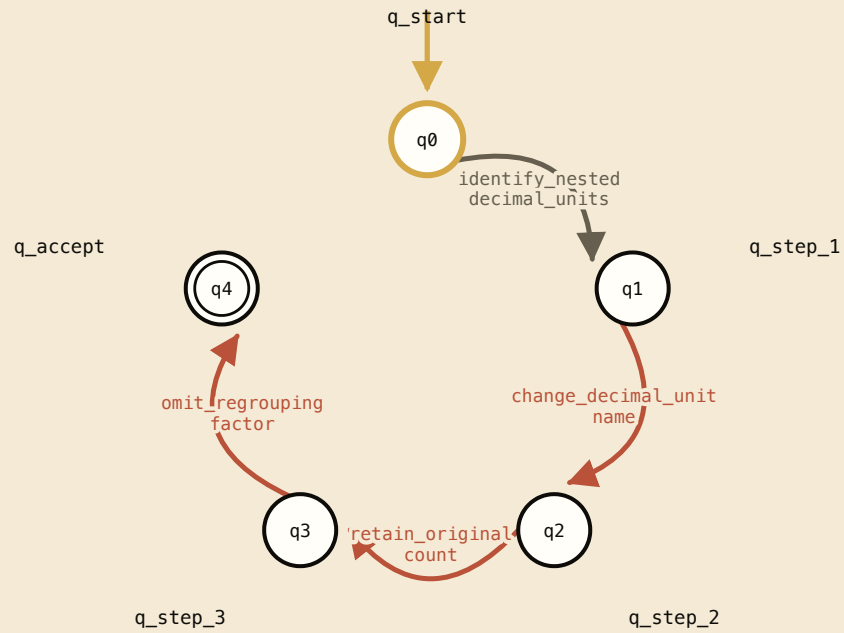
1. `change_decimal_place_name_without_regrouping`

1. `change_decimal_place_name_without_regrouping`



The run produced `equivalent_decimal_units(7,unit_fraction(1,100))`; the recorded expected result is `equivalent_decimal_units(70,unit_fraction(1,100))`.

The run produced `equivalent_decimal_units(7,unit_fraction(1,100))`; the recorded expected result is `equivalent_decimal_units(70,unit_fraction(1,100))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{change_decimal_unit_name, identify_nested_decimal_units, omit_regrouping_factor, retain_original_count (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

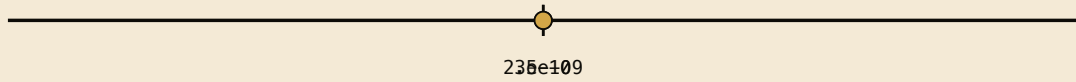
Input contract: {"count": "integer", "from_scale": "positive_integer", "kind": "decimal_unit_conversion", "to_scale": "positive_integer"}; example {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_nested_decimal_units	q_step_1	decimal_action_pairs.pl:386 observed on {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100}
q_step_1	change_decimal_unit_name	q_step_2	decimal_action_pairs.pl:386 observed on {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100}
q_step_2	retain_original_count	q_step_3	decimal_action_pairs.pl:386 observed on {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100}
q_step_3	omit_regrouping_factor	q_accept	decimal_action_pairs.pl:386 observed on {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100}

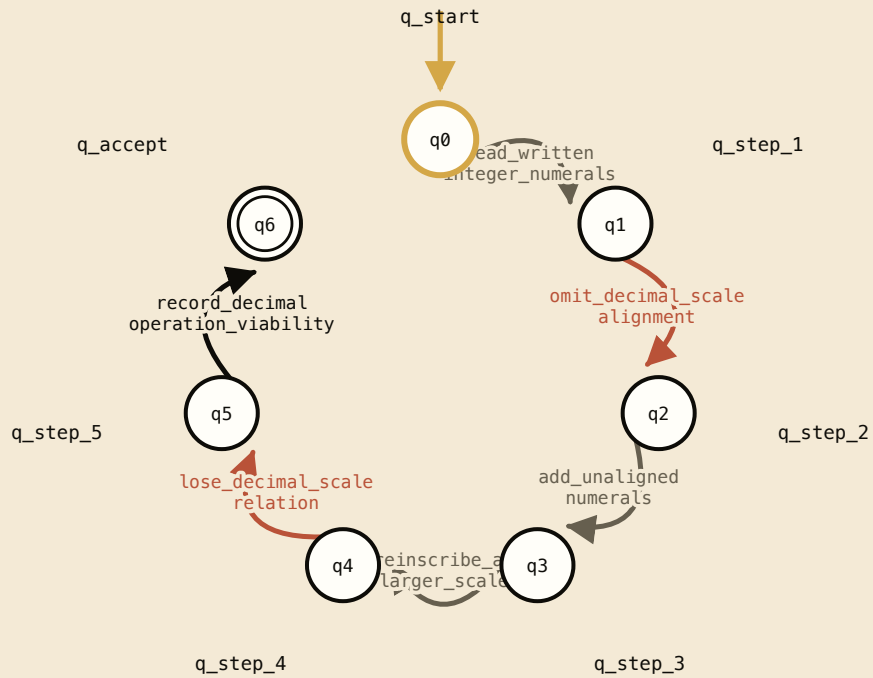
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

2. decimal_add_unaligned_numerals



The run produced `decimal(2,fractional_digits(8,1),tenths)`; the recorded expected result is `decimal(2`

The run produced `decimal(2,fractional_digits(8,1),tenths)`; the recorded expected result is `decimal(2,fractional_digits(8,1),tenths)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \text{add_unaligned_numerals, lose_decimal_scale_relation, omit_decimal_scale_alignment, read_written_integer_numerals, record_decimal_operation_viability, reinscribe_at_larger_scale (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

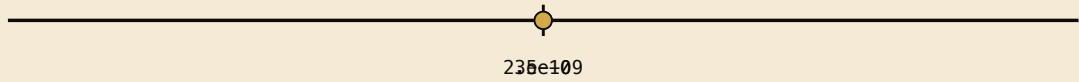
δ : recorded transitions

From	Action	To	Provenance
q_start	read_written_integer_numerals	q_step_1	decimal_action_pairs.pl:261 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_1	omit_decimal_scale_alignment	q_step_2	decimal_action_pairs.pl:261 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_2	add_unaligned_numerals	q_step_3	decimal_action_pairs.pl:261 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_step_3	reinscribe_at_larger_scale	q_step_4	decimal_action_pairs.pl:261 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_4	lose_decimal_scale_relation	q_step_5	decimal_action_pairs.pl:261 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_5	record_decimal_operation_viability	q_accept	decimal_action_pairs.pl:261 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

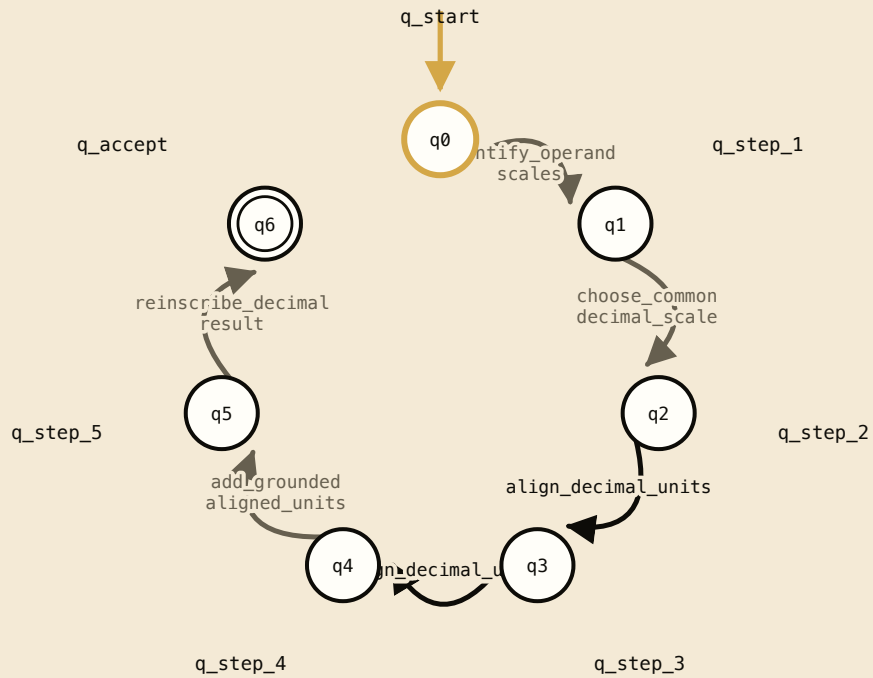
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

3. decimal_addition_by_aligned_units



The run produced `decimal(2,fractional_digits(8,1),tenths)`.

The run produced `decimal(2,fractional_digits(8,1),tenths)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ add_grounded_aligned_units, align_decimal_units, choose_common_decimal_scale, identify_operand_scales, reinscribe_decimal_result (5 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

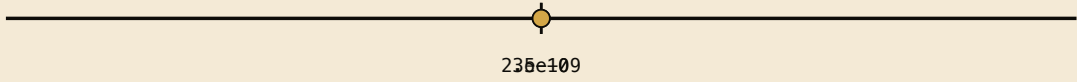
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_operands_scales	q_step_1	decimal_action_pairs.pl:228 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_1	choose_common_decimal_scale	q_step_2	decimal_action_pairs.pl:228 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_2	align_decimal_units	q_step_3	decimal_action_pairs.pl:228 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_step_3	align_decimal_units	q_step_4	decimal_action_pairs.pl:228 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_4	add_grounded_aligned_units	q_step_5	decimal_action_pairs.pl:228 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_5	reinscribe_decimal_result	q_accept	decimal_action_pairs.pl:228 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

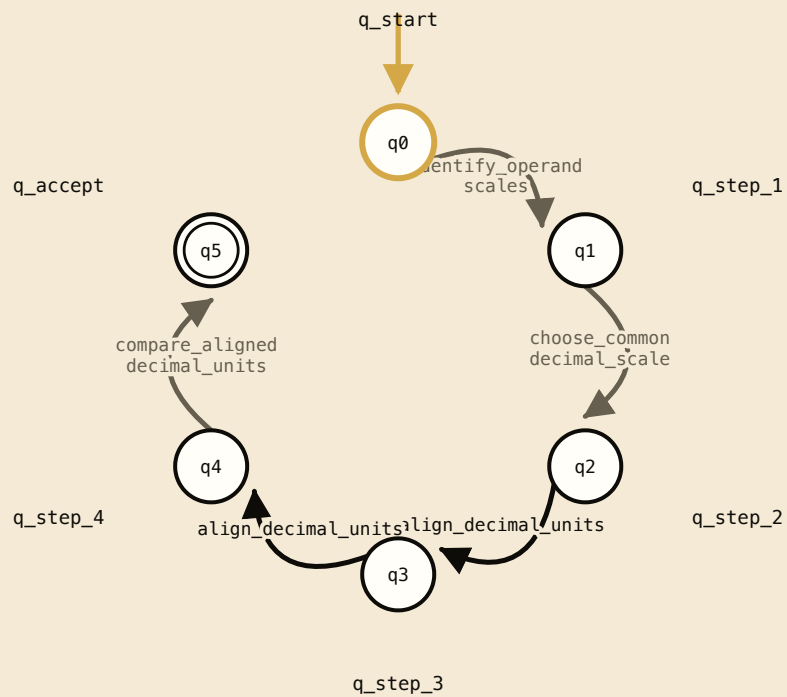
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 5 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

4. decimal_comparison_by_aligned_units



The run produced more.

The run produced more. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ align_decimal_units, choose_common_decimal_scale, compare_aligned_decimal_units, identify_operand_scales (4 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

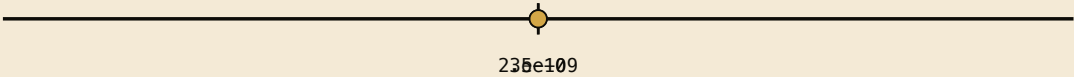
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_operand_scales	q_step_1	decimal_action_pairs.pl:121 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_1	choose_common_decimal_scale	q_step_2	decimal_action_pairs.pl:121 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_2	align_decimal_units	q_step_3	decimal_action_pairs.pl:121 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_step_3	align_decimal_units	q_step_4	decimal_action_pairs.pl:121 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_4	compare_aligned_decimal_units	q_accept	decimal_action_pairs.pl:121 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

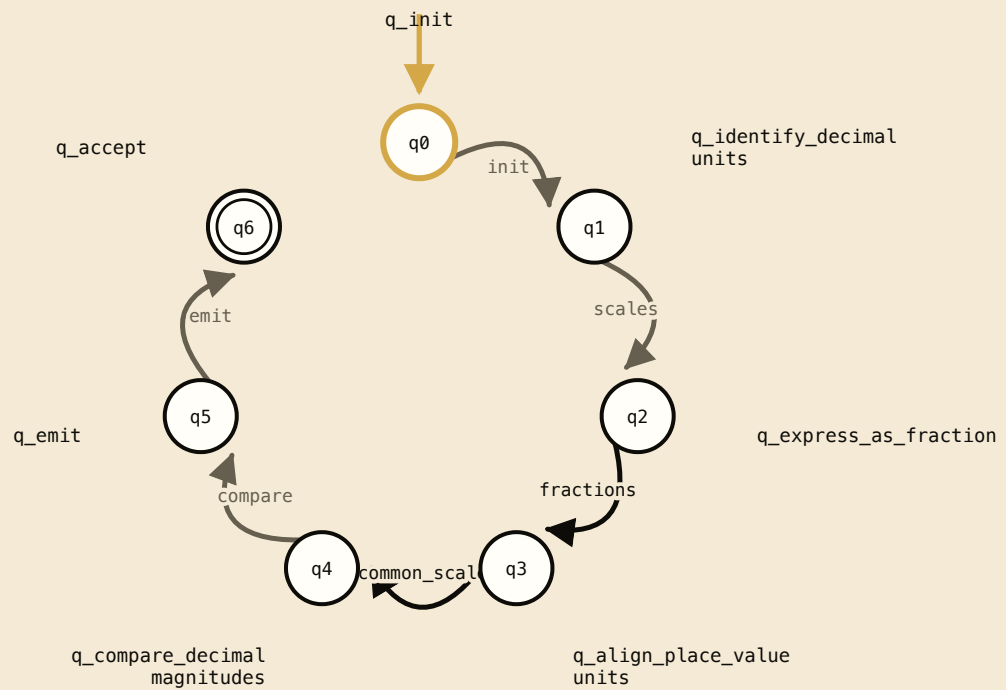
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 4 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

5. decimal_fraction_place_value_comparison



The run produced `greater_than`.

The run produced `greater_than`. The scene is derived from the contract run (7 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \text{common_scale, compare, emit, fractions, init, scales (6 distinct)}$$

$$q_0 = q_{\text{init}}$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_identify_decimal_units	<i>refers to a whole</i> (asktm_g4q2; ASKTM G4Q2 coded legend, A2/A3)
q_2	q_express_as_fraction	<i>converts decimals to fractions</i> (asktm_g4q2; ASKTM G4Q2 coded legend, A2/A3)
q_3	q_align_place_value_units	<i>place value reasoning</i> (asktm_g4q2; ASKTM G4Q2 coded legend, A2/A3)
q_4	q_compare_decimal_magnitudes	<i>uses < or ></i> (asktm_g4q2; ASKTM G4Q2 coded legend, A1/A2)
q_5	q_emit	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

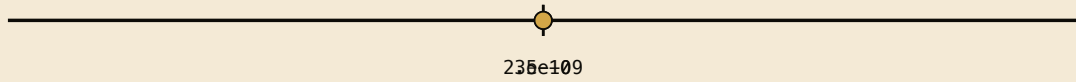
δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_identify_decimal_units	smr_decimal_fraction_compare.pl:11 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_identify_decimal_units	scales	q_express_as_fraction	smr_decimal_fraction_compare.pl:11 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_express_as_fraction	fraction_s	q_align_place_value_units	smr_decimal_fraction_compare.pl:11 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_align_place_value_units	common_scale	q_compare_decimal_magnitudes	smr_decimal_fraction_compare.pl:11 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_compare_decimal_magnitudes	compare	q_emit	smr_decimal_fraction_compare.pl:11 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_emit	emit	q_accept	smr_decimal_fraction_compare.pl:11 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

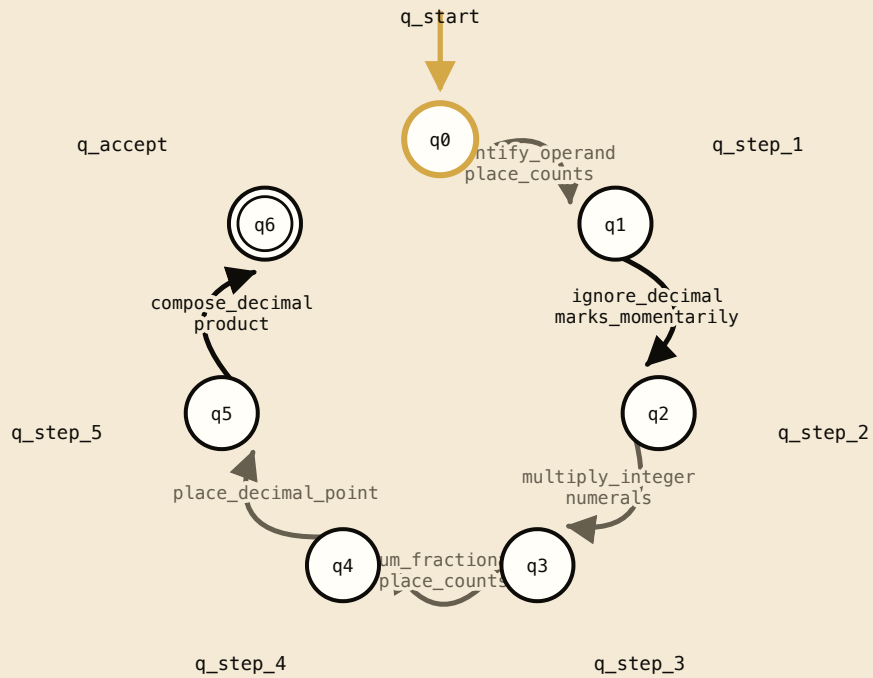
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

6. decimal_multiplication_rule



The run produced `decimal(0,fractional_digits(75,2),hundredths)`.

The run produced `decimal(0,fractional_digits(75,2),hundredths)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ compose_decimal_product, identify_operand_place_counts,
ignore_decimal_marks_momentarily, multiply_integer_numerals, place_decimal_point,
sum_fractional_place_counts (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

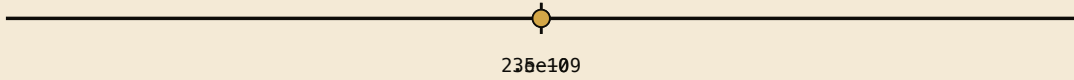
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_operand_place_counts	q_step_1	decimal_action_pairs.pl:412 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_1	ignore_decimal_marks_momentarily	q_step_2	decimal_action_pairs.pl:412 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_2	multiply_integer_numerals	q_step_3	decimal_action_pairs.pl:412 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_step_3	sum_fractional_place_counts	q_step_4	decimal_action_pairs.pl:412 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_4	place_decimal_point	q_step_5	decimal_action_pairs.pl:412 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_5	compose_decimal_product	q_accept	decimal_action_pairs.pl:412 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

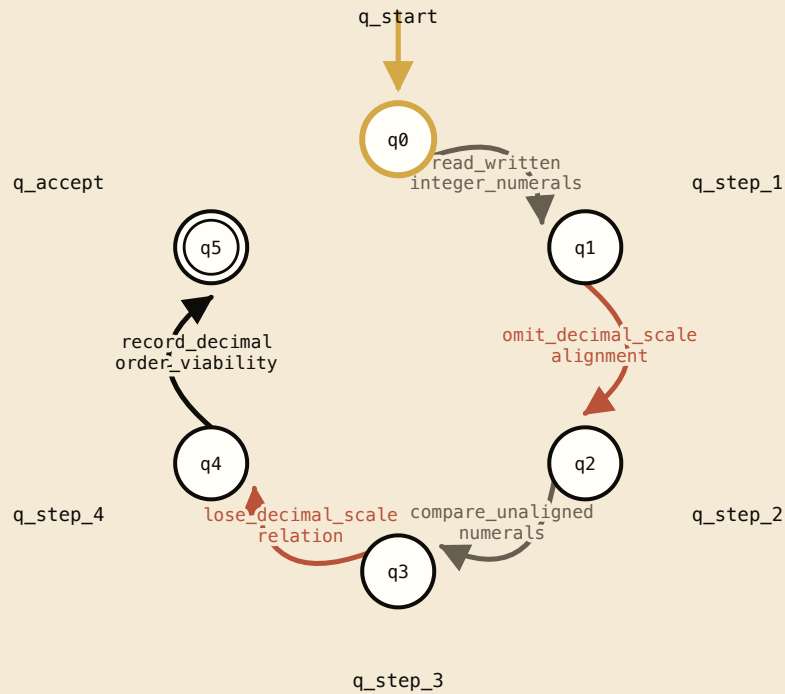
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

7. decimal_numeral_comparison_without_scale_alignment



The run produced more; the recorded expected result is more.

The run produced more; the recorded expected result is more. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ `compare_unaligned_numerals`, `lose_decimal_scale_relation`,
`omit_decimal_scale_alignment`, `read_written_integer_numerals`,
`record_decimal_order_viability` (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

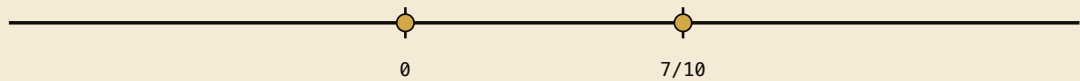
δ : recorded transitions

From	Action	To	Provenance
q_start	read_written_integer_numerals	q_step_1	decimal_action_pairs.pl:153 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_1	omit_decimal_scale_alignment	q_step_2	decimal_action_pairs.pl:153 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_2	compare_unaligned_numerals	q_step_3	decimal_action_pairs.pl:153 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_step_3	lose_decimal_scale_relation	q_step_4	decimal_action_pairs.pl:153 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_4	record_decimal_order_viability	q_accept	decimal_action_pairs.pl:153 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

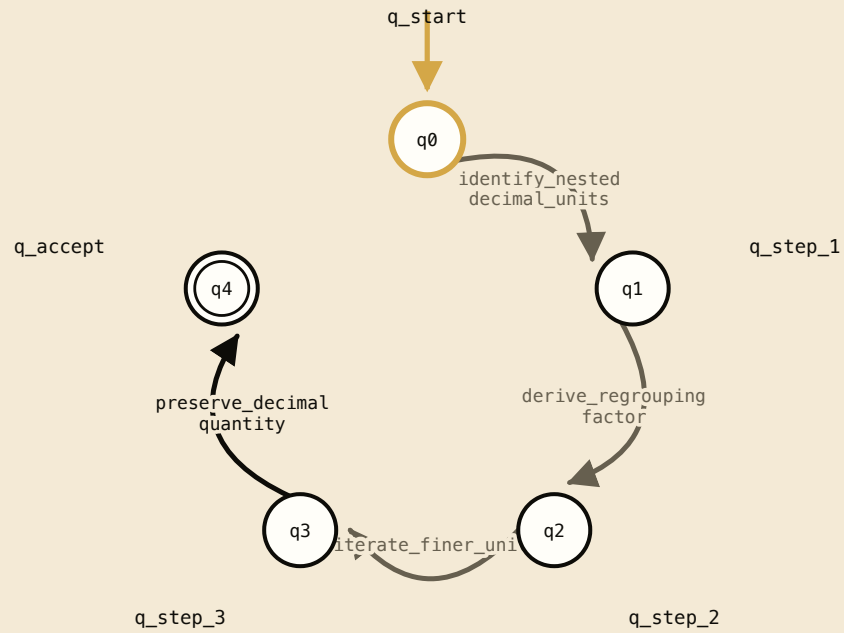
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

8. decimal_place_unit_regrouping



The run produced `equivalent_decimal_units(70,unit_fraction(1,100))`.

The run produced `equivalent_decimal_units(70,unit_fraction(1,100))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ derive_regrouping_factor, identify_nested_decimal_units, iterate_finer_unit, preserve_decimal_quantity (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

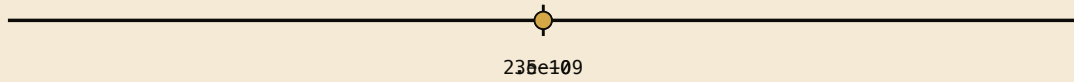
Input contract: {"count": "integer", "from_scale": "positive_integer", "kind": "decimal_unit_conversion", "to_scale": "positive_integer"}; example {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_nested_decimal_units	q_step_1	decimal_action_pairs.pl:356 observed on {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100}
q_step_1	derive_regrouping_factor	q_step_2	decimal_action_pairs.pl:356 observed on {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100}
q_step_2	iterate_finer_unit	q_step_3	decimal_action_pairs.pl:356 observed on {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100}
q_step_3	preserve_decimal_quantity	q_accept	decimal_action_pairs.pl:356 observed on {"count": 7, "from_scale": 10, "kind": "decimal_unit_conversion", "to_scale": 100}

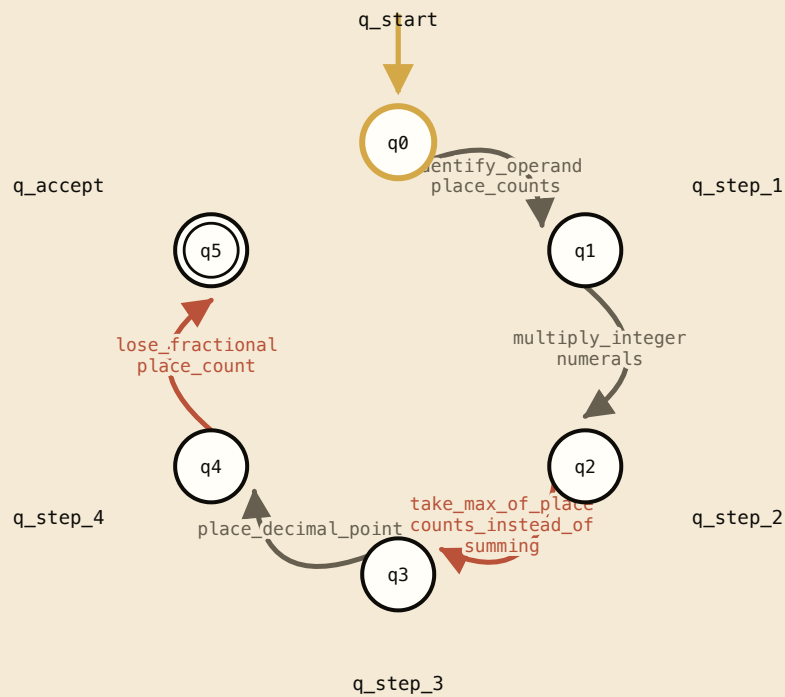
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

9. decimal_point_rule_misapplication



The run produced `decimal(7,fractional_digits(5,1),tenths)`; the recorded expected result is `decimal(0`

The run produced `decimal(7,fractional_digits(5,1),tenths)`; the recorded expected result is `decimal(0,fractional_digits(75,2),hundredths)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ identify_operand_place_counts, lose_fractional_place_count,
multiply_integer_numerals, place_decimal_point,
take_max_of_place_counts_instead_of_summing (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

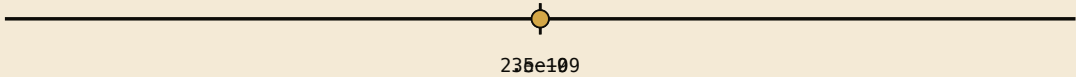
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_operand_place_counts	q_step_1	decimal_action_pairs.pl:442 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_1	multiply_integer_numerals	q_step_2	decimal_action_pairs.pl:442 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_2	take_max_of_place_counts_instead_of_summing	q_step_3	decimal_action_pairs.pl:442 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_step_3	place_decimal_point	q_step_4	decimal_action_pairs.pl:442 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_4	lose_fractional_place_count	q_accept	decimal_action_pairs.pl:442 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

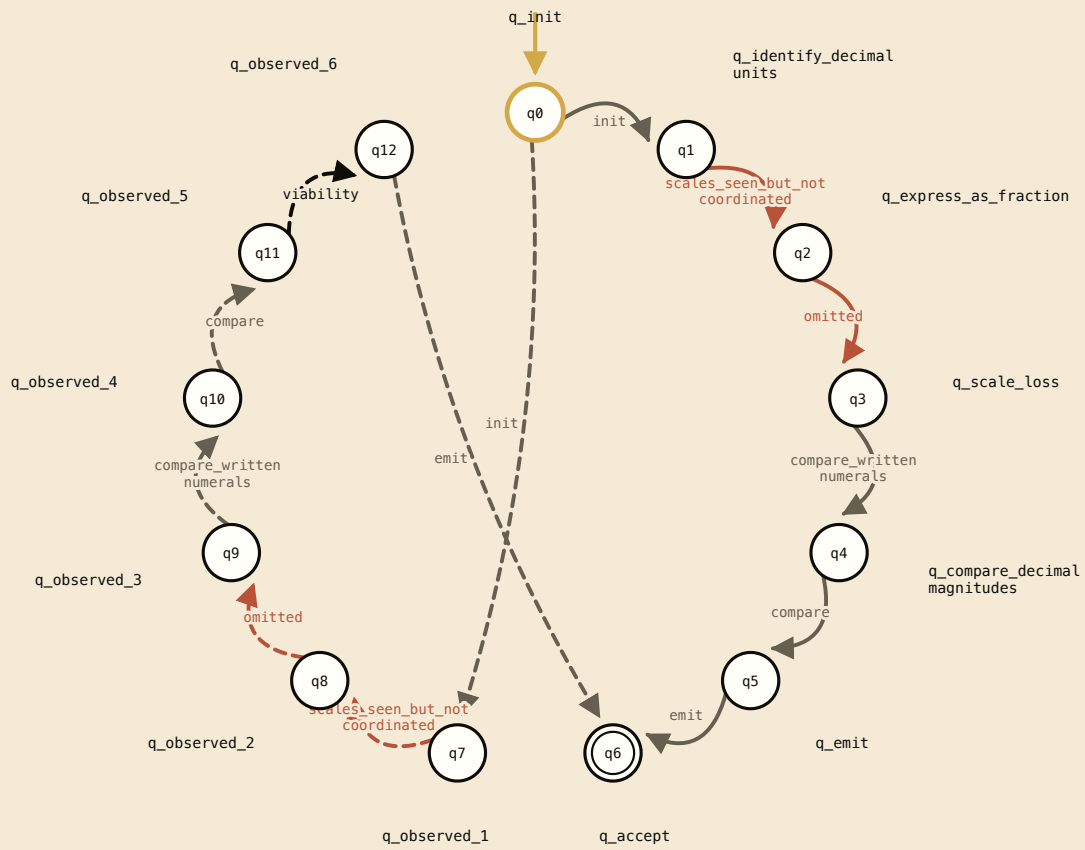
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

10. decimal_scale_loss_comparison



The run produced less_than; the recorded expected result is greater_than.

The run produced less_than; the recorded expected result is greater_than. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}\}$

Σ compare, compare_written_numerals, emit, init, omitted, scales_seen_but_not_coordinated, viability (7 distinct)

q_0 $q_0 = q_init$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_identify_decimal_units	<i>refers to a whole</i> (asktm_g4q2; ASKTM G4Q2 coded legend, A2/A3)
q_2	q_express_as_fraction	<i>converts decimals to fractions</i> (asktm_g4q2; ASKTM G4Q2 coded legend, A2/A3)
q_3	q_scale_loss	<i>misnomer of quantity</i> (asktm_g4q2; ASKTM G4Q2 coded legend, A2/B2)
q_4	q_compare_decimal_magnitudes	<i>uses < or ></i> (asktm_g4q2; ASKTM G4Q2 coded legend, A1/A2)
q_5	q_emit	none recorded
q_6	q_accept	none recorded
q_7	q_observed_1	none recorded
q_8	q_observed_2	none recorded
q_9	q_observed_3	none recorded
q_{10}	q_observed_4	none recorded
q_{11}	q_observed_5	none recorded
q_{12}	q_observed_6	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 3, "scale": 10}, "right": {"numeral": 25, "scale": 100}}; verified(strategy_trace_ok).

δ : recorded transitions

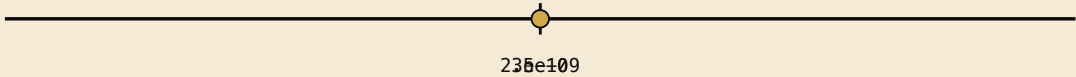
From	Action	To	Provenance
q_init	init	q_identify_decimal_units	smr_decimal_fraction_compare.pl:48
q_identify_decimal_units	scales_seen_but_not_coordinated	q_express_as_fraction	smr_decimal_fraction_compare.pl:48

From	Action	To	Provenance
q_express_as_fraction	omitted	q_scale_loss	smr_decimal_fraction_compare.pl:48
q_scale_loss	compare_written_numerals	q_compare_decimal_magnitudes	smr_decimal_fraction_compare.pl:48
q_compare_decimal_magnitudes	compare	q_emit	smr_decimal_fraction_compare.pl:48
q_emit	emit	q_accept	smr_decimal_fraction_compare.pl:48
q_init	init	q_observed_1	observed on {"kind": "decimal_pair", "left": {"numeral": 3, "scale": 10}, "right": {"numeral": 25, "scale": 100}} observed on {"kind": "decimal_pair", "left": {"numeral": 4, "scale": 10}, "right": {"numeral": 25, "scale": 100}} (derived template)
q_observed_1	scales_seen_but_not_coordinated	q_observed_2	observed on {"kind": "decimal_pair", "left": {"numeral": 3, "scale": 10}, "right": {"numeral": 25, "scale": 100}} observed on {"kind": "decimal_pair", "left": {"numeral": 4, "scale": 10}, "right": {"numeral": 25, "scale": 100}} (derived template)
q_observed_2	omitted	q_observed_3	observed on {"kind": "decimal_pair", "left": {"numeral": 3, "scale": 10}, "right": {"numeral": 25, "scale": 100}} observed on {"kind": "decimal_pair", "left": {"numeral": 4, "scale": 10}, "right": {"numeral": 25, "scale": 100}} (derived template)
q_observed_3	compare_written_numerals	q_observed_4	observed on {"kind": "decimal_pair", "left": {"numeral": 3, "scale": 10}, "right": {"numeral": 25, "scale": 100}} observed on {"kind": "decimal_pair", "left": {"numeral": 4, "scale": 10}, "right": {"numeral": 25, "scale": 100}} (derived template)
q_observed_4	compare	q_observed_5	observed on {"kind": "decimal_pair", "left": {"numeral": 3, "scale": 10}, "right": {"numeral": 25, "scale": 100}} observed on {"kind": "decimal_pair", "left": {"numeral": 4, "scale": 10}, "right": {"numeral": 25, "scale": 100}} (derived template)

From	Action	To	Provenance
q_observed_5	viability	q_observed_6	observed on {"kind": "decimal_pair", "left": {"numeral": 3, "scale": 10}, "right": {"numeral": 25, "scale": 100}} observed on {"kind": "decimal_pair", "left": {"numeral": 4, "scale": 10}, "right": {"numeral": 25, "scale": 100}} (derived template)
q_observed_6	emit	q_accept	observed on {"kind": "decimal_pair", "left": {"numeral": 3, "scale": 10}, "right": {"numeral": 25, "scale": 100}} observed on {"kind": "decimal_pair", "left": {"numeral": 4, "scale": 10}, "right": {"numeral": 25, "scale": 100}} (derived template)

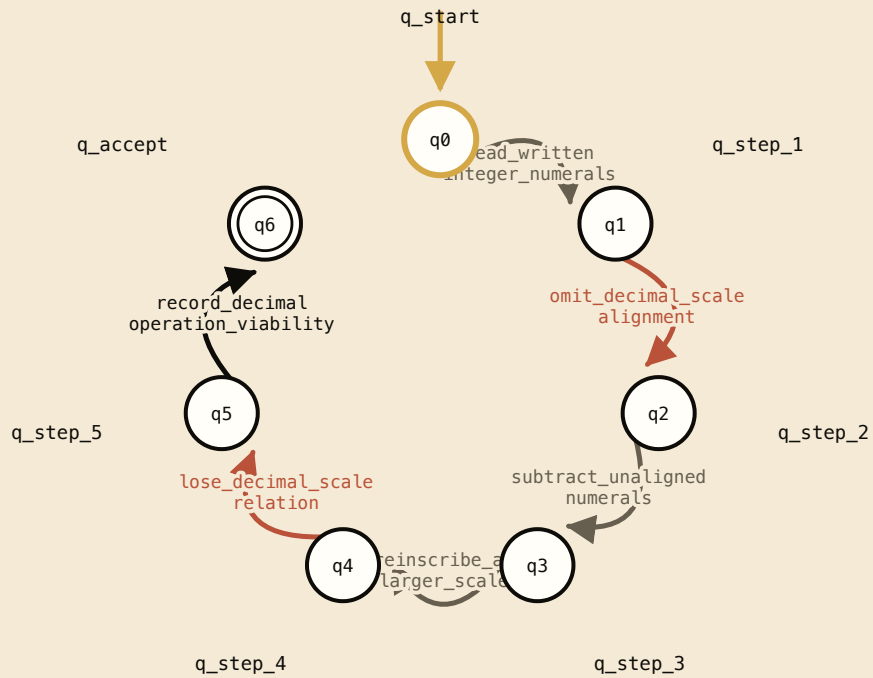
Computed structure: branching. one or more recorded states have multiple outgoing transitions; no loop edge is recorded. 13 states, 7 distinct actions, 1 branching states, 0 loop edges; 6 static rows and 14 observed rows.

11. decimal_subtract_unaligned_numerals



The run produced decimal(2,fractional_digits(2,1),tenths); the recorded expected result is decimal(2

The run produced decimal(2,fractional_digits(2,1),tenths); the recorded expected result is decimal(2,fractional_digits(2,1),tenths). The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \{lose_decimal_scale_relation, omit_decimal_scale_alignment, read_written_integer_numerals, record_decimal_operation_viability, reinscribe_at_larger_scale, subtract_unaligned_numerals\} \text{ (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

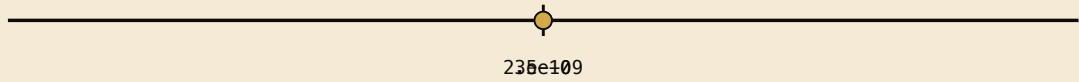
δ : recorded transitions

From	Action	To	Provenance
q_start	read_written_integer_numerals	q_step_1	decimal_action_pairs.pl:326 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_1	omit_decimal_scale_alignment	q_step_2	decimal_action_pairs.pl:326 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_2	subtract_unaligned_numerals	q_step_3	decimal_action_pairs.pl:326 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_step_3	reinscribe_at_larger_scale	q_step_4	decimal_action_pairs.pl:326 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_4	lose_decimal_scale_relation	q_step_5	decimal_action_pairs.pl:326 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_5	record_decimal_operation_viability	q_accept	decimal_action_pairs.pl:326 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

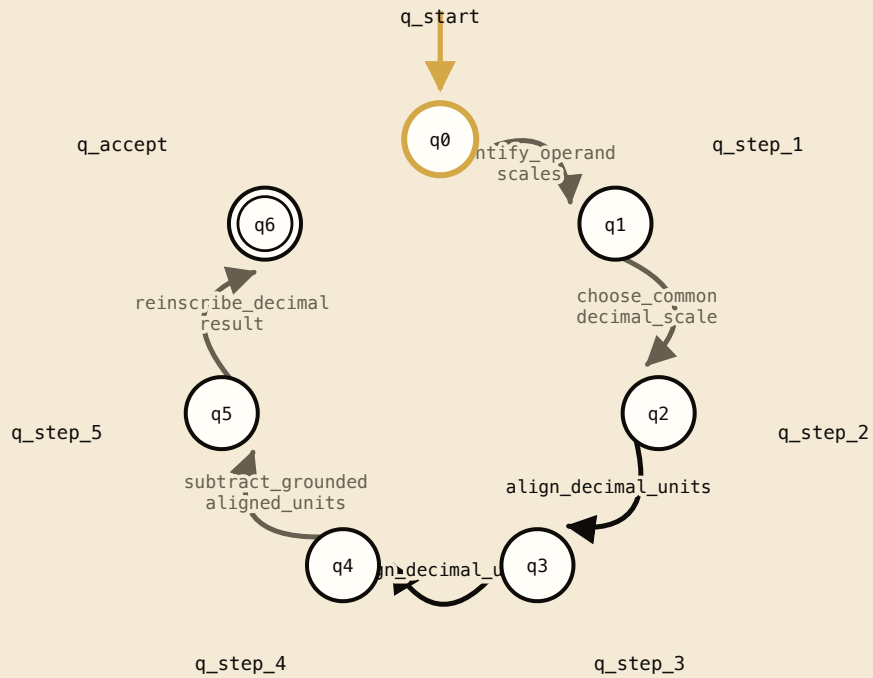
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

12. decimal_subtraction_by_aligned_units



The run produced `decimal(2,fractional_digits(2,1),tenths)`.

The run produced `decimal(2,fractional_digits(2,1),tenths)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ align_decimal_units, choose_common_decimal_scale, identify_operand_scales, reinscribe_decimal_result, subtract_grounded_aligned_units (5 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}}; verified(strategy_trace_ok).

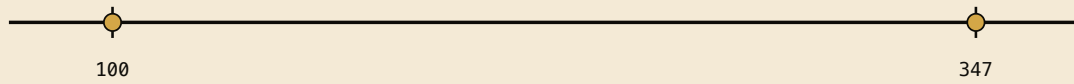
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_operand_scales	q_step_1	decimal_action_pairs.pl:291 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_1	choose_common_decimal_scale	q_step_2	decimal_action_pairs.pl:291 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_2	align_decimal_units	q_step_3	decimal_action_pairs.pl:291 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

From	Action	To	Provenance
q_step_3	align_decimal_units	q_step_4	decimal_action_pairs.pl:291 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_4	subtract_groundaligned_units	q_step_5	decimal_action_pairs.pl:291 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)
q_step_5	reinscribe_decimal_result	q_accept	decimal_action_pairs.pl:291 observed on {"kind": "decimal_pair", "left": {"numeral": 25, "scale": 10}, "right": {"numeral": 3, "scale": 10}} observed on {"kind": "decimal_pair", "left": {"numeral": 26, "scale": 10}, "right": {"numeral": 3, "scale": 10}} (derived template)

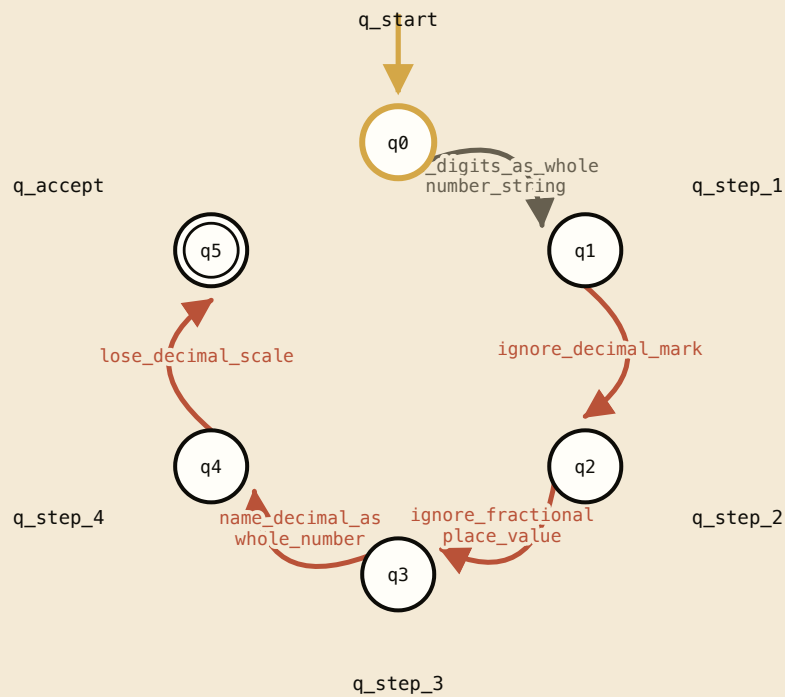
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 5 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

13. decimal_whole_number_reading



The run produced `whole_number(347)`; the recorded expected result is `decimal(3,fractional_digits(47,2`

The run produced `whole_number(347)`; the recorded expected result is `decimal(3,fractional_digits(47,2),hundredths)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ ignore_decimal_mark, ignore_fractional_place_value, lose_decimal_scale, name_decimal_as_whole_number, see_digits_as_whole_number_string (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

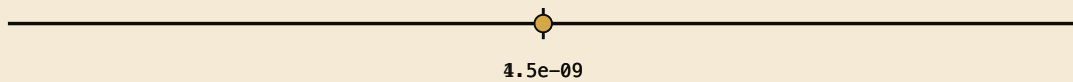
Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 347, "b": 100}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	see_digits_as_whole_number_string	q_step_1	decimal_action_pairs.pl:97 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)
q_step_1	ignore_decimal_mark	q_step_2	decimal_action_pairs.pl:97 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)
q_step_2	ignore_fractional_place_value	q_step_3	decimal_action_pairs.pl:97 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)
q_step_3	name_decimal_as_whole_number	q_step_4	decimal_action_pairs.pl:97 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)
q_step_4	lose_decimal_scale	q_accept	decimal_action_pairs.pl:97 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)

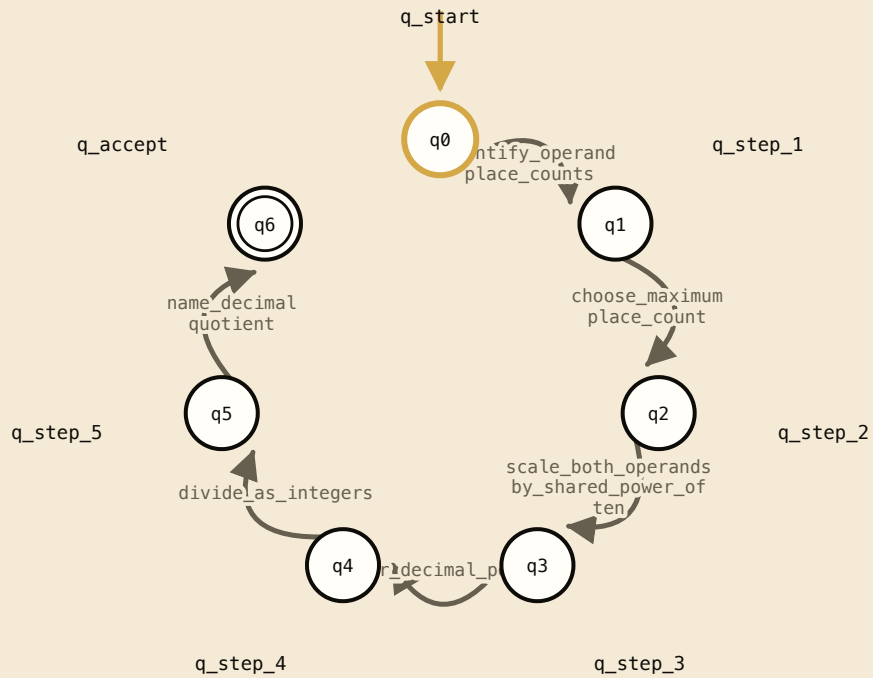
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

14. ecuadorian_decimal_long_division



The run produced `decimal_quotient(3,0)`.

The run produced `decimal_quotient(3,0)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ choose_maximum_place_count, clear_decimal_points, divide_as_integers, identify_operand_place_counts, name_decimal_quotient, scale_both_operands_by_shared_power_of_ten (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "decimal_pair", "left": {"numeral": "integer", "scale": "positive_integer"}, "right": {"numeral": "positive_integer", "scale": "positive_integer"}}; example {"kind": "decimal_pair", "left": {"numeral": 45, "scale": 10}, "right": {"numeral": 15, "scale": 10}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_operand_place_counts	q_step_1	decimal_action_pairs.pl:481 observed on {"kind": "decimal_pair", "left": {"numeral": 45, "scale": 10}, "right": {"numeral": 15, "scale": 10}}
q_step_1	choose_maximum_place_count	q_step_2	decimal_action_pairs.pl:481 observed on {"kind": "decimal_pair", "left": {"numeral": 45, "scale": 10}, "right": {"numeral": 15, "scale": 10}}
q_step_2	scale_both_operands_by_shared_power_of_ten	q_step_3	decimal_action_pairs.pl:481 observed on {"kind": "decimal_pair", "left": {"numeral": 45, "scale": 10}, "right": {"numeral": 15, "scale": 10}}
q_step_3	clear_decimal_points	q_step_4	decimal_action_pairs.pl:481 observed on {"kind": "decimal_pair", "left": {"numeral": 45, "scale": 10}, "right": {"numeral": 15, "scale": 10}}

From	Action	To	Provenance
q_step_4	divide_as_integers	q_step_5	decimal_action_pairs.pl:481 observed on {"kind": "decimal_pair", "left": {"numeral": 45, "scale": 10}, "right": {"numeral": 15, "scale": 10}}
q_step_5	name_decimal_quotient	q_accepted	decimal_action_pairs.pl:481 observed on {"kind": "decimal_pair", "left": {"numeral": 45, "scale": 10}, "right": {"numeral": 15, "scale": 10}}

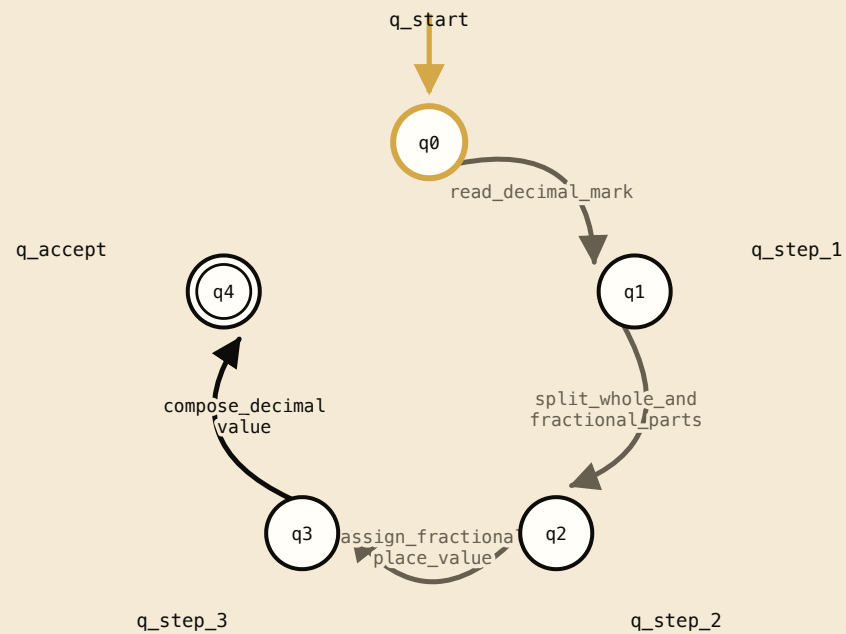
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 6 observed rows.

15. positional_decimal_reading



The run produced `decimal(3,fractional_digits(47,2),hundredths)`.

The run produced `decimal(3,fractional_digits(47,2),hundredths)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ assign_fractional_place_value, compose_decimal_value, read_decimal_mark, split_whole_and_fractional_parts (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

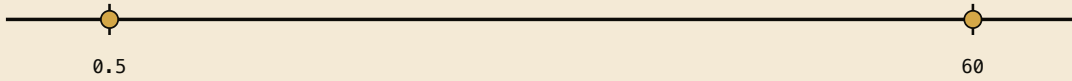
Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 347, "b": 100}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	read_decimal_mark	q_step_1	decimal_action_pairs.pl:77 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)
q_step_1	split_whole_and_fractional_parts	q_step_2	decimal_action_pairs.pl:77 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)
q_step_2	assign_fractional_place_value	q_step_3	decimal_action_pairs.pl:77 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)
q_step_3	compose_decimal_value	q_accept	decimal_action_pairs.pl:77 observed on {"a": 347, "b": 100} observed on {"a": 348, "b": 100} (derived template)

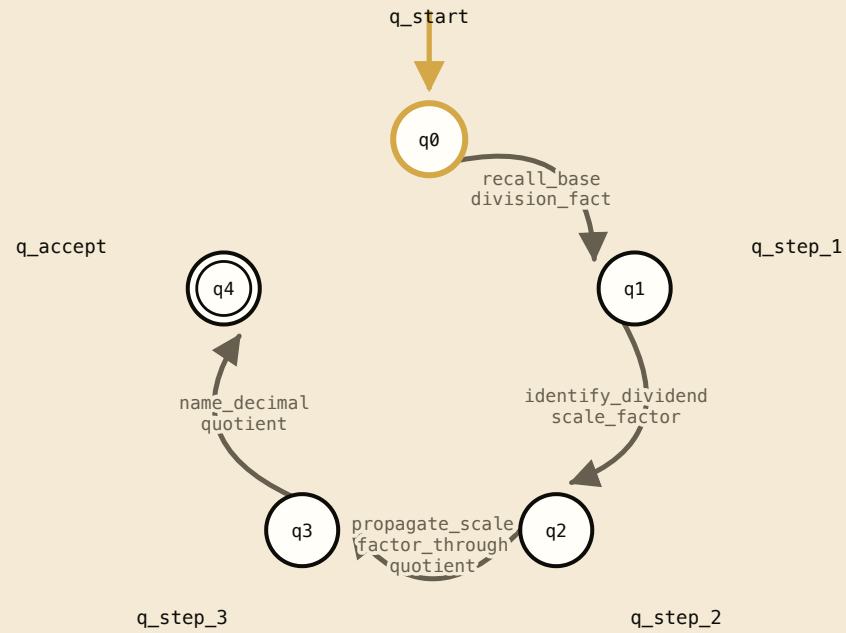
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

16. recalled_result_scaling



The run produced `decimal_quotient(120,0)`.

The run produced `decimal_quotient(120,0)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{identify_dividend_scale_factor, name_decimal_quotient, propagate_scale_factor_through_quotient, recall_base_division_fact\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

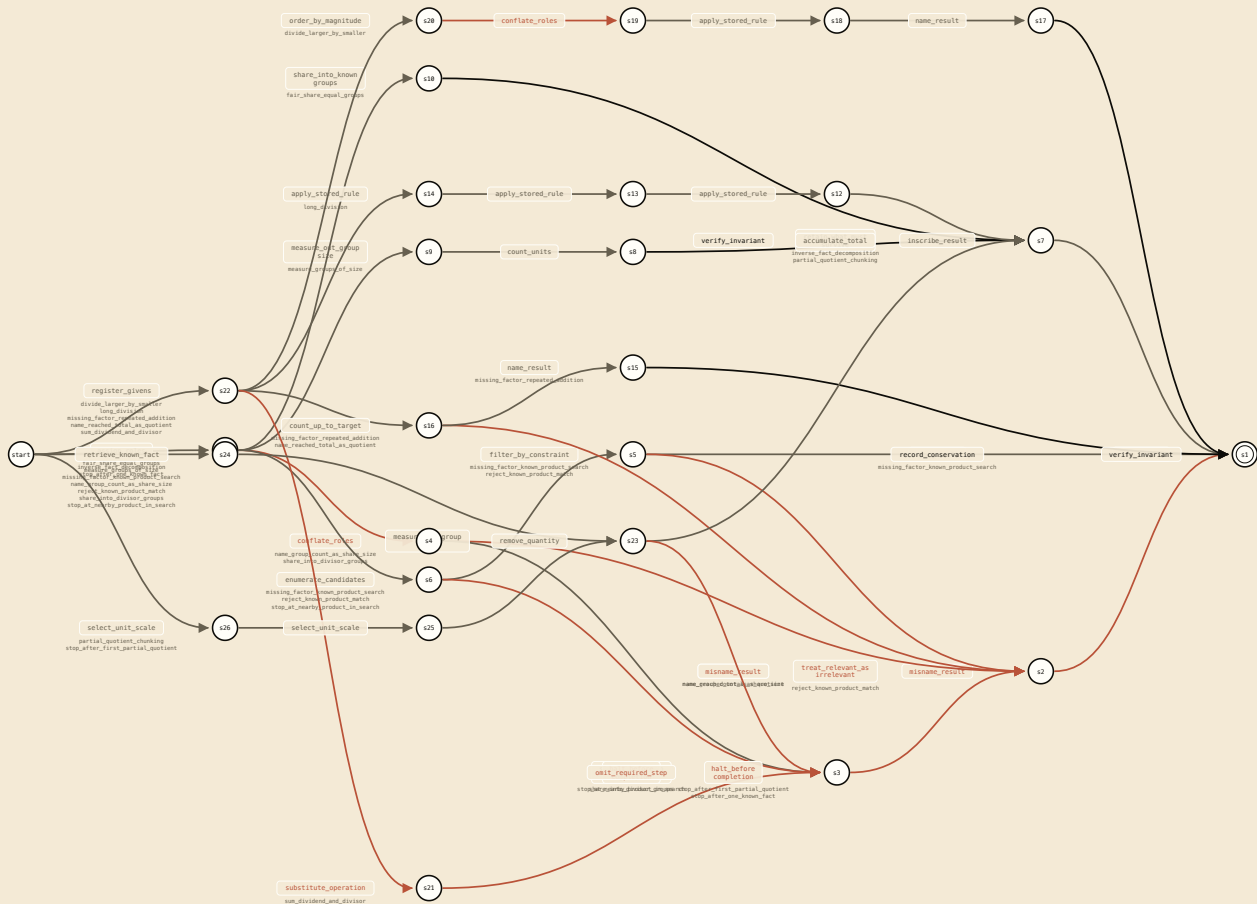
Input contract: {"kind": "decimal_pair", "left": {"numeral": "positive_integer", "scale": 1}, "right": {"numeral": "positive_integer", "scale": 1}}; example {"kind": "decimal_pair", "left": {"numeral": 600, "scale": 1}, "right": {"numeral": 5, "scale": 1}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	recall_base_division_fact	q_step_1	decimal_action_pairs.pl:513 observed on {"kind": "decimal_pair", "left": {"numeral": 600, "scale": 1}, "right": {"numeral": 5, "scale": 1}}
q_step_1	identify_dividend_scale_factor	q_step_2	decimal_action_pairs.pl:513 observed on {"kind": "decimal_pair", "left": {"numeral": 600, "scale": 1}, "right": {"numeral": 5, "scale": 1}}
q_step_2	propagate_scale_factor_through_quotient	q_step_3	decimal_action_pairs.pl:513 observed on {"kind": "decimal_pair", "left": {"numeral": 600, "scale": 1}, "right": {"numeral": 5, "scale": 1}}
q_step_3	name_decimal_quotient	q_accept	decimal_action_pairs.pl:513 observed on {"kind": "decimal_pair", "left": {"numeral": 600, "scale": 1}, "right": {"numeral": 5, "scale": 1}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

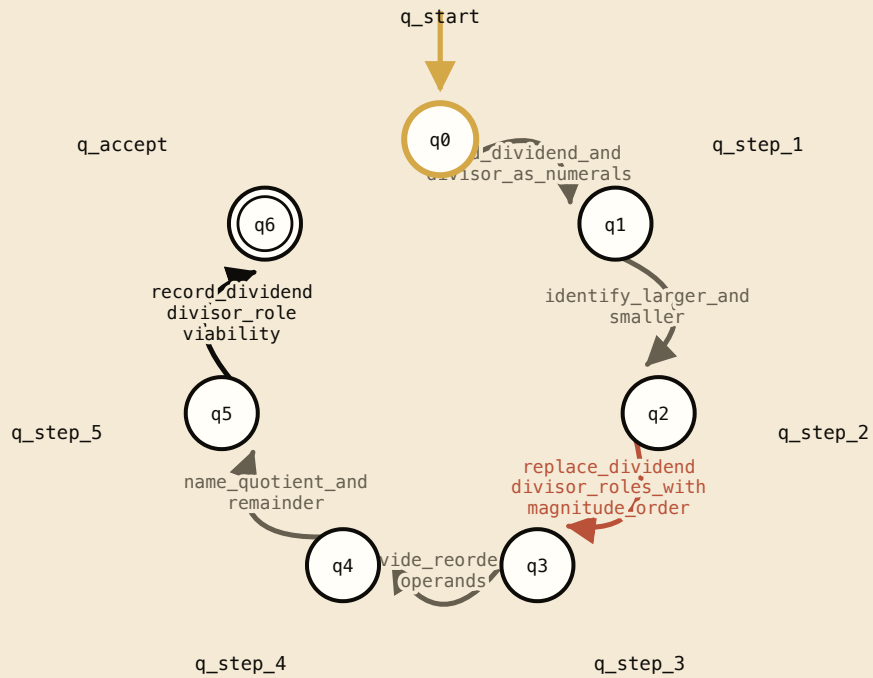
division



Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. divide_larger_by_smaller

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ divide_reordered_operands, identify_larger_and_smaller,
name_quotient_and_remainder, read_dividend_and_divisor_as_numerals,
record_dividend_divisor_role_viability,
replace_dividend_divisor_roles_with_magnitude_order (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "positive_integer", "b": "positive_integer"}; example {"a": 15, "b": 100}; verified(strategy_trace_ok).

δ : recorded transitions

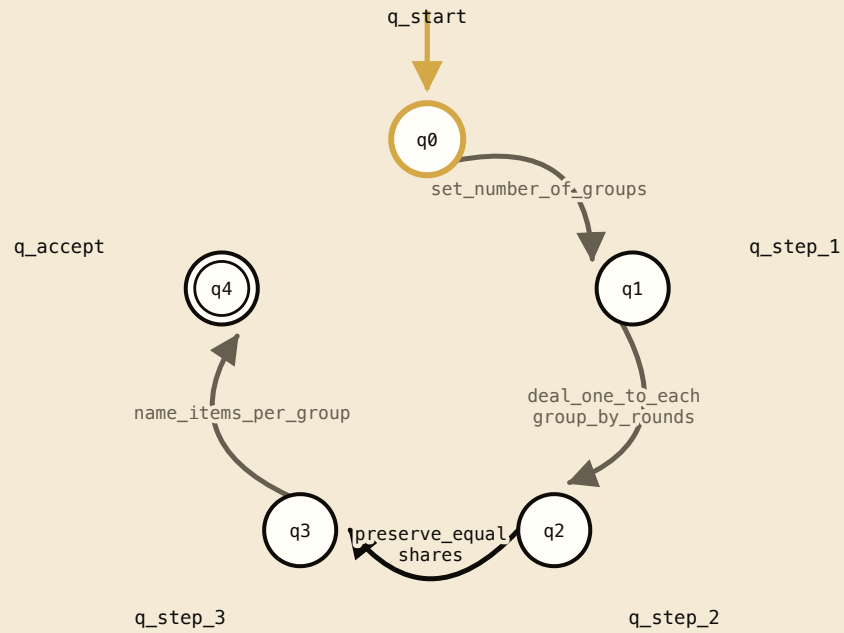
From	Action	To	Provenance
q_start	read_dividend_and_divisor_as_numerals	q_step_1	smr_div_action_pairs.pl:94 observed on {"a": 15, "b": 100} observed on {"a": 16, "b": 100} (derived template)
q_step_1	identify_larger_and_smaller	q_step_2	smr_div_action_pairs.pl:94 observed on {"a": 15, "b": 100} observed on {"a": 16, "b": 100} (derived template)
q_step_2	replace_dividend_divisor_roles_with_magnitude_order	q_step_3	smr_div_action_pairs.pl:94 observed on {"a": 15, "b": 100} observed on {"a": 16, "b": 100} (derived template)
q_step_3	divide_reordered_operands	q_step_4	smr_div_action_pairs.pl:94 observed on {"a": 15, "b": 100} observed on {"a": 16, "b": 100} (derived template)
q_step_4	name_quotient_and_remainder	q_step_5	smr_div_action_pairs.pl:94 observed on {"a": 15, "b": 100} observed on {"a": 16, "b": 100} (derived template)

From	Action	To	Provenance
q_step_5	record_dividend_divisor_role_viability	q_accept	smr_div_action_pairs.pl:94 observed on {"a": 15, "b": 100} observed on {"a": 16, "b": 100} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

2. fair_share_equal_groups

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{deal_one_to_each_group_by_rounds}, \text{name_items_per_group}, \text{preserve_equal_shares}, \text{set_number_of_groups}\} \text{ (4 distinct)}$$

$$q_0 = q_{\text{start}}$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 12, "b": 3};
verified(strategy_trace_ok).

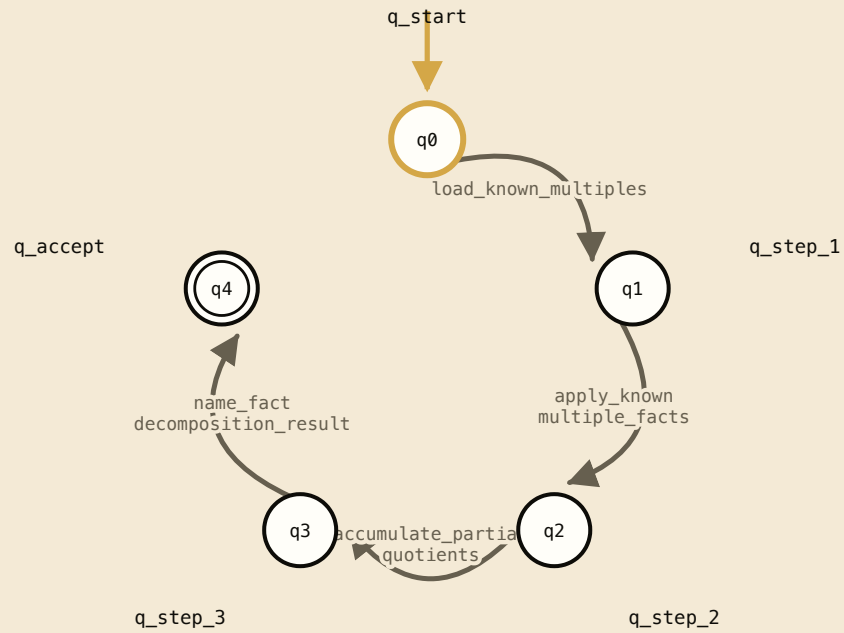
δ : recorded transitions

From	Action	To	Provenance
q_start	set_number_of_groups	q_step_1	smr_div_action_pairs.pl:158 observed on {"a": 12, "b": 3}
q_step_1	deal_one_to_each_group_by_rounds	q_step_2	smr_div_action_pairs.pl:158 observed on {"a": 12, "b": 3}
q_step_2	preserve_equal_shares	q_step_3	smr_div_action_pairs.pl:158 observed on {"a": 12, "b": 3}
q_step_3	name_items_per_group	q_accept	smr_div_action_pairs.pl:158 observed on {"a": 12, "b": 3}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

3. inverse_fact_decomposition

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{accumulate_partial_quotients}, \text{apply_known_multiple_facts}, \text{load_known_multiples}, \text{name_fact_decomposition_result}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 47, "b": 3};
verified(strategy_trace_ok).

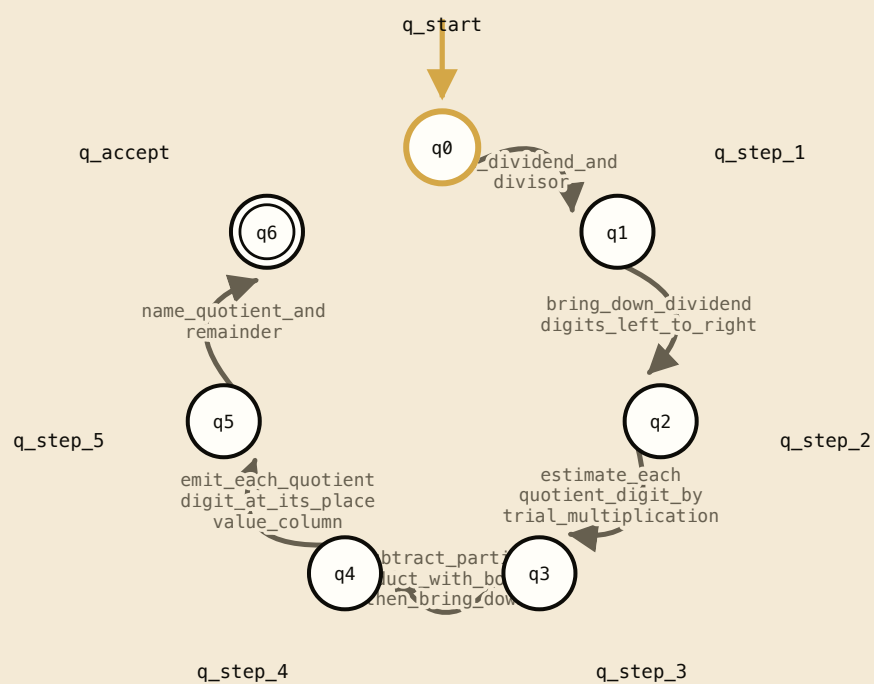
δ : recorded transitions

From	Action	To	Provenance
q_start	load_known_multiples	q_step_1	smr_div_action_pairs.pl:263 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)
q_step_1	apply_known_multiple_facts	q_step_2	smr_div_action_pairs.pl:263 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)
q_step_2	accumulate_partial_quotients	q_step_3	smr_div_action_pairs.pl:263 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)
q_step_3	name_fact_decomposition_result	q_accept	smr_div_action_pairs.pl:263 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

4. long_division

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

Σ bring_down_dividend_digits_left_to_right,
 emit_each_quotient_digit_at_its_place_value_column,
 estimate_each_quotient_digit_by_trial_multiplication,
 name_quotient_and_remainder, set_dividend_and_divisor,
 subtract_partial_product_with_borrow_then_bring_down (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
 verified(strategy_trace_ok).

δ : recorded transitions

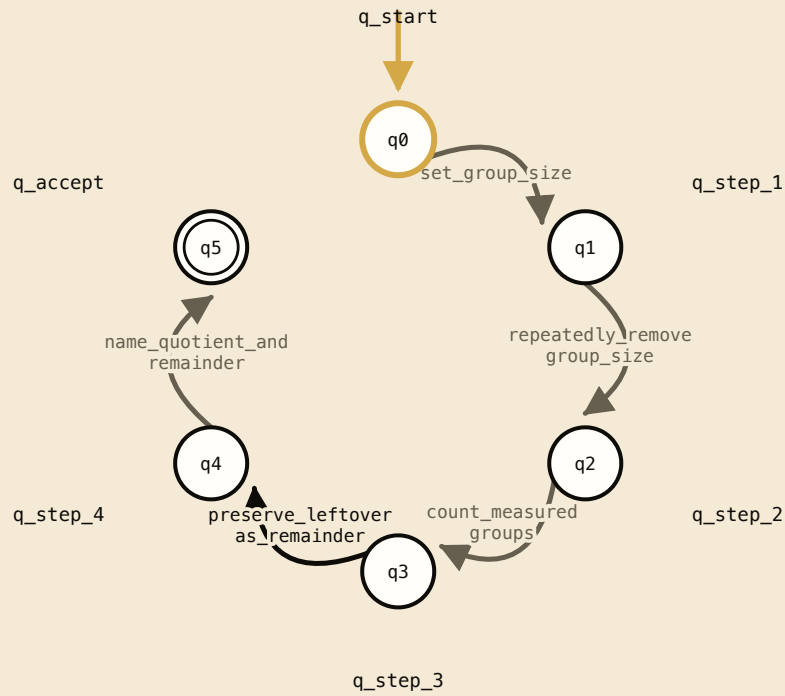
From	Action	To	Provenance
q_start	set_dividend_and_divisor	q_step_1	smr_div_action_pairs.pl:448 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	bring_down_dividend_digits_left_to_right	q_step_2	smr_div_action_pairs.pl:448 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	estimate_each_quotient_digit_by_trial_multiplication	q_step_3	smr_div_action_pairs.pl:448 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_3	subtract_partial_product_with_borrow_then_bring_down	q_step_4	smr_div_action_pairs.pl:448 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	emit_each_quotient_digit_at_its_place_value_column	q_step_5	smr_div_action_pairs.pl:448 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_5	name_quotient_and_remainder	q_accept	smr_div_action_pairs.pl:448 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

5. measure_groups_of_size

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{\text{count_measured_groups}, \text{name_quotient_and_remainder}, \text{preserve_leftover_as_remainder}, \text{repeatedly_remove_group_size}, \text{set_group_size}\} \text{ (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

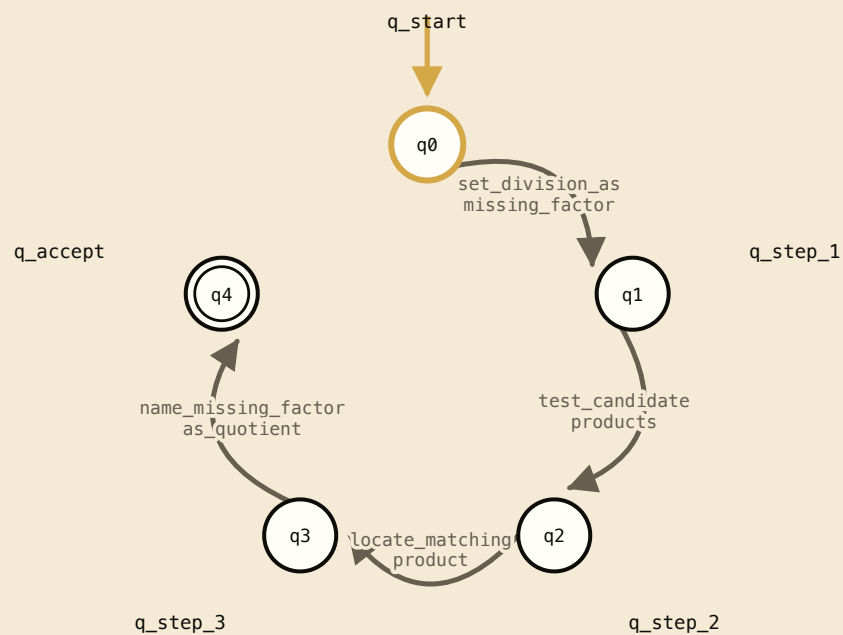
δ : recorded transitions

From	Action	To	Provenance
q_start	set_group_size	q_step_1	smr_div_action_pairs.pl:70 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	repeatedly_remove_group_size	q_step_2	smr_div_action_pairs.pl:70 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	count_measured_groups	q_step_3	smr_div_action_pairs.pl:70 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	preserve_leftover_as_remainder	q_step_4	smr_div_action_pairs.pl:70 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	name_quotient_and_remainder	q_accept	smr_div_action_pairs.pl:70 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

6. missing_factor_known_product_search

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$M = (Q, \Sigma, \delta, q_0, F)$

$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma \quad \text{locate_matching_product, name_missing_factor_as_quotient,}$
 $\text{set_division_as_missing_factor, test_candidate_products (4 distinct)}$

$q_0 \quad q_0 = \text{q_start}$

$F \quad \{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 48, "b": 6};
verified(strategy_trace_ok).

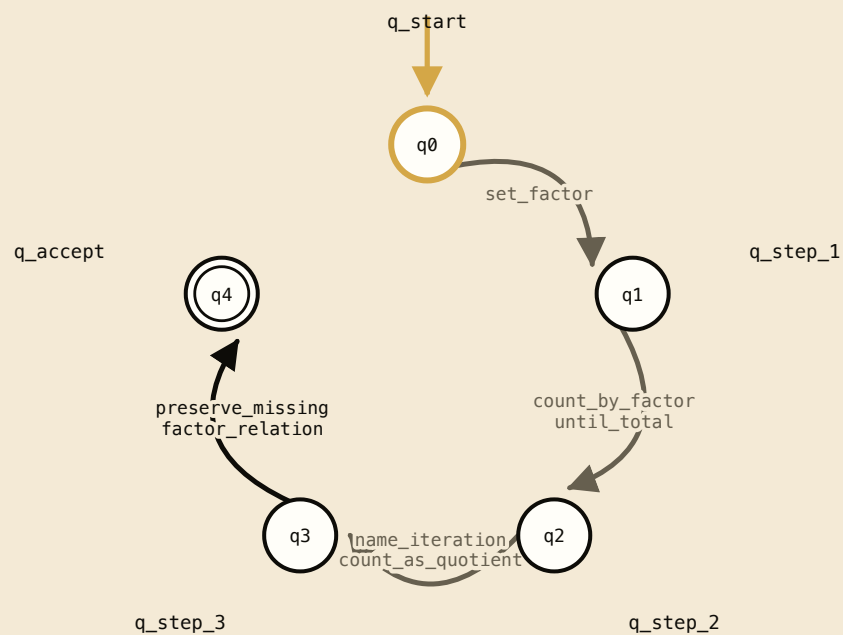
δ : recorded transitions

From	Action	To	Provenance
q_start	set_division_as_missing_factor	q_step_1	smr_div_action_pairs.pl:376 observed on {"a": 48, "b": 6}
q_step_1	test_candidate_products	q_step_2	smr_div_action_pairs.pl:376 observed on {"a": 48, "b": 6}
q_step_2	locate_matching_product	q_step_3	smr_div_action_pairs.pl:376 observed on {"a": 48, "b": 6}
q_step_3	name_missing_factor_as_quotient	q_accept	smr_div_action_pairs.pl:376 observed on {"a": 48, "b": 6}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

7. missing_factor_repeated_addition

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ count_by_factor_until_total, name_iteration_count_as_quotient,
preserve_missing_factor_relation, set_factor (4 distinct)

q_0 $q_0 = q_{\text{start}}$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 12, "b": 3};
verified(strategy_trace_ok).

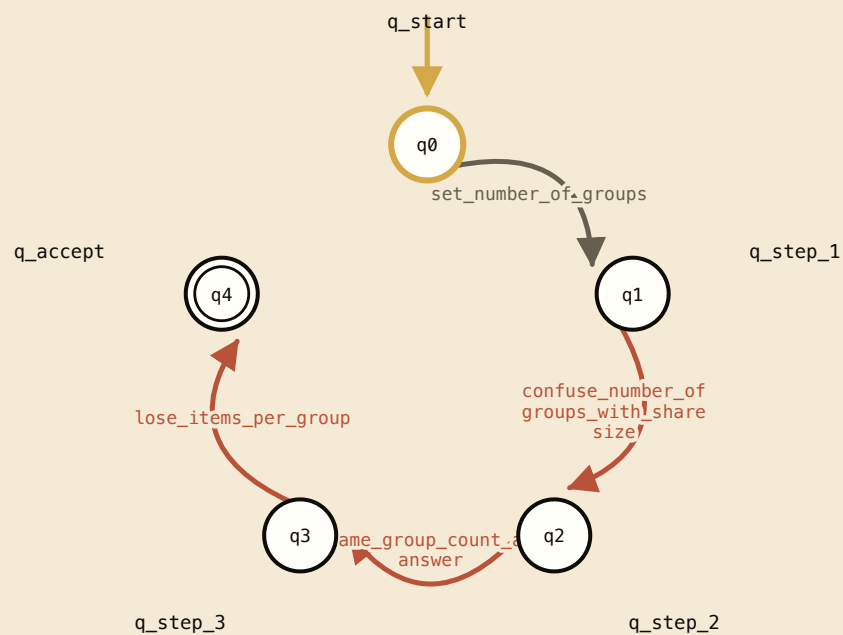
δ : recorded transitions

From	Action	To	Provenance
q_start	set_factor	q_step_1	smr_div_action_pairs.pl:209 observed on {"a": 12, "b": 3}
q_step_1	count_by_factor_until_total	q_step_2	smr_div_action_pairs.pl:209 observed on {"a": 12, "b": 3}
q_step_2	name_iteration_count_as_quotient	q_step_3	smr_div_action_pairs.pl:209 observed on {"a": 12, "b": 3}
q_step_3	preserve_missing_factor_relation	q_accept	smr_div_action_pairs.pl:209 observed on {"a": 12, "b": 3}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

8. name_group_count_as_share_size

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ confuse_number_of_groups_with_share_size, lose_items_per_group,
name_group_count_as_answer, set_number_of_groups (4 distinct)

q_0 $q_0 = q_{\text{start}}$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 12, "b": 3};
verified(strategy_trace_ok).

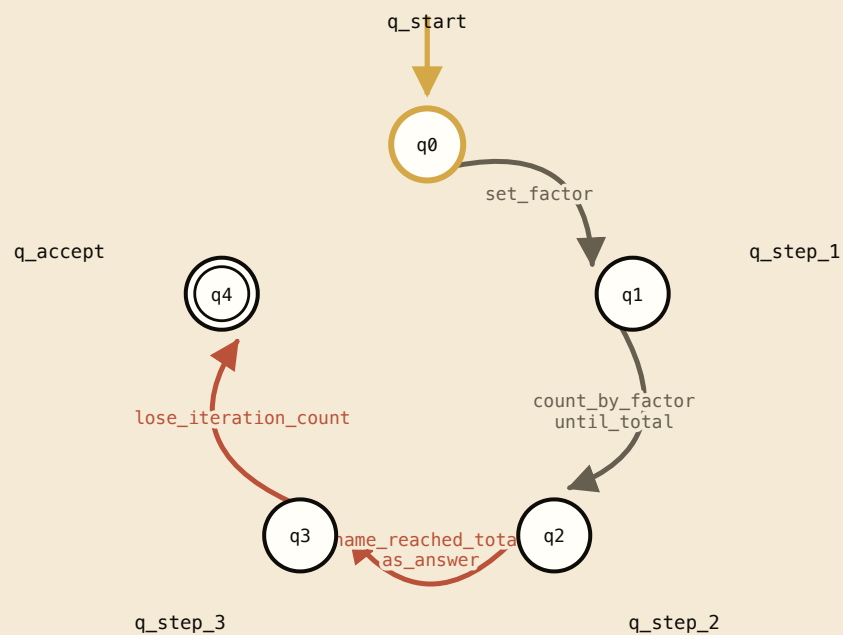
δ : recorded transitions

From	Action	To	Provenance
q_start	set_number_of_groups	q_step_1	smr_div_action_pairs.pl:181 observed on {"a": 12, "b": 3}
q_step_1	confuse_number_of_groups_with_share_size	q_step_2	smr_div_action_pairs.pl:181 observed on {"a": 12, "b": 3}
q_step_2	name_group_count_as_answer	q_step_3	smr_div_action_pairs.pl:181 observed on {"a": 12, "b": 3}
q_step_3	lose_items_per_group	q_accept	smr_div_action_pairs.pl:181 observed on {"a": 12, "b": 3}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

9. name_reached_total_as_quotient

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ count_by_factor_until_total, lose_iteration_count, name_reached_total_as_answer, set_factor (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 12, "b": 3};
verified(strategy_trace_ok).

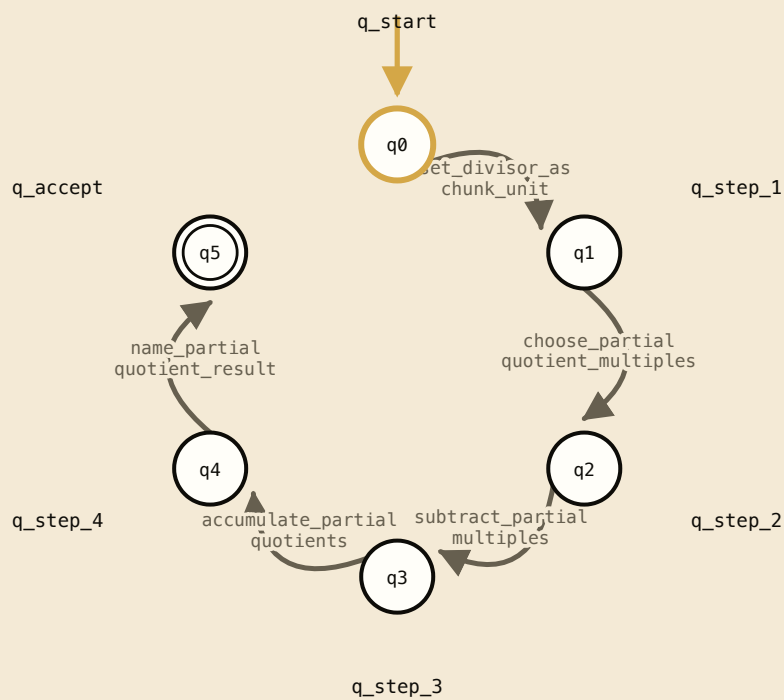
δ : recorded transitions

From	Action	To	Provenance
q_start	set_factor	q_step_1	smr_div_action_pairs.pl:233 observed on {"a": 12, "b": 3}
q_step_1	count_by_factor_until_total	q_step_2	smr_div_action_pairs.pl:233 observed on {"a": 12, "b": 3}
q_step_2	name_reached_total_as_answer	q_step_3	smr_div_action_pairs.pl:233 observed on {"a": 12, "b": 3}
q_step_3	lose_iteration_count	q_accept	smr_div_action_pairs.pl:233 observed on {"a": 12, "b": 3}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

10. partial_quotient_chunking

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$M = (Q, \Sigma, \delta, q_0, F)$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ accumulate_partial_quotients, choose_partial_quotient_multiples,
name_partial_quotient_result, set_divisor_as_chunk_unit,
subtract_partial_multiples (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

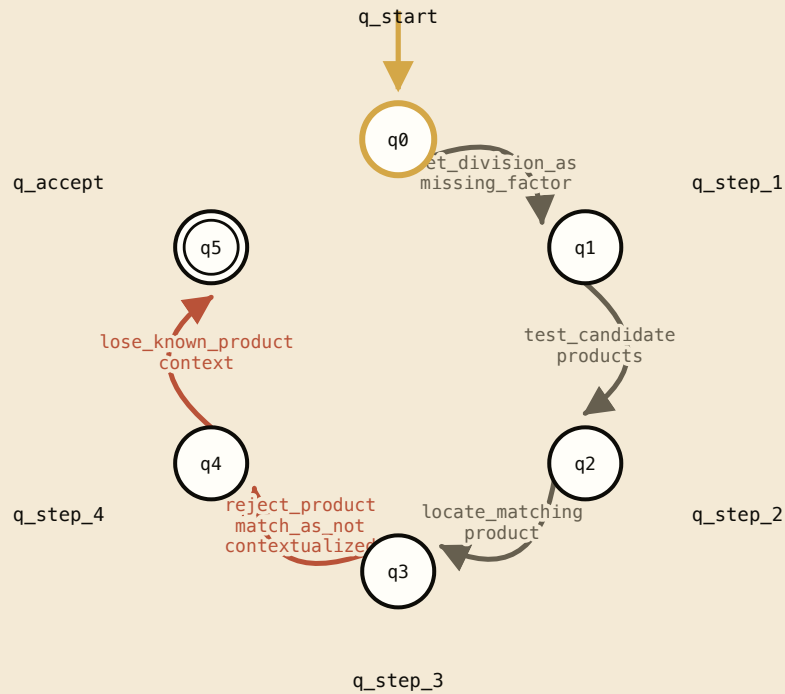
From	Action	To	Provenance
q_start	set_divisor_as_chunk_unit	q_step_1	smr_div_action_pairs.pl:320 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	choose_partial_quotient_multiples	q_step_2	smr_div_action_pairs.pl:320 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_2	subtract_partial_multiples	q_step_3	smr_div_action_pairs.pl:320 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	accumulate_partial_quotients	q_step_4	smr_div_action_pairs.pl:320 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	name_partial_quotient_result	q_accept	smr_div_action_pairs.pl:320 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

11. reject_known_product_match

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{\text{locate_matching_product}, \text{lose_known_product_context}, \text{reject_product_match_as_not_contextualized}, \text{set_division_as_missing_factor}, \text{test_candidate_products}\} \text{ (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 48, "b": 6};
verified(strategy_trace_ok).

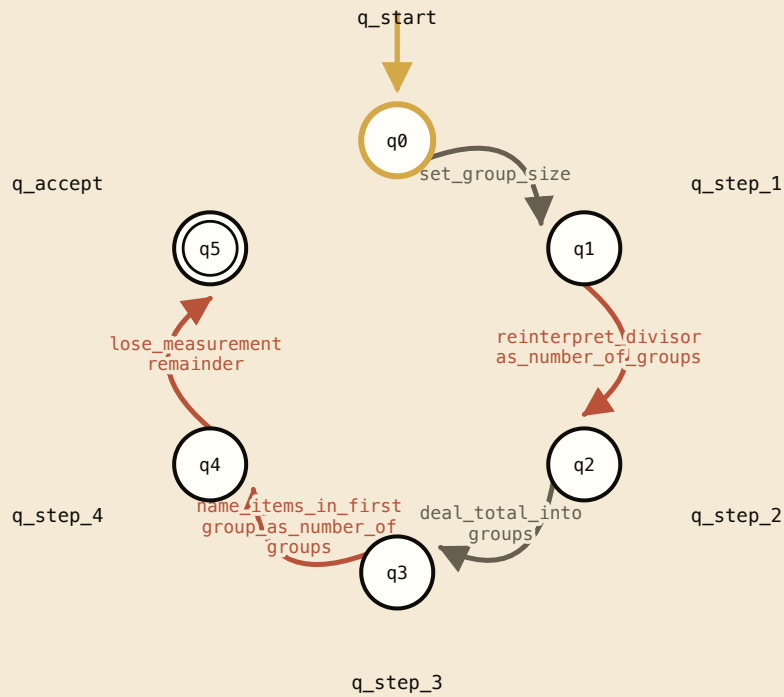
δ : recorded transitions

From	Action	To	Provenance
q_start	set_division_as_missing_factor	q_step_1	smr_div_action_pairs.pl:424 observed on {"a": 48, "b": 6}
q_step_1	test_candidate_products	q_step_2	smr_div_action_pairs.pl:424 observed on {"a": 48, "b": 6}
q_step_2	locate_matching_product	q_step_3	smr_div_action_pairs.pl:424 observed on {"a": 48, "b": 6}
q_step_3	reject_product_match_as_not_contextualized	q_step_4	smr_div_action_pairs.pl:424 observed on {"a": 48, "b": 6}
q_step_4	lose_known_product_context	q_accept	smr_div_action_pairs.pl:424 observed on {"a": 48, "b": 6}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

12. share_into_divisor_groups

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ deal_total_into_groups, lose_measurement_remainder,
name_items_in_first_group_as_number_of_groups,
reinterpret_divisor_as_number_of_groups, set_group_size (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

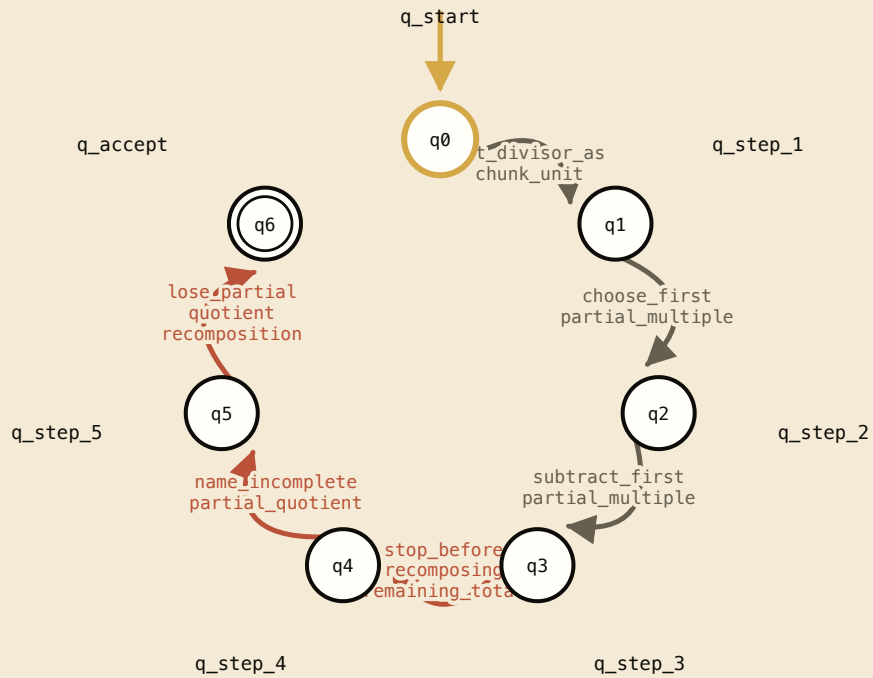
From	Action	To	Provenance
q_start	set_group_size	q_step_1	smr_div_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	reinterpret_divisor_as_number_of_groups	q_step_2	smr_div_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	deal_total_into_groups	q_step_3	smr_div_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	name_items_in_first_group_as_number_of_groups	q_step_4	smr_div_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_4	lose_measurement_remainder	q_accept	smr_div_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

13. stop_after_first_partial_quotient

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ choose_first_partial_multiple, lose_partial_quotient_recomposition, name_incomplete_partial_quotient, set_divisor_as_chunk_unit, stop_before_recomposing_remaining_total, subtract_first_partial_multiple (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 96, "b": 4};
verified(strategy_trace_ok).

δ : recorded transitions

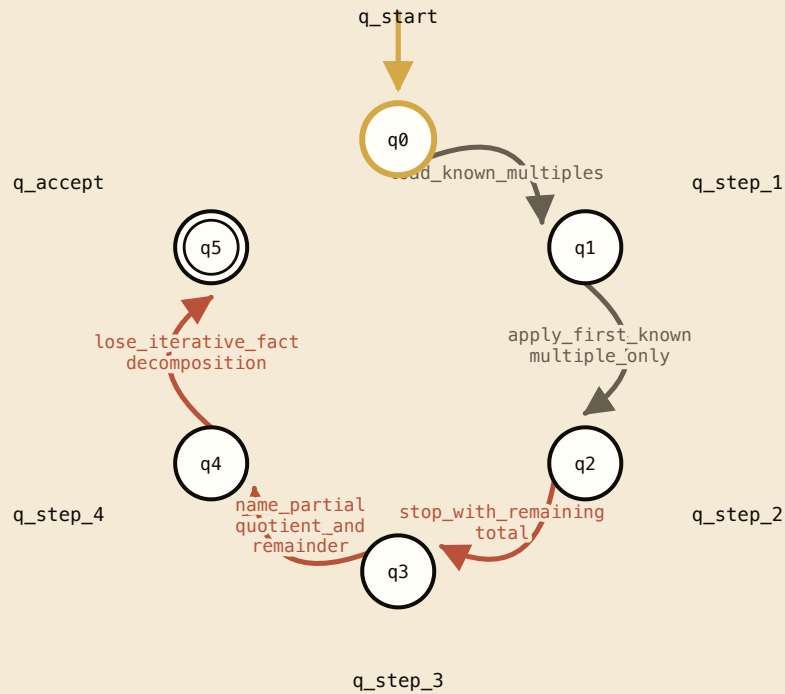
From	Action	To	Provenance
q_start	set_divisor_as_chunk_unit	q_step_1	smr_div_action_pairs.pl:342 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)
q_step_1	choose_first_partial_multiple	q_step_2	smr_div_action_pairs.pl:342 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)
q_step_2	subtract_first_partial_multiple	q_step_3	smr_div_action_pairs.pl:342 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)
q_step_3	stop_before_recomposing_remaining_total	q_step_4	smr_div_action_pairs.pl:342 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)
q_step_4	name_incomplete_partial_quotient	q_step_5	smr_div_action_pairs.pl:342 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)

From	Action	To	Provenance
q_step_5	lose_partial_quotient_recompositi on	q_accep t	smr_div_action_pairs.pl:342 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

14. stop_after_one_known_fact

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{\text{apply_first_known_multiple_only}, \text{load_known_multiples}, \text{lose_iterative_fact_decomposition}, \text{name_partial_quotient_and_remainder}, \text{stop_with_remaining_total}\} \text{ (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 47, "b": 3};
verified(strategy_trace_ok).

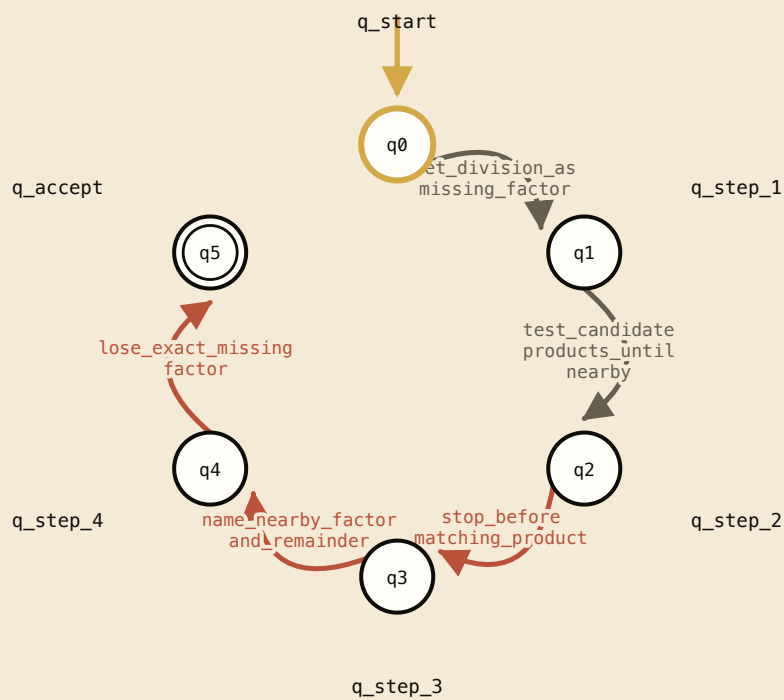
δ : recorded transitions

From	Action	To	Provenance
q_start	load_known_multiples	q_step_1	smr_div_action_pairs.pl:286 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)
q_step_1	apply_first_known_multiple_only	q_step_2	smr_div_action_pairs.pl:286 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)
q_step_2	stop_with_remaining_total	q_step_3	smr_div_action_pairs.pl:286 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)
q_step_3	name_partial_quotient_and_remainder	q_step_4	smr_div_action_pairs.pl:286 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)
q_step_4	lose_iterative_fact_decomposition	q_accept	smr_div_action_pairs.pl:286 observed on {"a": 47, "b": 3} observed on {"a": 48, "b": 3} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

15. stop_at_nearby_product_in_search

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ lose_exact_missing_factor, name_nearby_factor_and_remainder,
set_division_as_missing_factor, stop_before_matching_product,
test_candidate_products_until_nearby (5 distinct)

q_0 $q_0 = q_{\text{start}}$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 48, "b": 6};
verified(strategy_trace_ok).

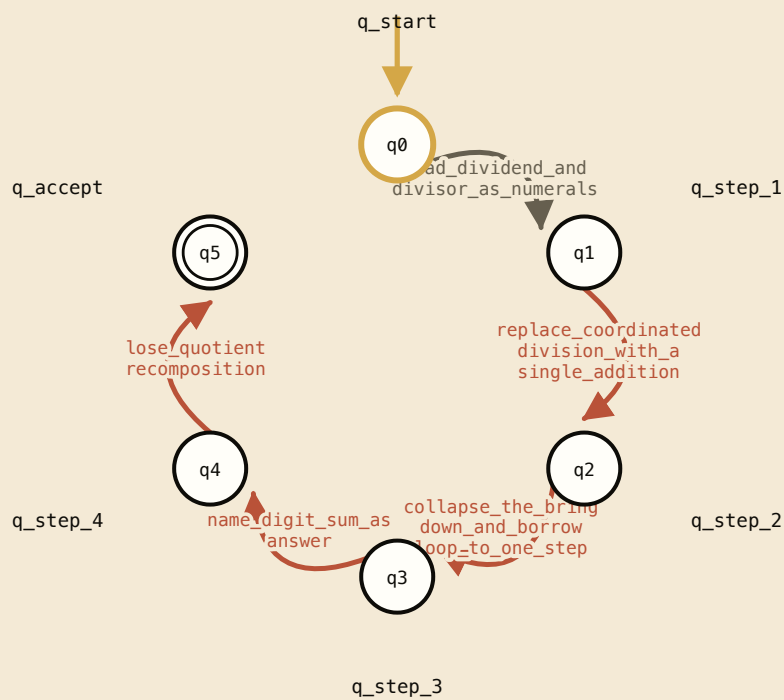
δ : recorded transitions

From	Action	To	Provenance
q_start	set_division_as_missing_factor	q_step_1	smr_div_action_pairs.pl:397 observed on {"a": 48, "b": 6}
q_step_1	test_candidate_products_until_nearby	q_step_2	smr_div_action_pairs.pl:397 observed on {"a": 48, "b": 6}
q_step_2	stop_before_matching_product	q_step_3	smr_div_action_pairs.pl:397 observed on {"a": 48, "b": 6}
q_step_3	name_nearby_factor_and_remainder	q_step_4	smr_div_action_pairs.pl:397 observed on {"a": 48, "b": 6}
q_step_4	lose_exact_missing_factor	q_accept	smr_div_action_pairs.pl:397 observed on {"a": 48, "b": 6}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

16. sum_dividend_and_divisor

No domain scene: the static scene emitter has no division form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$M = (Q, \Sigma, \delta, q_0, F)$

$Q \quad \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma \quad \text{collapse_the_bring_down_and_borrow_loop_to_one_step, lose_quotient_recomposition,}$
 $\text{name_digit_sum_as_answer, read_dividend_and_divisor_as_numerals,}$
 $\text{replace_coordinated_division_with_a_single_addition (5 distinct)}$

$q_0 \quad q_0 = \text{q_start}$

$F \quad \{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 96, "b": 4};
verified(strategy_trace_ok).

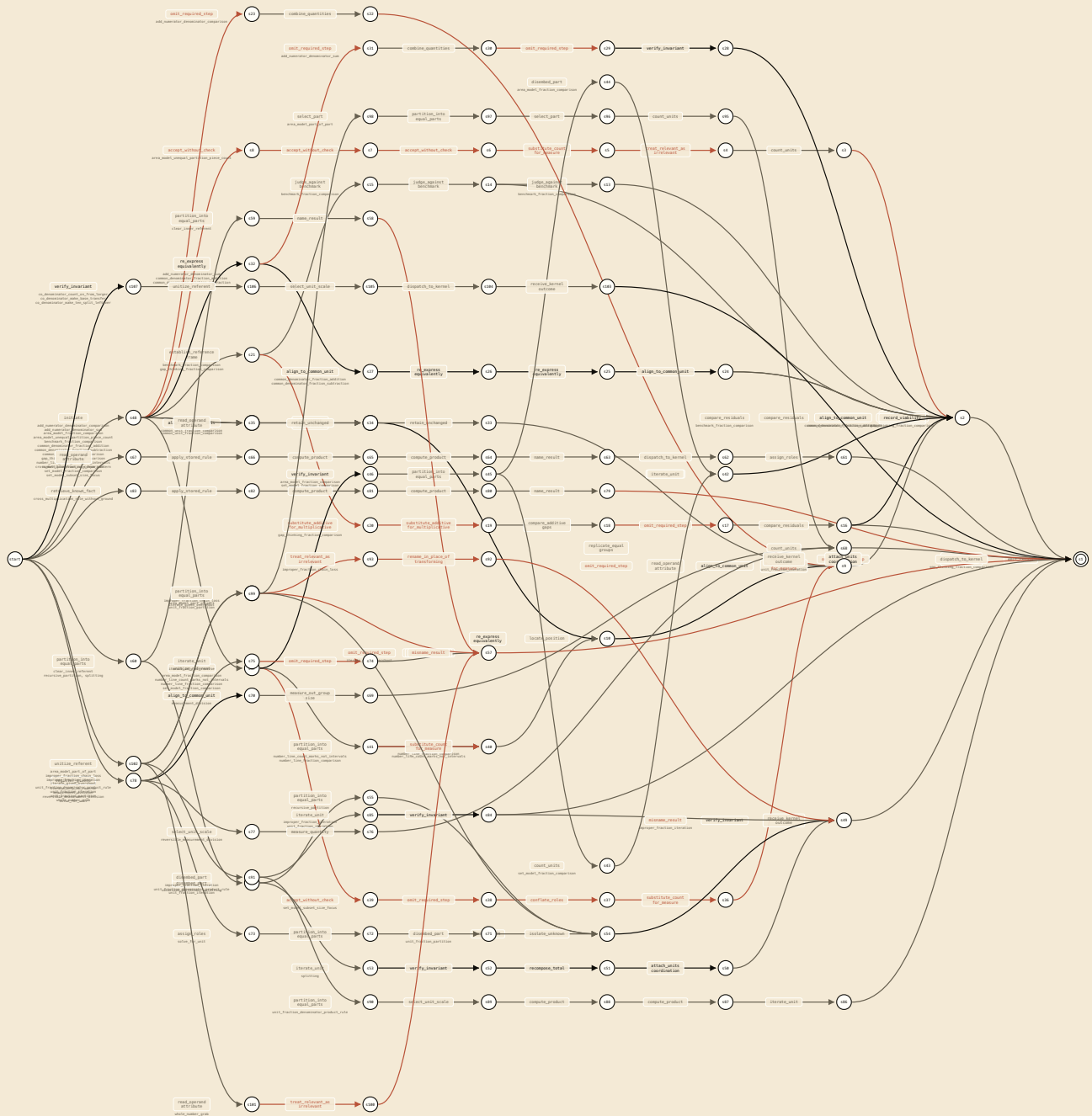
δ : recorded transitions

From	Action	To	Provenance
q_start	read_dividend_and_divisor_as_numerals	q_step_1	smr_div_action_pairs.pl:480 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)
q_step_1	replace_coordinated_division_with_a_single_addition	q_step_2	smr_div_action_pairs.pl:480 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)

From	Action	To	Provenance
q_step_2	collapse_the_bring_down_and_borrow_loop_to_one_step	q_step_3	smr_div_action_pairs.pl:480 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)
q_step_3	name_digit_sum_as_answer	q_step_4	smr_div_action_pairs.pl:480 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)
q_step_4	lose_quotient_recomposition	q_accept	smr_div_action_pairs.pl:480 observed on {"a": 96, "b": 4} observed on {"a": 97, "b": 4} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

fraction



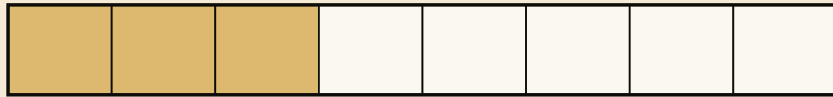
Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. add_numerator_denominator_comparison

left: 2/3

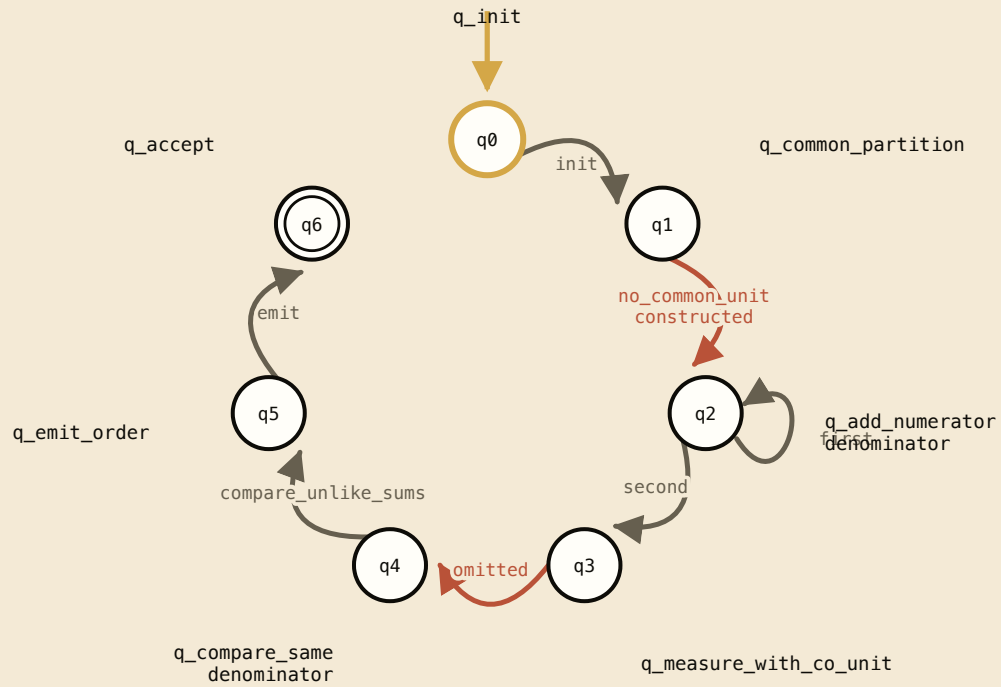


right: 3/8



The run produced less_than; the recorded expected result is greater_than.

The run produced less_than; the recorded expected result is greater_than. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma \quad \text{compare_unlike_sums, emit, first, init, no_common_unit_constructed, omitted, second (7 distinct)}$$

$$q_0 \quad q_0 = q_init$$

$$F \quad \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_common_partition	<i>common partitioning</i> (constructivist; Shin & Lee 2018) <i>find a common denominator</i> (van_de_walle; Van de Walle, ch. 15) <i>converts to common denominator</i> (clarke_roche; Clarke & Roche 2009)
q_2	q_add_numerator_denominator	<i>addition</i> (behr_post_lesch; Behr, Wachsmuth, Post & Lesh 1984)
q_3	q_measure_with_co_unit	<i>co-measurement unit</i> (constructivist; Shin & Lee 2018; Nabors 2003)
q_4	q_compare_same_denominator	<i>same denominator strategy</i> (van_de_walle; Van de Walle, ch. 15) <i>numerator and denominator strategy</i> (behr_post_lesch; Behr, Wachsmuth, Post & Lesh 1984) <i>denominator the same and compares numerator</i> (clarke_roche; Clarke & Roche 2009) <i>same denominator</i> (cramer_post_delmas; Cramer, Post & delMas 2002)
q_5	q_emit_order	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_common_partition	smr_frac_common_unit_compare.pl:43 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)

From	Action	To	Provenance
q_common_partition	no_common_unit_constructed	q_add_numerator_denominator	smr_frac_common_unit_compare.pl:43 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_add_numerator_denominator	first	q_add_numerator_denominator	smr_frac_common_unit_compare.pl:43 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_add_numerator_denominator	second	q_measure_with_co_unit	smr_frac_common_unit_compare.pl:43 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_measure_with_co_unit	omitted	q_compare_same_denominator	smr_frac_common_unit_compare.pl:43 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_compare_same_denominator	compare_unlike_sums	q_emit_order	smr_frac_common_unit_compare.pl:43 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_emit_order	emit	q_accept	smr_frac_common_unit_compare.pl:43 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)

Computed structure: branching_looping. the recorded graph contains both a branch state and a loop edge. 7 states, 7 distinct actions, 1 branching states, 1 loop edges; 7 static rows

and 14 observed rows.

2. add_numerator_denominator_sum

left: $\frac{3}{4}$

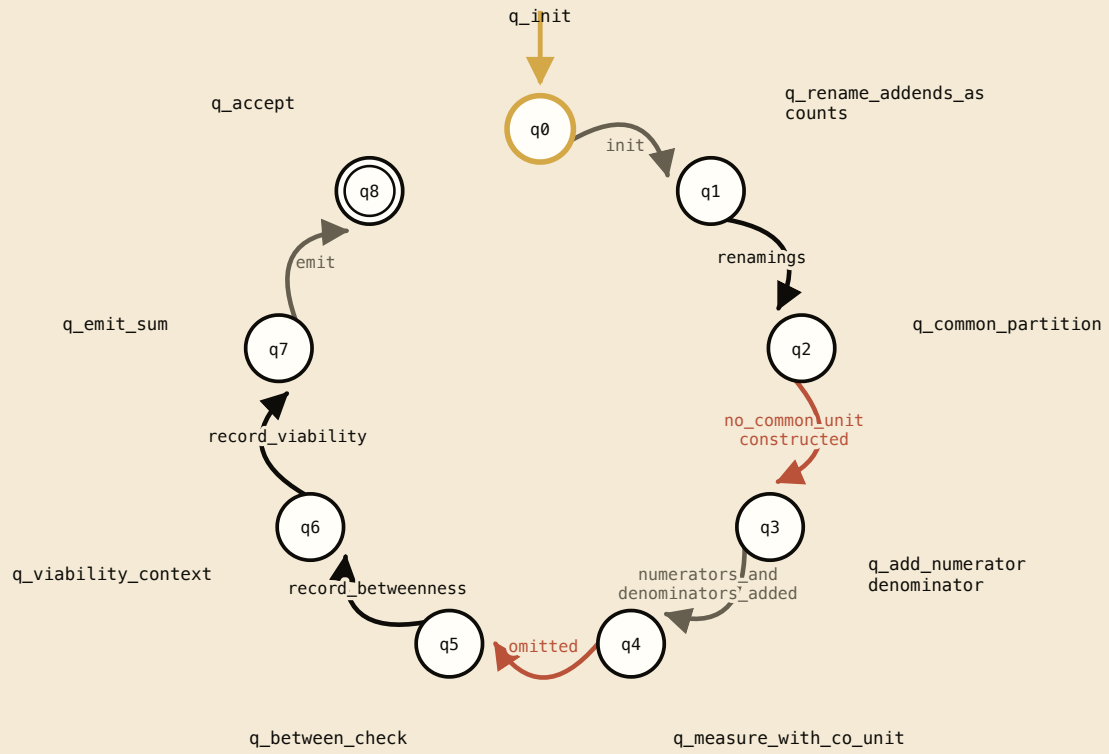


right: $\frac{7}{8}$



The run produced `fraction(10,12)`; the recorded expected result is `fraction(13,8)`.

The run produced `fraction(10,12)`; the recorded expected result is `fraction(13,8)`. The scene is derived from the contract run (9 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ emit, init, no_common_unit_constructed, numerators_and_denominators_added, omitted, record_betweenness, record_viability, renamings (8 distinct)

q_0 $q_0 = q_{\text{init}}$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_rename_addends_as_counts	none recorded
q_2	q_common_partition	<i>common partitioning</i> (constructivist; Shin & Lee 2018) <i>find a common denominator</i> (van_de_walle; Van de Walle, ch. 15) <i>converts to common denominator</i> (clarke_roche; Clarke & Roche 2009)
q_3	q_add_numerator_denominator	<i>addition</i> (behr_post_lesh; Behr, Wachsmuth, Post & Lesh 1984)
q_4	q_measure_with_co_unit	<i>co-measurement unit</i> (constructivist; Shin & Lee 2018; Nabors 2003)
q_5	q_between_check	none recorded
q_6	q_viability_context	none recorded
q_7	q_emit_sum	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_addend_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_rename_addends_as_counts	smr_frac_common_unit_add.pl:128 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_rename_addends_as_counts	renamings	q_common_partition	smr_frac_common_unit_add.pl:128 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}

From	Action	To	Provenance
q_common_partition	no_common_unit_constructed	q_add_numerator_denominator	smr_frac_common_unit_add.pl:128 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_add_numerator_denominator	numerators_and_denominators_added	q_measure_width_co_unit	smr_frac_common_unit_add.pl:128 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_measure_width_co_unit	omitted	q_between_check	smr_frac_common_unit_add.pl:128 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_between_check	record_betweenness	q_viability_context	smr_frac_common_unit_add.pl:128 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_viability_context	record_viability	q_emit_sum	smr_frac_common_unit_add.pl:128 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_emit_sum	emit	q_accept	smr_frac_common_unit_add.pl:128 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 8 observed rows.

3. area_model_fraction_comparison

left: 3/4

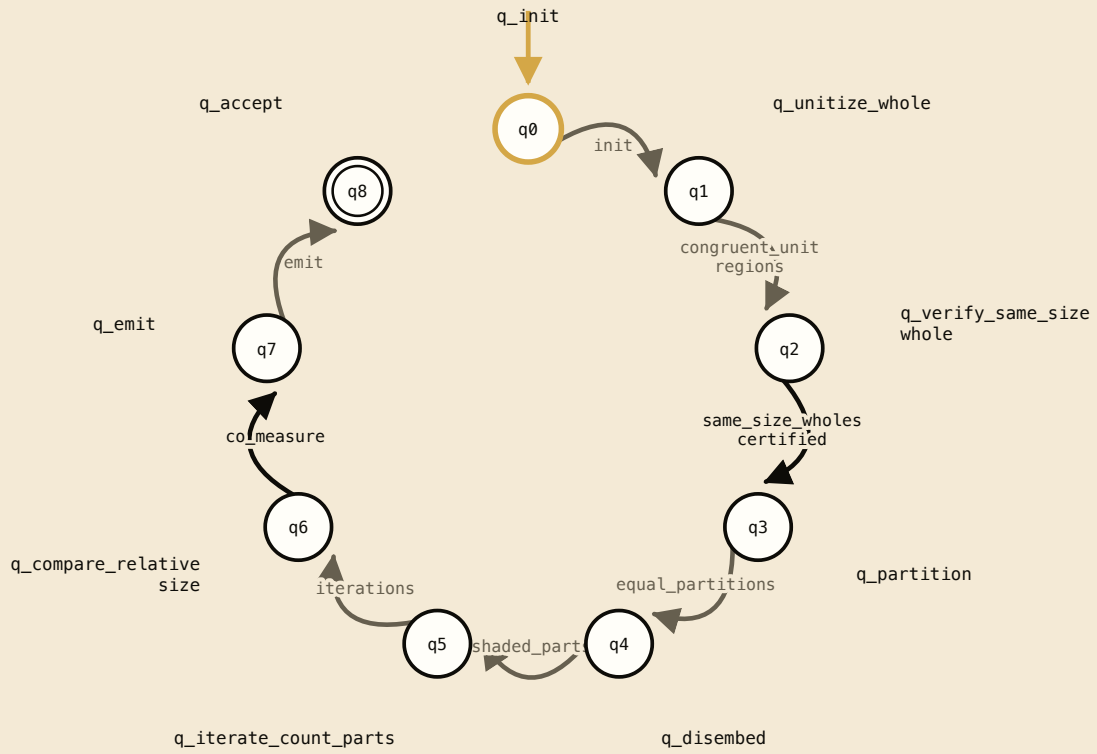


right: 2/3



The run produced greater_than.

The run produced greater_than. The scene is derived from the contract run (9 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ co_measure, congruent_unit_regions, emit, equal_partitions, init, iterations, same_size_wholes_certified, shaded_parts (8 distinct)

q_0 $q_0 = q_init$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_unitize_whole	<i>unitizing</i> (constructivist; Olive 1999; Norton & Wilkins 2009) <i>the whole or unit</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_2	q_verify_same_size_whole	<i>fraction size is relative</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts) <i>same-size whole</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions)
q_3	q_partition	<i>equi-partitioning</i> (constructivist; Steffe 2001, via Boyce & Norton 2017) <i>partitioning</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts)
q_4	q_disembed	<i>disembedding</i> (constructivist; Hackenberg 2013; Steffe & Olive 2010)
q_5	q_iterate_count_parts	<i>iterating</i> (constructivist; Steffe/Olive/Hackenberg fraction-scheme vocabulary) <i>counting fractional parts, or iterating</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts)
q_6	q_compare_relative_size	<i>reasoning about the relative size of the fractions</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions) <i>numerator and denominator strategy</i> (behr_post_lesh; Behr, Wachsmuth, Post & Lesh 1984)
q_7	q_emit	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_unitize_whole	smr_frac_area_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_unitize_whole	congruent_unit_regions	q_verify_same_size_whole	smr_frac_area_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_verify_same_size_whole	same_size_wholes_certified	q_partition	smr_frac_area_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_partition	equal_partitions	q_disembed	smr_frac_area_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_disembed	shaded_parts	q_iterate_count_parts	smr_frac_area_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_iterate_count_parts	iterations	q_compare_relative_size	smr_frac_area_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_compare_relative_size	co_measure	q_emit	smr_frac_area_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_emit	emit	q_accept	smr_frac_area_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

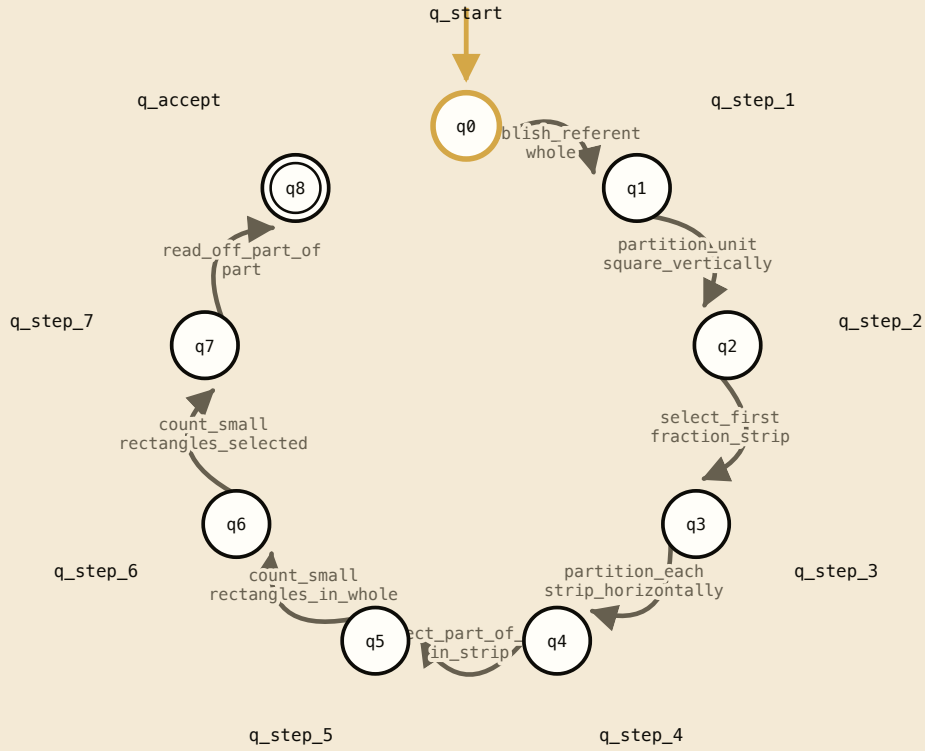
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 16 observed rows.

4. area_model_part_of_part



The run produced `fraction(6,12)`.

The run produced `fraction(6,12)`. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ count_small_rectangles_in_whole, count_small_rectangles_selected, establish_referent_whole, partition_each_strip_horizontally, partition_unit_square_vertically, read_off_part_of_part, select_first_fraction_strip, select_part_of_part_in_strip (8 distinct)

q_0 $q_0 = q_start$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_step_6	none recorded
q_7	q_step_7	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_referent_whole	q_step_1	fraction_action_pairs.pl:722 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_1	partition_unit_squares_vertically	q_step_2	fraction_action_pairs.pl:722 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_2	select_first_fraction_strip	q_step_3	fraction_action_pairs.pl:722 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_step_3	partition_each_strip_horizontally	q_step_4	fraction_action_pairs.pl:722 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_4	select_part_of_part_in_strip	q_step_5	fraction_action_pairs.pl:722 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_5	count_small_rectangles_in_whole	q_step_6	fraction_action_pairs.pl:722 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_6	count_small_rectangles_selected	q_step_7	fraction_action_pairs.pl:722 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_7	read_off_part_of_part	q_accept	fraction_action_pairs.pl:722 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

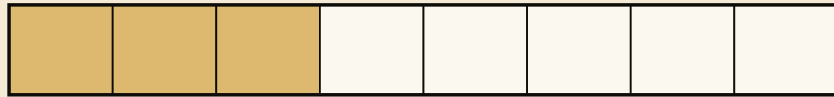
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 16 observed rows.

5. area_model_unequal_partition_piece_count

left: 2/3

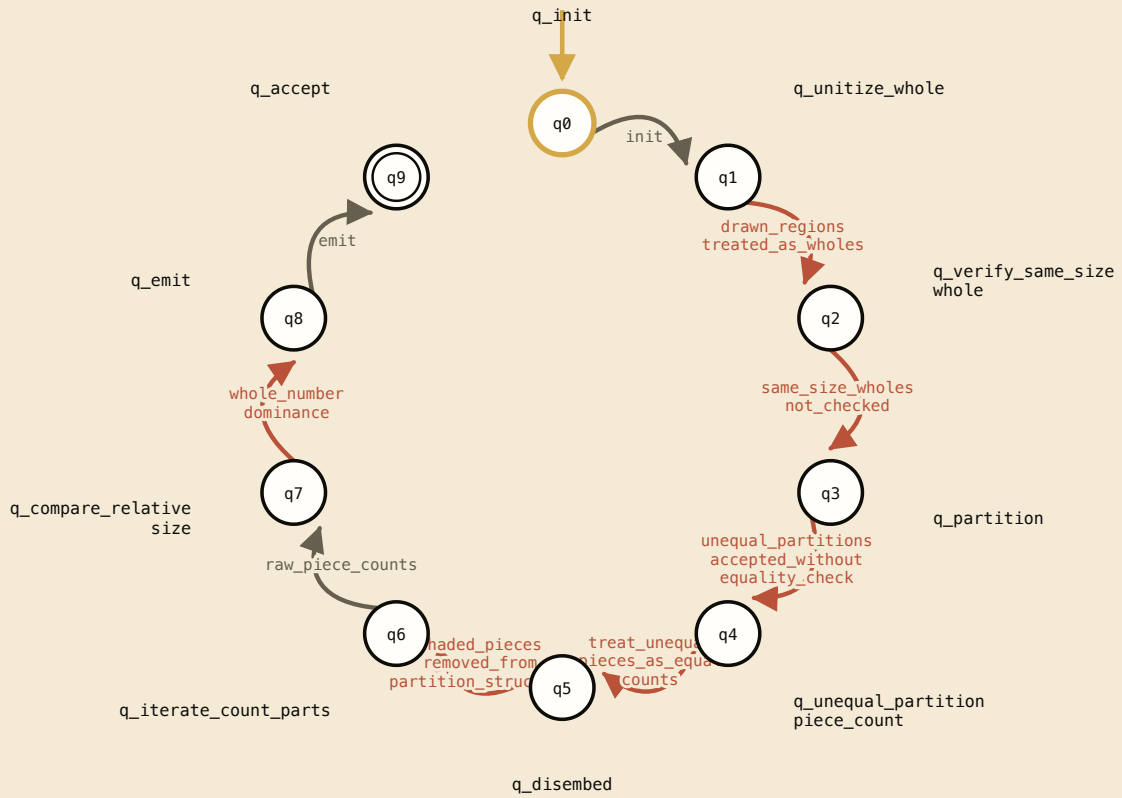


right: 3/8



The run produced less_than; the recorded expected result is greater_than.

The run produced less_than; the recorded expected result is greater_than. The scene is derived from the contract run (10 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9\}$

Σ drawn_regions_treated_as_wholes, emit, init, raw_piece_counts, same_size_wholes_not_checked, shaded_pieces_removed_from_partition_structure, treat_unequal_pieces_as_equal_counts, unequal_partitions_accepted_without_equality_check, whole_number_dominance (9 distinct)

q_0 $q_0 = q_init$

$F \quad \{q_9\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_unitize_whole	<i>unitizing</i> (constructivist; Olive 1999; Norton & Wilkins 2009) <i>the whole or unit</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_2	q_verify_same_size_whole	<i>fraction size is relative</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts) <i>same-size whole</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions)
q_3	q_partition	<i>equi-partitioning</i> (constructivist; Steffe 2001, via Boyce & Norton 2017) <i>partitioning</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts)
q_4	q_unequal_partition_piece_count	<i>unequal-sized parts do not name fractional parts</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts) <i>whole number dominance</i> (behr_post_lesh; Behr et al. 1984)
q_5	q_disembed	<i>disembedding</i> (constructivist; Hackenberg 2013; Steffe & Olive 2010)
q_6	q_iterate_count_parts	<i>iterating</i> (constructivist; Steffe/Olive/Hackenberg fraction-scheme vocabulary) <i>counting fractional parts, or iterating</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts)
q_7	q_compare_relative_size	<i>reasoning about the relative size of the fractions</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions) <i>numerator and denominator strategy</i> (behr_post_lesh; Behr, Wachsmuth, Post & Lesh 1984)
q_8	q_emit	none recorded
q_9	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_unitize_whole	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_unitize_whole	drawn_regions_treated_as_wholes	q_verify_same_size_whole	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_verify_same_size_whole	same_size_wholes_not_checked	q_partition	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_partition	unequal_partitions_accepted_without_equality_check	q_unequal_partition_piece_count	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_unequal_partition_piece_count	treat_unequal_pieces_as_equal_counts	q_disembed	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)

From	Action	To	Provenance
q_disembed	shaded_pieces_remove d_from_partition_structure	q_iterate_count_parts	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_iterate_count_parts	raw_piece_counts	q_compare_relative_size	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_compare_relative_size	whole_number_dominance	q_emit	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_emit	emit	q_accept	smr_frac_area_compare.pl:33 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 10 states, 9 distinct actions, 0 branching states, 0 loop edges; 9 static rows and 18 observed rows.

6. benchmark_fraction_comparison

left: 3/4

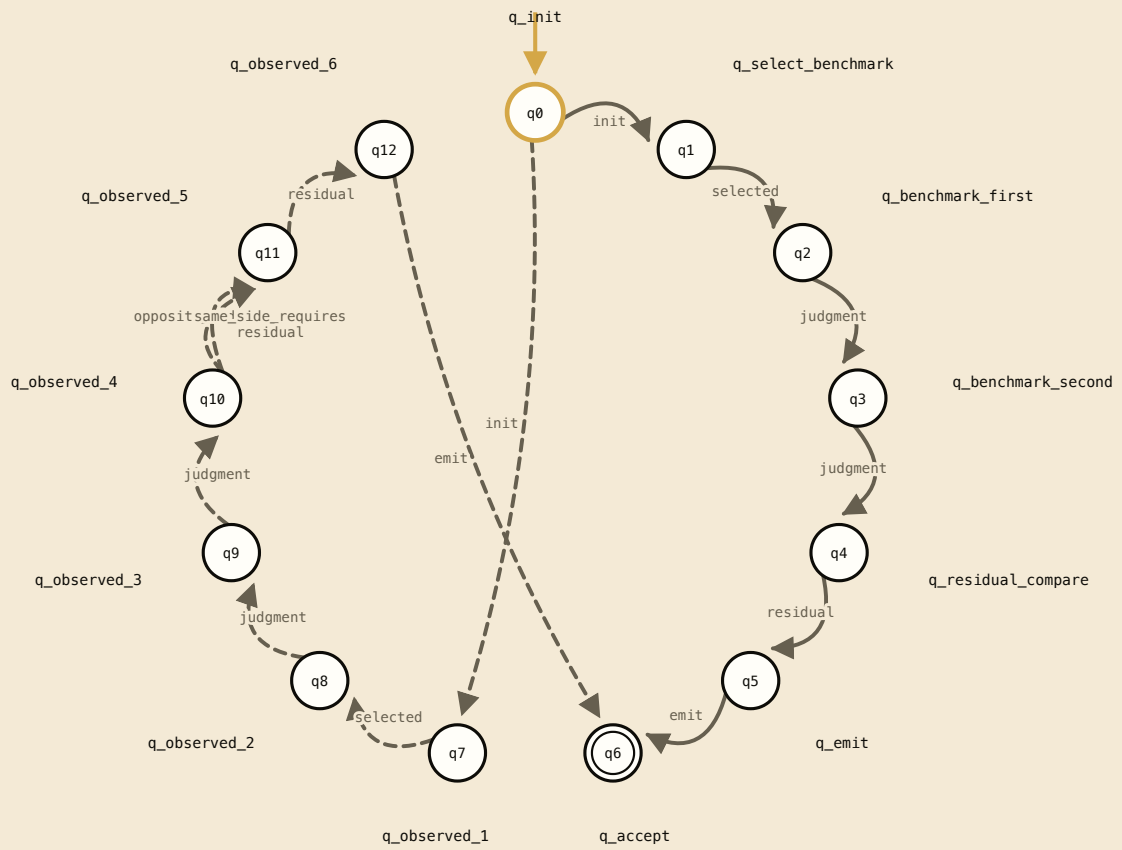


right: 2/3



The run produced greater_than.

The run produced greater_than. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}\}$$

$$\Sigma = \text{emit, init, judgment, opposite_sides, residual, same_side_requires_residual, selected} \\ (7 \text{ distinct})$$

$$q_0 = q_init$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_select_benchmark	<i>reference point</i> (behr_post_lesh; Behr, Wachsmuth, Post & Lesh 1984) <i>benchmarking</i> (clarke_roche; Clarke & Roche 2009)
q_2	q_benchmark_first	<i>more than or less than one-half or one</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions)
q_3	q_benchmark_second	<i>more than or less than one-half or one</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions)
q_4	q_residual_compare	<i>closeness to one-half or one</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions) <i>residual thinking</i> (clarke_roche; Clarke & Roche 2009; Post & Cramer 2002) <i>residual</i> (cramer_post_delmas; Cramer, Post & delMas 2002) <i>filling up the whole</i> (riddle_rodzwel; Riddle & Rodzwell 2000, via Clarke & Roche 2009)
q_5	q_emit	none recorded
q_6	q_accept	none recorded
q_7	q_observed_1	none recorded
q_8	q_observed_2	none recorded
q_9	q_observed_3	none recorded
q_{10}	q_observed_4	none recorded
q_{11}	q_observed_5	none recorded
q_{12}	q_observed_6	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_select_benchmark	smr_frac_benchmark_compare.pl:11

From	Action	To	Provenance
q_select_benchmark	selected	q_benchmark_first	smr_frac_benchmark_compare.pl:11
q_benchmark_first	judgment	q_benchmark_second	smr_frac_benchmark_compare.pl:11
q_benchmark_second	judgment	q_residual_compare	smr_frac_benchmark_compare.pl:11
q_residual_compare	residual	q_emit	smr_frac_benchmark_compare.pl:11
q_emit	emit	q_accept	smr_frac_benchmark_compare.pl:11
q_init	init	q_observe_d_1	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_1	selected	q_observe_d_2	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_2	judgment	q_observe_d_3	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_3	judgment	q_observe_d_4	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_4	same_side_requires_residual	q_observe_d_5	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}
q_observed_5	residual	q_observe_d_6	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_observed_6	emit	q_accept	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_4	opposite_sides	q_observe_d_5	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

Computed structure: branching. one or more recorded states have multiple outgoing transitions; no loop edge is recorded. 13 states, 7 distinct actions, 2 branching states, 0 loop edges; 6 static rows and 14 observed rows.

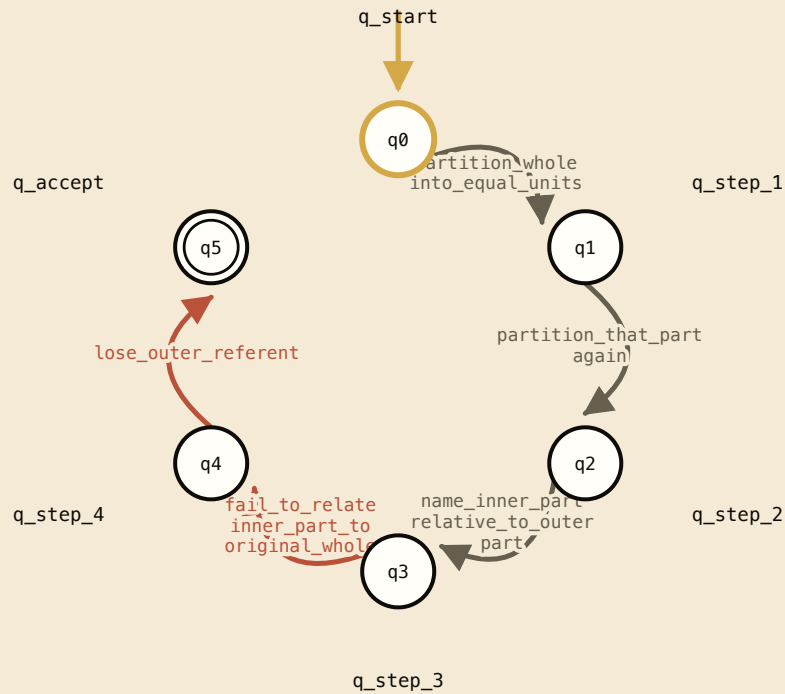
7. clear_inner_referent

iterated unit: 3/4



The run produced fraction(1,4); the recorded expected result is fraction(1,12).

The run produced fraction(1,4); the recorded expected result is fraction(1,12). The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ $fail_to_relate_inner_part_to_original_whole, lose_outer_referent, name_inner_part_relative_to_outer_part, partition_that_part_again, partition_whole_into_equal_units$ (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "positive_integer", "b": "positive_integer"}; example {"a": 3, "b": 4}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	partition_whole_into_equal_units	q_step_1	<code>fraction_action_pairs.pl:417</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_1	partition_that_part_again	q_step_2	<code>fraction_action_pairs.pl:417</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_2	name_inner_part_relative_to_outer_part	q_step_3	<code>fraction_action_pairs.pl:417</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_3	fail_to_relate_inner_part_to_origin_al_whole	q_step_4	<code>fraction_action_pairs.pl:417</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_4	lose_outer_referent	q_accept	<code>fraction_action_pairs.pl:417</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)

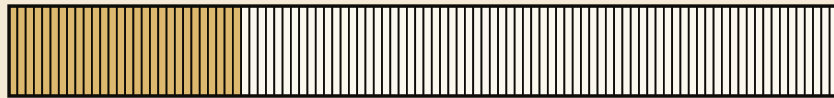
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

8. co_denominator_count_on_from_larger

left: 47/100

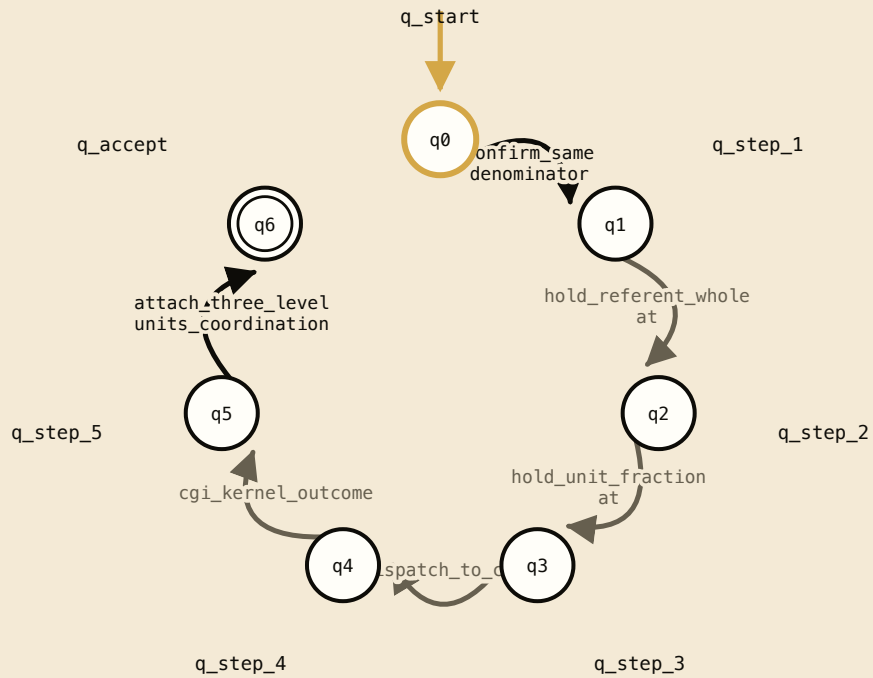


right: 28/100



The run produced 75.

The run produced 75. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ $attach_three_level_units_coordination, cgi_kernel_outcome,$
 $confirm_same_denominator, dispatch_to_cgi, hold_referent_whole_at,$
 $hold_unit_fraction_at$ (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}}; verified(strategy_trace_ok).

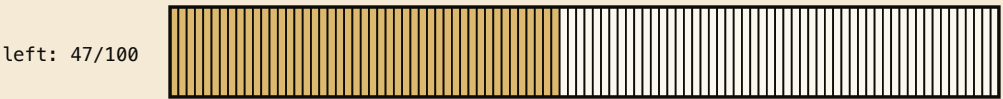
δ : recorded transitions

From	Action	To	Provenance
q_start	confirm_same_denominator	q_step_1	fraction_action_pairs.pl:861 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)
q_step_1	hold_referent_whole_at	q_step_2	fraction_action_pairs.pl:861 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)
q_step_2	hold_unit_fraction_at	q_step_3	fraction_action_pairs.pl:861 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)

From	Action	To	Provenance
q_step_3	dispatch_to_cgi	q_step_4	fraction_action_pairs.pl:861 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)
q_step_4	cgi_kernel_outcome	q_step_5	fraction_action_pairs.pl:861 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)
q_step_5	attach_three_level_units_coordination	q_accept	fraction_action_pairs.pl:861 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)

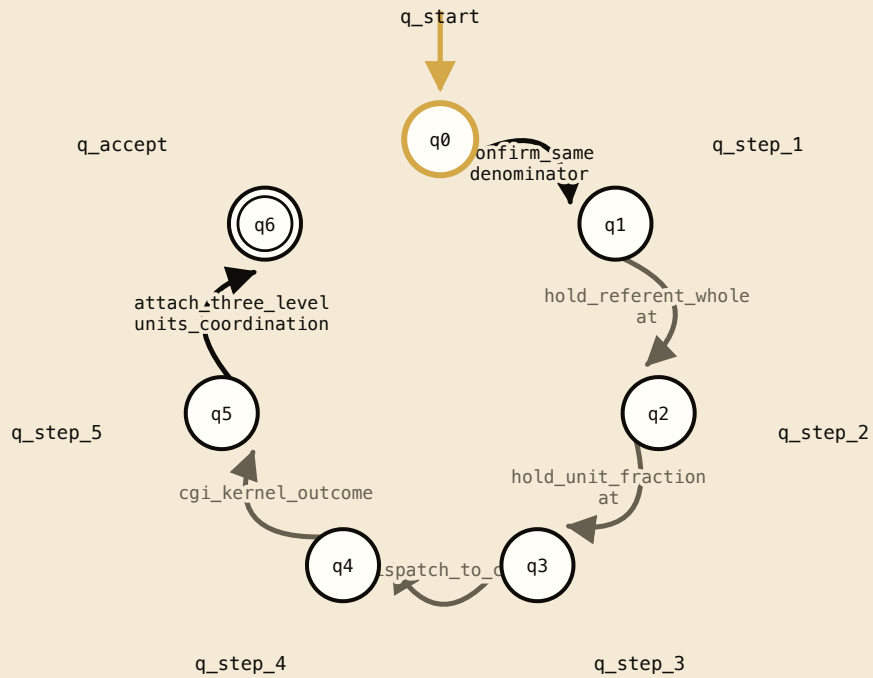
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

9. co_denominator_make_base_transfer



The run produced 75.

The run produced 75. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ $attach_three_level_units_coordination, cgi_kernel_outcome, confirm_same_denominator, dispatch_to_cgi, hold_referent_whole_at, hold_unit_fraction_at$ (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}}; verified(strategy_trace_ok).

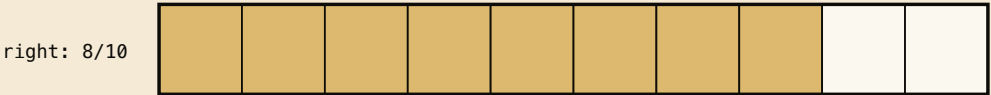
δ : recorded transitions

From	Action	To	Provenance
q_start	confirm_same_denominator	q_step_1	fraction_action_pairs.pl:925 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)
q_step_1	hold_referent_whole_at	q_step_2	fraction_action_pairs.pl:925 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)
q_step_2	hold_unit_fraction_at	q_step_3	fraction_action_pairs.pl:925 observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)

From	Action	To	Provenance
q_step_3	dispatch_to_cgi	q_step_4	<code>fraction_action_pairs.pl:925</code> observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)
q_step_4	cgi_kernel_outcome	q_step_5	<code>fraction_action_pairs.pl:925</code> observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)
q_step_5	attach_three_level_units_coordination	q_accept	<code>fraction_action_pairs.pl:925</code> observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 47}, "right": {"d": 100, "n": 28}} observed on {"kind": "fraction_pair", "left": {"d": 100, "n": 48}, "right": {"d": 100, "n": 28}} (derived template)

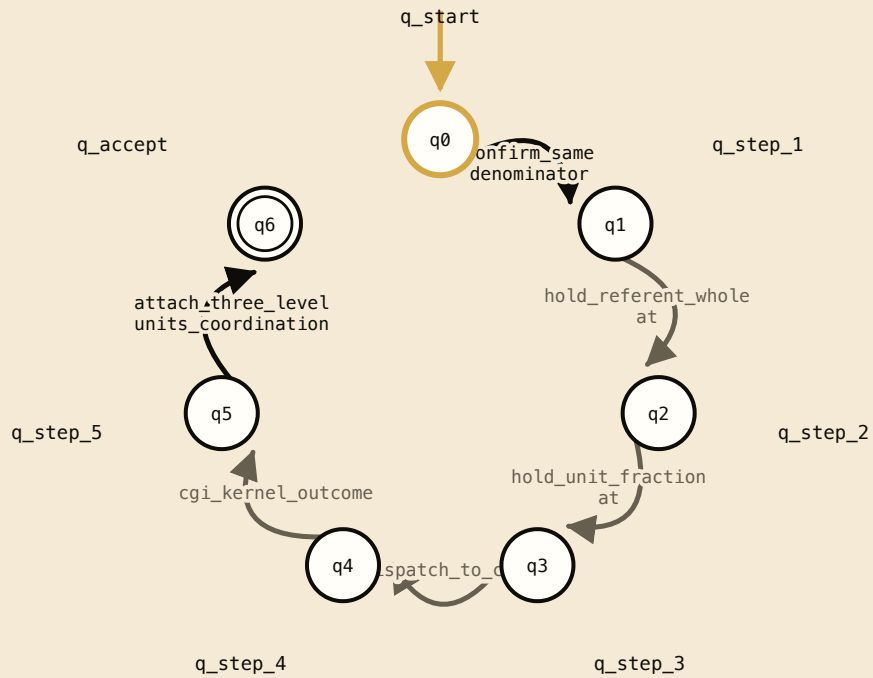
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

10. co_denominator_make_ten_split_leftover



The run produced 15.

The run produced 15. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ $attach_three_level_units_coordination, cgi_kernel_outcome,$
 $confirm_same_denominator, dispatch_to_cgi, hold_referent_whole_at,$
 $hold_unit_fraction_at$ (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 10, "n": 7}, "right": {"d": 10, "n": 8}}; verified(strategy_trace_ok).

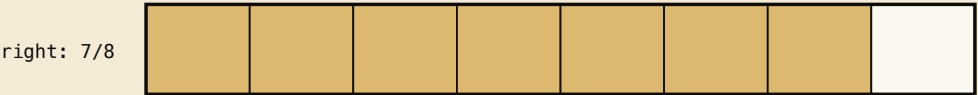
δ : recorded transitions

From	Action	To	Provenance
q_start	confirm_same_denominator	q_step_1	fraction_action_pairs.pl:893 observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 7}, "right": {"d": 10, "n": 8}} observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 8}, "right": {"d": 10, "n": 8}} (derived template)
q_step_1	hold_referent_whole_at	q_step_2	fraction_action_pairs.pl:893 observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 7}, "right": {"d": 10, "n": 8}} observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 8}, "right": {"d": 10, "n": 8}} (derived template)
q_step_2	hold_unit_fraction_at	q_step_3	fraction_action_pairs.pl:893 observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 7}, "right": {"d": 10, "n": 8}} observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 8}, "right": {"d": 10, "n": 8}} (derived template)

From	Action	To	Provenance
q_step_3	dispatch_to_cgi	q_step_4	<code>fraction_action_pairs.pl:893</code> observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 7}, "right": {"d": 10, "n": 8}} observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 8}, "right": {"d": 10, "n": 8}} (derived template)
q_step_4	cgi_kernel_outcome	q_step_5	<code>fraction_action_pairs.pl:893</code> observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 7}, "right": {"d": 10, "n": 8}} observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 8}, "right": {"d": 10, "n": 8}} (derived template)
q_step_5	attach_three_level_units_coordination	q_accept	<code>fraction_action_pairs.pl:893</code> observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 7}, "right": {"d": 10, "n": 8}} observed on {"kind": "fraction_pair", "left": {"d": 10, "n": 8}, "right": {"d": 10, "n": 8}} (derived template)

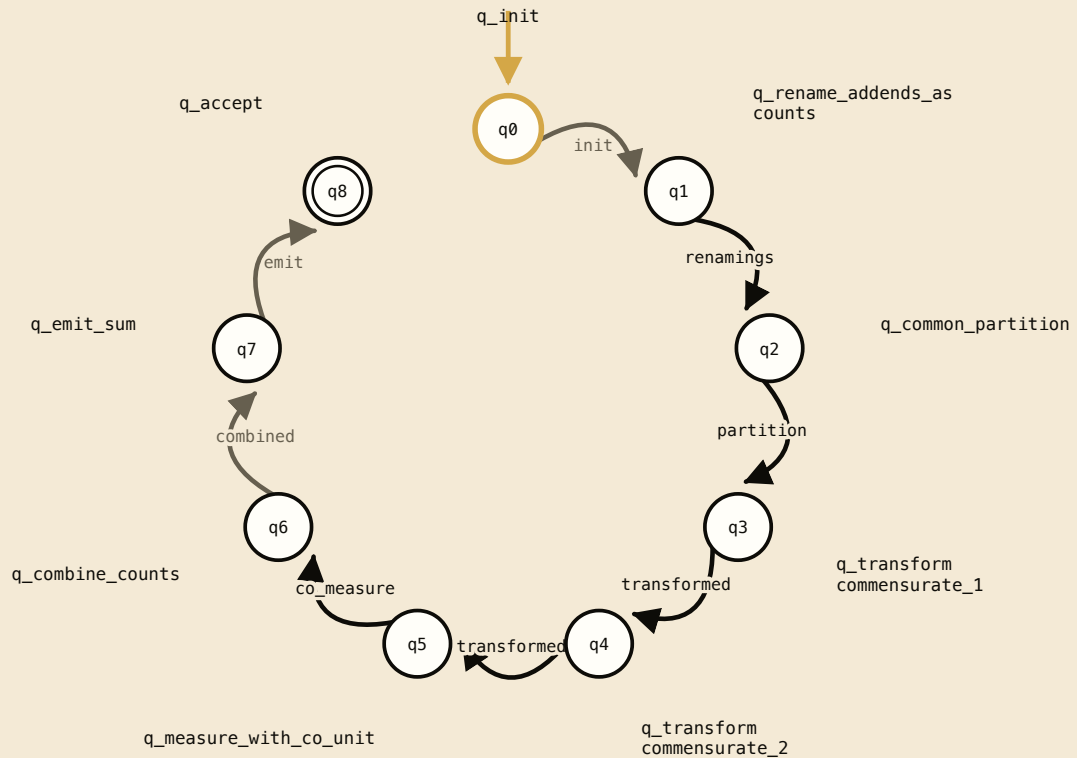
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

11. common_denominator_fraction_addition



The run produced `fraction(13,8)`.

The run produced `fraction(13,8)`. The scene is derived from the contract run (9 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ co_measure, combined, emit, init, partition, renamings, transformed (7 distinct)

q_0 $q_0 = q_{\text{init}}$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_rename_addends_as_counts	none recorded
q_2	q_common_partition	<i>common partitioning</i> (constructivist; Shin & Lee 2018) <i>find a common denominator</i> (van_de_walle; Van de Walle, ch. 15) <i>converts to common denominator</i> (clarke_roche; Clarke & Roche 2009)
q_3	q_transform_commensurate_1	<i>commensurate</i> (constructivist; Steffe 2003) <i>equivalent-fraction procedure</i> (van_de_walle; Van de Walle, ch. 15)
q_4	q_transform_commensurate_2	<i>commensurate</i> (constructivist; Steffe 2003) <i>equivalent-fraction procedure</i> (van_de_walle; Van de Walle, ch. 15)
q_5	q_measure_with_co_unit	<i>co-measurement unit</i> (constructivist; Shin & Lee 2018; Nabors 2003)
q_6	q_combine_counts	none recorded
q_7	q_emit_sum	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_addend_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_rename_addends_as_counts	smr_frac_common_unit_add.pl:70 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}

From	Action	To	Provenance
q_rename_addends_as_counts	renamings	q_common_partition	smr_frac_common_unit_add.pl:70 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_common_partition	partition	q_transform_commensurate_1	smr_frac_common_unit_add.pl:70 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_transform_commensurate_1	transformed	q_transform_commensurate_2	smr_frac_common_unit_add.pl:70 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_transform_commensurate_2	transformed	q_measure_with_co_unit	smr_frac_common_unit_add.pl:70 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_measure_with_co_unit	co_measure	q_combine_counts	smr_frac_common_unit_add.pl:70 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_combine_counts	combined	q_emit_sum	smr_frac_common_unit_add.pl:70 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}
q_emit_sum	emit	q_accept	smr_frac_common_unit_add.pl:70 observed on {"kind": "fraction_addend_pair", "left": {"d": 4, "n": 3}, "right": {"d": 8, "n": 7}}

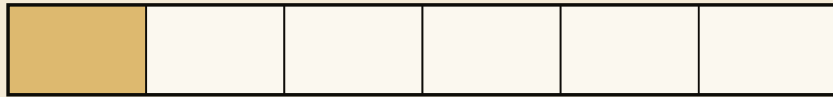
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 7 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 8 observed rows.

12. common_denominator_fraction_subtraction

left: $\frac{2}{3}$

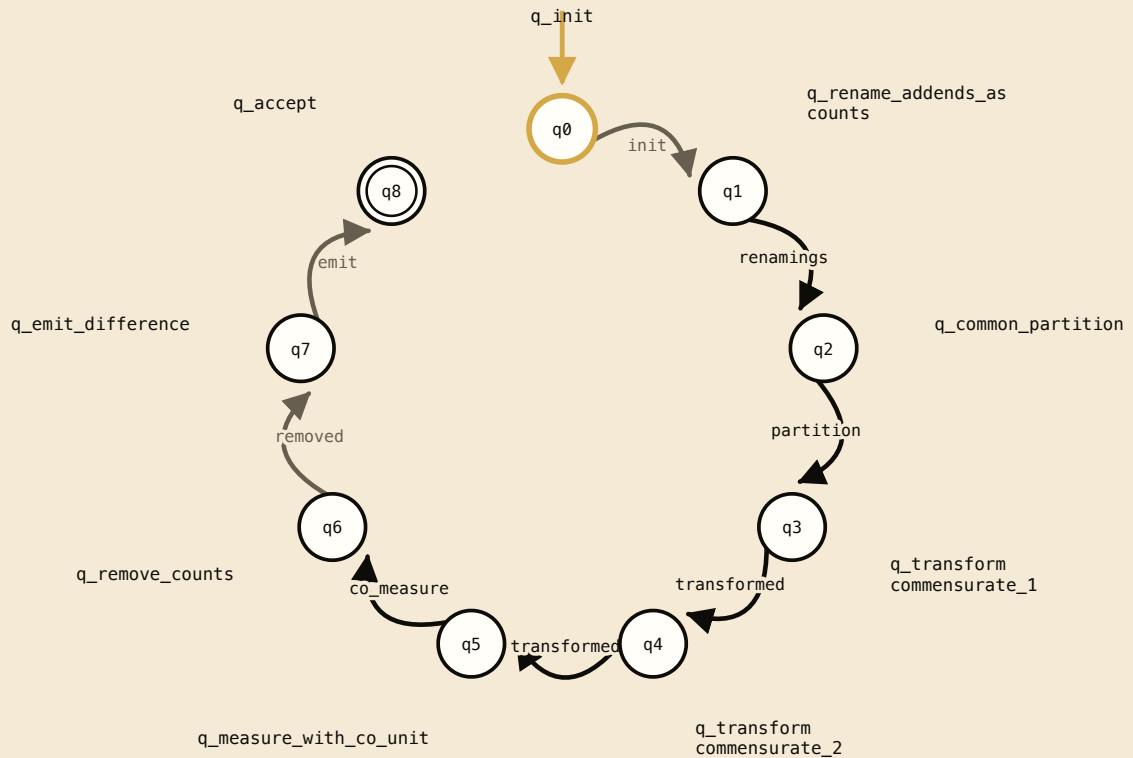


right: $\frac{1}{6}$



The run produced `fraction(3,6)`.

The run produced `fraction(3,6)`. The scene is derived from the contract run (9 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ co_measure, emit, init, partition, removed, renamings, transformed (7 distinct)

q_0 $q_0 = q_init$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_rename_addends_as_counts	none recorded
q_2	q_common_partition	<i>common partitioning</i> (constructivist; Shin & Lee 2018) <i>find a common denominator</i> (van_de_walle; Van de Walle, ch. 15) <i>converts to common denominator</i> (clarke_roche; Clarke & Roche 2009)
q_3	q_transform_commensurate_1	<i>commensurate</i> (constructivist; Steffe 2003) <i>equivalent-fraction procedure</i> (van_de_walle; Van de Walle, ch. 15)
q_4	q_transform_commensurate_2	<i>commensurate</i> (constructivist; Steffe 2003) <i>equivalent-fraction procedure</i> (van_de_walle; Van de Walle, ch. 15)
q_5	q_measure_with_co_unit	<i>co-measurement unit</i> (constructivist; Shin & Lee 2018; Nabors 2003)
q_6	q_remove_counts	none recorded
q_7	q_emit_difference	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_minuend_subtrahend", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_rename_addends_as_counts	smr_frac_common_unit_add.pl:98 observed on {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}

From	Action	To	Provenance
q_rename_addends_as_counts	renaming	q_common_partition	smr_frac_common_unit_add.pl:98 observed on {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}
q_common_partition	partition	q_transform_commensurate_1	smr_frac_common_unit_add.pl:98 observed on {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}
q_transform_commensurate_1	transformed	q_transform_commensurate_2	smr_frac_common_unit_add.pl:98 observed on {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}
q_transform_commensurate_2	transformed	q_measure_with_co_unit	smr_frac_common_unit_add.pl:98 observed on {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}
q_measure_with_co_unit	co_measure	q_remove_counts	smr_frac_common_unit_add.pl:98 observed on {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}
q_remove_counts	removed	q_emit_difference	smr_frac_common_unit_add.pl:98 observed on {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}
q_emit_difference	emit	q_accept	smr_frac_common_unit_add.pl:98 observed on {"kind": "fraction_minuend_subtrahend", "left": {"d": 3, "n": 2}, "right": {"d": 6, "n": 1}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 7 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 8 observed rows.

13. common_unit_fraction_comparison

left: 3/4

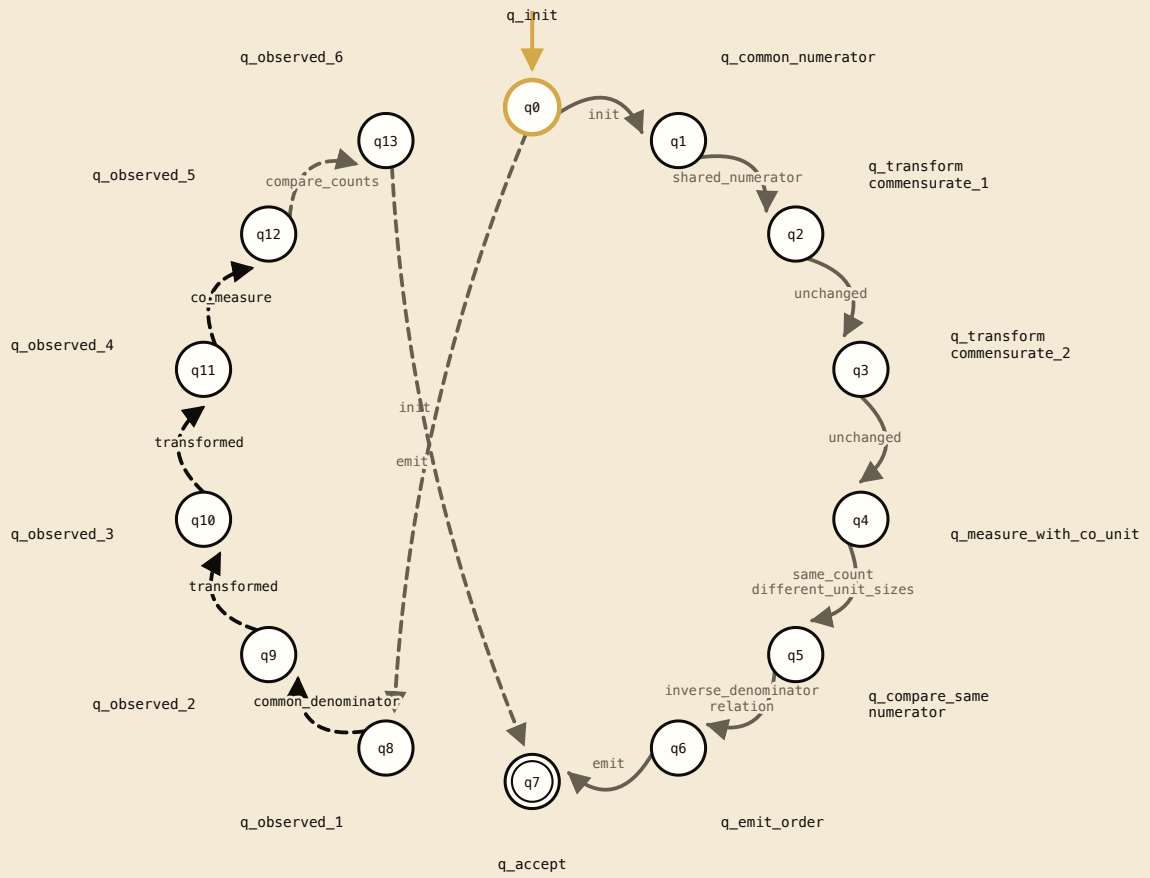


right: 2/3



The run produced greater_than.

The run produced greater_than. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}\}$

Σ co_measure, common_denominator, compare_counts, emit, init, inverse_denominator_relation, same_count_different_unit_sizes, shared_numerator, transformed, unchanged (10 distinct)

q_0 $q_0 = q_{\text{init}}$

F $\{q_7\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_common_numerator	<i>common numerator</i> (van_de_walle; Burns 1999, via Van de Walle, ch. 15)
q_2	q_transform_commensurate_1	<i>commensurate</i> (constructivist; Steffe 2003) <i>equivalent-fraction procedure</i> (van_de_walle; Van de Walle, ch. 15)
q_3	q_transform_commensurate_2	<i>commensurate</i> (constructivist; Steffe 2003) <i>equivalent-fraction procedure</i> (van_de_walle; Van de Walle, ch. 15)
q_4	q_measure_with_co_unit	<i>co-measurement unit</i> (constructivist; Shin & Lee 2018; Nabors 2003)
q_5	q_compare_same_numerator	<i>numerator the same and compares denominator</i> (clarke_roche; Clarke & Roche 2009) <i>same numerator</i> (cramer_post_delmas; Cramer, Post & delMas 2002)
q_6	q_emit_order	none recorded
q_7	q_accept	none recorded
q_8	q_observed_1	none recorded
q_9	q_observed_2	none recorded
q_{10}	q_observed_3	none recorded
q_{11}	q_observed_4	none recorded
q_{12}	q_observed_5	none recorded
q_{13}	q_observed_6	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

δ : recorded transitions

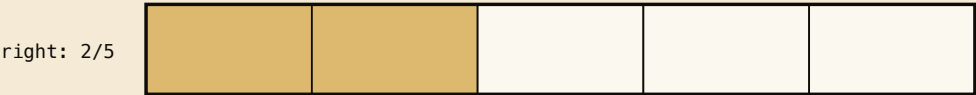
From	Action	To	Provenance
q_init	init	q_common_numerator	smr_frac_common_unit_compare.pl:11

From	Action	To	Provenance
q_common_numerator	shared_numerator	q_transform_commensurate_1	smr_frac_common_unit_compare.pl:11
q_transform_commensurate_1	unchanged	q_transform_commensurate_2	smr_frac_common_unit_compare.pl:11
q_transform_commensurate_2	unchanged	q_measure_with_co_unit	smr_frac_common_unit_compare.pl:11
q_measure_with_co_unit	same_count_different_unit_sizes	q_compare_same_numerator	smr_frac_common_unit_compare.pl:11
q_compare_same_numerator	inverse_denominator_relation	q_emit_order	smr_frac_common_unit_compare.pl:11
q_emit_order	emit	q_accept	smr_frac_common_unit_compare.pl:11
q_init	init	q_observed_1	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_1	common_denominator	q_observed_2	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_2	transformed	q_observed_3	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_3	transformed	q_observed_4	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_observed_4	co_measure	q_observed_5	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_5	compare_counts	q_observed_6	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_6	emit	q_accept	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

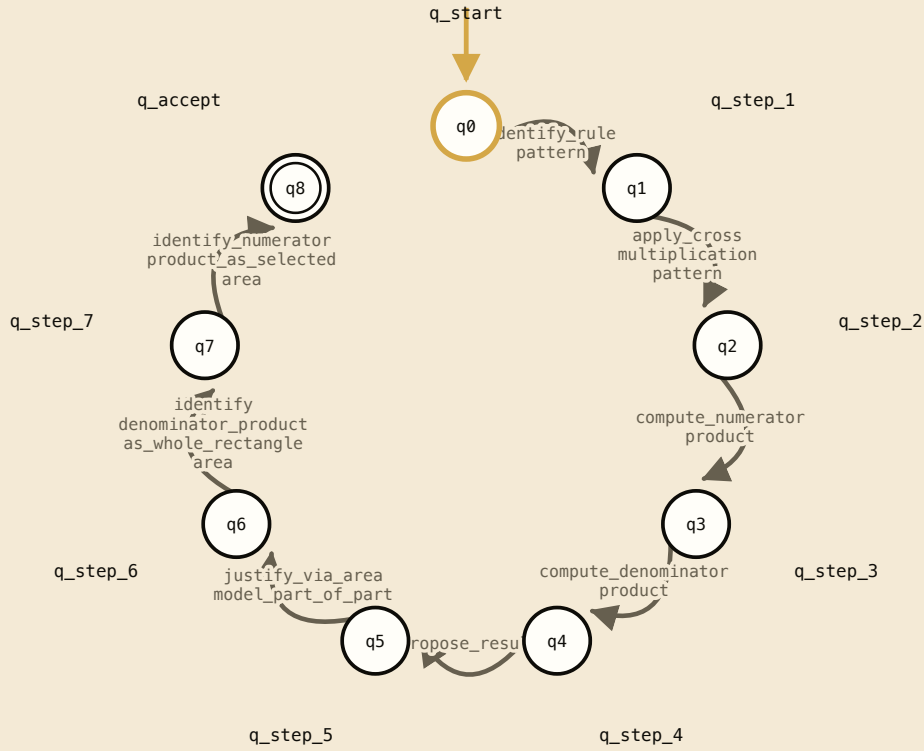
Computed structure: branching. one or more recorded states have multiple outgoing transitions; no loop edge is recorded. 14 states, 10 distinct actions, 1 branching states, 0 loop edges; 7 static rows and 14 observed rows.

14. cross_multiplication_rule_from_pattern



The run produced `fraction(6,20)`.

The run produced `fraction(6,20)`. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ apply_cross_multiplication_pattern, compute_denominator_product, compute_numerator_product, identify_denominator_product_as_whole_rectangle_area, identify_numerator_product_as_selected_area, identify_rule_pattern, justify_via_area_model_part_of_part, propose_result (8 distinct)

q_0 $q_0 = q_start$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_step_6	none recorded
q_7	q_step_7	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_rule_pattern	q_step_1	<code>fraction_action_pairs.pl:786</code> observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_1	apply_cross_multiplication_pattern	q_step_2	<code>fraction_action_pairs.pl:786</code> observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)

From	Action	To	Provenance
q_step_2	compute_numerator_product	q_step_3	fraction_action_pairs.pl:786 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_3	compute_denominator_product	q_step_4	fraction_action_pairs.pl:786 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_4	propose_result	q_step_5	fraction_action_pairs.pl:786 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_5	justify_via_area_model_part_of_part	q_step_6	fraction_action_pairs.pl:786 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_6	identify_denominator_product_as_whole_rectangle_area	q_step_7	fraction_action_pairs.pl:786 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_7	identify_numerator_product_as_selected_area	q_accept	fraction_action_pairs.pl:786 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and

16 observed rows.

15. cross_multiplication_rule_without_ground

left: $\frac{3}{4}$

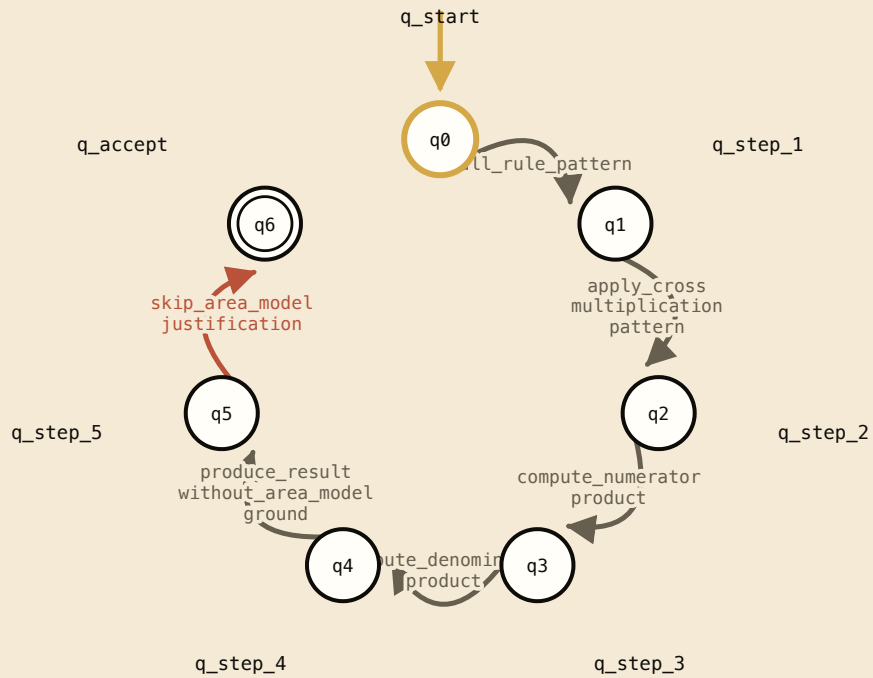


right: $\frac{2}{5}$



The run produced `fraction(6,20)`; the recorded expected result is `fraction(6,20)`.

The run produced `fraction(6,20)`; the recorded expected result is `fraction(6,20)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \{\text{apply_cross_multiplication_pattern}, \text{compute_denominator_product}, \text{compute_numerator_product}, \text{produce_result_without_area_model_ground}, \text{recall_rule_pattern}, \text{skip_area_model_justification}\} \text{ (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}}; verified(strategy_trace_ok).

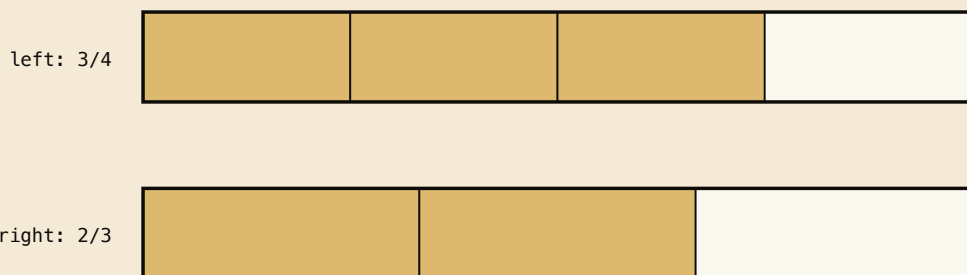
δ : recorded transitions

From	Action	To	Provenance
q_start	recall_rule_pattern	q_step_1	fraction_action_pairs.pl:816 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_1	apply_cross_multiplication_pattern	q_step_2	fraction_action_pairs.pl:816 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_2	compute_numerator_product	q_step_3	fraction_action_pairs.pl:816 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)

From	Action	To	Provenance
q_step_3	compute_denominator_product	q_step_4	fraction_action_pairs.pl:816 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_4	produce_result_without_area_model_ground	q_step_5	fraction_action_pairs.pl:816 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)
q_step_5	skip_area_model_justification	q_accepted	fraction_action_pairs.pl:816 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 5, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 5, "n": 2}} (derived template)

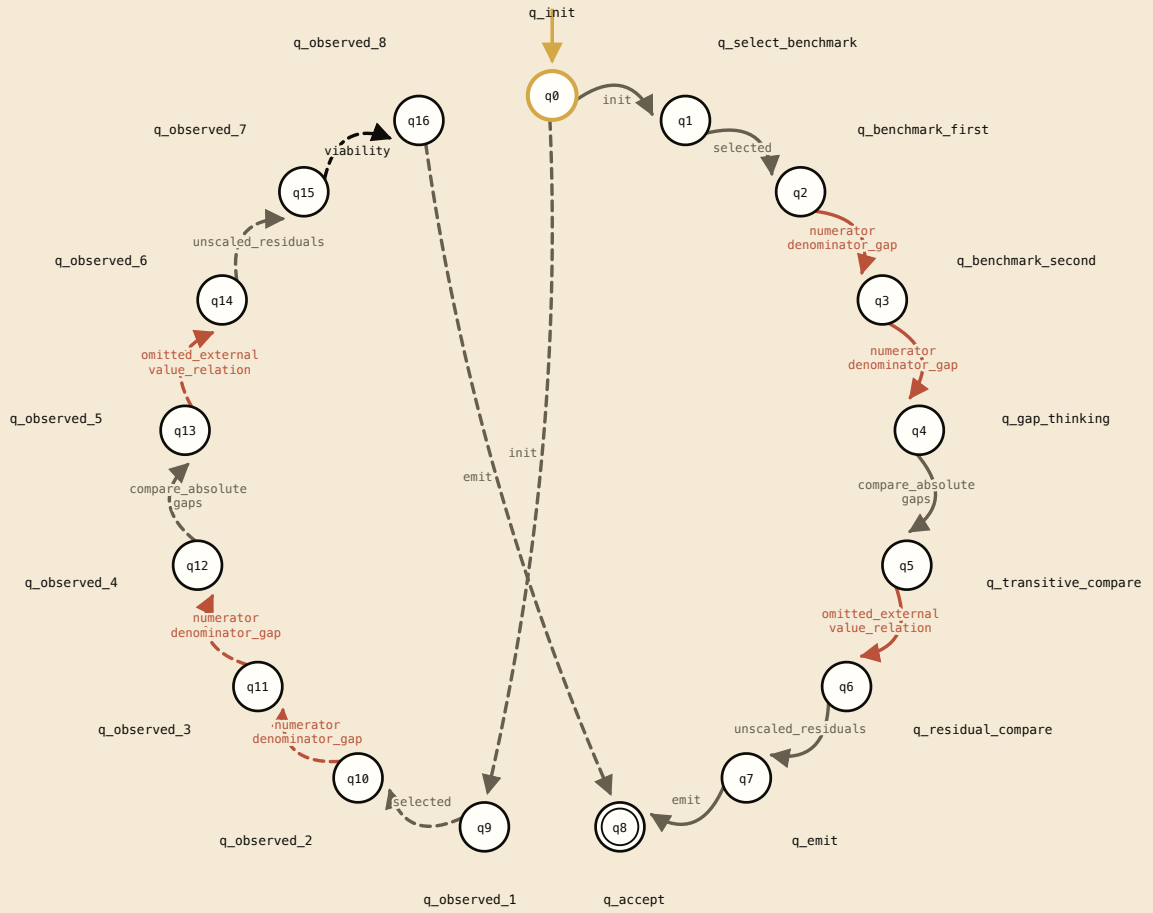
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

16. gap_thinking_fraction_comparison



The run produced equivalent; the recorded expected result is greater_than.

The run produced equivalent; the recorded expected result is greater_than. The scene is derived from the contract run (10 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15}, q_{16}\}$

Σ compare_absolute_gaps, emit, init, numerator_denominator_gap, omitted_external_value_relation, selected, unscaled_residuals, viability (8 distinct)

q_0 $q_0 = q_init$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_select_benchmark	<i>reference point</i> (behr_post_lesh; Behr, Wachsmuth, Post & Lesh 1984) <i>benchmarking</i> (clarke_roche; Clarke & Roche 2009)
q_2	q_benchmark_first	<i>more than or less than one-half or one</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions)
q_3	q_benchmark_second	<i>more than or less than one-half or one</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions)
q_4	q_gap_thinking	<i>gap thinking</i> (clarke_roche; Pearn & Stephens 2004, via Clarke & Roche 2009)
q_5	q_transitive_compare	<i>transitive</i> (post_cramer; Post et al. 1986; Cramer, Post & delMas 2002)
q_6	q_residual_compare	<i>closeness to one-half or one</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions) <i>residual thinking</i> (clarke_roche; Clarke & Roche 2009; Post & Cramer 2002) <i>residual</i> (cramer_post_delmas; Cramer, Post & delMas 2002) <i>filling up the whole</i> (riddle_rodzwell; Riddle & Rodzwell 2000, via Clarke & Roche 2009)
q_7	q_emit	none recorded
q_8	q_accept	none recorded
q_9	q_observed_1	none recorded
q_{10}	q_observed_2	none recorded
q_{11}	q_observed_3	none recorded
q_{12}	q_observed_4	none recorded
q_{13}	q_observed_5	none recorded
q_{14}	q_observed_6	none recorded
q_{15}	q_observed_7	none recorded
q_{16}	q_observed_8	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind":

```
"fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}};
verified(strategy_trace_ok).
```

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_select_benchmark	smr_frac_benchmark_compare.pl:35
q_select_benchmark	selected	q_benchmark_first	smr_frac_benchmark_compare.pl:35
q_benchmark_first	numerator_denominator_gap	q_benchmark_second	smr_frac_benchmark_compare.pl:35
q_benchmark_second	numerator_denominator_gap	q_gap_thinking	smr_frac_benchmark_compare.pl:35
q_gap_thinking	compare_absolute_gaps	q_transitive_compare	smr_frac_benchmark_compare.pl:35
q_transitive_compare	omitted_external_value_relation	q_residual_compare	smr_frac_benchmark_compare.pl:35
q_residual_compare	unscaled_residuals	q_emit	smr_frac_benchmark_compare.pl:35
q_emit	emit	q_accept	smr_frac_benchmark_compare.pl:35
q_init	init	q_observed_1	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_1	selected	q_observed_2	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_2	numerator_denominator_gap	q_observed_3	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_observed_3	numerator_denominator_gap	q_observe_d_4	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_4	compare_absolute_gaps	q_observe_d_5	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_5	omitted_external_value_relation	q_observe_d_6	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_6	unscaled_residuals	q_observe_d_7	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_7	viability	q_observe_d_8	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_observed_8	emit	q_accept	observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

Computed structure: branching. one or more recorded states have multiple outgoing transitions; no loop edge is recorded. 17 states, 8 distinct actions, 1 branching states, 0 loop edges; 8 static rows and 18 observed rows.

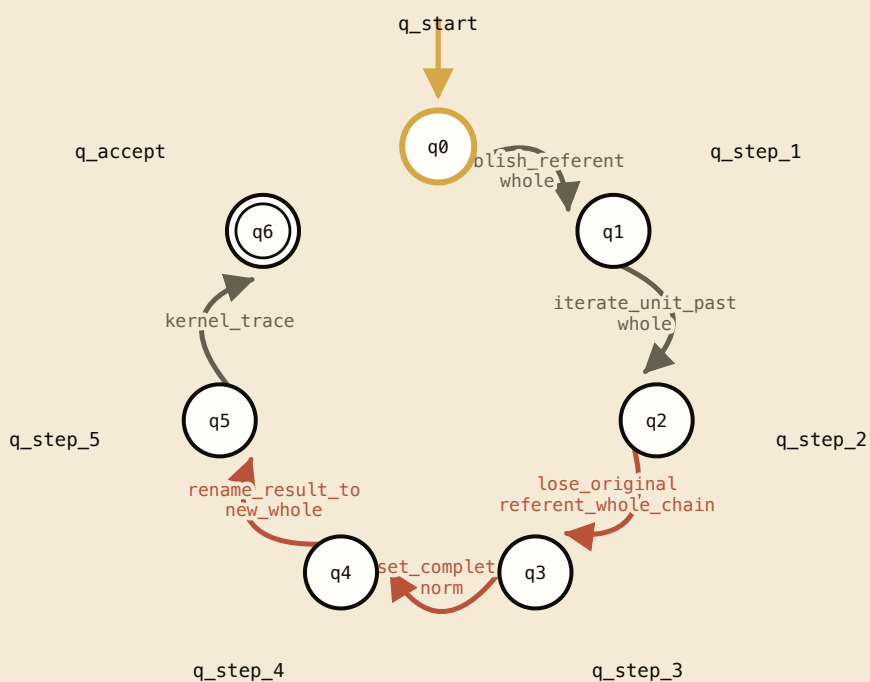
17. improper_fraction_chain_loss

iterated unit: 47/28



The run produced `reset_fraction(fraction(47,47))`; the recorded expected result is `fraction(47,28)`.

The run produced `reset_fraction(fraction(47,47))`; the recorded expected result is `fraction(47,28)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma \quad \text{establish_referent_whole, iterate_unit_past_whole, kernel_trace,} \\ \text{lose_original_referent_whole_chain, rename_result_to_new_whole,} \\ \text{reset_completion_norm (6 distinct)}$$

$$q_0 \quad q_0 = \text{q_start}$$

$$F \quad \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
 verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_referent_whole	q_step_1	<code>fraction_action_pairs.pl:338</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	iterate_unit_past_whole	q_step_2	<code>fraction_action_pairs.pl:338</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_ 2	lose_original_referent_whole_c hain	q_step_ 3	<code>fraction_action_pairs.pl:338</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 3	reset_completion_norm	q_step_ 4	<code>fraction_action_pairs.pl:338</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 4	rename_result_to_new_whole	q_step_ 5	<code>fraction_action_pairs.pl:338</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 5	kernel_trace	q_accep t	<code>fraction_action_pairs.pl:338</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

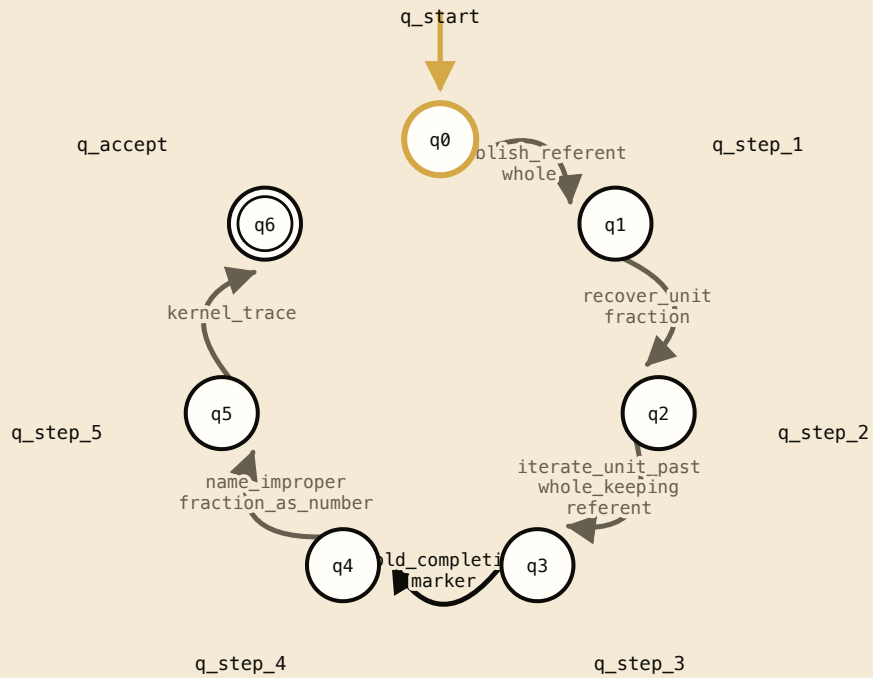
18. improper_fraction_iteration

iterated unit: 7/4



The run produced `fraction(7,4)`.

The run produced `fraction(7,4)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ establish_referent_whole, hold_completion_marker,
iterate_unit_past_whole_keeping_referent, kernel_trace,
name_improper_fraction_as_number, recover_unit_fraction (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "positive_integer", "b": "positive_integer"}; example {"a": 7, "b": 4}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_referent_whole	q_step_1	<code>fraction_action_pairs.pl:307</code> observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)
q_step_1	recover_unit_fraction	q_step_2	<code>fraction_action_pairs.pl:307</code> observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)
q_step_2	iterate_unit_past_whole_keeping_referent	q_step_3	<code>fraction_action_pairs.pl:307</code> observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)
q_step_3	hold_completion_marker	q_step_4	<code>fraction_action_pairs.pl:307</code> observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)
q_step_4	name_improper_fraction_as_number	q_step_5	<code>fraction_action_pairs.pl:307</code> observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)

From	Action	To	Provenance
q_step_5	kernel_trace	q_accept	<code>fraction_action_pairs.pl:307</code> observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)

Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

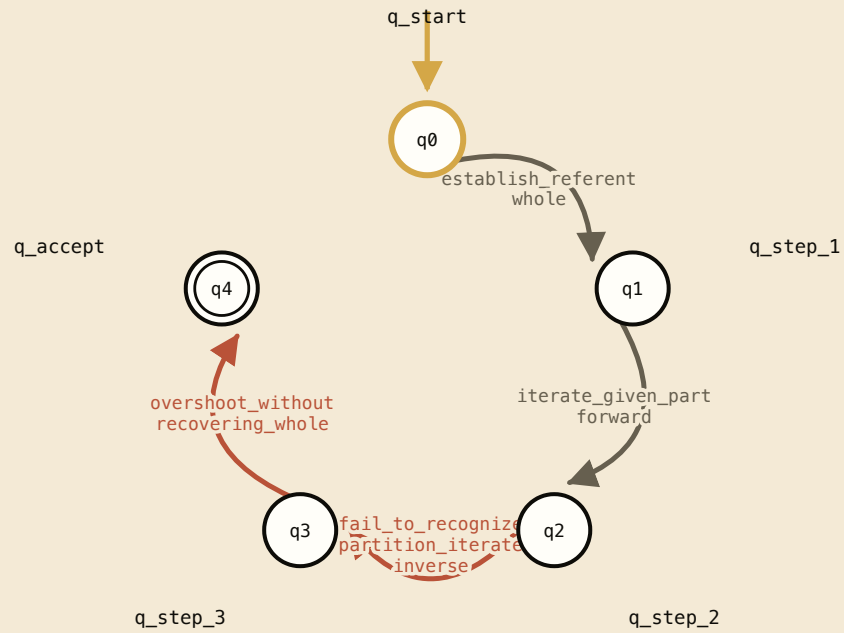
19. `iterate_given_overshoot`

iterated unit: 7/4



The run produced `overshot(fraction(7,4))`; the recorded expected result is `whole_recovered(unit(whole))`.

The run produced `overshot(fraction(7,4))`; the recorded expected result is `whole_recovered(unit(whole))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ $establish_referent_whole, fail_to_recognize_partition_iterate_inverse, iterate_given_part_forward, overshoot_without_recovering_whole$ (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "positive_integer", "b": "positive_integer"}; example {"a": 7, "b": 4}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_referent_whole	q_step_1	fraction_action_pairs.pl:267 observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)
q_step_1	iterate_given_part_forward	q_step_2	fraction_action_pairs.pl:267 observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)
q_step_2	fail_to_recognize_partition_iterate_inverse	q_step_3	fraction_action_pairs.pl:267 observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)
q_step_3	overshoot_without_recovering_whole	q_accept	fraction_action_pairs.pl:267 observed on {"a": 7, "b": 4} observed on {"a": 8, "b": 4} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

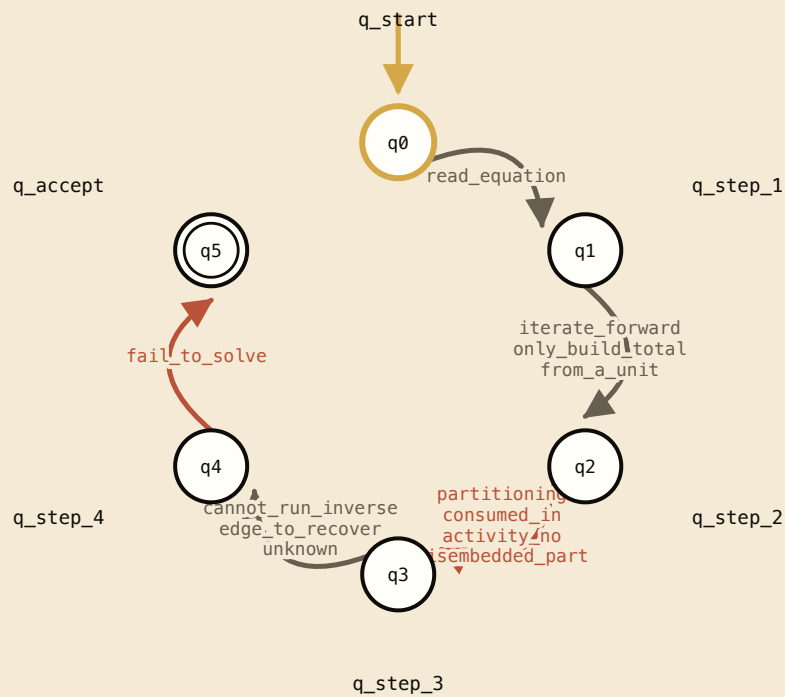
20. iterate_only_no_reverse

coefficient: $\frac{2}{3}$



The run produced `unknown_unrecovered`; the recorded expected result is `solved_unknown(coefficient(frac`

The run produced `unknown_unrecovered`; the recorded expected result is `solved_unknown(coefficient(fraction(2,3)),total(12),reciprocal_operator(fraction(3,2)))`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{\text{cannot_run_inverse_edge_to_recover_unknown}, \text{fail_to_solve}, \text{iterate_forward_only_build_total_from_a_unit}, \text{partitioning_consumed_in_activity_no_disembedded_part}, \text{read_equation}\} \text{ (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"coefficient": {"d": "positive_integer", "n": "positive_integer"}, "kind": "fraction_solve", "total": "positive_integer"}; example {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12};
verified(strategy_trace_ok).

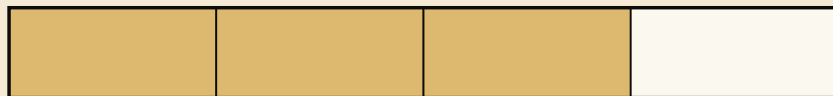
δ : recorded transitions

From	Action	To	Provenance
q_start	read_equation	q_step_1	fraction_action_pairs.pl:504 observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_1	iterate_forward_only_build_total_from_a_unit	q_step_2	fraction_action_pairs.pl:504 observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_2	partitioning_consumed_in_activity_no_disembedded_part	q_step_3	fraction_action_pairs.pl:504 observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_3	cannot_run_inverse_edge_to_recover_unknown	q_step_4	fraction_action_pairs.pl:504 observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_4	fail_to_solve	q_accept	fraction_action_pairs.pl:504 observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

21. measurement_division

left: $3/4$

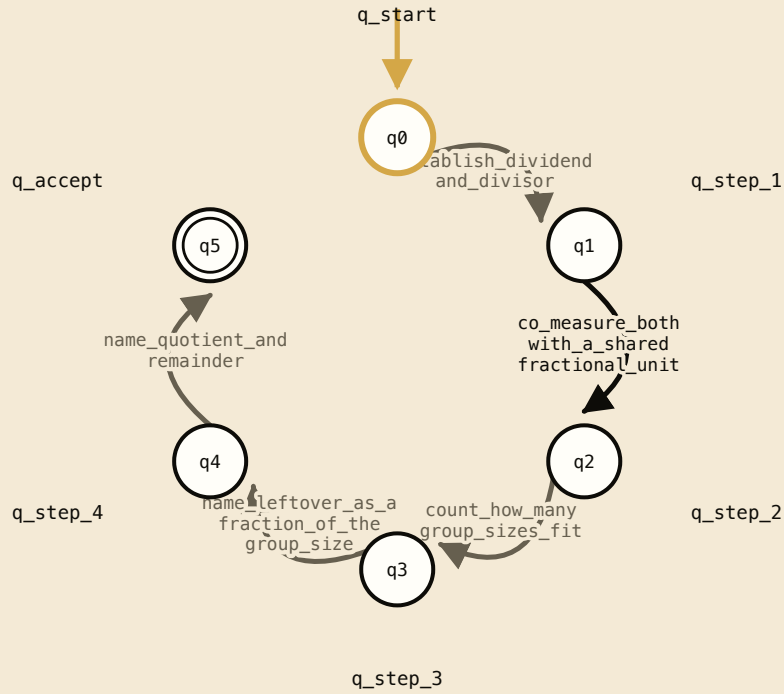


right: $2/3$



The run produced `fraction_division_quotient(fraction(3,4),fraction(2,3),whole_groups(1),remainder(fraction(1,8)))`.

The run produced `fraction_division_quotient(fraction(3,4),fraction(2,3),whole_groups(1),remainder(fraction(1,8)))`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ co_measure_both_with_a_shared_fractional_unit, count_how_many_group_sizes_fit, establish_dividend_and_divisor, name_leftover_as_a_fraction_of_the_group_size, name_quotient_and_remainder (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

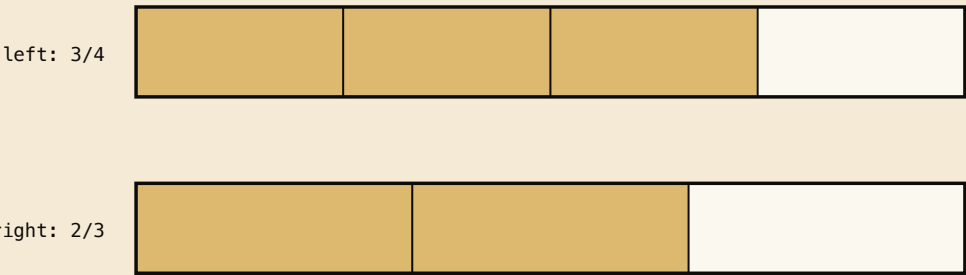
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_dividend_and_divisor	q_step_1	fraction_action_pairs.pl:1046 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_1	co_measure_both_with_a_shared_fractional_unit	q_step_2	fraction_action_pairs.pl:1046 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_2	count_how_many_group_sizes_fit	q_step_3	fraction_action_pairs.pl:1046 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_step_3	name_leftover_as_a_fraction_of_the_group_size	q_step_4	<code>fraction_action_pairs.pl:1046</code> observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_4	name_quotient_and_remainder	q_accept	<code>fraction_action_pairs.pl:1046</code> observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

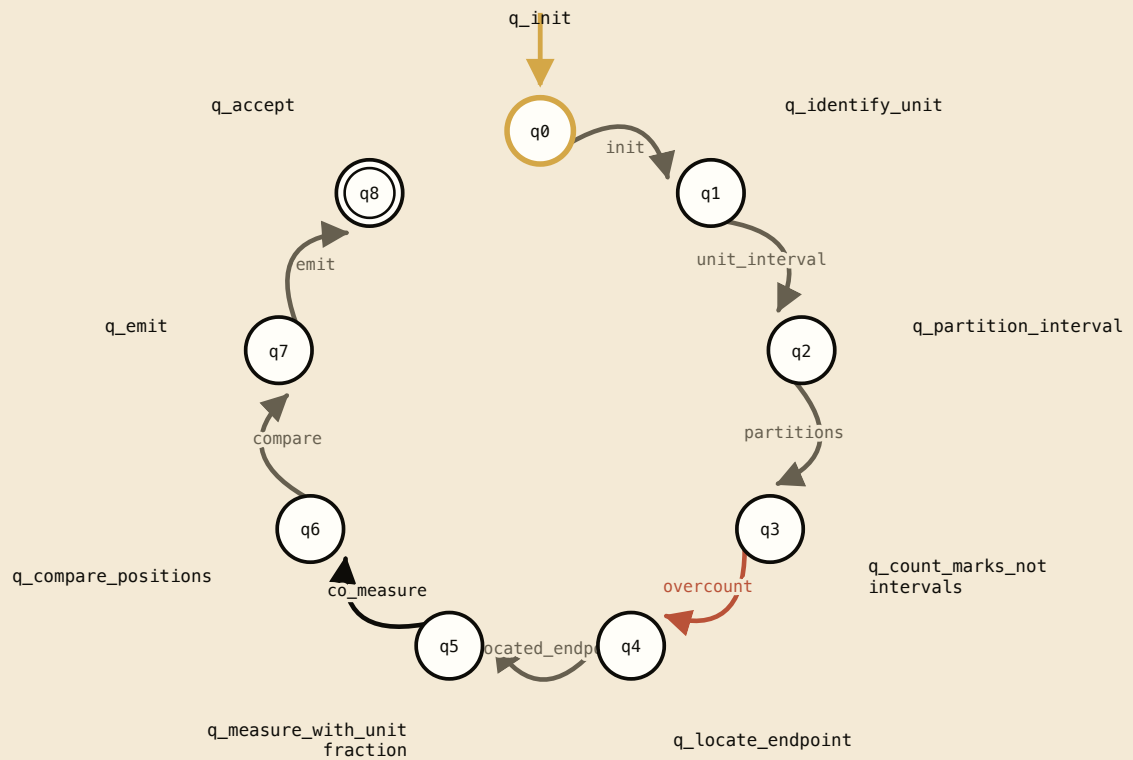
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

22. number_line_count_marks_not_intervals



The run produced equivalent; the recorded expected result is `greater_than`.

The run produced equivalent; the recorded expected result is `greater_than`. The scene is derived from the contract run (9 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ co_measure, compare, emit, init, misplaced_endpoints, overcount, partitions, unit_interval (8 distinct)

q_0 $q_0 = q_{\text{init}}$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_identify_unit	<i>the unit (whole)</i> (van_de_walle; Van de Walle, ch. 15; Shaughnessy 2011) <i>the unit</i> (bright_behr_post_wachsmuth; Bright, Behr, Post & Wachsmuth 1988)
q_2	q_partition_interval	<i>partition a number line into equal parts</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts) <i>partition a number line into fourths</i> (ccss; CCSS 3.NF.A.2b, via Van de Walle)
q_3	q_count_marks_not_intervals	<i>count marks instead of intervals</i> (bright_behr_post_wachsmuth; Bright, Behr, Post & Wachsmuth 1988) <i>count the tick marks rather than the space between the marks</i> (shaughnessy; Shaughnessy 2011, via Van de Walle)
q_4	q_locate_endpoint	<i>locating a fractional value on a number line</i> (van_de_walle; Van de Walle, ch. 15) <i>its endpoint locates the number a/b</i> (ccss; CCSS 3.NF.A.2b, via Van de Walle)
q_5	q_measure_with_unit_fraction	<i>fraction as a measure</i> (constructivist; Simon et al. 2018)
q_6	q_compare_positions	<i>compare the relative size of numbers on a number line</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_7	q_emit	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_identify_u nit	smr_frac_nl_compare.pl:65 observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 3}, "right": { "d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 4}, "right": { "d": 3, "n": 2}} (derived template)
q_identify_uni t	unit_interv al	q_partition_ interval	smr_frac_nl_compare.pl:65 observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 3}, "right": { "d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 4}, "right": { "d": 3, "n": 2}} (derived template)
q_partition_in terval	partitions	q_count_mark s_not_interv als	smr_frac_nl_compare.pl:65 observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 3}, "right": { "d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 4}, "right": { "d": 3, "n": 2}} (derived template)
q_count_marks_ not_intervals	overcount	q_locate_end point	smr_frac_nl_compare.pl:65 observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 3}, "right": { "d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 4}, "right": { "d": 3, "n": 2}} (derived template)
q_locate_endpo int	mislocated_ endpoints	q_measure_wi th_unit_frac tion	smr_frac_nl_compare.pl:65 observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 3}, "right": { "d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 4}, "right": { "d": 3, "n": 2}} (derived template)
q_measure_with _unit_fraction	co_measure	q_compare_po sitions	smr_frac_nl_compare.pl:65 observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 3}, "right": { "d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": { "d": 4, "n": 4}, "right": { "d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_compare_posi tions	compare	q_emit	smr_frac_nl_compare.pl:65 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_emit	emit	q_accept	smr_frac_nl_compare.pl:65 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

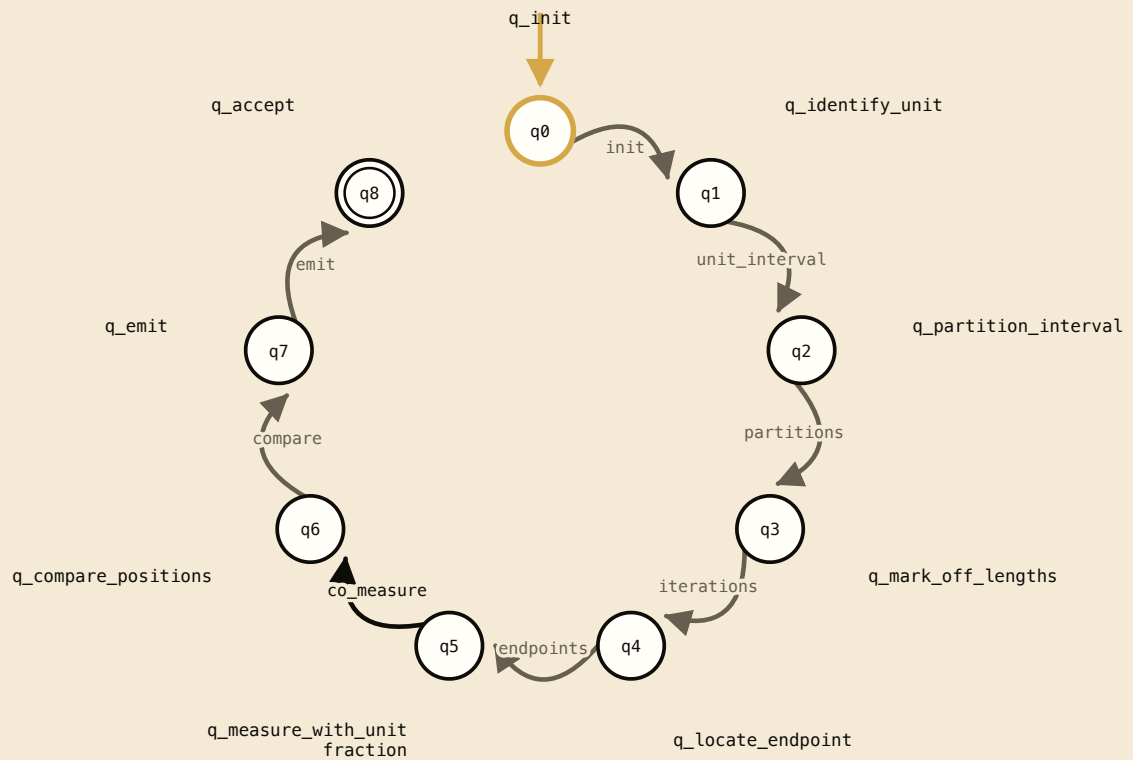
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 16 observed rows.

23. number_line_fraction_comparison



The run produced greater_than.

The run produced greater_than. The scene is derived from the contract run (9 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ co_measure, compare, emit, endpoints, init, iterations, partitions, unit_interval (8 distinct)

q_0 $q_0 = q_{\text{init}}$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_identify_unit	<i>the unit (whole)</i> (van_de_walle; Van de Walle, ch. 15; Shaughnessy 2011) <i>the unit</i> (bright_behr_post_wachsmuth; Bright, Behr, Post & Wachsmuth 1988)
q_2	q_partition_interval	<i>partition a number line into equal parts</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts) <i>partition a number line into fourths</i> (ccss; CCSS 3.NF.A.2b, via Van de Walle)
q_3	q_mark_off_lengths	<i>iterating</i> (constructivist; Steffe/Olive/Hackenberg fraction-scheme vocabulary; Van de Walle, ch. 15) <i>marking off lengths 1/b from 0</i> (ccss; CCSS 3.NF.A.2b, via Van de Walle)
q_4	q_locate_endpoint	<i>locating a fractional value on a number line</i> (van_de_walle; Van de Walle, ch. 15) <i>its endpoint locates the number a/b</i> (ccss; CCSS 3.NF.A.2b, via Van de Walle)
q_5	q_measure_with_unit_fraction	<i>fraction as a measure</i> (constructivist; Simon et al. 2018)
q_6	q_compare_positions	<i>compare the relative size of numbers on a number line</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_7	q_emit	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_identify_unit	smr_frac_nl_compare.pl:29 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_identify_unit	unit_interval	q_partition_interval	smr_frac_nl_compare.pl:29 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_partition_interval	partitions	q_mark_off_lengths	smr_frac_nl_compare.pl:29 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_mark_off_lengths	iterations	q_locate_endpoint	smr_frac_nl_compare.pl:29 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_locate_endpoint	endpoints	q_measure_with_unit_fraction	smr_frac_nl_compare.pl:29 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_measure_with_unit_fraction	co_measure	q_compare_positions	smr_frac_nl_compare.pl:29 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_compare_posi tions	compare	q_emit	smr_frac_nl_compare.pl:29 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_emit	emit	q_accept	smr_frac_nl_compare.pl:29 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

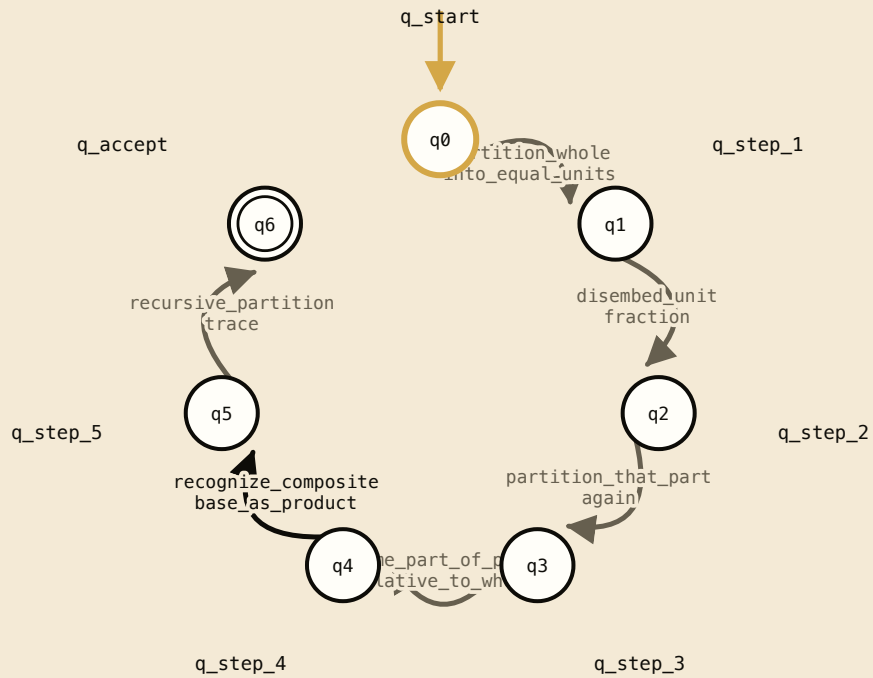
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 16 observed rows.

24. recursive_partition



The run produced fraction(1,12).

The run produced fraction(1,12). The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ disembed_unit_fraction, name_part_of_part_relative_to_whole, partition_that_part_again, partition_whole_into_equal_units, recognize_composite_base_as_product, recursive_partition_trace (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "positive_integer", "b": "positive_integer"}; example {"a": 3, "b": 4}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	partition_whole_into_equal_units	q_step_1	fraction_action_pairs.pl:382 observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_1	disembed_unit_fraction	q_step_2	fraction_action_pairs.pl:382 observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_2	partition_that_part_again	q_step_3	fraction_action_pairs.pl:382 observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_3	name_part_of_part_relative_to_whole	q_step_4	fraction_action_pairs.pl:382 observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_4	recognize_composite_base_as_product	q_step_5	fraction_action_pairs.pl:382 observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)

From	Action	To	Provenance
q_step_5	recursive_partition_trace	q_accept	fraction_action_pairs.pl:382 observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

25. reversible_measurement_division

left: $\frac{3}{4}$

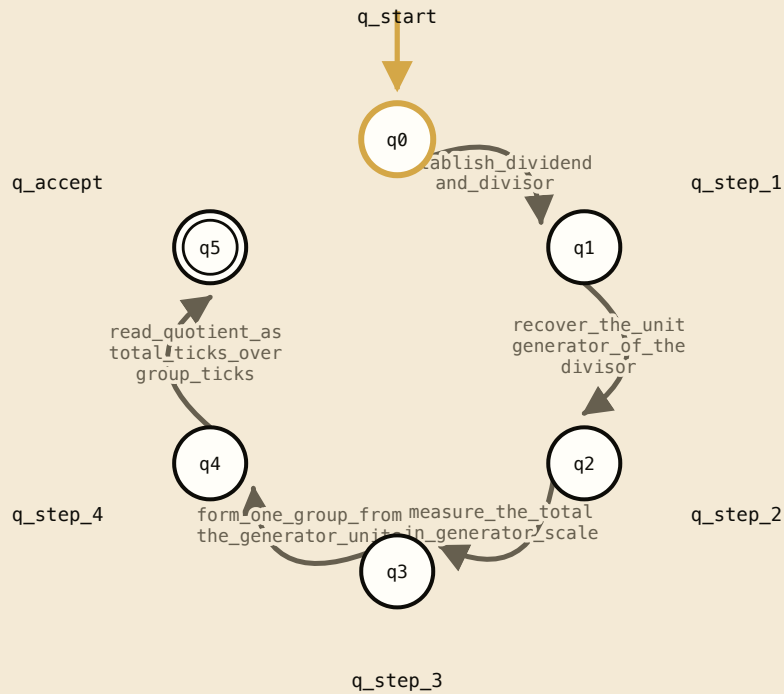


right: $\frac{2}{3}$



The run produced `fraction_division_quotient(fraction(3,4),fraction(2,3),quotient(fraction(9,8)))`.

The run produced `fraction_division_quotient(fraction(3,4),fraction(2,3),quotient(fraction(9,8)))`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ `establish_dividend_and_divisor`, `form_one_group_from_the_generator_units`,
`measure_the_total_in_generator_scale`,
`read_quotient_as_total_ticks_over_group_ticks`,
`recover_the_unit_generator_of_the_divisor` (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

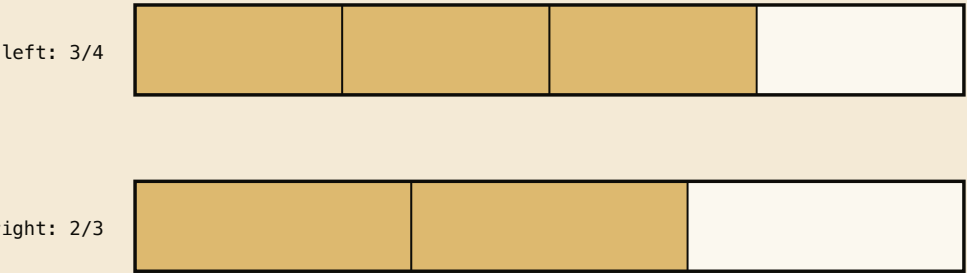
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_dividend_and_divisor	q_step_1	fraction_action_pairs.pl:1085 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_1	recover_the_unit_generator_of_the_divisor	q_step_2	fraction_action_pairs.pl:1085 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_2	measure_the_total_in_generator_scale	q_step_3	fraction_action_pairs.pl:1085 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_step_3	form_one_group_from_the_generator_units	q_step_4	<code>fraction_action_pairs.pl:1085</code> observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_step_4	read_quotient_as_total_ticks_over_group_ticks	q_accept	<code>fraction_action_pairs.pl:1085</code> observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

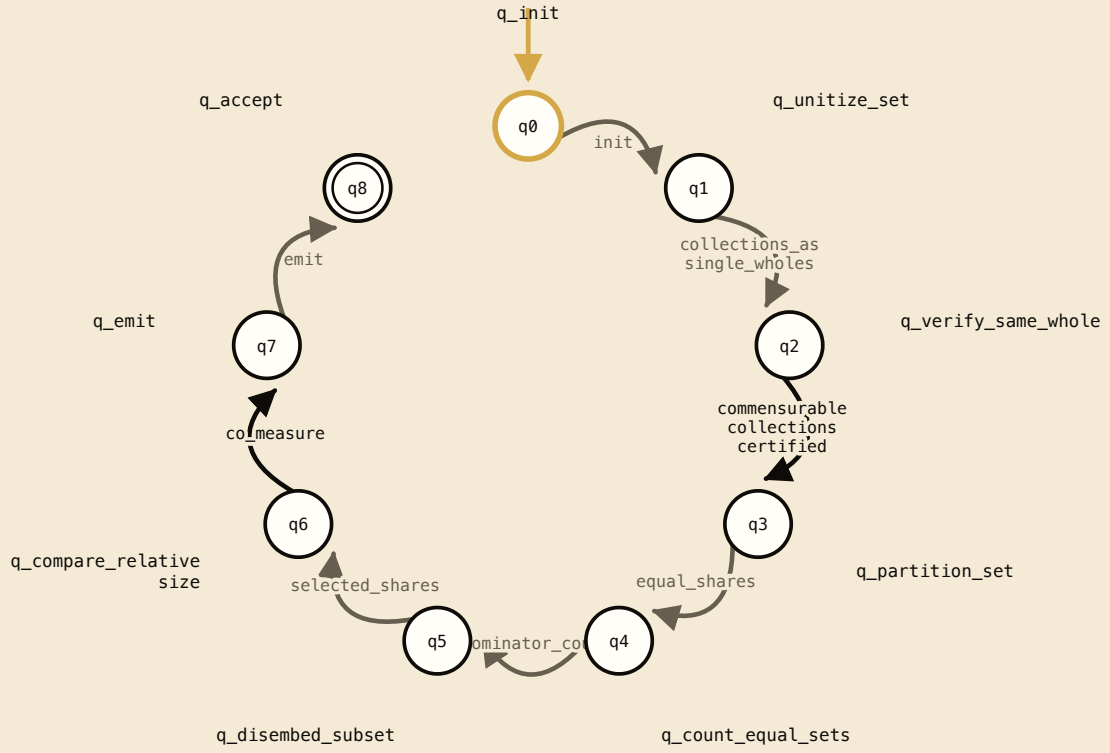
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

26. set_model_fraction_comparison



The run produced `greater_than`.

The run produced `greater_than`. The scene is derived from the contract run (9 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ co_measure, collections_as_single_wholes, commensurable_collections_certified, denominator_counts, emit, equal_shares, init, selected_shares (8 distinct)

q_0 $q_0 = q_{\text{init}}$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_unitize_set	<i>unitizing</i> (constructivist; Steffe/Olive/Hackenberg fraction-scheme vocabulary) <i>the whole is understood to be a set of objects</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_2	q_verify_same_whole	<i>fraction size is relative</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts)
q_3	q_partition_set	<i>partition sets of objects</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts)
q_4	q_count_equal_sets	<i>the number of equal sets in the whole</i> (van_de_walle; Van de Walle, ch. 15)
q_5	q_disembed_subset	<i>disembedding</i> (constructivist; Hackenberg 2014) <i>subsets of the whole make up fractional parts</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_6	q_compare_relative_size	<i>reasoning about the relative size of the fractions</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions) <i>numerator and denominator strategy</i> (behr_post_lesh; Behr, Wachsmuth, Post & Lesh 1984)
q_7	q_emit	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_unitize_set	smr_frac_set_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_unitize_set	collections_as_single_wholes	q_verify_same_whole	smr_frac_set_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_verify_same_whole	commensurable_collections_certified	q_partition_set	smr_frac_set_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_partition_set	equal_shares	q_count_equal_sets	smr_frac_set_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_count_equal_sets	denominator_counts	q_disembed_subset	smr_frac_set_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_disembed_subset	selected_shares	q_compare_relative_size	smr_frac_set_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)
q_compare_relative_size	co_measure	q_emit	smr_frac_set_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

From	Action	To	Provenance
q_emit	emit	q_accept	smr_frac_set_compare.pl:13 observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 3}, "right": {"d": 3, "n": 2}} observed on {"kind": "fraction_pair", "left": {"d": 4, "n": 4}, "right": {"d": 3, "n": 2}} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 16 observed rows.

27. set_model_subset_size_focus

left: 2/3

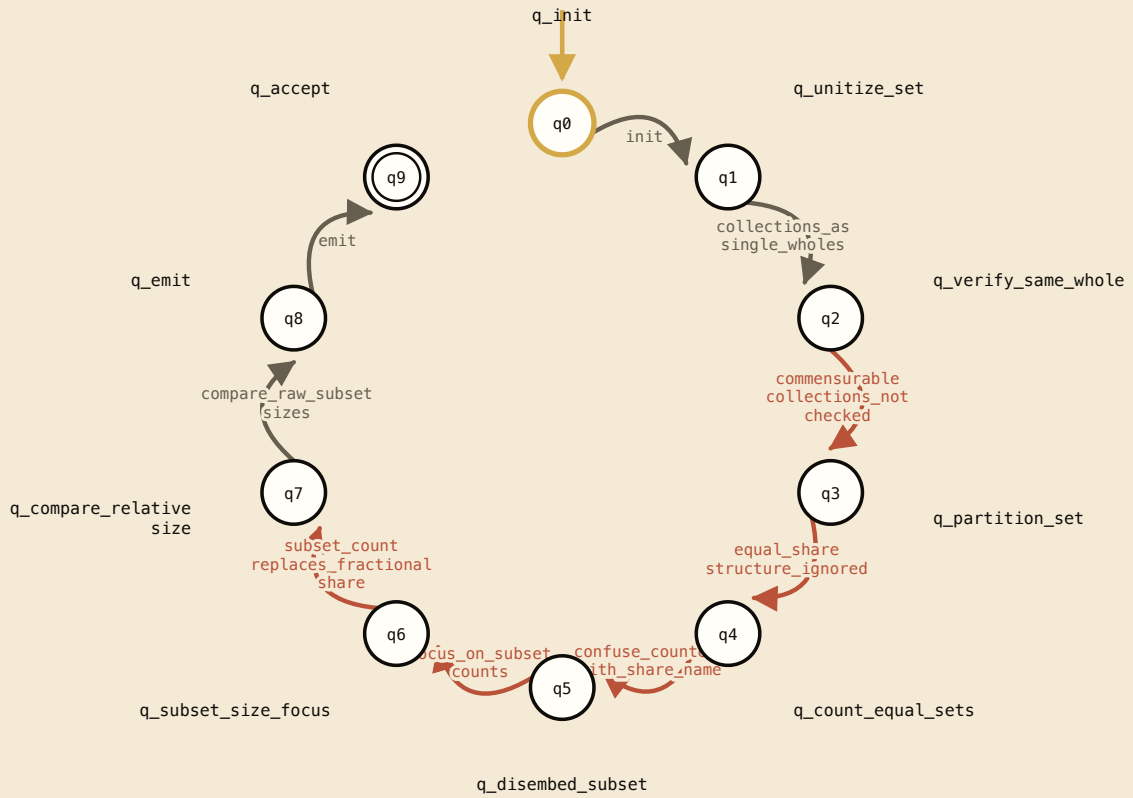


right: 3/8



The run produced less_than; the recorded expected result is greater_than.

The run produced less_than; the recorded expected result is greater_than. The scene is derived from the contract run (10 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9\}$

Σ collections_as_single_wholes, commensurable_collections_not_checked, compare_raw_subset_sizes, confuse_counters_with_share_name, emit, equal_share_structure_ignored, focus_on_subset_counts, init, subset_count_replaces_fractional_share (9 distinct)

q_0 $q_0 = q_init$

F $\{q_9\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_init	none recorded
q_1	q_unitize_set	<i>unitizing</i> (constructivist; Steffe/Olive/Hackenberg fraction-scheme vocabulary) <i>the whole is understood to be a set of objects</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_2	q_verify_same_whole	<i>fraction size is relative</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts)
q_3	q_partition_set	<i>partition sets of objects</i> (van_de_walle; Van de Walle, ch. 15, Fractional Parts)
q_4	q_count_equal_sets	<i>the number of equal sets in the whole</i> (van_de_walle; Van de Walle, ch. 15)
q_5	q_disembed_subset	<i>disembedding</i> (constructivist; Hackenberg 2014) <i>subsets of the whole make up fractional parts</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_6	q_subset_size_focus	<i>focus on the size of a subset rather than the number of equal sets</i> (van_de_walle; Van de Walle, ch. 15, Models for Fractions)
q_7	q_compare_relative_size	<i>reasoning about the relative size of the fractions</i> (van_de_walle; Van de Walle, ch. 15, Comparing Fractions) <i>numerator and denominator strategy</i> (behr_post_lesh; Behr, Wachsmuth, Post & Lesh 1984)
q_8	q_emit	none recorded
q_9	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_init	init	q_unitize_set	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_unitize_set	collections_as_single_wholes	q_verify_same_whole	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_verify_same_whole	commensurable_collections_not_checked	q_partition_set	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_partition_set	equal_share_structure_ignored	q_count_equal_sets	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_count_equal_sets	confuse_counters_with_share_name	q_disembed_subset	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_disembed_subset	focus_on_subset_counts	q_subset_size_focus	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)

From	Action	To	Provenance
q_subset_size_focus	subset_count_replaces_fractional_share	q_compare_relative_size	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_compare_relative_size	compare_raw_subset_sizes	q_emit	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)
q_emit	emit	q_accept	smr_frac_set_compare.pl:32 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 8, "n": 3}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 3}, "right": {"d": 8, "n": 3}} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 10 states, 9 distinct actions, 0 branching states, 0 loop edges; 9 static rows and 18 observed rows.

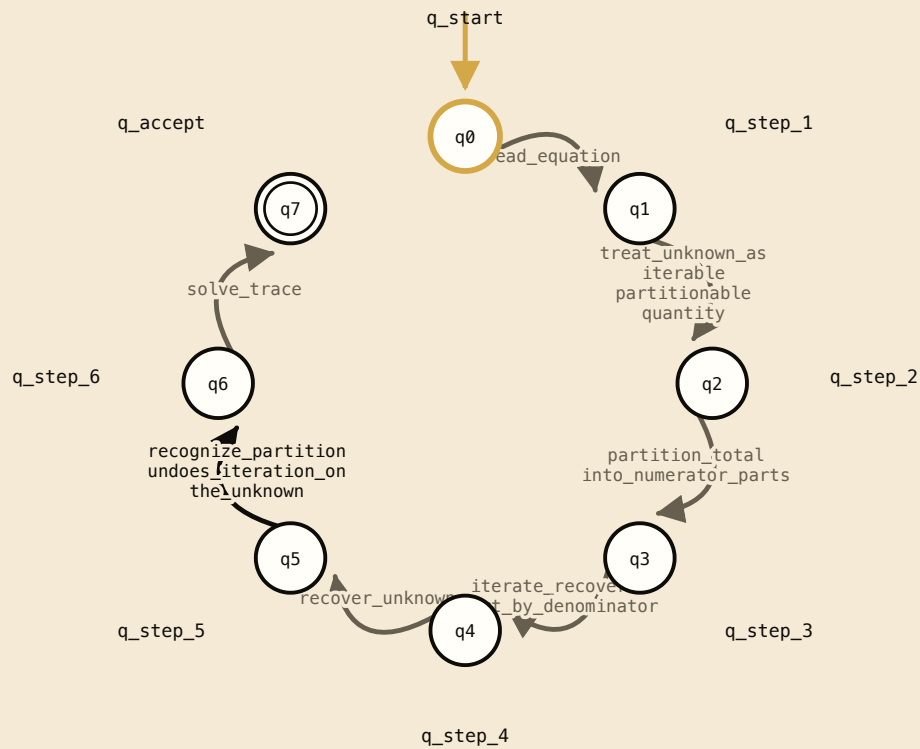
28. solve_for_unit

coefficient: $\frac{2}{3}$



The run produced `solved_unknown(coefficient(fraction(2,3)),total(12),reciprocal_operator(fraction(3,2)))`.

The run produced `solved_unknown(coefficient(fraction(2,3)),total(12),reciprocal_operator(fraction(3,2)))`. The scene is derived from the contract run (7 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}$

Σ $\text{iterate_recovered_part_by_denominator, partition_total_into_numerator_parts, read_equation, recognize_partition_undoes_iteration_on_the_unknown, recover_unknown, solve_trace, treat_unknown_as_iterable_partitionable_quantity}$ (7 distinct)

q_0 $q_0 = q_start$

F $\{q_7\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_step_6	none recorded
q_7	q_accept	none recorded

Input contract: {"coefficient": {"d": "positive_integer", "n": "positive_integer"}, "kind": "fraction_solve", "total": "positive_integer"}; example {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	read_equation	q_step_1	<code>fraction_action_pairs.pl:469</code> observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_1	treat_unknown_as_iterable_partitionable_quantity	q_step_2	<code>fraction_action_pairs.pl:469</code> observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_2	partition_total_into_numerator_parts	q_step_3	<code>fraction_action_pairs.pl:469</code> observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_3	iterate_recovered_part_by_denominator	q_step_4	<code>fraction_action_pairs.pl:469</code> observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_4	recover_unknown	q_step_5	<code>fraction_action_pairs.pl:469</code> observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}

From	Action	To	Provenance
q_step_5	recognize_partition_undoes_iteration_on_the_unknown	q_step_6	<code>fraction_action_pairs.pl:469</code> observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}
q_step_6	solve_trace	q_accept	<code>fraction_action_pairs.pl:469</code> observed on {"coefficient": {"d": 3, "n": 2}, "kind": "fraction_solve", "total": 12}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 8 states, 7 distinct actions, 0 branching states, 0 loop edges; 7 static rows and 7 observed rows.

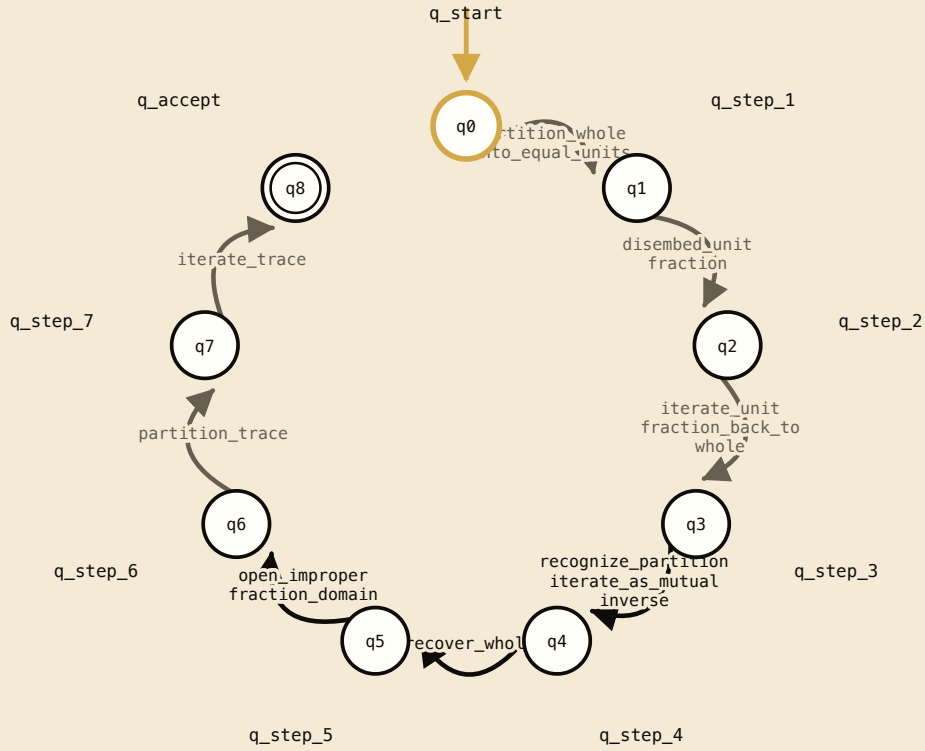
29. splitting

iterated unit: 1/4



The run produced `whole_recovered(unit(whole))`.

The run produced `whole_recovered(unit(whole))`. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ disembed_unit_fraction, iterate_trace, iterate_unit_fraction_back_to_whole, open_improper_fraction_domain, partition_trace, partition_whole_into_equal_units, recognize_partition_iterate_as_mutual_inverse, recover_whole (8 distinct)

q_0 $q_0 = q_{start}$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_step_6	none recorded
q_7	q_step_7	none recorded
q_8	q_accept	none recorded

Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 1, "b": 4};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	partition_whole_into_equal_units	q_step_1	<code>fraction_action_pairs.pl:228</code> observed on {"a": 1, "b": 4}
q_step_1	disembed_unit_fraction	q_step_2	<code>fraction_action_pairs.pl:228</code> observed on {"a": 1, "b": 4}
q_step_2	iterate_unit_fraction_back_to_whole	q_step_3	<code>fraction_action_pairs.pl:228</code> observed on {"a": 1, "b": 4}
q_step_3	recognize_partition_iterate_as_mutual_inverse	q_step_4	<code>fraction_action_pairs.pl:228</code> observed on {"a": 1, "b": 4}
q_step_4	recover_whole	q_step_5	<code>fraction_action_pairs.pl:228</code> observed on {"a": 1, "b": 4}

From	Action	To	Provenance
q_step_5	open_improper_fraction_domain	q_step_6	<code>fraction_action_pairs.pl:22</code> 8 observed on {"a": 1, "b": 4}
q_step_6	partition_trace	q_step_7	<code>fraction_action_pairs.pl:22</code> 8 observed on {"a": 1, "b": 4}
q_step_7	iterate_trace	q_accept	<code>fraction_action_pairs.pl:22</code> 8 observed on {"a": 1, "b": 4}

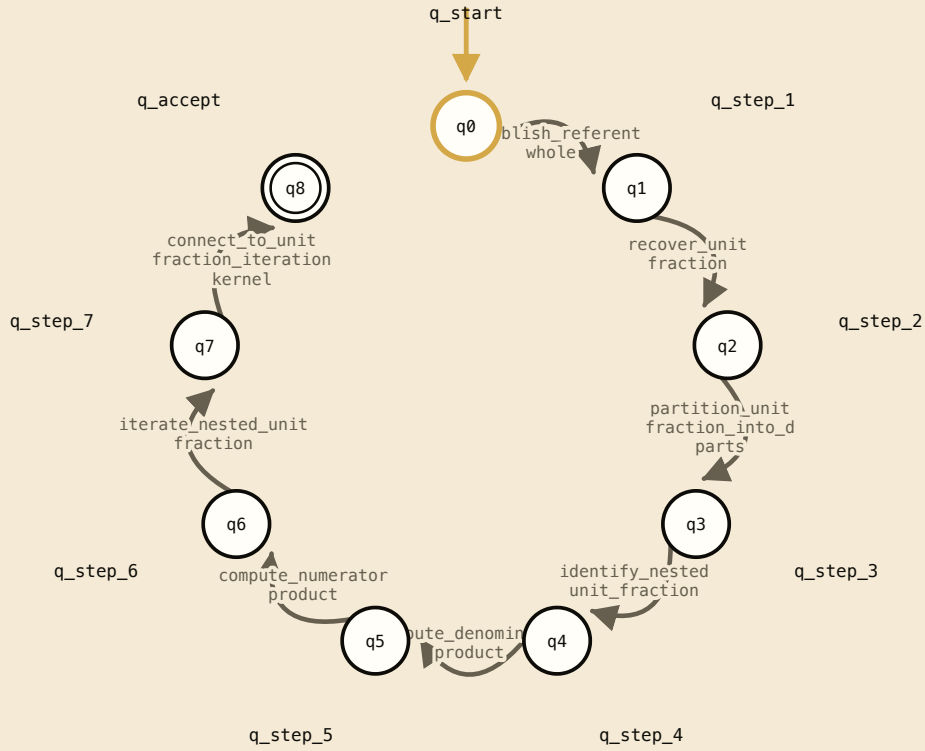
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 8 observed rows.

30. unit_fraction_denominator_product_rule



The run produced `fraction(1,12)`.

The run produced `fraction(1,12)`. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ compute_denominator_product, compute_numerator_product, connect_to_unit_fraction_iteration_kernel, establish_referent_whole, identify_nested_unit_fraction, iterate_nested_unit_fraction, partition_unit_fraction_into_d_parts, recover_unit_fraction (8 distinct)

q_0 $q_0 = q_start$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_step_6	none recorded
q_7	q_step_7	none recorded
q_8	q_accept	none recorded

Input contract: {"kind": "fraction_pair", "left": {"d": "positive_integer", "n": "integer"}, "right": {"d": "positive_integer", "n": "integer"}}; example {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_referent_whole	q_step_1	<code>fraction_action_pairs.pl:752</code> observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 4, "n": 1}} (derived template)
q_step_1	recover_unit_fraction	q_step_2	<code>fraction_action_pairs.pl:752</code> observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 4, "n": 1}} (derived template)
q_step_2	partition_unit_fraction_into_d_parts	q_step_3	<code>fraction_action_pairs.pl:752</code> observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 4, "n": 1}} (derived template)

From	Action	To	Provenance
q_step_3	identify_nested_unit_fraction	q_step_4	fraction_action_pairs.pl:752 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 4, "n": 1}} (derived template)
q_step_4	compute_denominator_product	q_step_5	fraction_action_pairs.pl:752 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 4, "n": 1}} (derived template)
q_step_5	compute_numerator_product	q_step_6	fraction_action_pairs.pl:752 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 4, "n": 1}} (derived template)
q_step_6	iterate_nested_unit_fraction	q_step_7	fraction_action_pairs.pl:752 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 4, "n": 1}} (derived template)
q_step_7	connect_to_unit_fraction_iteration_kernel	q_accept	fraction_action_pairs.pl:752 observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 1}, "right": {"d": 4, "n": 1}} observed on {"kind": "fraction_pair", "left": {"d": 3, "n": 2}, "right": {"d": 4, "n": 1}} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 16 observed rows.

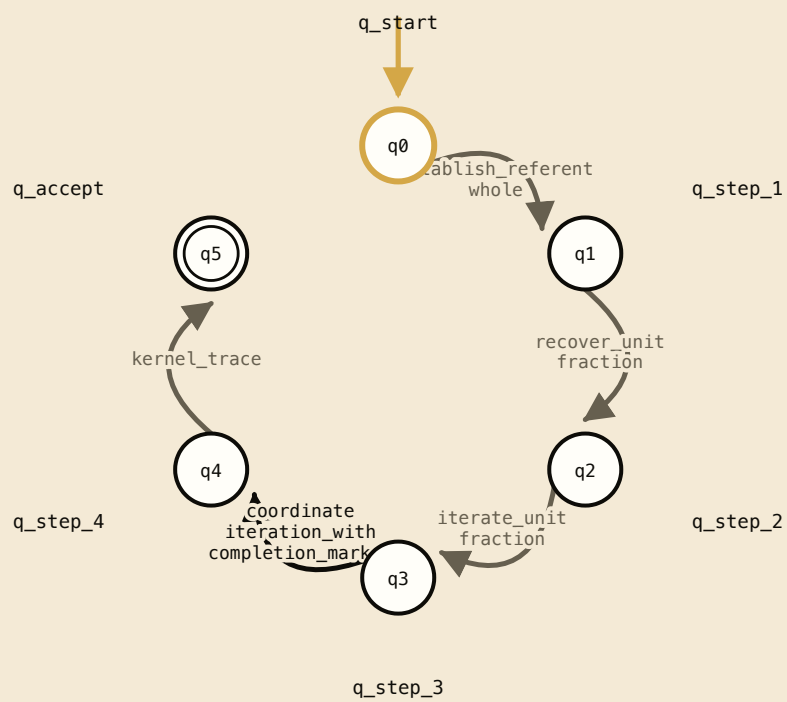
31. unit_fraction_iteration

iterated unit: 47/28



The run produced `fraction(47,28)`.

The run produced `fraction(47,28)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ coordinate_iteration_with_completion_marker, establish_referent_whole, iterate_unit_fraction, kernel_trace, recover_unit_fraction (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_referent_whole	q_step_1	fraction_action_pairs.pl:157 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	recover_unit_fraction	q_step_2	fraction_action_pairs.pl:157 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	iterate_unit_fraction	q_step_3	fraction_action_pairs.pl:157 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_3	coordinate_iteration_with_completion_marker	q_step_4	<code>fraction_action_pairs.pl:157</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	kernel_trace	q_accept	<code>fraction_action_pairs.pl:157</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

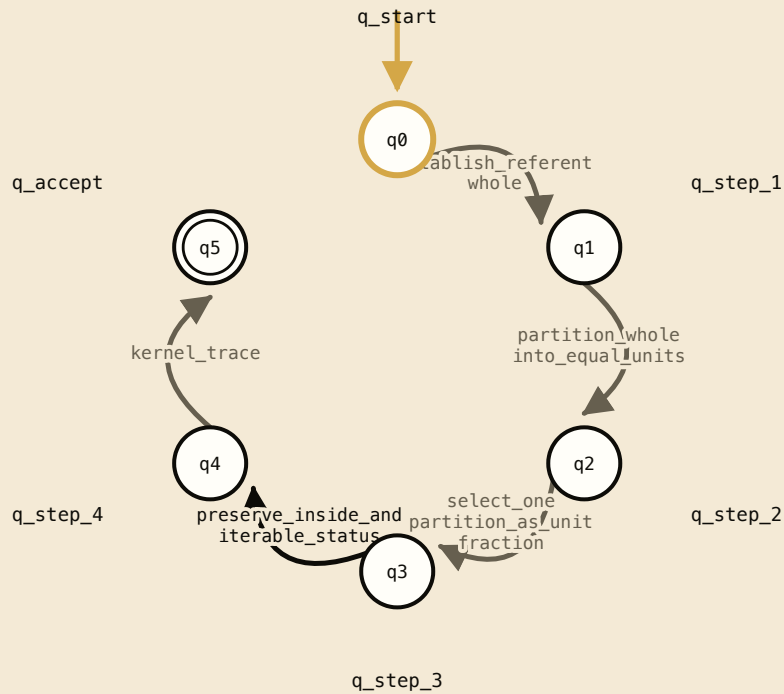
32. unit_fraction_partition

iterated unit: 1/4



The run produced `fraction(1,4)`.

The run produced `fraction(1,4)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ establish_referent_whole, kernel_trace, partition_whole_into_equal_units, preserve_inside_and_iterable_status, select_one_partition_as_unit_fraction (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "positive_integer"}; example {"a": 1, "b": 4};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_referent_whole	q_step_1	<code>fraction_action_pairs.pl:126</code> observed on {"a": 1, "b": 4}
q_step_1	partition_whole_into_equal_units	q_step_2	<code>fraction_action_pairs.pl:126</code> observed on {"a": 1, "b": 4}
q_step_2	select_one_partition_as_unit_fraction	q_step_3	<code>fraction_action_pairs.pl:126</code> observed on {"a": 1, "b": 4}
q_step_3	preserve_inside_and_iterable_status	q_step_4	<code>fraction_action_pairs.pl:126</code> observed on {"a": 1, "b": 4}
q_step_4	kernel_trace	q_accept	<code>fraction_action_pairs.pl:126</code> observed on {"a": 1, "b": 4}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

Authored class: finite_state_automaton, attested at
knowledge/strategies/math/fraction_partitioning.pl:3. This claim comes from module
prose, not from the table graph.

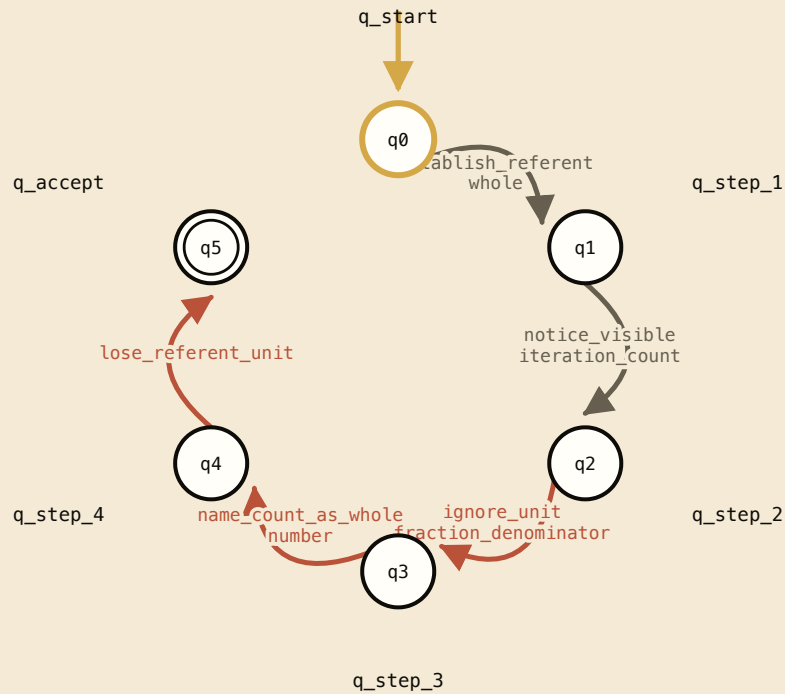
33. whole_number_grab

iterated unit: $\frac{3}{4}$



The run produced `whole_number(3)`; the recorded expected result is `fraction(3,4)`.

The run produced `whole_number(3)`; the recorded expected result is `fraction(3,4)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ establish_referent_whole, ignore_unit_fraction_denominator, lose_referent_unit, name_count_as_whole_number, notice_visible_iteration_count (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

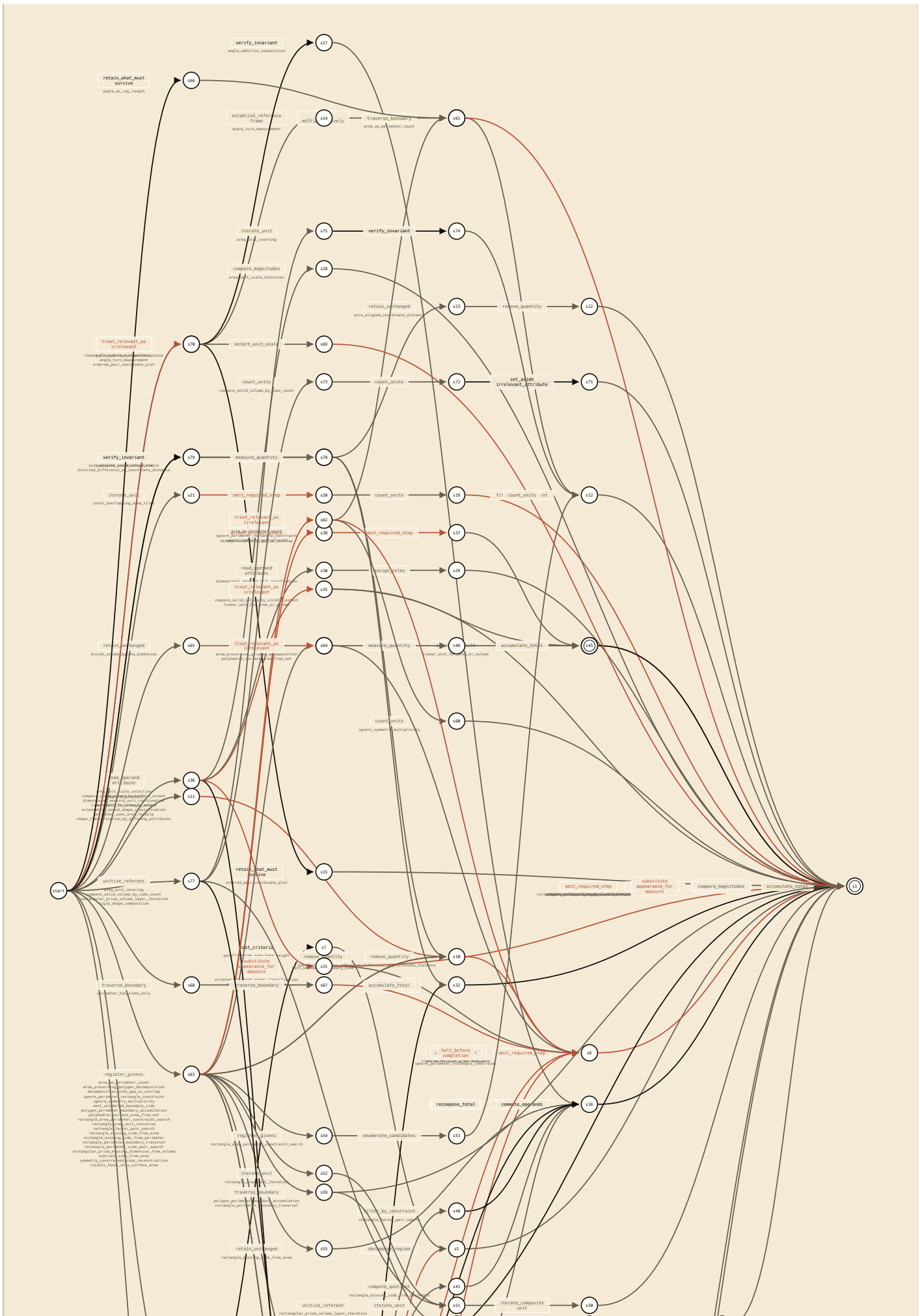
Input contract: {"a": "positive_integer", "b": "positive_integer"}; example {"a": 3, "b": 4}; verified(strategy_trace_ok).

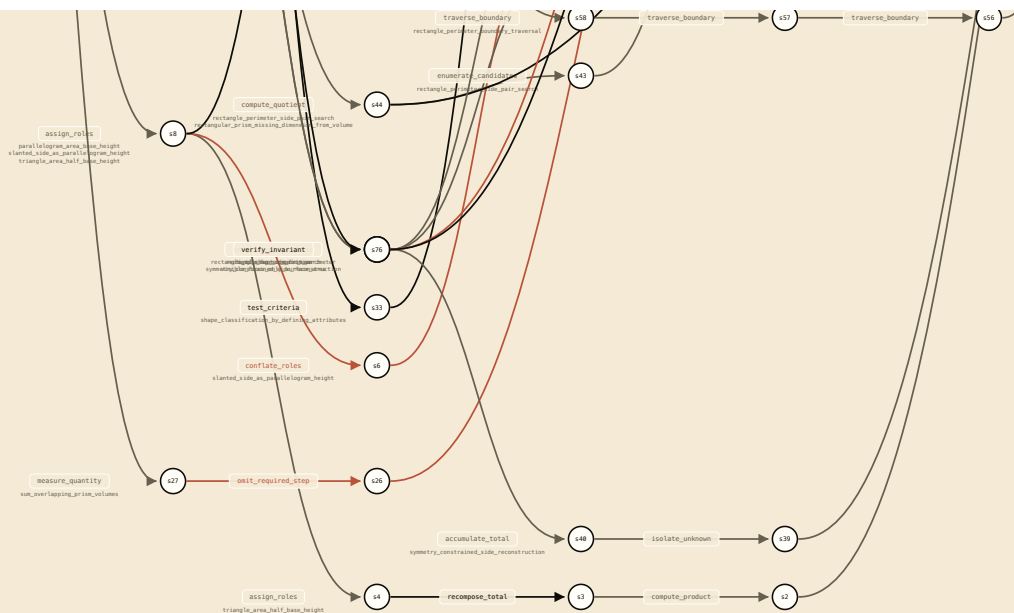
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_referent_whole	q_step_1	<code>fraction_action_pairs.pl:185</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_1	notice_visible_iteration_count	q_step_2	<code>fraction_action_pairs.pl:185</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_2	ignore_unit_fraction_denominator	q_step_3	<code>fraction_action_pairs.pl:185</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_3	name_count_as_whole_number	q_step_4	<code>fraction_action_pairs.pl:185</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)
q_step_4	lose_referent_unit	q_accept	<code>fraction_action_pairs.pl:185</code> observed on {"a": 3, "b": 4} observed on {"a": 4, "b": 4} (derived template)

Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

geometry





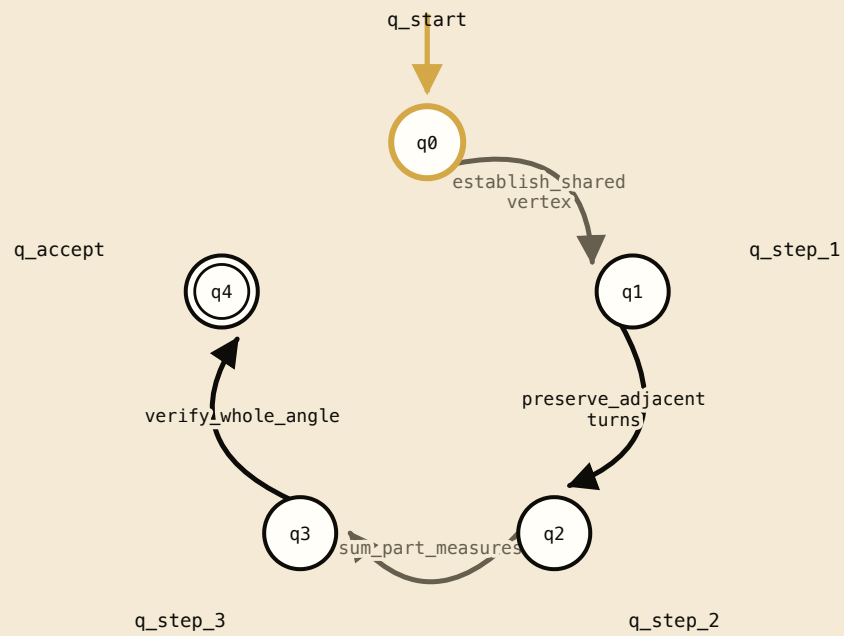
Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. angle_additive_composition



The run produced `angle_measure(130)`.

The run produced `angle_measure(130)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ establish_shared_vertex, preserve_adjacent_turns, sum_part_measures, verify_whole_angle (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

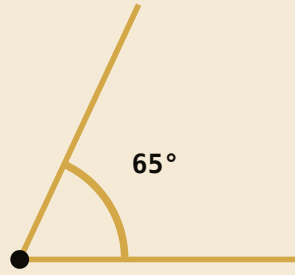
Input contract: {"kind": "angle_parts", "parts": ["positive_integer"], "whole": "positive_integer"}; example {"kind": "angle_parts", "parts": [35, 50, 45], "whole": 130}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_shared_vertex	q_step_1	geometry_action_pairs.pl:1168 observed on {"kind": "angle_parts", "parts": [35, 50, 45], "whole": 130}
q_step_1	preserve_adjacent_turns	q_step_2	geometry_action_pairs.pl:1168 observed on {"kind": "angle_parts", "parts": [35, 50, 45], "whole": 130}
q_step_2	sum_part_measures	q_step_3	geometry_action_pairs.pl:1168 observed on {"kind": "angle_parts", "parts": [35, 50, 45], "whole": 130}
q_step_3	verify_whole_angle	q_accept	geometry_action_pairs.pl:1168 observed on {"kind": "angle_parts", "parts": [35, 50, 45], "whole": 130}

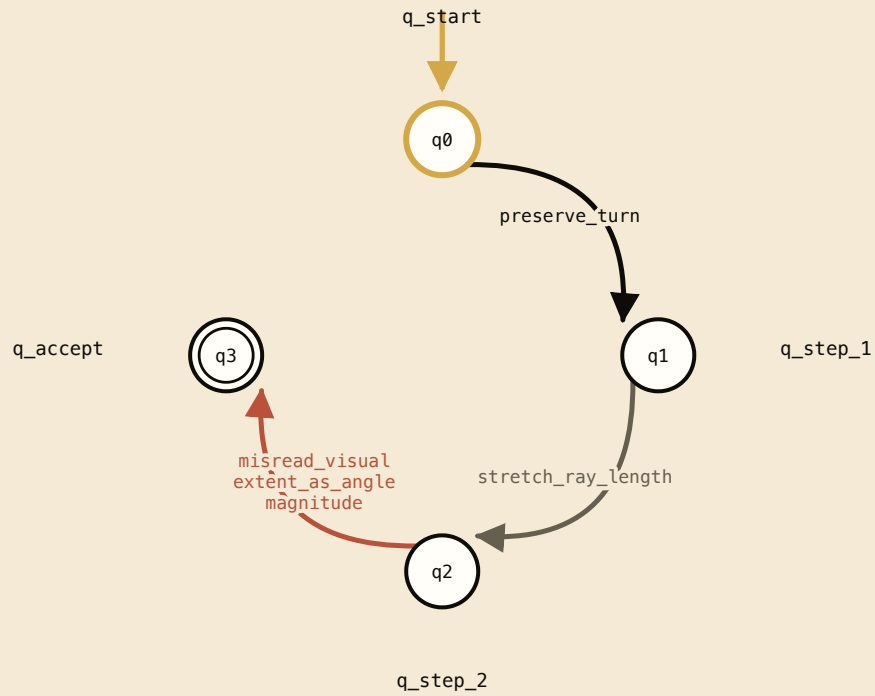
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

2. angle_as_ray_length



The run produced `misread_as_larger(angle_measure(65))`; the recorded expected result is `angle_measure(`

The run produced `misread_as_larger(angle_measure(65))`; the recorded expected result is `angle_measure(65)`. The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \text{misread_visual_extent_as_angle_magnitude, preserve_turn, stretch_ray_length (3 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_3\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

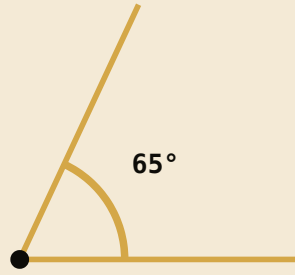
Input contract: {"degrees": "positive_integer", "kind": "angle_measure"}; example {"degrees": 65, "kind": "angle_measure"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_turn	q_step_1	geometry_action_pairs.pl:1146 observed on {"degrees": 65, "kind": "angle_measure"}
q_step_1	stretch_ray_length	q_step_2	geometry_action_pairs.pl:1146 observed on {"degrees": 65, "kind": "angle_measure"}
q_step_2	misread_visual_extent_as_angle_magnitude	q_accept	geometry_action_pairs.pl:1146 observed on {"degrees": 65, "kind": "angle_measure"}

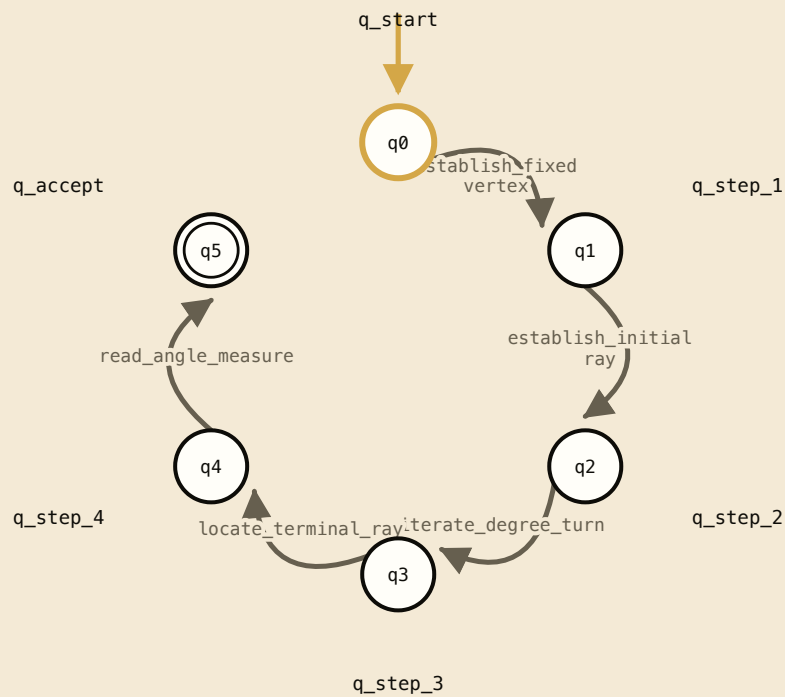
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

3. angle_turn_measurement



The run produced `angle_measure(65)`.

The run produced `angle_measure(65)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ establish_fixed_vertex, establish_initial_ray, iterate_degree_turn, locate_terminal_ray, read_angle_measure (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"degrees": "positive_integer", "kind": "angle_measure"}; example {"degrees": 65, "kind": "angle_measure"}; verified(strategy_trace_ok).

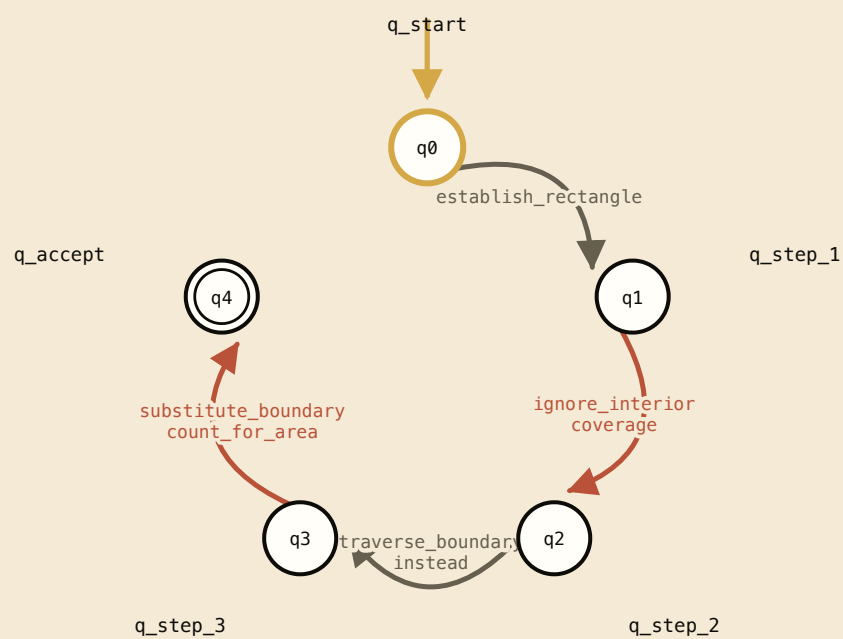
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_fixed_vertex	q_step_1	geometry_action_pairs.pl:1122 observed on {"degrees": 65, "kind": "angle_measure"}
q_step_1	establish_initial_ray	q_step_2	geometry_action_pairs.pl:1122 observed on {"degrees": 65, "kind": "angle_measure"}
q_step_2	iterate_degree_turn	q_step_3	geometry_action_pairs.pl:1122 observed on {"degrees": 65, "kind": "angle_measure"}
q_step_3	locate_terminal_ray	q_step_4	geometry_action_pairs.pl:1122 observed on {"degrees": 65, "kind": "angle_measure"}
q_step_4	read_angle_measure	q_accept	geometry_action_pairs.pl:1122 observed on {"degrees": 65, "kind": "angle_measure"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

4. area_as_perimeter_count

No domain scene: the static scene emitter has no geometry form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{establish_rectangle, ignore_interior_coverage, substitute_boundary_count_for_area, traverse_boundary_instead (4 distinct)}$$

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

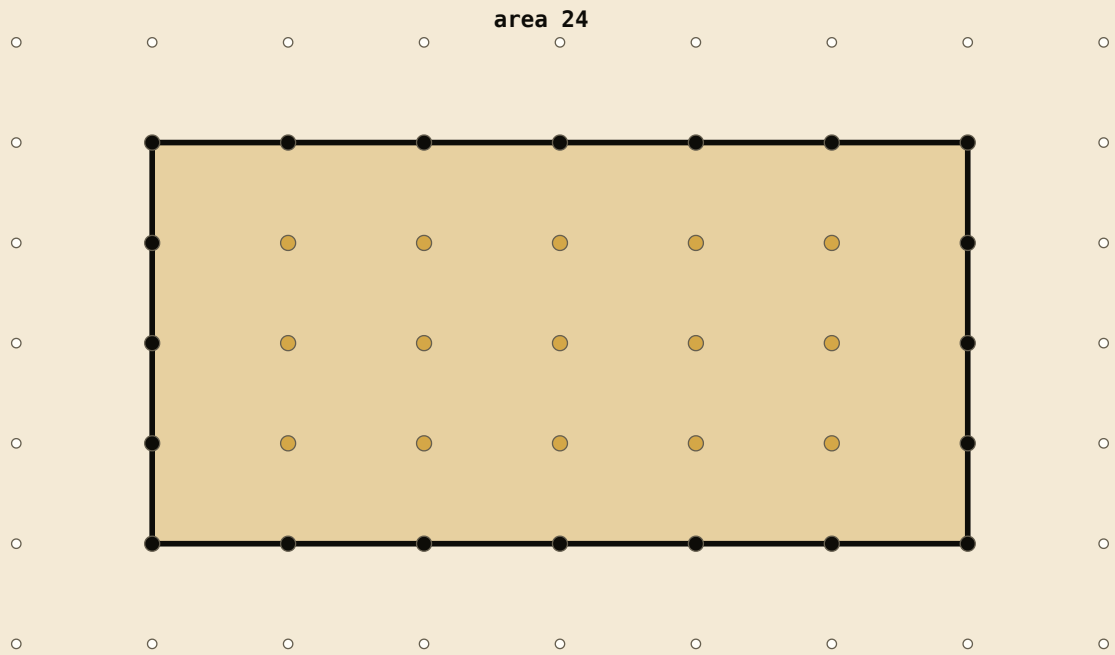
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_rectangle	q_step_1	geometry_action_pairs.pl:65 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	ignore_interior_coverage	q_step_2	geometry_action_pairs.pl:65 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	traverse_boundary_instead	q_step_3	geometry_action_pairs.pl:65 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	substitute_boundary_count_for_area	q_accept	geometry_action_pairs.pl:65 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and

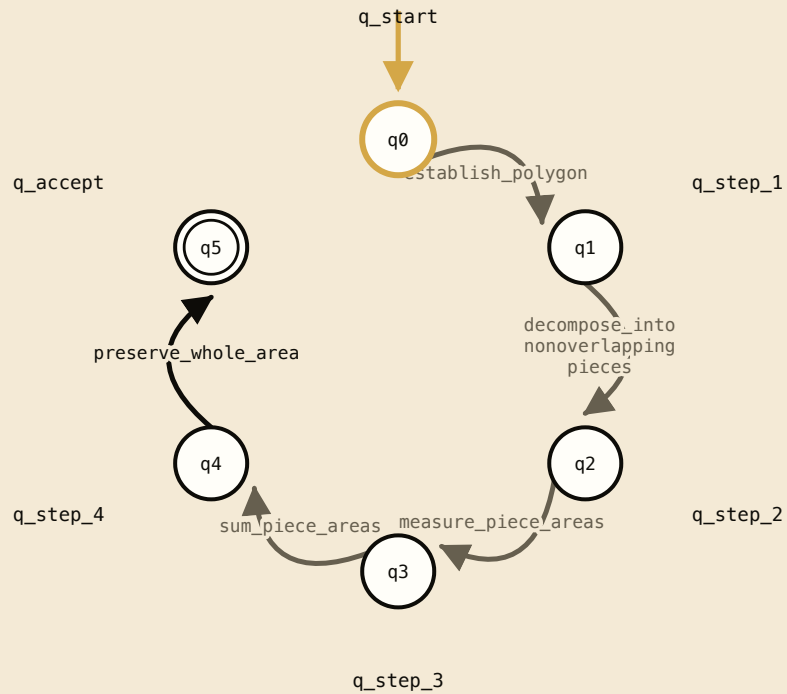
8 observed rows.

5. area_preserving_polygon_decomposition



The run produced `square_units(24)`.

The run produced `square_units(24)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ $decompose_into_nonoverlapping_pieces, establish_polygon, measure_piece_areas, preserve_whole_area, sum_piece_areas$ (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"certification": "atom", "kind": "polygon_partition", "pieces": [[{"x": "integer", "y": "integer"}]], "vertices": [{"x": "integer", "y": "integer"}]}; example {"certification": "certified", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], [{"x": 0, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}; verified(strategy_trace_ok).

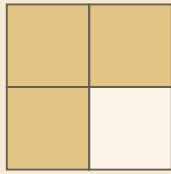
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_polygon	q_step_1	geometry_action_pairs.pl:816 observed on {"certification": "certified", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], [{"x": 0, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}
q_step_1	decompose_into_nonoverlapping_pieces	q_step_2	geometry_action_pairs.pl:816 observed on {"certification": "certified", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], [{"x": 0, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}
q_step_2	measure_piece_areas	q_step_3	geometry_action_pairs.pl:816 observed on {"certification": "certified", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], [{"x": 0, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}

From	Action	To	Provenance
q_ste p_3	sum_piece_ar eas	q_s tep _4	geometry_action_pairs.pl:816 observed on {"certification": "certified", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], [{"x": 0, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}
q_ste p_4	preserve_who le_area	q_a cce pt	geometry_action_pairs.pl:816 observed on {"certification": "certified", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], [{"x": 0, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}

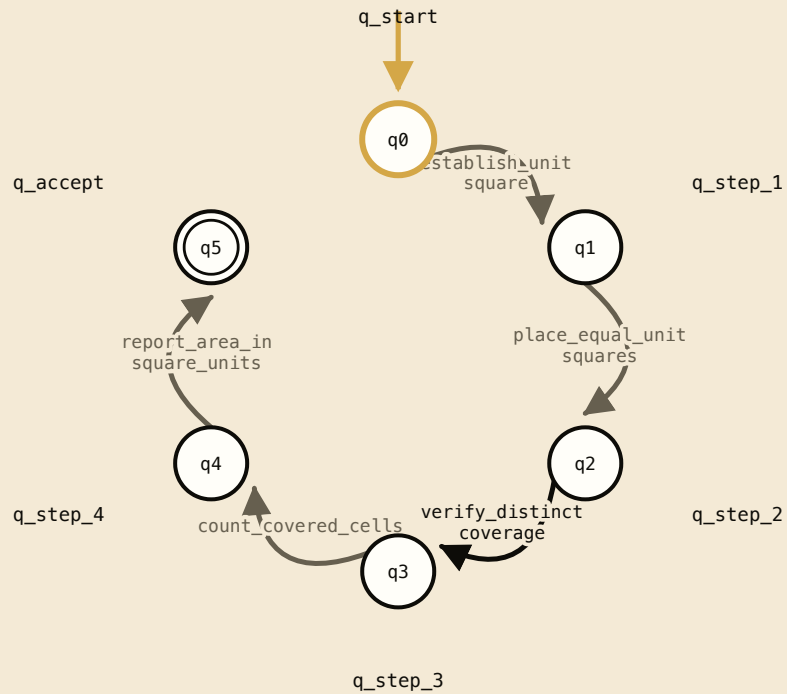
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

6. area_unit_covering



The run produced `area(3,square(centimeter))`.

The run produced `area(3,square(centimeter))`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ count_covered_cells, establish_unit_square, place_equal_unit_squares, report_area_in_square_units, verify_distinct_coverage (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"cells": [{"x": "integer", "y": "integer"}], "kind": "covered_cells", "unit": "atom"}; example {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 1}], "kind": "covered_cells", "unit": "centimeter"}; verified(strategy_trace_ok).

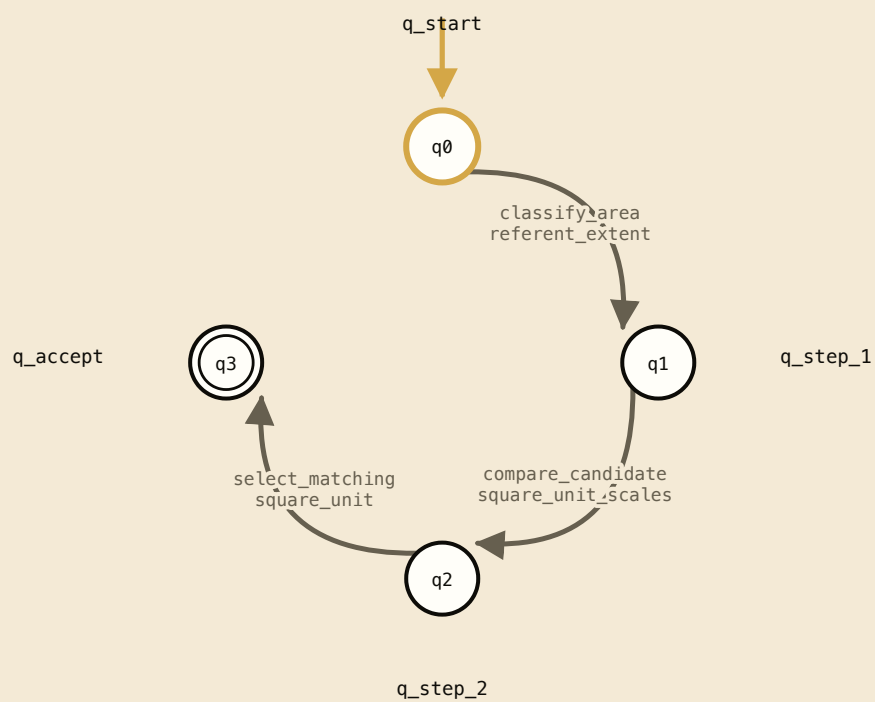
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_unit_square	q_step_1	geometry_action_pairs.pl:92 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 1}], "kind": "covered_cells", "unit": "centimeter"}
q_step_1	place_equal_unit_squares	q_step_2	geometry_action_pairs.pl:92 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 1}], "kind": "covered_cells", "unit": "centimeter"}
q_step_2	verify_distinct_coverage	q_step_3	geometry_action_pairs.pl:92 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 1}], "kind": "covered_cells", "unit": "centimeter"}
q_step_3	count_covered_cells	q_step_4	geometry_action_pairs.pl:92 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 1}], "kind": "covered_cells", "unit": "centimeter"}
q_step_4	report_area_in_square_units	q_accept	geometry_action_pairs.pl:92 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 1}], "kind": "covered_cells", "unit": "centimeter"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

7. area_unit_scale_selection

No domain scene: the static scene emitter has no geometry form for the area_unit_candidates contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3\}$

Σ `classify_area_referent_extent`, `compare_candidate_square_unit_scales`,
`select_matching_square_unit` (3 distinct)

q_0 $q_0 = q_start$

F $\{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	<code>q_start</code>	none recorded
q_1	<code>q_step_1</code>	none recorded
q_2	<code>q_step_2</code>	none recorded
q_3	<code>q_accept</code>	none recorded

Input contract: `{"candidates": [{"extent": "atom", "unit": "atom"}], "extent": "atom", "kind": "area_unit_candidates"}`; example `{"candidates": [{"extent": "small", "unit": "square_inch"}, {"extent": "medium", "unit": "square_foot"}, {"extent": "large", "unit": "square_meter"}], "extent": "medium", "kind": "area_unit_candidates"}`; `verified(strategy_trace_ok)`.

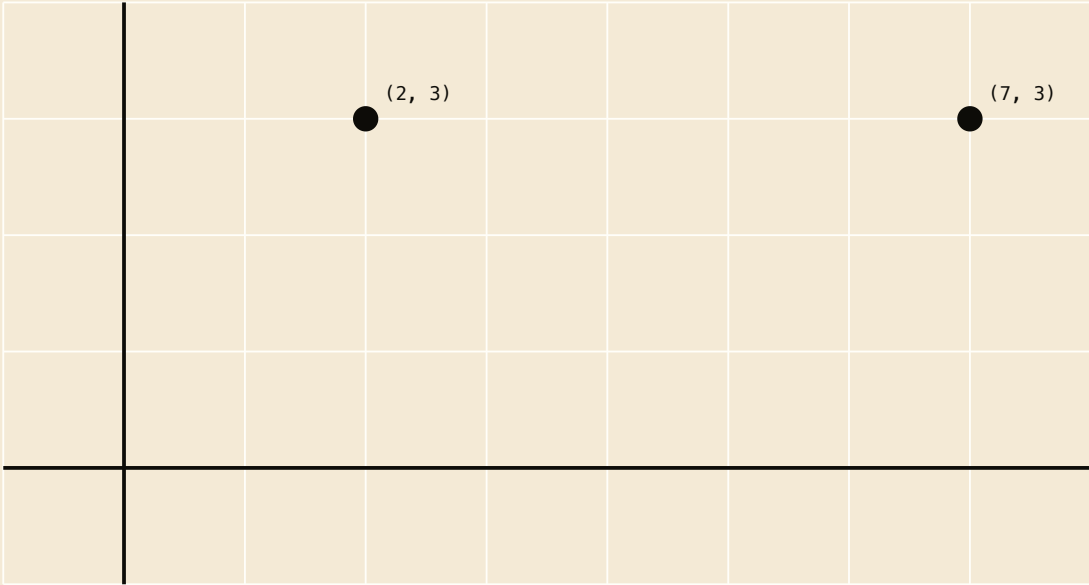
δ : recorded transitions

From	Action	To	Provenance
<code>q_start</code>	<code>classify_area_referent_extent</code>	<code>q_step_1</code>	<code>geometry_action_pairs.pl:147</code> observed on <code>{"candidates": [{"extent": "small", "unit": "square_inch"}, {"extent": "medium", "unit": "square_foot"}, {"extent": "large", "unit": "square_meter"}], "extent": "medium", "kind": "area_unit_candidates"}</code>
<code>q_step_1</code>	<code>compare_candidate_square_unit_scales</code>	<code>q_step_2</code>	<code>geometry_action_pairs.pl:147</code> observed on <code>{"candidates": [{"extent": "small", "unit": "square_inch"}, {"extent": "medium", "unit": "square_foot"}, {"extent": "large", "unit": "square_meter"}], "extent": "medium", "kind": "area_unit_candidates"}</code>

From	Action	To	Provenance
q_ste p_2	select_matching_square_unit	q_accept	<code>geometry_action_pairs.pl:147</code> observed on {"candidates": [{"extent": "small", "unit": "square_inch"}, {"extent": "medium", "unit": "square_foot"}, {"extent": "large", "unit": "square_meter"}], "extent": "medium", "kind": "area_unit_candidates"}

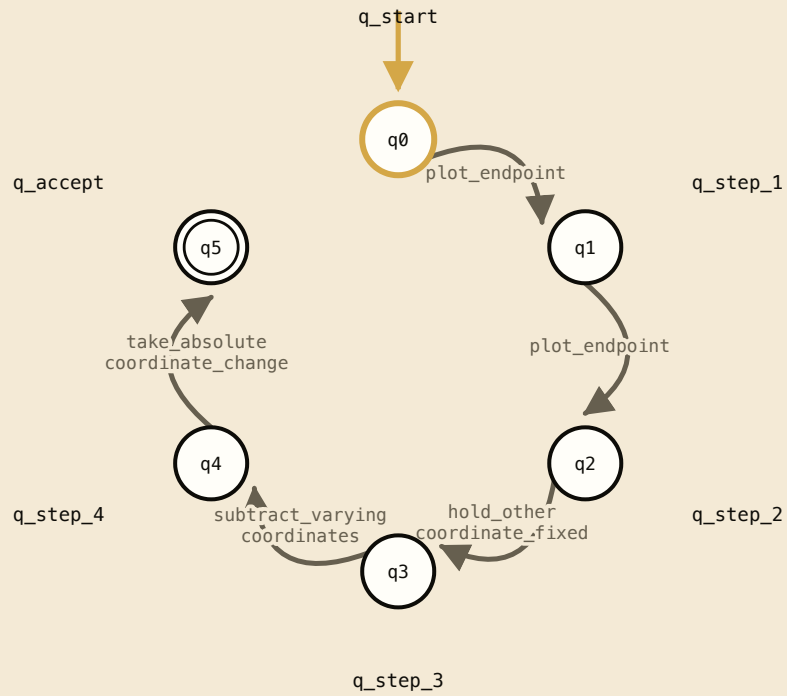
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

8. axis_aligned_coordinate_distance



The run produced `length(5,centimeter)`.

The run produced `length(5,centimeter)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ hold_other_coordinate_fixed, plot_endpoint, subtract_varying_coordinates, take_absolute_coordinate_change (4 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"first": {"x": "number", "y": "number"}, "kind": "coordinate_point_pair", "second": {"x": "number", "y": "number"}, "unit": "atom"}; example {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}; verified(strategy_trace_ok).

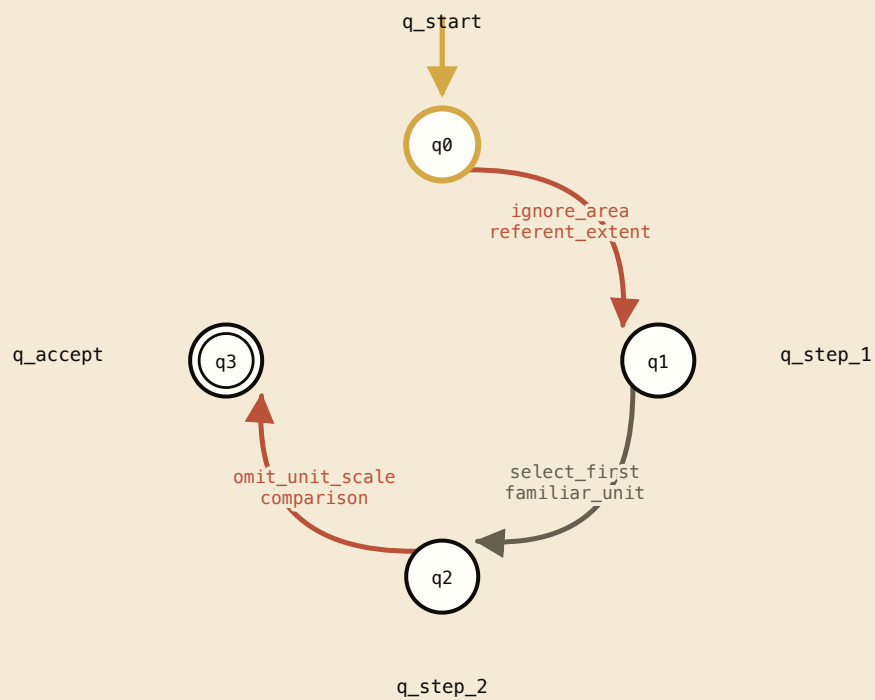
δ : recorded transitions

From	Action	To	Provenance
q_start	plot_endpoint	q_step_1	geometry_action_pairs.pl:758 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}
q_step_1	plot_endpoint	q_step_2	geometry_action_pairs.pl:758 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}
q_step_2	hold_other_coordinate_fixed	q_step_3	geometry_action_pairs.pl:758 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}
q_step_3	subtract_varying_coordinates	q_step_4	geometry_action_pairs.pl:758 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}
q_step_4	take_absolute_coordinate_change	q_accept	geometry_action_pairs.pl:758 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 4 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

9. choose_first_area_unit_without_scale

No domain scene: the static scene emitter has no geometry form for the area_unit_candidates contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3\}$

Σ ignore_area_referent_extent, omit_unit_scale_comparison,
select_first_familiar_unit (3 distinct)

q_0 $q_0 = q_{\text{start}}$

F $\{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

Input contract: {"candidates": [{"extent": "atom", "unit": "atom"}], "extent": "atom", "kind": "area_unit_candidates"}; example {"candidates": [{"extent": "small", "unit": "square_inch"}, {"extent": "medium", "unit": "square_foot"}, {"extent": "large", "unit": "square_meter"}], "extent": "medium", "kind": "area_unit_candidates"}; verified(strategy_trace_ok).

δ : recorded transitions

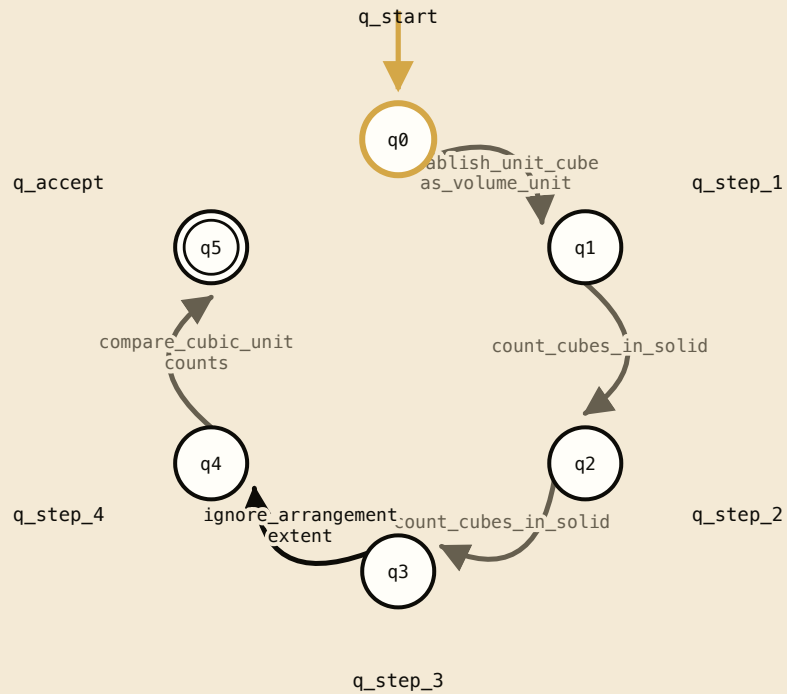
From	Action	To	Provenance
q_start	ignore_area_referent_extent	q_step_1	geometry_action_pairs.pl:171 observed on {"candidates": [{"extent": "small", "unit": "square_inch"}, {"extent": "medium", "unit": "square_foot"}, {"extent": "large", "unit": "square_meter"}], "extent": "medium", "kind": "area_unit_candidates"}
q_step_1	select_first_familiar_unit	q_step_2	geometry_action_pairs.pl:171 observed on {"candidates": [{"extent": "small", "unit": "square_inch"}, {"extent": "medium", "unit": "square_foot"}, {"extent": "large", "unit": "square_meter"}], "extent": "medium", "kind": "area_unit_candidates"}

From	Action	To	Provenance
q_ste p_2	omit_unit_s cale_compar ison	q_a cce pt	geometry_action_pairs.pl:171 observed on {"candidates": [{"extent": "small", "unit": "square_inch"}, {"extent": "medium", "unit": "square_foot"}, {"extent": "large", "unit": "square_meter"}], "extent": "medium", "kind": "area_unit_candidates"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

10. compare_solid_volume_by_cube_count

No domain scene: the static scene emitter has no geometry form for the solid_volume_comparison contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ compare_cubic_unit_counts, count_cubes_in_solid,
establish_unit_cube_as_volume_unit, ignore_arrangement_extent (4 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"count_a": "positive_integer", "count_b": "positive_integer", "extent_a": "positive_number", "extent_b": "positive_number", "kind": "solid_volume_comparison"}; example {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}; verified(strategy_trace_ok).

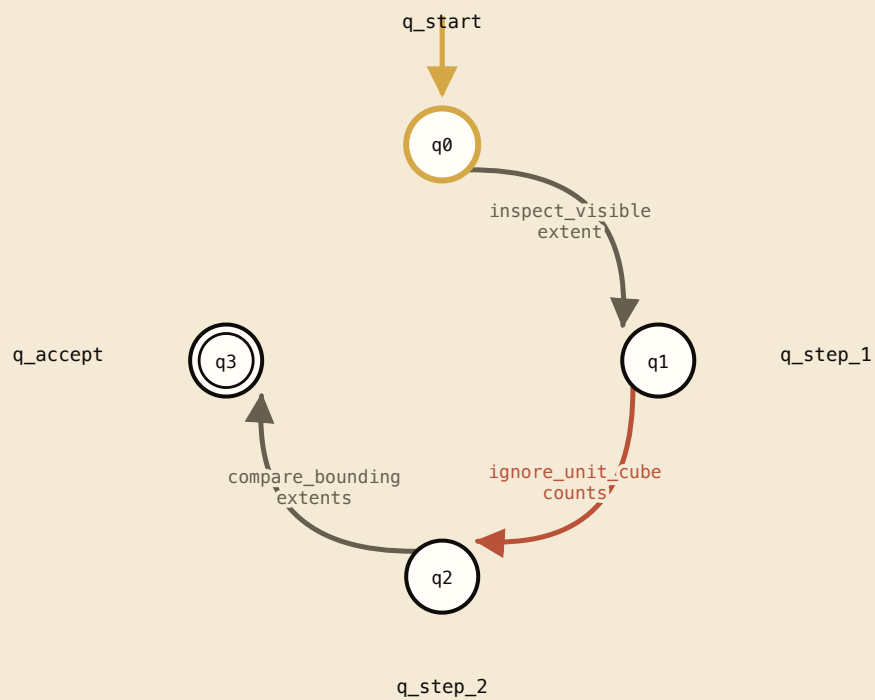
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_unit_cube_as_volume_unit	q_step_1	geometry_action_pairs.pl:1223 observed on {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}
q_step_1	count_cubes_in_solid	q_step_2	geometry_action_pairs.pl:1223 observed on {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}
q_step_2	count_cubes_in_solid	q_step_3	geometry_action_pairs.pl:1223 observed on {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}
q_step_3	ignore_arrangement_extent	q_step_4	geometry_action_pairs.pl:1223 observed on {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}
q_step_4	compare_cubic_unit_counts	q_accept	geometry_action_pairs.pl:1223 observed on {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 4 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

11. compare_solid_volume_by_visible_extent

No domain scene: the static scene emitter has no geometry form for the solid_volume_comparison contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$M = (Q, \Sigma, \delta, q_0, F)$

$Q \quad \{q_0, q_1, q_2, q_3\}$

$\Sigma \quad \text{compare_bounding_extents, ignore_unit_cube_counts, inspect_visible_extent (3 distinct)}$

$q_0 \quad q_0 = \text{q_start}$

$F \quad \{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

Input contract: {"count_a": "positive_integer", "count_b": "positive_integer", "extent_a": "positive_number", "extent_b": "positive_number", "kind": "solid_volume_comparison"}; **example** {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}; **verified**(strategy_trace_ok).

δ : recorded transitions

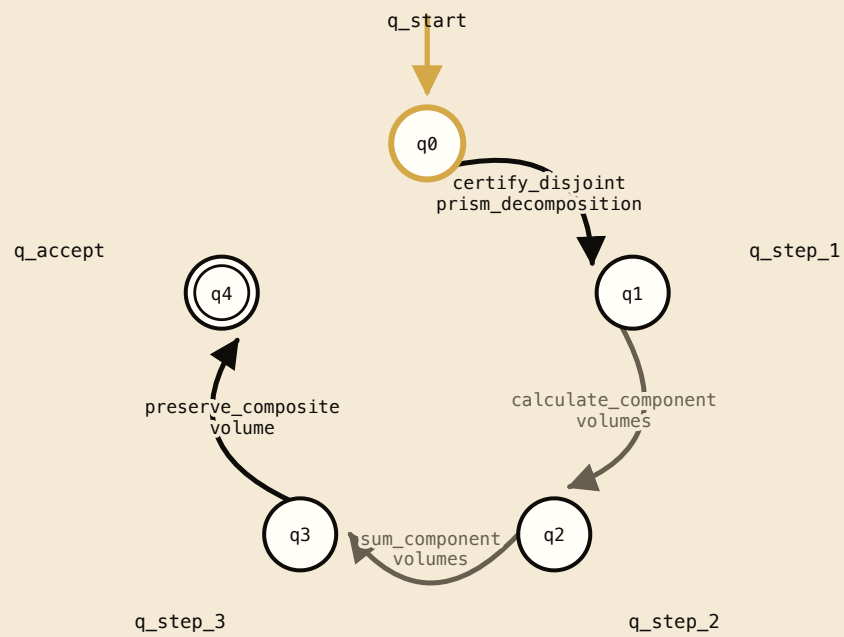
From	Action	To	Provenance
q_start	inspect_visible_extent	q_step_1	geometry_action_pairs.pl:1252 observed on {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}
q_step_1	ignore_unit_cube_counts	q_step_2	geometry_action_pairs.pl:1252 observed on {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}
q_step_2	compare_bounding_extents	q_accept	geometry_action_pairs.pl:1252 observed on {"count_a": 12, "count_b": 10, "extent_a": 5, "extent_b": 8, "kind": "solid_volume_comparison"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and

3 observed rows.

12. composite_prism_volume_sum

No domain scene: the static scene emitter has no geometry form for the disjoint_prisms contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ calculate_component_volumes, certify_disjoint_prism_decomposition, preserve_composite_volume, sum_component_volumes (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"kind": "disjoint_prisms", "prisms": [{"height": "positive_integer", "length": "positive_integer", "width": "positive_integer"}], "unit": "atom"}; example {"kind": "disjoint_prisms", "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

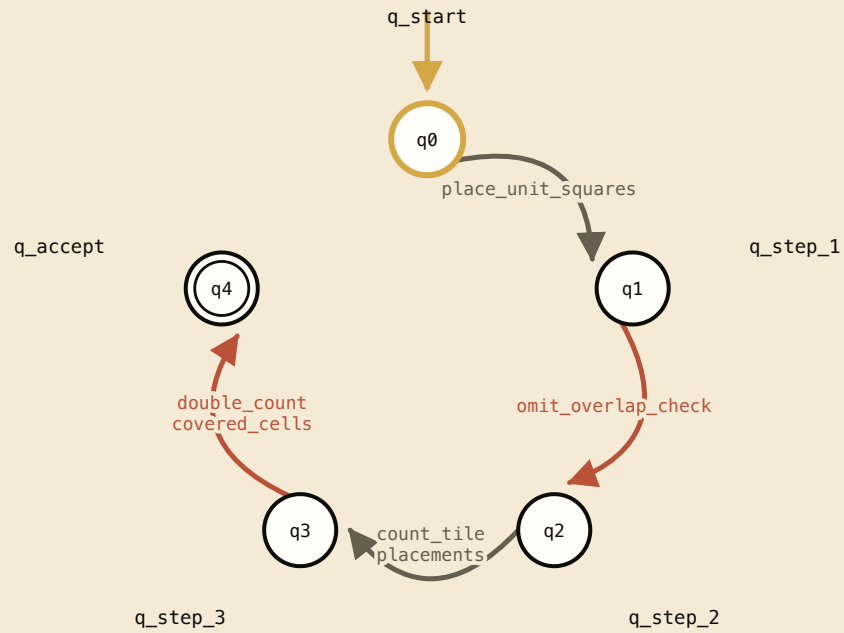
From	Action	To	Provenance
q_start	certify_disjoint_prism_decomposition	q_step_1	geometry_action_pairs.pl:681 observed on {"kind": "disjoint_prisms", "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}
q_step_1	calculate_component_volumes	q_step_2	geometry_action_pairs.pl:681 observed on {"kind": "disjoint_prisms", "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}
q_step_2	sum_component_volumes	q_step_3	geometry_action_pairs.pl:681 observed on {"kind": "disjoint_prisms", "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}

From	Action	To	Provenance
q_step_3	preserve_composite_volume	q_accept	geometry_action_pairs.pl:681 observed on {"kind": "disjoint_prisms", "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

13. count_overlapping_area_tiles

No domain scene: the static scene emitter has no geometry form for the placed_tiles contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{count_tile_placements, double_count_covered_cells, omit_overlap_check, place_unit_squares (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

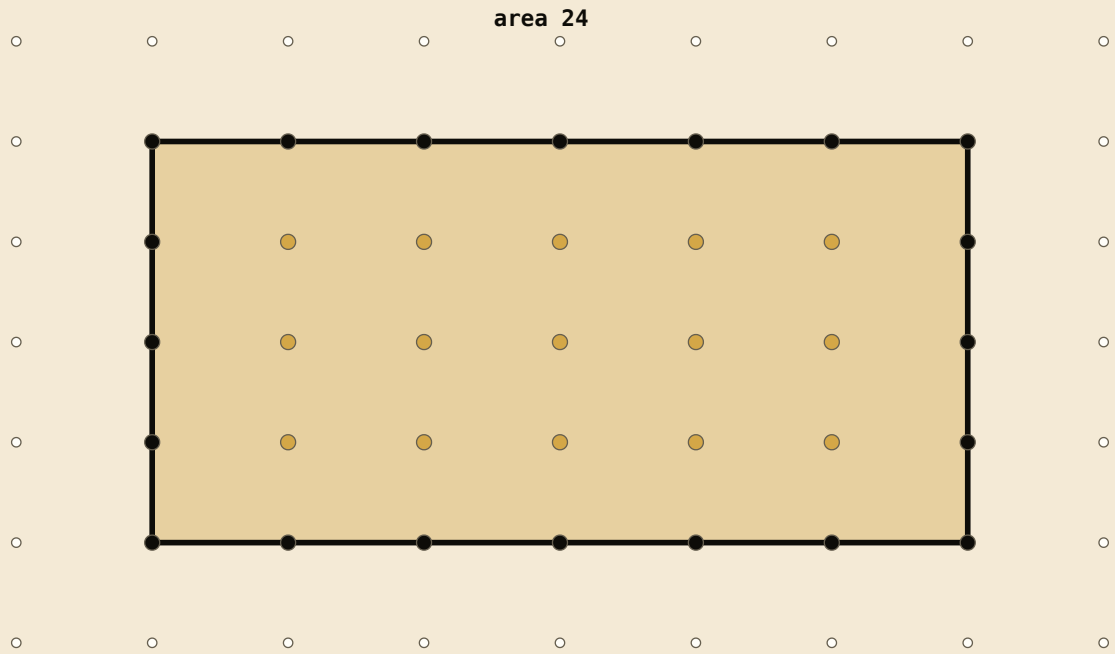
Input contract: {"cells": [{"x": "integer", "y": "integer"}], "kind": "placed_tiles", "unit": "atom"}; example {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 0}], "kind": "placed_tiles", "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	place_unit_squares	q_step_1	geometry_action_pairs.pl:118 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 0}], "kind": "placed_tiles", "unit": "centimeter"}
q_step_1	omit_overlap_check	q_step_2	geometry_action_pairs.pl:118 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 0}], "kind": "placed_tiles", "unit": "centimeter"}
q_step_2	count_tile_placements	q_step_3	geometry_action_pairs.pl:118 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 0}], "kind": "placed_tiles", "unit": "centimeter"}
q_step_3	double_count_covered_cells	q_accept	geometry_action_pairs.pl:118 observed on {"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 0, "y": 0}], "kind": "placed_tiles", "unit": "centimeter"}

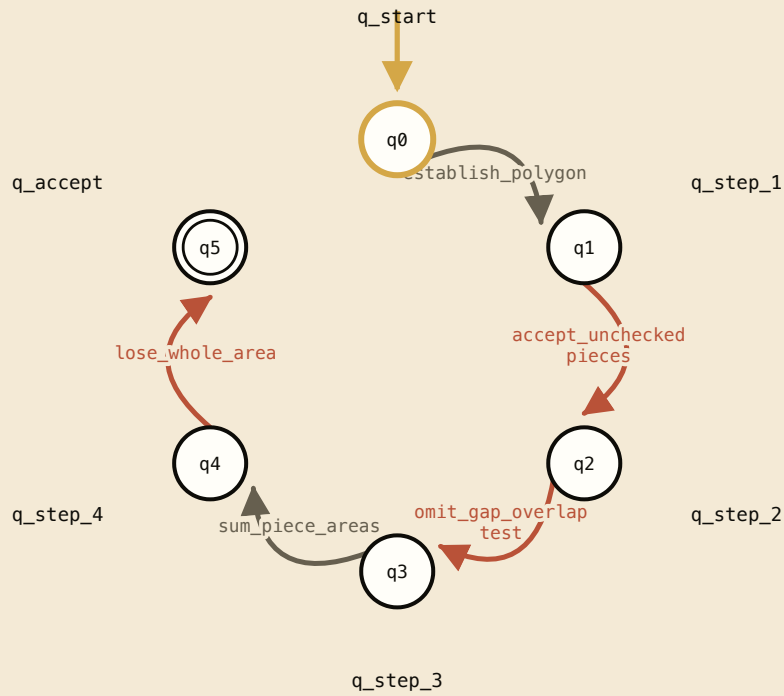
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

14. decomposition_with_gap_or_overlap



The run produced `square_units(12)`; the recorded expected result is `square_units(24)`.

The run produced `square_units(12)`; the recorded expected result is `square_units(24)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{\text{accept_unchecked_pieces, establish_polygon, lose_whole_area, omit_gap_overlap_test, sum_piece_areas}\} \text{ (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"certification": "atom", "kind": "polygon_partition", "pieces": [[{"x": "integer", "y": "integer"}]], "vertices": [{"x": "integer", "y": "integer"}]}; example {"certification": "unchecked", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}; verified(strategy_trace_ok).

δ : recorded transitions

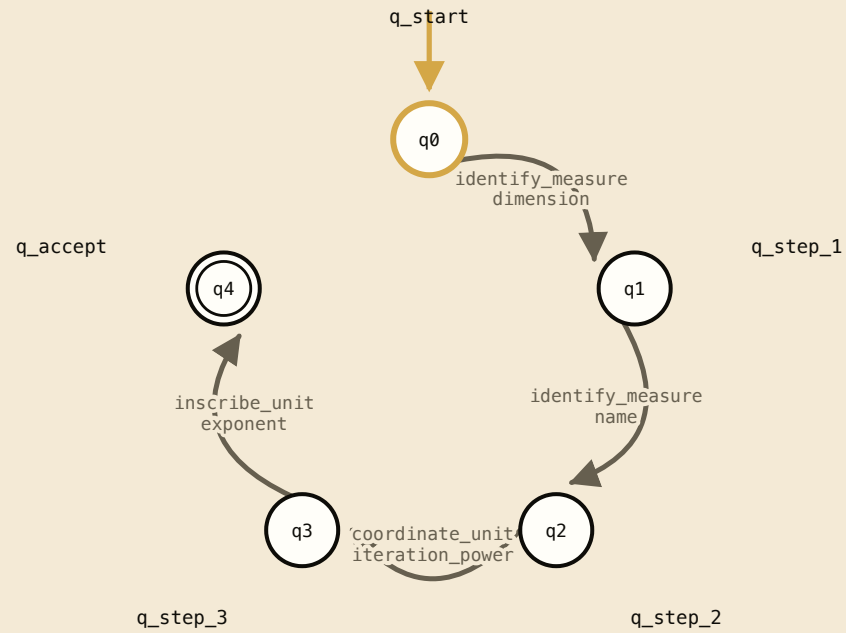
From	Action	To	Provenance
q_start	establish_polygon	q_step_1	geometry_action_pairs.pl:842 observed on {"certification": "unchecked", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}
q_step_1	accept_unchecked_pieces	q_step_2	geometry_action_pairs.pl:842 observed on {"certification": "unchecked", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}
q_step_2	omit_gap_overlap_test	q_step_3	geometry_action_pairs.pl:842 observed on {"certification": "unchecked", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}

From	Action	To	Provenance
q_ste p_3	sum_piece_ areas	q_s tep _4	geometry_action_pairs.pl:842 observed on {"certification": "unchecked", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}
q_ste p_4	lose_whole _area	q_a cce pt	geometry_action_pairs.pl:842 observed on {"certification": "unchecked", "kind": "polygon_partition", "pieces": [[{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}], "vertices": [{"x": 0, "y": 0}, {"x": 6, "y": 0}, {"x": 6, "y": 4}, {"x": 0, "y": 4}]}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

15. dimensional_measure_unit_coordination

No domain scene: the static scene emitter has no geometry form for the dimensional_measure contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ coordinate_unit_iteration_power, identify_measure_dimension,
identify_measure_name, inscribe_unit_exponent (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

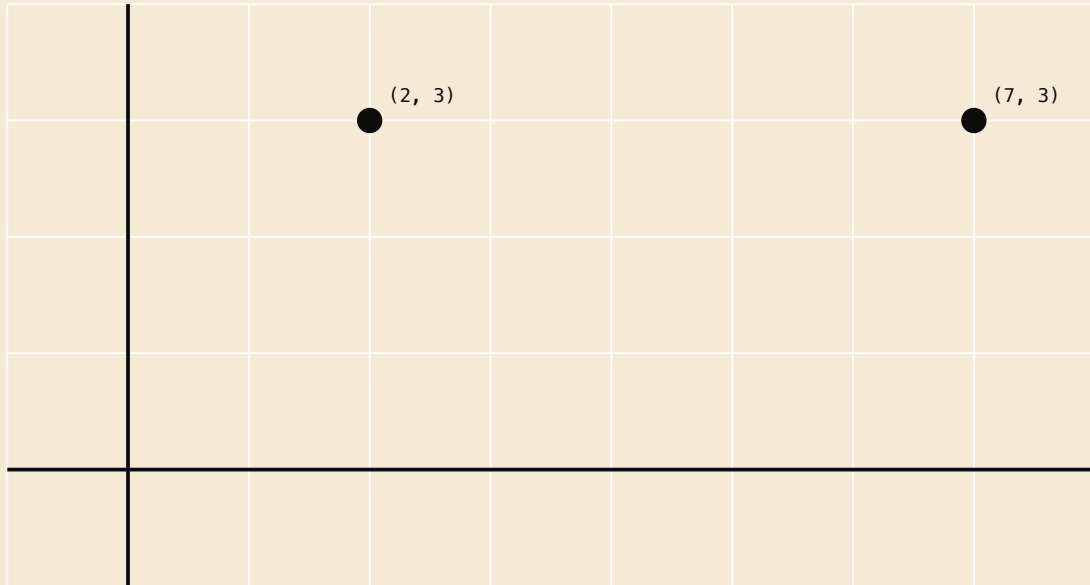
Input contract: {"dimension": "atom", "kind": "dimensional_measure", "unit": "atom", "value": "number"}; example {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_measure_dimension	q_step_1	geometry_action_pairs.pl:1022 observed on {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28}
q_step_1	identify_measure_name	q_step_2	geometry_action_pairs.pl:1022 observed on {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28}
q_step_2	coordinate_unit_iteration_power	q_step_3	geometry_action_pairs.pl:1022 observed on {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28}
q_step_3	inscribe_unit_exponent	q_accept	geometry_action_pairs.pl:1022 observed on {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28}

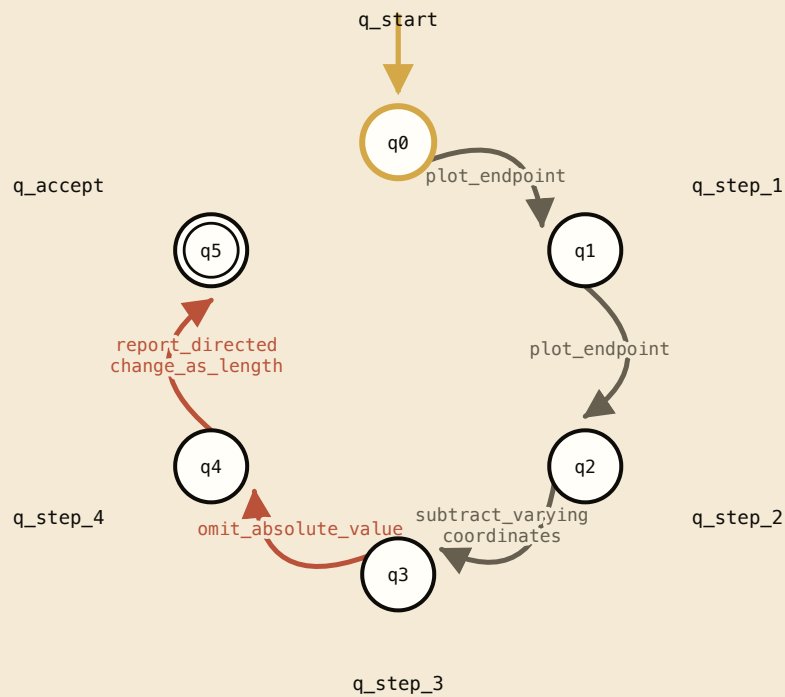
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

16. directed_difference_as_coordinate_distance



The run produced `length(-5,centimeter)`; the recorded expected result is `length(5,centimeter)`.

The run produced `length(-5,centimeter)`; the recorded expected result is `length(5,centimeter)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ omit_absolute_value, plot_endpoint, report_directed_change_as_length, subtract_varying_coordinates (4 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

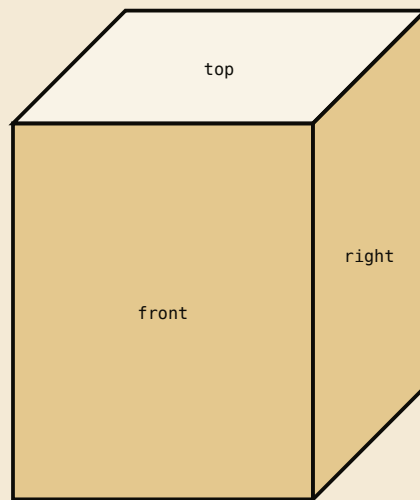
Input contract: {"first": {"x": "number", "y": "number"}, "kind": "coordinate_point_pair", "second": {"x": "number", "y": "number"}, "unit": "atom"}; example {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	plot_endpoint	q_step_1	geometry_action_pairs.pl:786 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}
q_step_1	plot_endpoint	q_step_2	geometry_action_pairs.pl:786 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}
q_step_2	subtract_varying_coordinates	q_step_3	geometry_action_pairs.pl:786 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}
q_step_3	omit_absolute_value	q_step_4	geometry_action_pairs.pl:786 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}
q_step_4	report_directed_change_as_length	q_accept	geometry_action_pairs.pl:786 observed on {"first": {"x": 7, "y": 3}, "kind": "coordinate_point_pair", "second": {"x": 2, "y": 3}, "unit": "centimeter"}

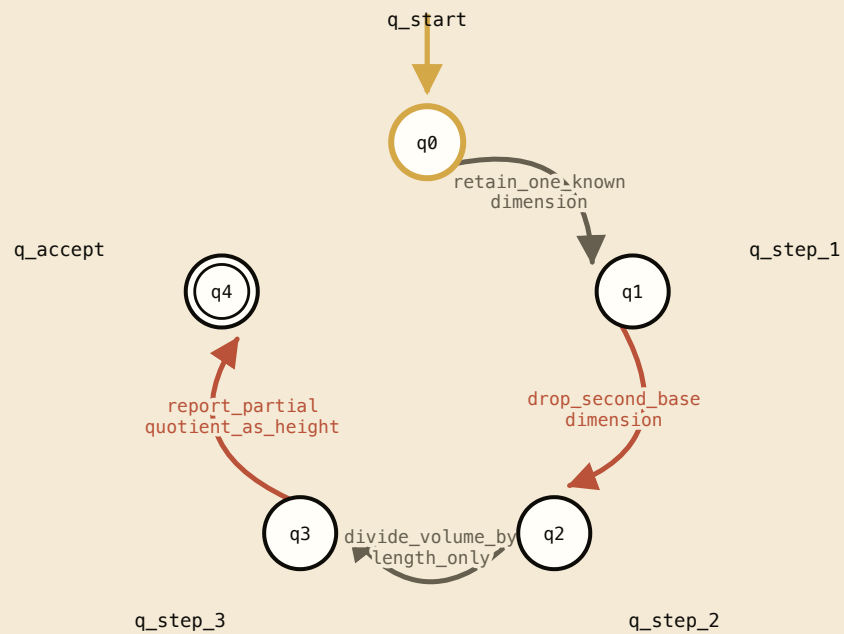
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 4 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

17. divide_volume_by_one_dimension



The run produced `dimension(15)`; the recorded expected result is `dimension(5)`.

The run produced `dimension(15)`; the recorded expected result is `dimension(5)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ divide_volume_by_length_only, drop_second_base_dimension,
report_partial_quotient_as_height, retain_one_known_dimension (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

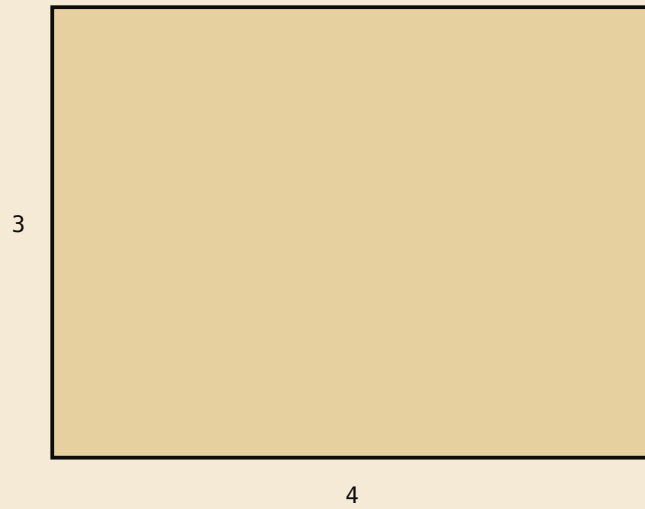
Input contract: {"kind": "volume_known_base", "length": "positive_integer", "volume": "positive_integer", "width": "positive_integer"}; example {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	retain_one_known_dimension	q_step_1	geometry_action_pairs.pl:649 observed on {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3}
q_step_1	drop_second_base_dimension	q_step_2	geometry_action_pairs.pl:649 observed on {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3}
q_step_2	divide_volume_by_length_only	q_step_3	geometry_action_pairs.pl:649 observed on {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3}
q_step_3	report_partial_quotient_as_height	q_accept	geometry_action_pairs.pl:649 observed on {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3}

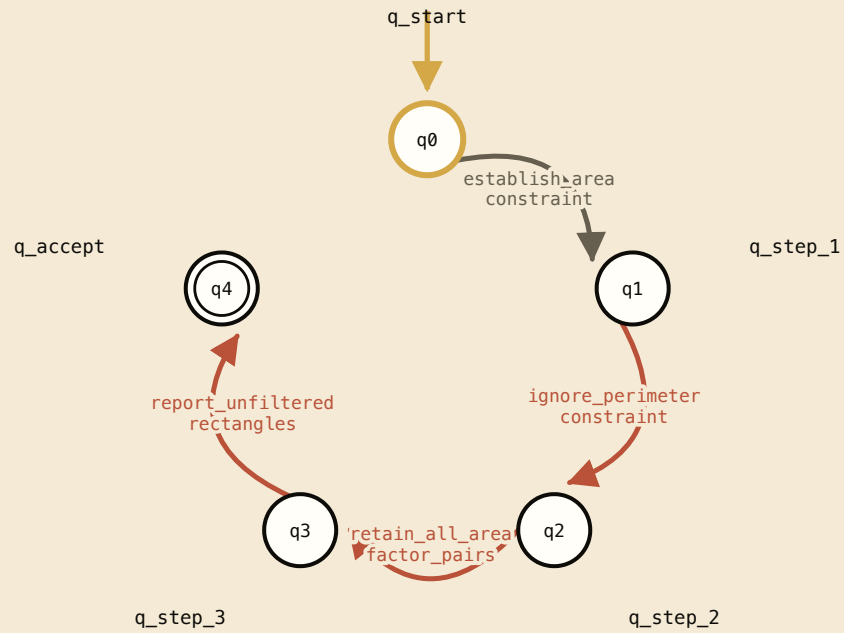
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

18. ignore_perimeter_rectangle_constraint



The run produced `rectangles([1-24,2-12,3-8,4-6])`; the recorded expected result is `rectangles`

The run produced `rectangles([1-24,2-12,3-8,4-6])`; the recorded expected result is `rectangles([3-8])`.
The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ establish_area_constraint, ignore_perimeter_constraint,
report_unfiltered_rectangles, retain_all_area_factor_pairs (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"area": "positive_integer", "kind": "rectangle_constraints", "perimeter": "positive_integer"}; example {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}; verified(strategy_trace_ok).

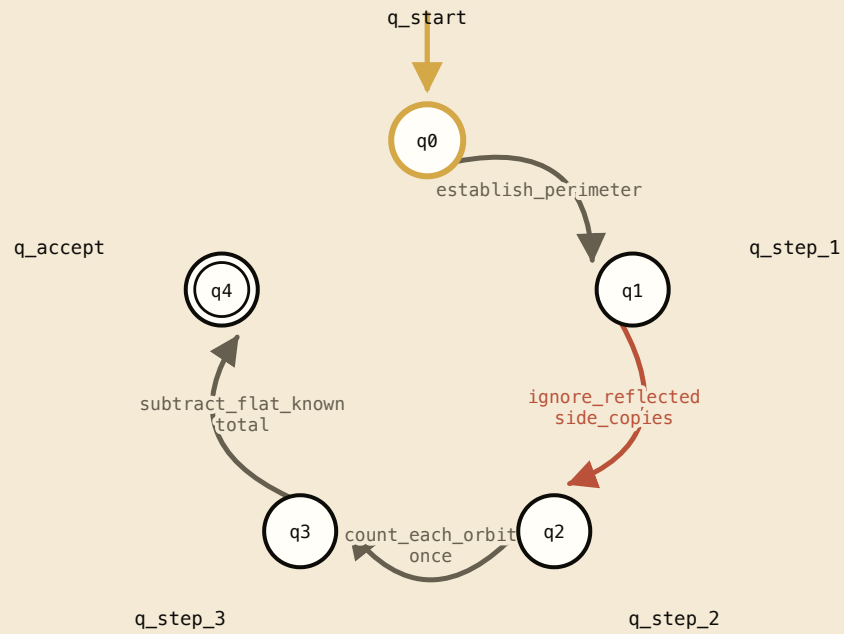
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_area_constraint	q_step_1	geometry_action_pairs.pl:511 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}
q_step_1	ignore_perimeter_constraint	q_step_2	geometry_action_pairs.pl:511 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}
q_step_2	retain_all_area_factor_pairs	q_step_3	geometry_action_pairs.pl:511 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}
q_step_3	report_unfiltered_rectangles	q_accept	geometry_action_pairs.pl:511 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

19. ignore_symmetry_multiplicity

No domain scene: the static scene emitter has no geometry form for the symmetry_side_orbits contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{count_each_orbit_once, establish_perimeter, ignore_reflected_side_copies, subtract_flat_known_total (4 distinct)}$$

$$q_0 = q_start$$

$F \quad \{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": "positive_integer", "length": "positive_integer"}], "perimeter": "positive_integer", "unit": "atom", "unknown_orbit": {"copies": "positive_integer", "name": "atom"}}; example {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}; verified(strategy_trace_ok).

δ : recorded transitions

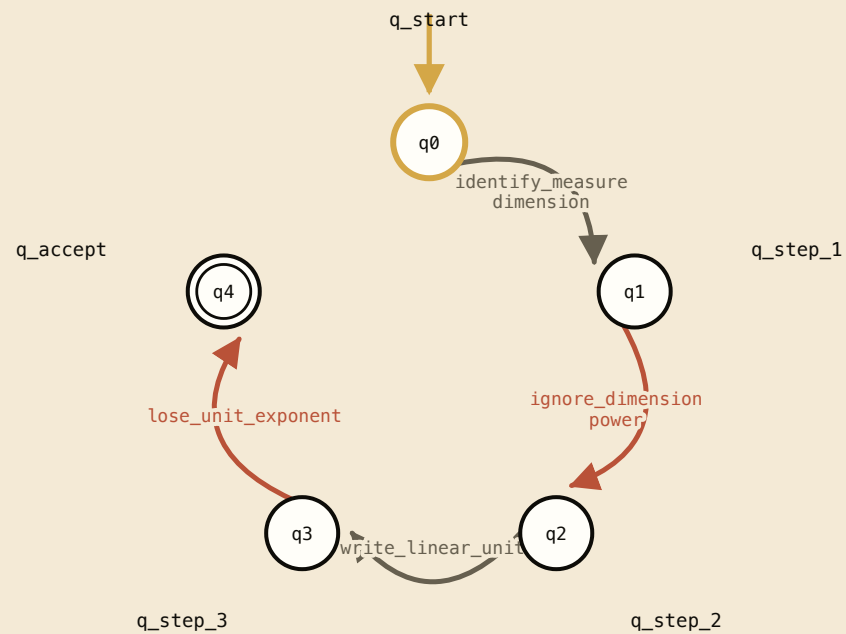
From	Action	To	Provenance
q_start	establish_perimeter	q_step_1	geometry_action_pairs.pl:362 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}
q_step_1	ignore_reflected_side_copies	q_step_2	geometry_action_pairs.pl:362 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}
q_step_2	count_each_orbit_once	q_step_3	geometry_action_pairs.pl:362 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}

From	Action	To	Provenance
q_ste p_3	subtract_flat_known_total	q_accept	geometry_action_pairs.pl:362 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

20. linear_unit_for_area_or_volume

No domain scene: the static scene emitter has no geometry form for the dimensional_measure contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{identify_measure_dimension}, \text{ignore_dimension_power}, \text{lose_unit_exponent}, \text{write_linear_unit}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

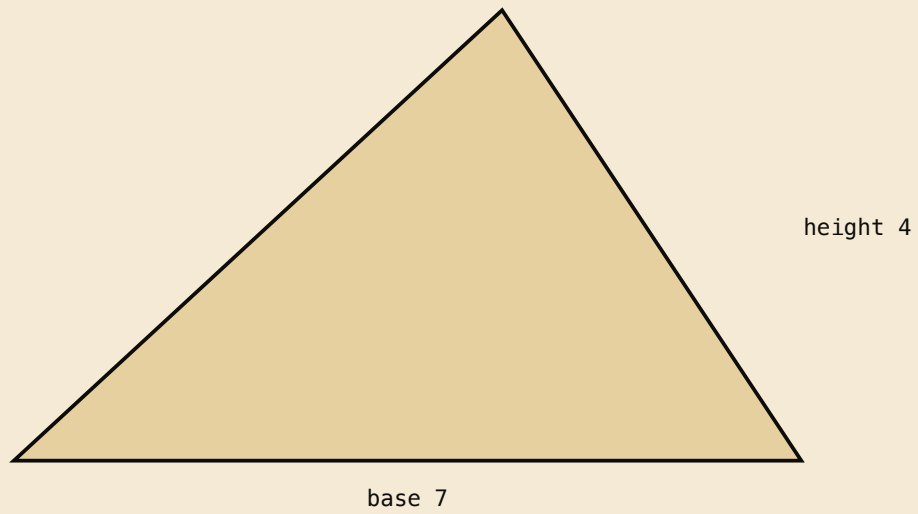
Input contract: {"dimension": "atom", "kind": "dimensional_measure", "unit": "atom", "value": "number"}; example {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_measure_dimension	q_step_1	geometry_action_pairs.pl:1045 observed on {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28}
q_step_1	ignore_dimension_power	q_step_2	geometry_action_pairs.pl:1045 observed on {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28}
q_step_2	write_linear_unit	q_step_3	geometry_action_pairs.pl:1045 observed on {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28}
q_step_3	lose_unit_exponent	q_accept	geometry_action_pairs.pl:1045 observed on {"dimension": "two_dimensional", "kind": "dimensional_measure", "unit": "centimeter", "value": 28}

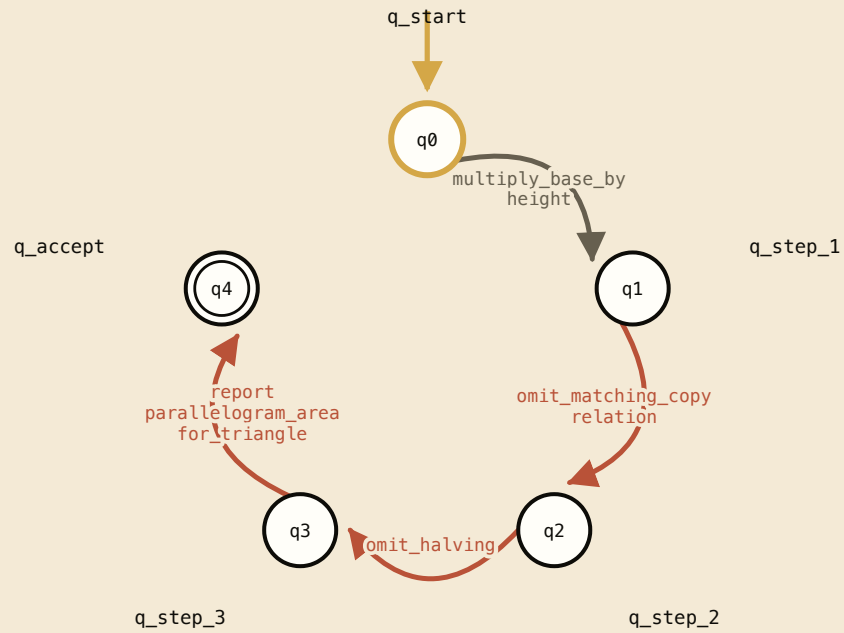
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

21. omit_half_in_triangle_area



The run produced `area(28,square(centimeter))`; the recorded expected result is `area(14,square`

`area(14,square(centimeter))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{multiply_base_by_height, omit_halving, omit_matching_copy_relation, report_parallelogram_area_for_triangle (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"base": "positive_integer", "height": "positive_integer", "kind": "triangle_with_unit", "unit": "atom"}; example {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}; verified(strategy_trace_ok).

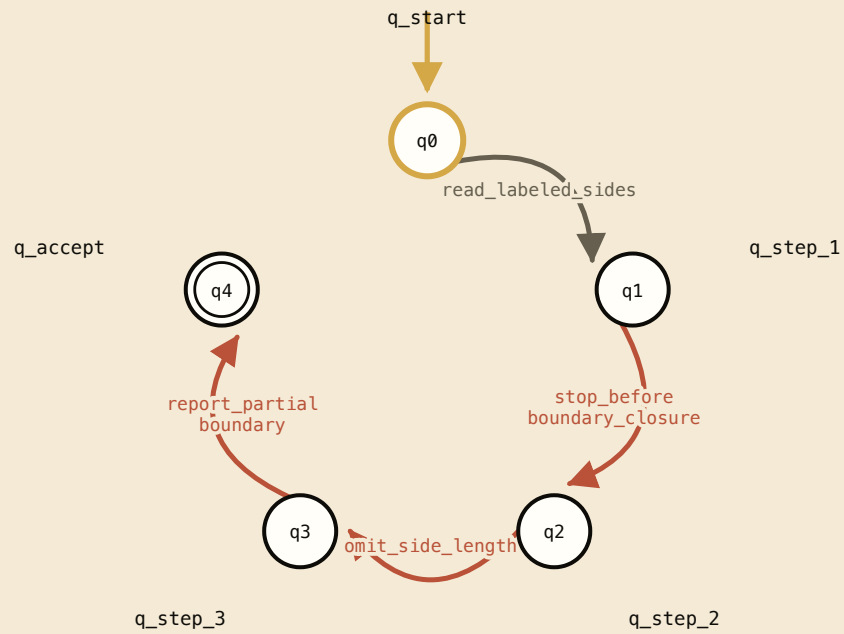
δ : recorded transitions

From	Action	To	Provenance
q_start	multiply_base_by_height	q_step_1	geometry_action_pairs.pl:948 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}
q_step_1	omit_matching_copy_relation	q_step_2	geometry_action_pairs.pl:948 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}
q_step_2	omit_halving	q_step_3	geometry_action_pairs.pl:948 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}
q_step_3	report_parallelogram_area_for_triangle	q_accept	geometry_action_pairs.pl:948 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

22. omit_unlabeled_boundary_side

No domain scene: the static scene emitter has no geometry form for the polygon_sides_with_unit contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{omit_side_length, read_labeled_sides, report_partial_boundary, stop_before_boundary_closure (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$F \quad \{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"kind": "polygon_sides_with_unit", "sides": ["positive_number"], "unit": "atom"}; example {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}; verified(strategy_trace_ok).

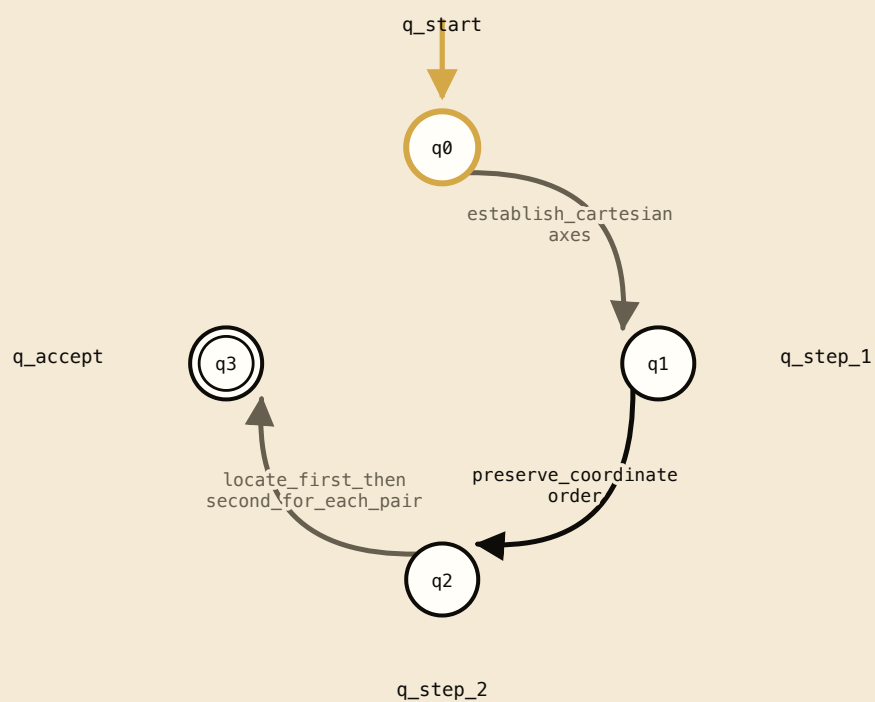
δ : recorded transitions

From	Action	To	Provenance
q_start	read_labeled_sides	q_step_1	geometry_action_pairs.pl:303 observed on {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}
q_step_1	stop_before_boundary_closure	q_step_2	geometry_action_pairs.pl:303 observed on {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}
q_step_2	omit_side_length	q_step_3	geometry_action_pairs.pl:303 observed on {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}
q_step_3	report_partial_boundary	q_accept	geometry_action_pairs.pl:303 observed on {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

23. ordered_pair_coordinate_plot

No domain scene: the static scene emitter has no geometry form for the coordinate_points contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3\}$

Σ `establish_cartesian_axes`, `locate_first_then_second_for_each_pair`,
`preserve_coordinate_order` (3 distinct)

q_0 $q_0 = q_start$

F $\{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

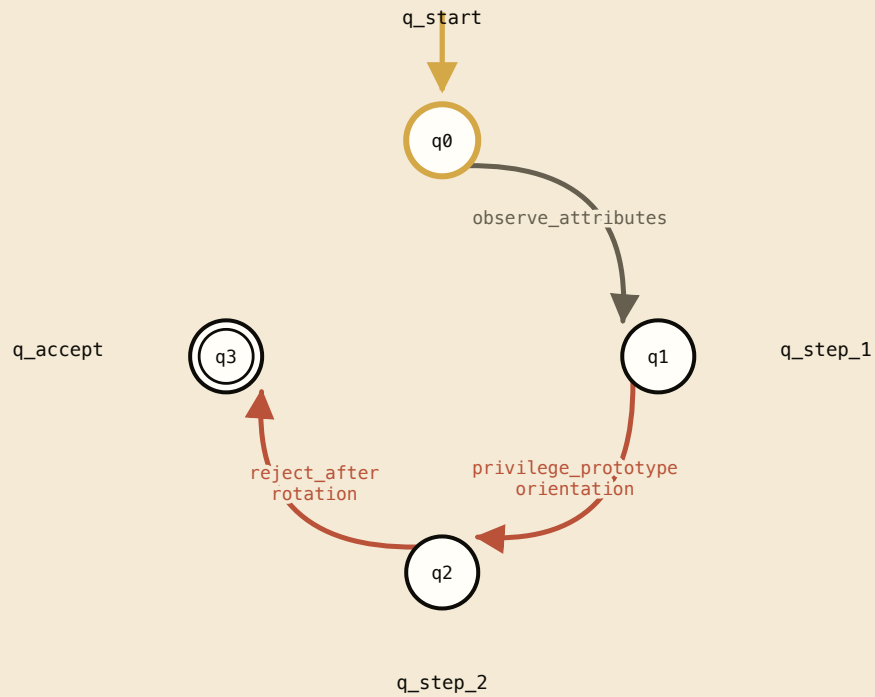
Input contract: {"kind": "coordinate_points", "points": [{"x": "number", "y": "number"}]}; example {"kind": "coordinate_points", "points": [{"x": -3, "y": 2}, {"x": 1, "y": 4}, {"x": 5, "y": -1}]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_cartesian_axes	q_step_1	geometry_action_pairs.pl:735 observed on {"kind": "coordinate_points", "points": [{"x": -3, "y": 2}, {"x": 1, "y": 4}, {"x": 5, "y": -1}]}
q_step_1	preserve_coordinate_order	q_step_2	geometry_action_pairs.pl:735 observed on {"kind": "coordinate_points", "points": [{"x": -3, "y": 2}, {"x": 1, "y": 4}, {"x": 5, "y": -1}]}
q_step_2	locate_first_then_second_for_each_pair	q_accept	geometry_action_pairs.pl:735 observed on {"kind": "coordinate_points", "points": [{"x": -3, "y": 2}, {"x": 1, "y": 4}, {"x": 5, "y": -1}]}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

No domain scene: the static scene emitter has no geometry form for the shape_attributes contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad \text{observe_attributes, privilege_prototype_orientation, reject_after_rotation (3 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$F \quad \{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

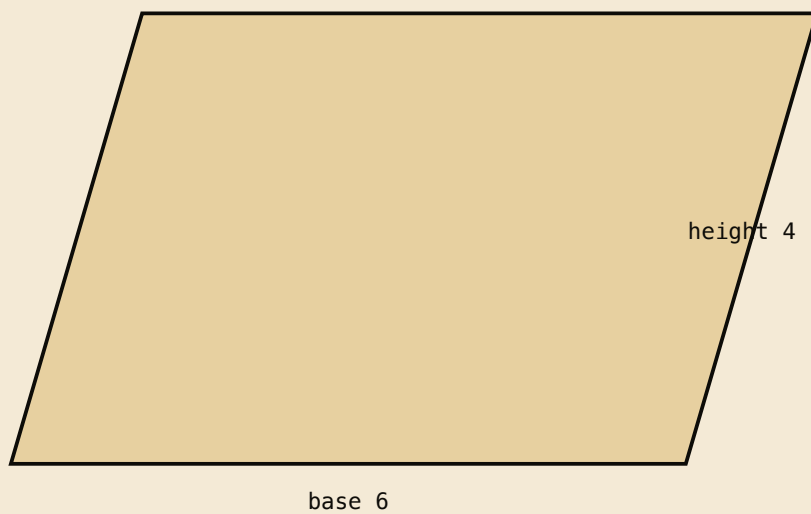
Input contract: {"attributes": [{"name": "atom", "value": "integer"}], "kind": "shape_attributes", "quarter_turns": "integer", "shape": "atom"}; example {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	observe_attributes	q_step_1	geometry_action_pairs.pl:1096 observed on {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}
q_step_1	privilege_prototype_orientation	q_step_2	geometry_action_pairs.pl:1096 observed on {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}
q_step_2	reject_after_rotation	q_accept	geometry_action_pairs.pl:1096 observed on {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}

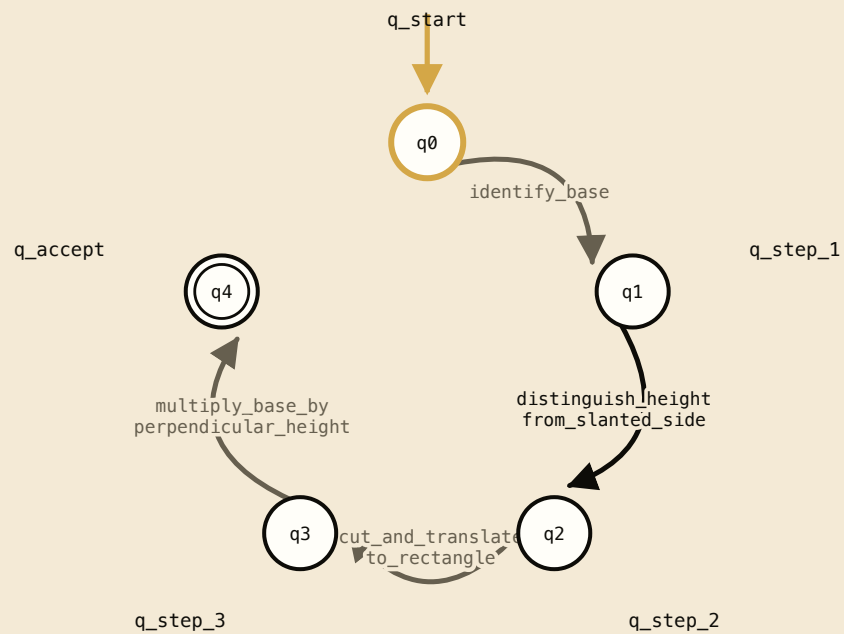
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

25. parallelogram_area_base_height



The run produced `area(24,square(centimeter))`.

The run produced `area(24,square(centimeter))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ cut_and_translate_to_rectangle, distinguish_height_from_slanted_side, identify_base, multiply_base_by_perpendicular_height (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

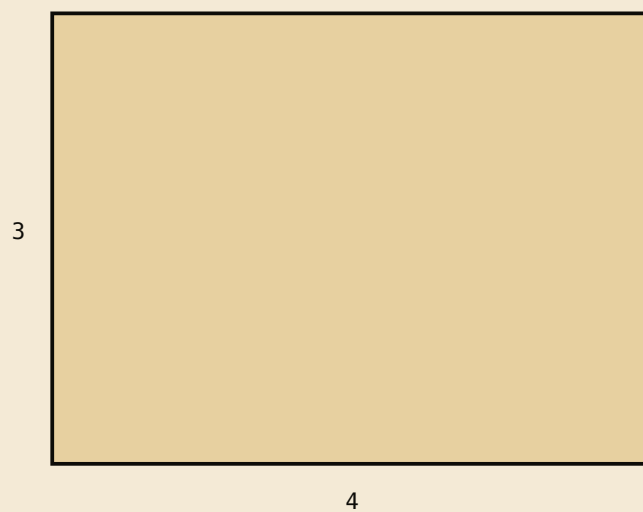
Input contract: {"base": "positive_integer", "height": "positive_integer", "kind": "parallelogram_with_unit", "offset": "integer", "slanted_side": "positive_integer", "unit": "atom"}; example {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_base	q_step_1	geometry_action_pairs.pl:872 observed on {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}
q_step_1	distinguish_height_from_slanted_side	q_step_2	geometry_action_pairs.pl:872 observed on {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}
q_step_2	cut_and_translate_to_rectangle	q_step_3	geometry_action_pairs.pl:872 observed on {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}
q_step_3	multiply_base_by_perpendicular_height	q_accept	geometry_action_pairs.pl:872 observed on {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}

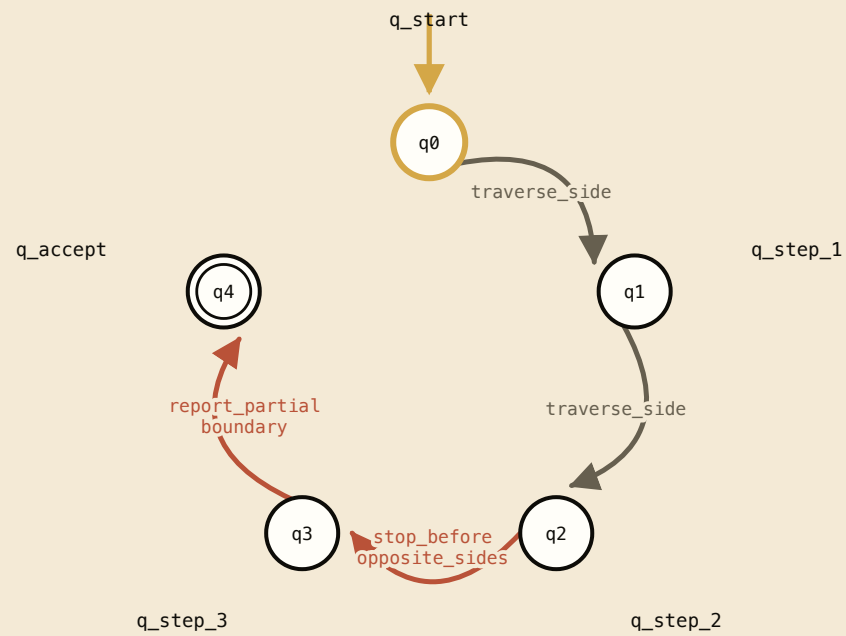
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

26. perimeter_two_sides_only



The run produced `length(11,centimeter)`; the recorded expected result is `length(22,centimeter)`

The run produced `length(11,centimeter)`; the recorded expected result is `length(22,centimeter)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{report_partial_boundary, stop_before_opposite_sides, traverse_side (3 distinct)}$$

$$q_0 \quad q_0 = \text{q_start}$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

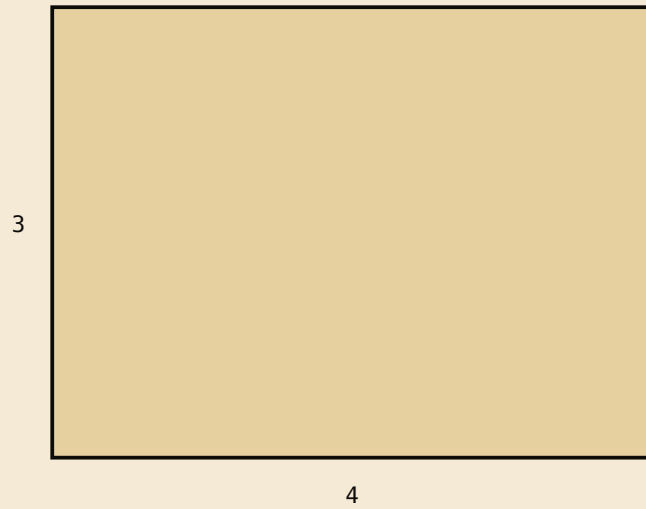
Input contract: {"kind": "rectangle_with_unit", "length": "positive_integer", "unit": "atom", "width": "positive_integer"}; example {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	traverse_side	q_step_1	geometry_action_pairs.pl:227 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_1	traverse_side	q_step_2	geometry_action_pairs.pl:227 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_2	stop_before_opposite_sides	q_step_3	geometry_action_pairs.pl:227 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_3	report_partial_boundary	q_accept	geometry_action_pairs.pl:227 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}

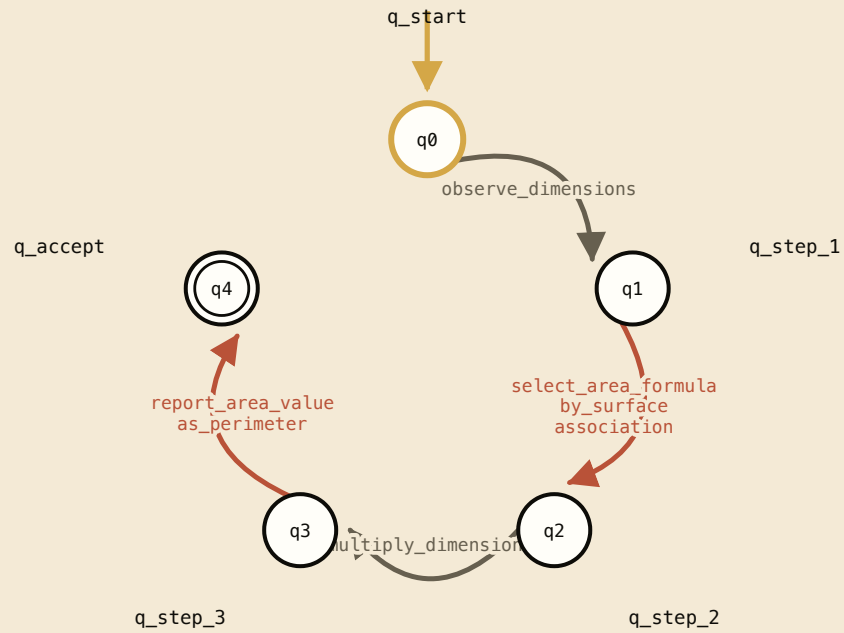
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 3 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

27. perimeter_uses_area_formula



The run produced `length(28,centimeter)`; the recorded expected result is `length(22,centimeter`

The run produced `length(28,centimeter)`; the recorded expected result is `length(22,centimeter)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ multiply_dimensions, observe_dimensions, report_area_value_as_perimeter, select_area_formula_by_surface_association (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"kind": "rectangle_with_unit", "length": "positive_integer", "unit": "atom", "width": "positive_integer"}; example {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}; verified(strategy_trace_ok).

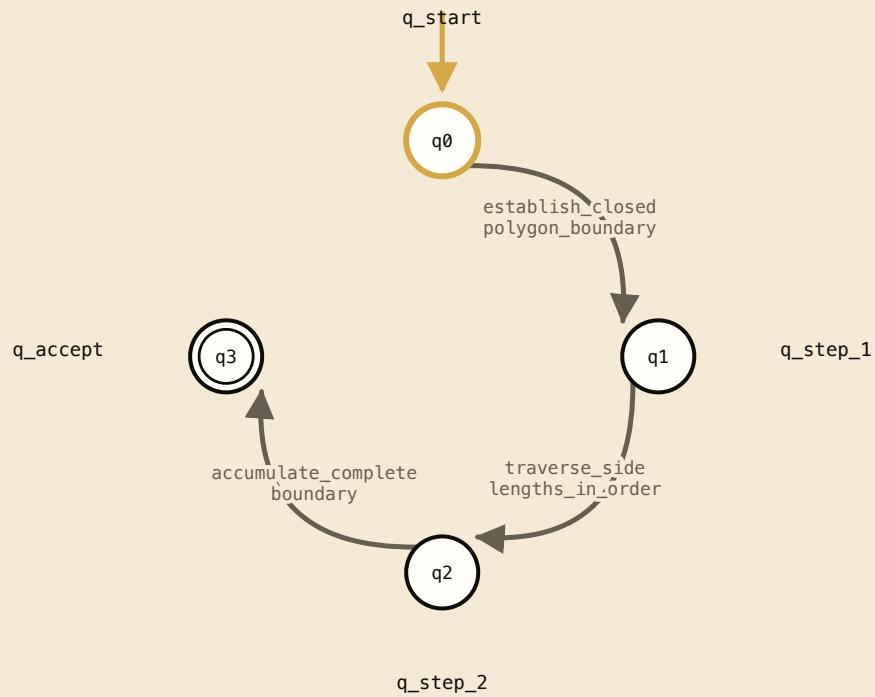
δ : recorded transitions

From	Action	To	Provenance
q_start	observe_dimensions	q_step_1	geometry_action_pairs.pl:252 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_1	select_area_formula_by_surface_association	q_step_2	geometry_action_pairs.pl:252 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_2	multiply_dimensions	q_step_3	geometry_action_pairs.pl:252 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_3	report_area_value_as_perimeter	q_accept	geometry_action_pairs.pl:252 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

28. polygon_perimeter_boundary_accumulation

No domain scene: the static scene emitter has no geometry form for the `polygon_sides_with_unit` contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad \text{accumulate_complete_boundary, establish_closed_polygon_boundary,} \\ \text{traverse_side_lengths_in_order (3 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$F \quad \{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

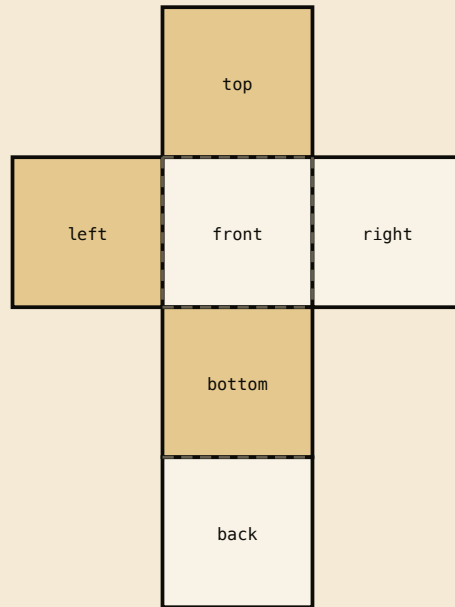
Input contract: {"kind": "polygon_sides_with_unit", "sides": ["positive_number"], "unit": "atom"}; example {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_closed_polygon_boundary	q_step_1	geometry_action_pairs.pl:279 observed on {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}
q_step_1	traverse_side_lengths_in_order	q_step_2	geometry_action_pairs.pl:279 observed on {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}
q_step_2	accumulate_complete_boundary	q_accept	geometry_action_pairs.pl:279 observed on {"kind": "polygon_sides_with_unit", "sides": [5, 7, 4, 6], "unit": "centimeter"}

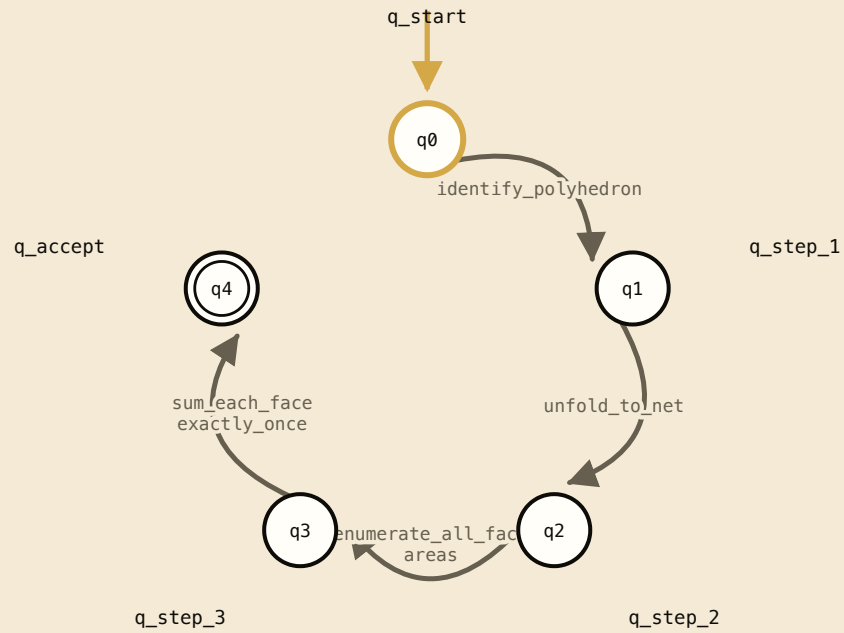
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

29. polyhedron_surface_area_from_net



The run produced `area(52,square(centimeter))`.

The run produced `area(52,square(centimeter))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{enumerate_all_face_areas, identify_polyhedron, sum_each_face_exactly_once, unfold_to_net (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

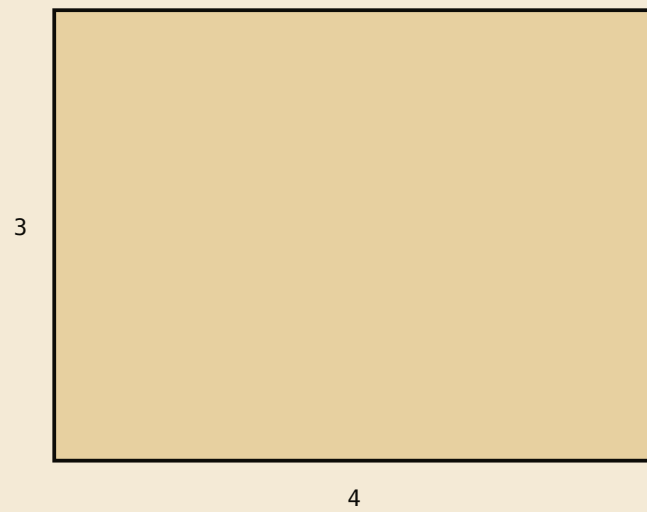
Input contract: {"face_areas": ["positive_integer"], "kind": "solid_net", "solid": "atom", "unit": "atom"}; example {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_polyhedron	q_step_1	geometry_action_pairs.pl:972 observed on {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}
q_step_1	unfold_to_net	q_step_2	geometry_action_pairs.pl:972 observed on {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}
q_step_2	enumerate_all_face_areas	q_step_3	geometry_action_pairs.pl:972 observed on {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}
q_step_3	sum_each_face_exactly_once	q_accept	geometry_action_pairs.pl:972 observed on {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}

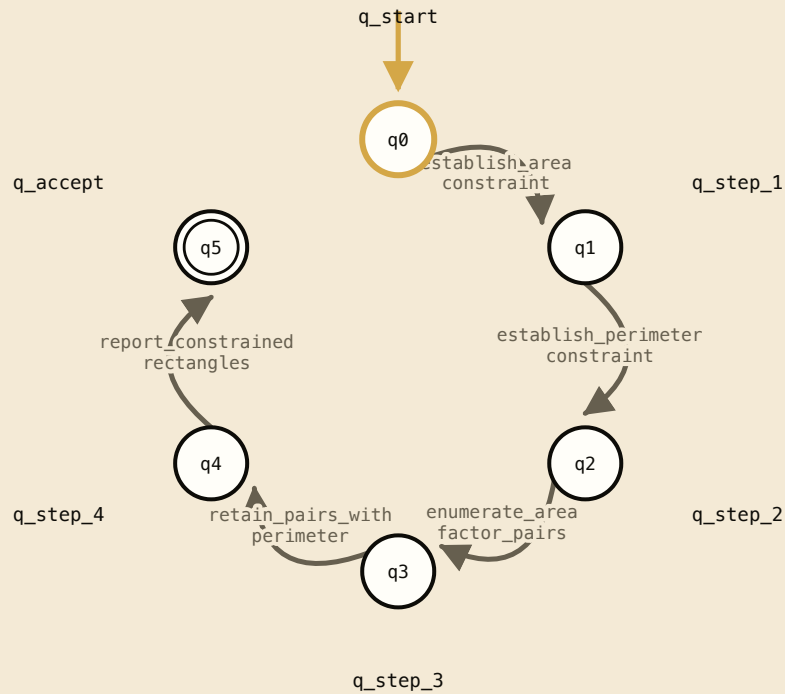
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

30. rectangle_area_perimeter_constraint_search



The run produced `rectangles([3-8])`.

The run produced `rectangles([3-8])`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ enumerate_area_factor_pairs, establish_area_constraint, establish_perimeter_constraint, report_constrained_rectangles, retain_pairs_with_perimeter (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"area": "positive_integer", "kind": "rectangle_constraints", "perimeter": "positive_integer"}; example {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}; verified(strategy_trace_ok).

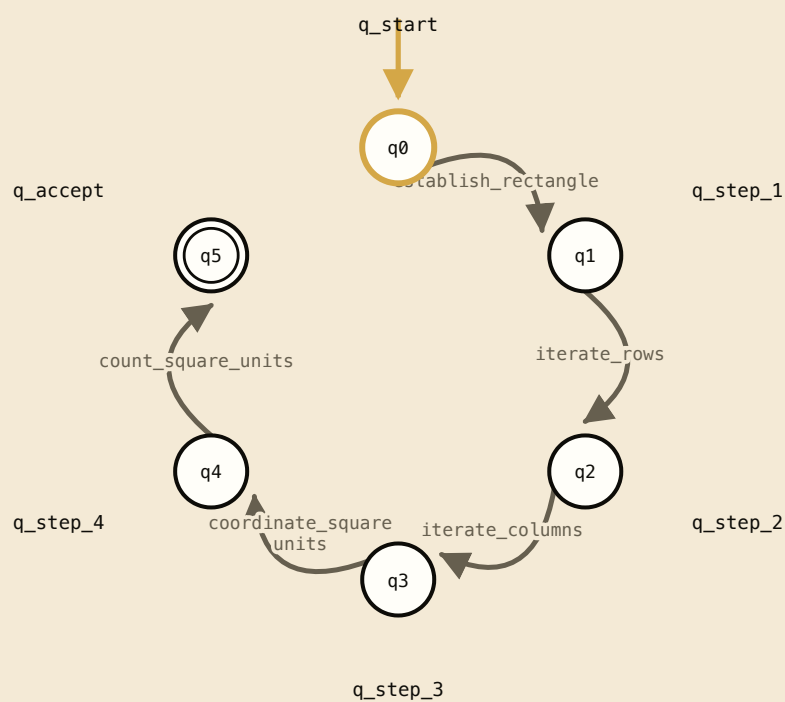
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_area_constraint	q_step_1	geometry_action_pairs.pl:482 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}
q_step_1	establish_perimeter_constraint	q_step_2	geometry_action_pairs.pl:482 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}
q_step_2	enumerate_area_factor_pairs	q_step_3	geometry_action_pairs.pl:482 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}
q_step_3	retain_pairs_with_perimeter	q_step_4	geometry_action_pairs.pl:482 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}
q_step_4	report_constrained_rectangles	q_accept	geometry_action_pairs.pl:482 observed on {"area": 24, "kind": "rectangle_constraints", "perimeter": 22}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

31. rectangle_area_unit_iteration

No domain scene: the static scene emitter has no geometry form for the two-operand contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ `coordinate_square_units`, `count_square_units`, `establish_rectangle`, `iterate_columns`, `iterate_rows` (5 distinct)

q_0 $q_0 = q_{\text{start}}$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

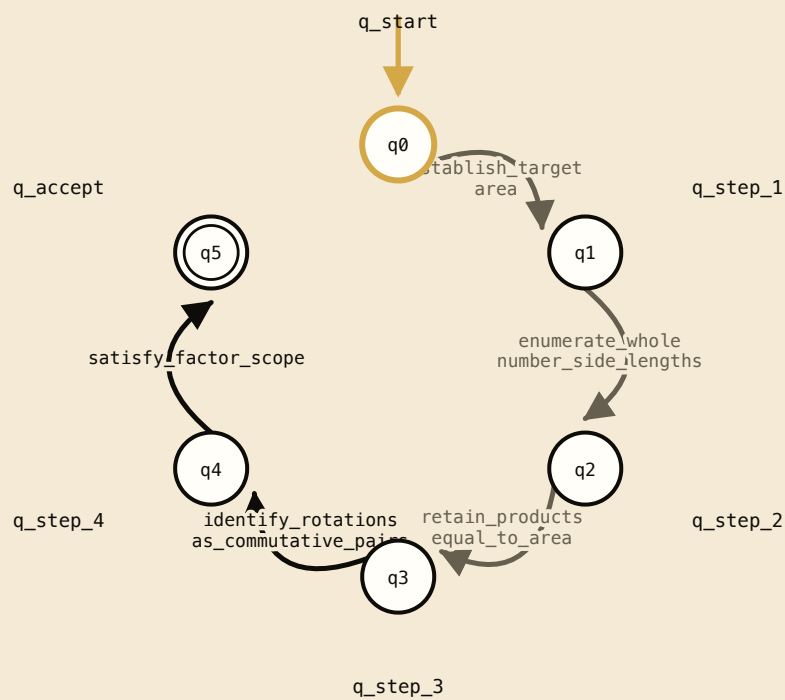
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_rectangle	q_step_1	geometry_action_pairs.pl:35 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	iterate_rows	q_step_2	geometry_action_pairs.pl:35 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	iterate_columns	q_step_3	geometry_action_pairs.pl:35 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	coordinate_square_units	q_step_4	geometry_action_pairs.pl:35 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	count_square_units	q_accept	geometry_action_pairs.pl:35 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

32. rectangle_factor_pair_search

No domain scene: the static scene emitter has no geometry form for the area_scope contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$M = (Q, \Sigma, \delta, q_0, F)$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ enumerate_whole_number_side_lengths, establish_target_area,
identify_rotations_as_commutative_pairs, retain_products_equal_to_area,
satisfy_factor_scope (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"area": "positive_integer", "kind": "area_scope", "scope": "atom"};
example {"area": 24, "kind": "area_scope", "scope": "all"};
verified(strategy_trace_ok).

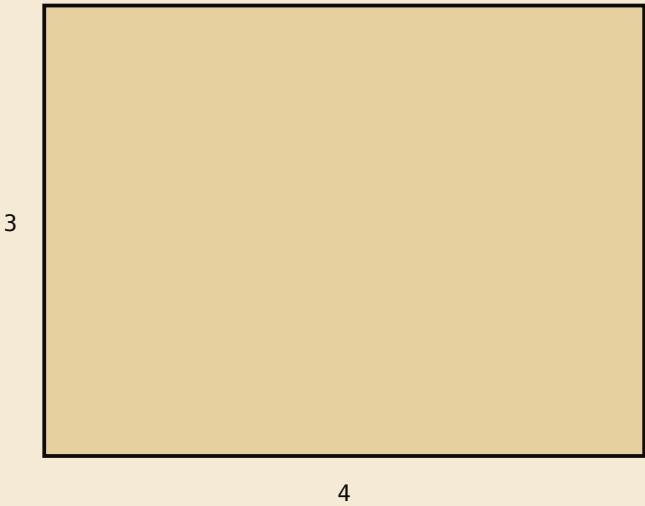
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_target_area	q_step_1	geometry_action_pairs.pl:451 observed on {"area": 24, "kind": "area_scope", "scope": "all"}
q_step_1	enumerate_whole_number_side_lengths	q_step_2	geometry_action_pairs.pl:451 observed on {"area": 24, "kind": "area_scope", "scope": "all"}
q_step_2	retain_products_equal_to_area	q_step_3	geometry_action_pairs.pl:451 observed on {"area": 24, "kind": "area_scope", "scope": "all"}

From	Action	To	Provenance
q_step_3	identify_rotations_as_commutative_pairs	q_step_4	geometry_action_pairs.pl:451 observed on {"area": 24, "kind": "area_scope", "scope": "all"}
q_step_4	satisfy_factor_scope	q_accept	geometry_action_pairs.pl:451 observed on {"area": 24, "kind": "area_scope", "scope": "all"}

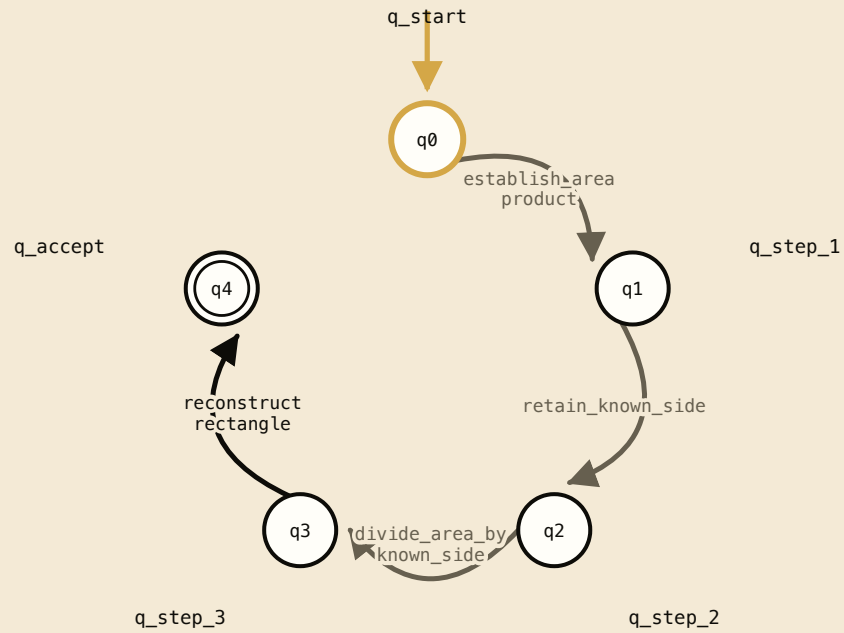
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

33. rectangle_missing_side_from_area



The run produced `side_length(7)`.

The run produced `side_length(7)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ divide_area_by_known_side, establish_area_product, reconstruct_rectangle, retain_known_side (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

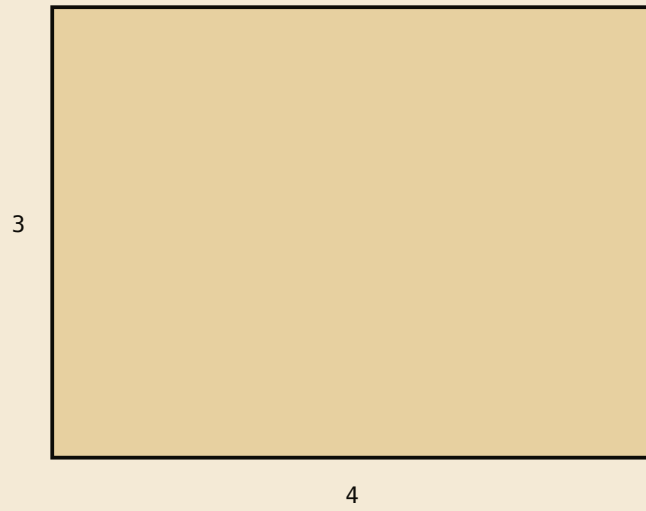
Input contract: {"area": "positive_integer", "kind": "area_known_side", "known_side": "positive_integer"}; example {"area": 28, "kind": "area_known_side", "known_side": 4}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_area_product	q_step_1	geometry_action_pairs.pl:540 observed on {"area": 28, "kind": "area_known_side", "known_side": 4}
q_step_1	retain_known_side	q_step_2	geometry_action_pairs.pl:540 observed on {"area": 28, "kind": "area_known_side", "known_side": 4}
q_step_2	divide_area_by_known_side	q_step_3	geometry_action_pairs.pl:540 observed on {"area": 28, "kind": "area_known_side", "known_side": 4}
q_step_3	reconstruct_rectangle	q_accept	geometry_action_pairs.pl:540 observed on {"area": 28, "kind": "area_known_side", "known_side": 4}

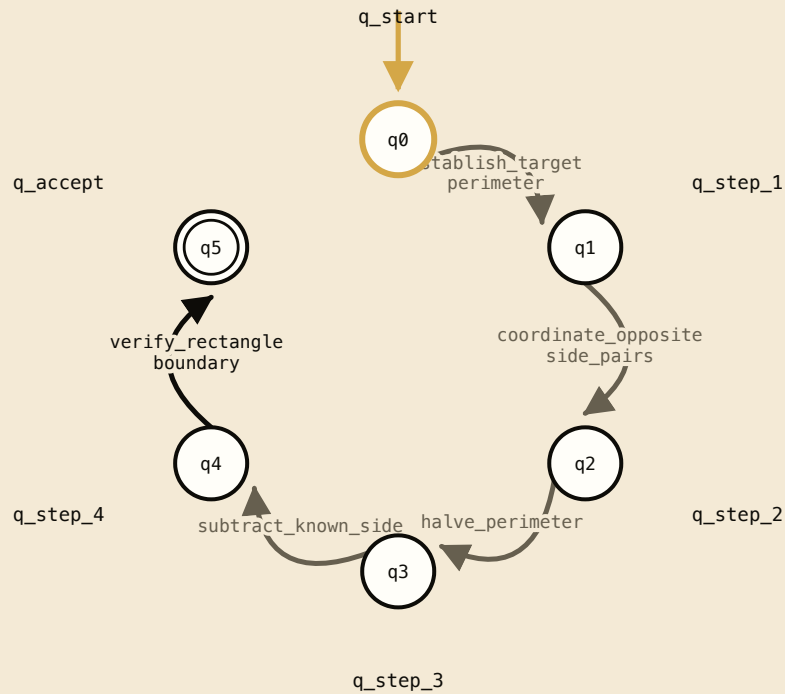
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

34. rectangle_missing_side_from_perimeter



The run produced `side_length(7)`.

The run produced `side_length(7)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ coordinate_opposite_side_pairs, establish_target_perimeter, halve_perimeter, subtract_known_side, verify_rectangle_boundary (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

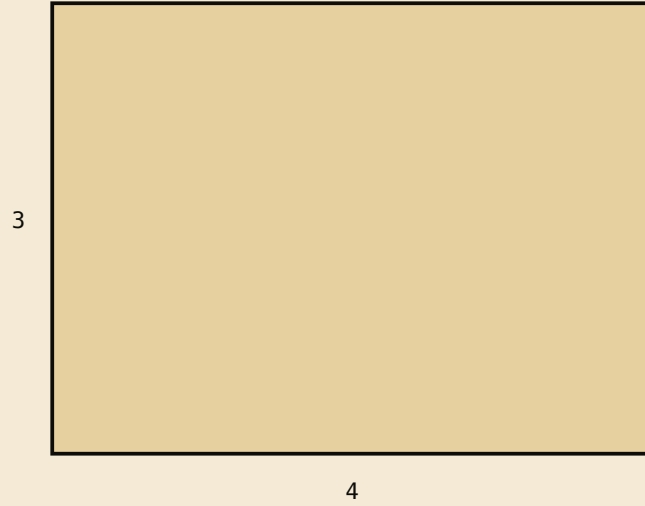
Input contract: {"kind": "perimeter_known_side", "known_side": "positive_integer", "perimeter": "positive_integer"}; example {"kind": "perimeter_known_side", "known_side": 4, "perimeter": 22}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_target_perimeter	q_step_1	geometry_action_pairs.pl:422 observed on {"kind": "perimeter_known_side", "known_side": 4, "perimeter": 22}
q_step_1	coordinate_opposite_side_pairs	q_step_2	geometry_action_pairs.pl:422 observed on {"kind": "perimeter_known_side", "known_side": 4, "perimeter": 22}
q_step_2	halve_perimeter	q_step_3	geometry_action_pairs.pl:422 observed on {"kind": "perimeter_known_side", "known_side": 4, "perimeter": 22}
q_step_3	subtract_known_side	q_step_4	geometry_action_pairs.pl:422 observed on {"kind": "perimeter_known_side", "known_side": 4, "perimeter": 22}
q_step_4	verify_rectangle_boundary	q_accept	geometry_action_pairs.pl:422 observed on {"kind": "perimeter_known_side", "known_side": 4, "perimeter": 22}

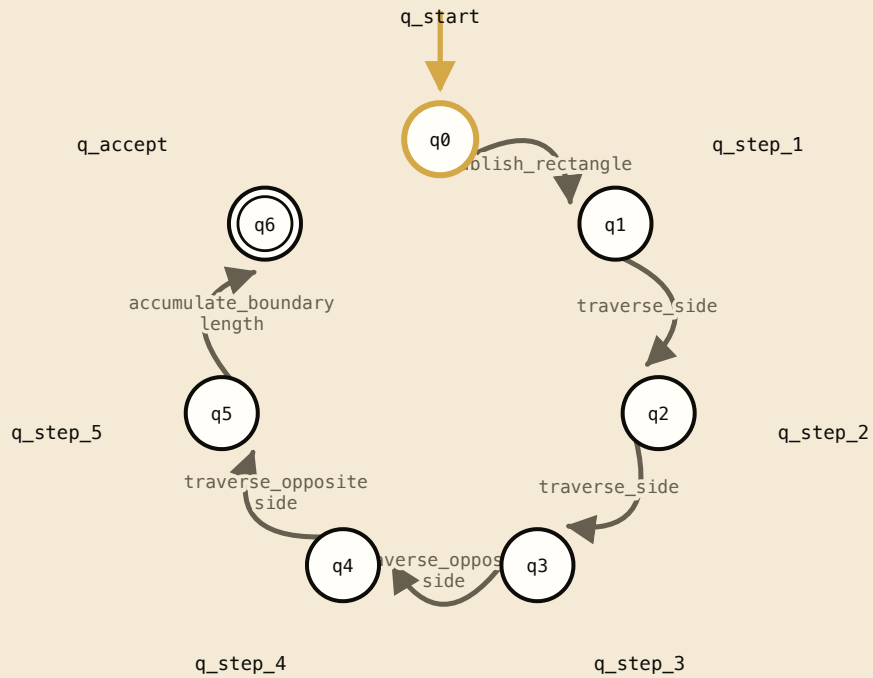
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

35. rectangle_perimeter_boundary_traversal



The run produced `length(22,centimeter)`.

The run produced `length(22,centimeter)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \text{accumulate_boundary_length, establish_rectangle, traverse_opposite_side, traverse_side (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"kind": "rectangle_with_unit", "length": "positive_integer", "unit": "atom", "width": "positive_integer"}; example {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4};
verified(strategy_trace_ok).

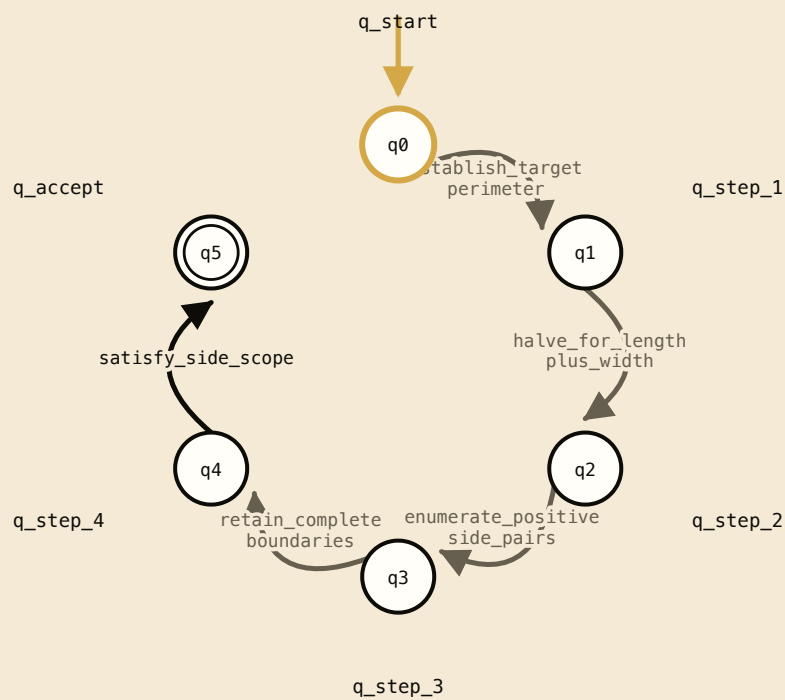
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_rectangle	q_step_1	geometry_action_pairs.pl:198 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_1	traverse_side	q_step_2	geometry_action_pairs.pl:198 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_2	traverse_side	q_step_3	geometry_action_pairs.pl:198 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_3	traverse_opposite_side	q_step_4	geometry_action_pairs.pl:198 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_4	traverse_opposite_side	q_step_5	geometry_action_pairs.pl:198 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}
q_step_5	accumulate_boundary_length	q_accept	geometry_action_pairs.pl:198 observed on {"kind": "rectangle_with_unit", "length": 7, "unit": "centimeter", "width": 4}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 4 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 6 observed rows.

36. rectangle_perimeter_side_pair_search

No domain scene: the static scene emitter has no geometry form for the perimeter_scope contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$M = (Q, \Sigma, \delta, q_0, F)$

$Q \quad \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma \quad$ enumerate_positive_side_pairs, establish_target_perimeter, halve_for_length_plus_width, retain_complete_boundaries, satisfy_side_scope (5 distinct)

$q_0 \quad q_0 = q_start$

$F \quad \{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"kind": "perimeter_scope", "perimeter": "positive_integer", "scope": "atom"}; example {"kind": "perimeter_scope", "perimeter": 22, "scope": "all"}; verified(strategy_trace_ok).

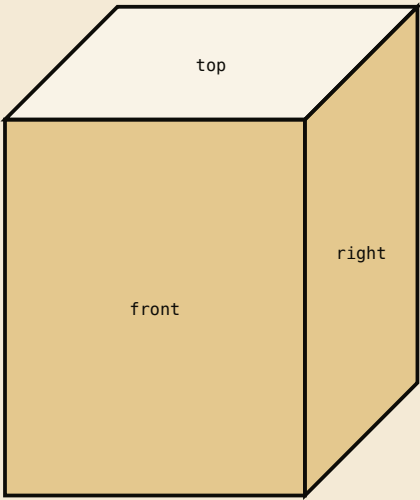
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_target_perimeter	q_step_1	geometry_action_pairs.pl:392 observed on {"kind": "perimeter_scope", "perimeter": 22, "scope": "all"}
q_step_1	halve_for_length_plus_width	q_step_2	geometry_action_pairs.pl:392 observed on {"kind": "perimeter_scope", "perimeter": 22, "scope": "all"}
q_step_2	enumerate_positive_side_pairs	q_step_3	geometry_action_pairs.pl:392 observed on {"kind": "perimeter_scope", "perimeter": 22, "scope": "all"}

From	Action	To	Provenance
q_step_3	retain_complete_boundaries	q_step_4	geometry_action_pairs.pl:392 observed on {"kind": "perimeter_scope", "perimeter": 22, "scope": "all"}
q_step_4	satisfy_side_scope	q_accept	geometry_action_pairs.pl:392 observed on {"kind": "perimeter_scope", "perimeter": 22, "scope": "all"}

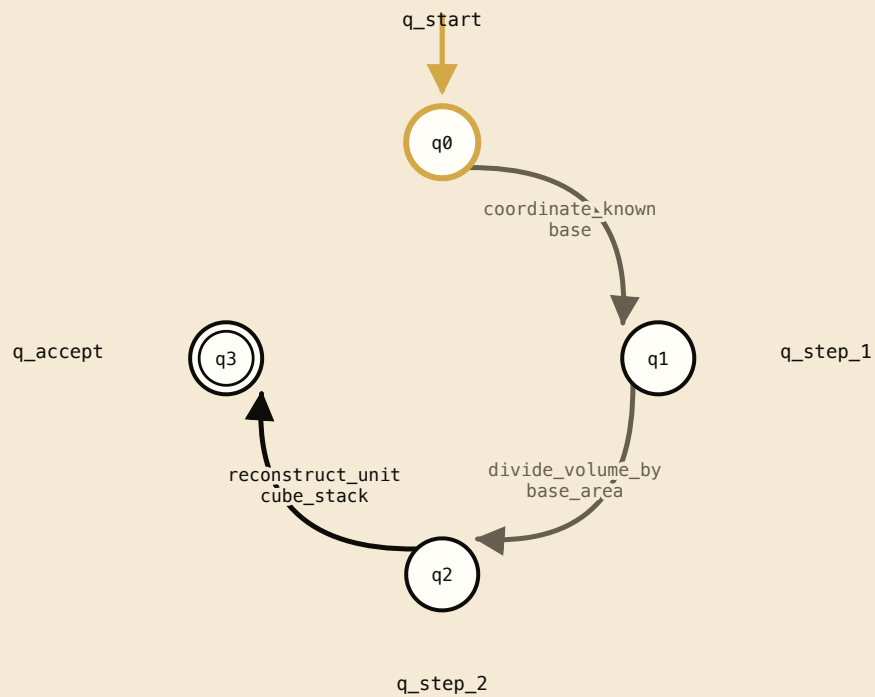
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

37. rectangular_prism_missing_dimension_from_volume



The run produced dimension(5).

The run produced dimension(5). The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad coordinate_known_base, divide_volume_by_base_area, reconstruct_unit_cube_stack (3 \text{ distinct})$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_3\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

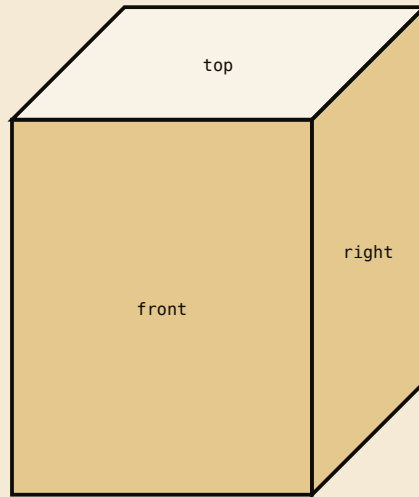
Input contract: {"kind": "volume_known_base", "length": "positive_integer", "volume": "positive_integer", "width": "positive_integer"}; example {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	coordinate_known_base	q_step_1	geometry_action_pairs.pl:621 observed on {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3}
q_step_1	divide_volume_by_base_area	q_step_2	geometry_action_pairs.pl:621 observed on {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3}
q_step_2	reconstruct_unit_cube_stack	q_accept	geometry_action_pairs.pl:621 observed on {"kind": "volume_known_base", "length": 4, "volume": 60, "width": 3}

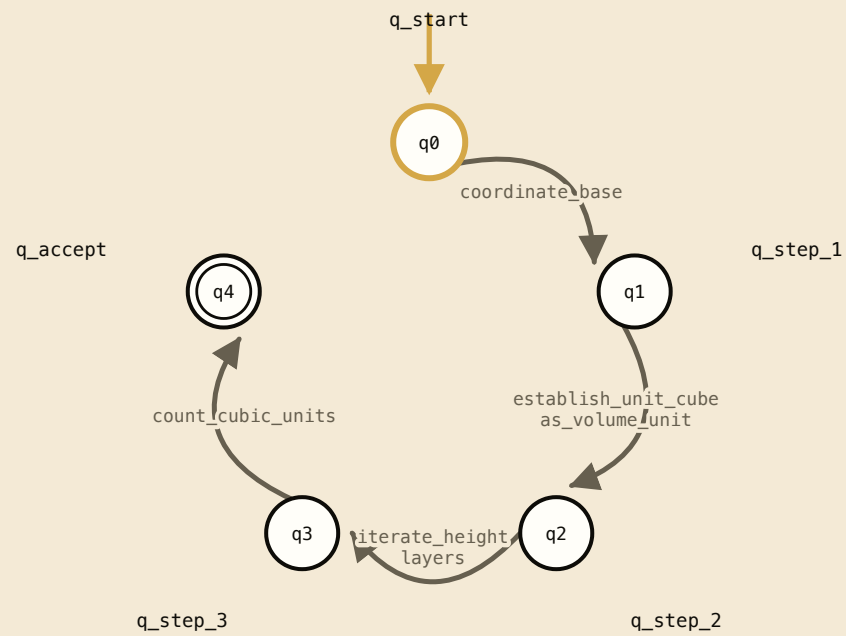
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

38. rectangular_prism_volume_layer_iteration



The run produced `cubic_units(60)`.

The run produced `cubic_units(60)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{coordinate_base, count_cubic_units, establish_unit_cube_as_volume_unit, iterate_height_layers (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

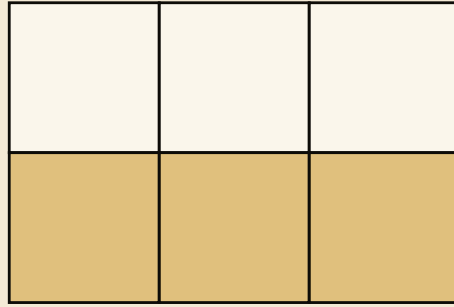
Input contract: {"height": "positive_integer", "kind": "rectangular_prism", "length": "positive_integer", "width": "positive_integer"}; example {"height": 5, "kind": "rectangular_prism", "length": 4, "width": 3}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	coordinate_base	q_step_1	geometry_action_pairs.pl:592 observed on {"height": 5, "kind": "rectangular_prism", "length": 4, "width": 3}
q_step_1	establish_unit_cube_as_volume_unit	q_step_2	geometry_action_pairs.pl:592 observed on {"height": 5, "kind": "rectangular_prism", "length": 4, "width": 3}
q_step_2	iterate_height_layers	q_step_3	geometry_action_pairs.pl:592 observed on {"height": 5, "kind": "rectangular_prism", "length": 4, "width": 3}
q_step_3	count_cubic_units	q_accept	geometry_action_pairs.pl:592 observed on {"height": 5, "kind": "rectangular_prism", "length": 4, "width": 3}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

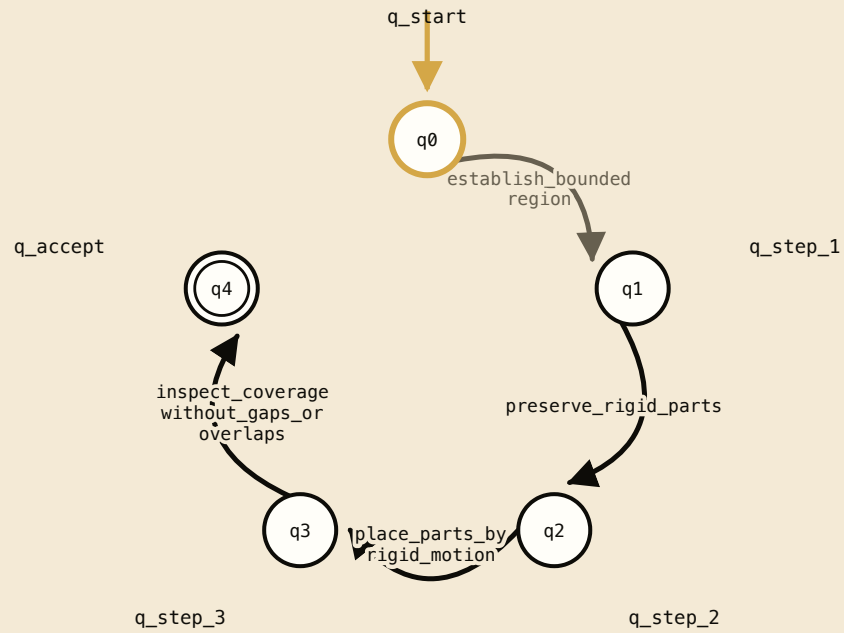
39. rigid_shape_composition



3x2 region

The run produced `composed_region(3,2)`.

The run produced `composed_region(3,2)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ establish_bounded_region, inspect_coverage_without_gaps_or_overlaps, place_parts_by_rigid_motion, preserve_rigid_parts (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"columns": "positive_integer", "kind": "rigid_shape_composition", "pieces": [{"cells": [{"x": "integer", "y": "integer"}], "id": "atom"}], "rows": "positive_integer"}; example {"columns": 3, "kind": "rigid_shape_composition", "pieces": [{"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 2, "y": 0}], "id": "top_row"}, {"cells": [{"x": 0, "y": 1}, {"x": 1, "y": 1}, {"x": 2, "y": 1}], "id": "bottom_row"}], "rows": 2}; verified(strategy_trace_ok).

δ : recorded transitions

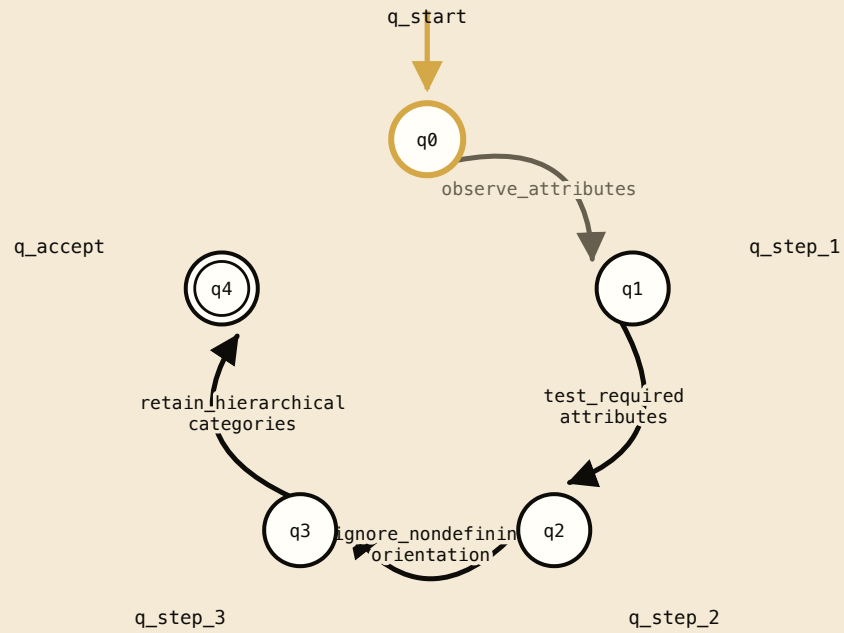
From	Action	To	Provenance
q_start	establish_bounded_region	q_step_1	geometry_action_pairs.pl:1194 observed on {"columns": 3, "kind": "rigid_shape_composition", "pieces": [{"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 2, "y": 0}], "id": "top_row"}, {"cells": [{"x": 0, "y": 1}, {"x": 1, "y": 1}, {"x": 2, "y": 1}], "id": "bottom_row"}], "rows": 2}
q_step_1	preserve_rigid_parts	q_step_2	geometry_action_pairs.pl:1194 observed on {"columns": 3, "kind": "rigid_shape_composition", "pieces": [{"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 2, "y": 0}], "id": "top_row"}, {"cells": [{"x": 0, "y": 1}, {"x": 1, "y": 1}, {"x": 2, "y": 1}], "id": "bottom_row"}], "rows": 2}
q_step_2	place_parts_beyond_rigid_motion	q_step_3	geometry_action_pairs.pl:1194 observed on {"columns": 3, "kind": "rigid_shape_composition", "pieces": [{"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 2, "y": 0}], "id": "top_row"}, {"cells": [{"x": 0, "y": 1}, {"x": 1, "y": 1}, {"x": 2, "y": 1}], "id": "bottom_row"}], "rows": 2}

From	Action	To	Provenance
q_ste p_3	inspect_cover age_without_g aps_or_overla ps	q_a cce pt	geometry_action_pairs.pl:1194 observed on {"columns": 3, "kind": "rigid_shape_composition", "pieces": [{"cells": [{"x": 0, "y": 0}, {"x": 1, "y": 0}, {"x": 2, "y": 0}], "id": "top_row"}, {"cells": [{"x": 0, "y": 1}, {"x": 1, "y": 1}, {"x": 2, "y": 1}], "id": "bottom_row"}], "rows": 2}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

40. shape_classification_by_defining_attributes

No domain scene: the static scene emitter has no geometry form for the shape_attributes contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{ignore_nondefining_orientation, observe_attributes, retain_hierarchical_categories, test_required_attributes (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

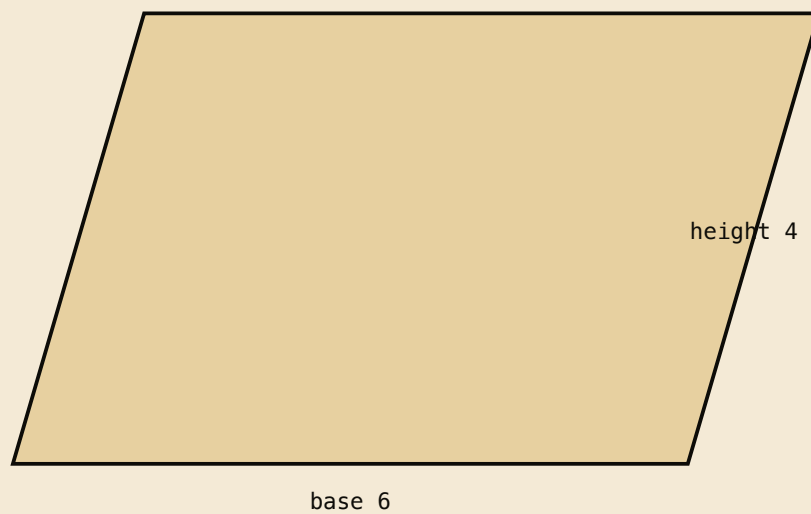
Input contract: {"attributes": [{"name": "atom", "value": "integer"}], "kind": "shape_attributes", "quarter_turns": "integer", "shape": "atom"}; example {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	observe_attributes	q_step_1	geometry_action_pairs.pl:1069 observed on {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}
q_step_1	test_required_attributes	q_step_2	geometry_action_pairs.pl:1069 observed on {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}
q_step_2	ignore_nondetermining_orientation	q_step_3	geometry_action_pairs.pl:1069 observed on {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}
q_step_3	retain_hierarchical_categories	q_accept	geometry_action_pairs.pl:1069 observed on {"attributes": [{"name": "straight_sides", "value": 4}, {"name": "vertices", "value": 4}, {"name": "right_angles", "value": 4}, {"name": "equal_sides", "value": 4}], "kind": "shape_attributes", "quarter_turns": 1, "shape": "square"}

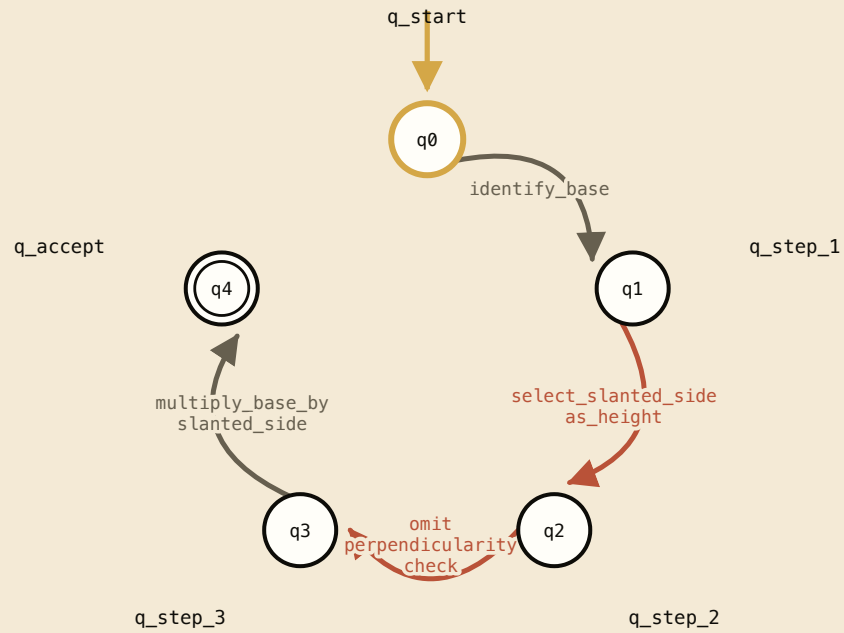
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

41. slanted_side_as_parallelogram_height



The run produced `area(30,square(centimeter))`; the recorded expected result is `area(24,square`

`area(24,square(centimeter))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ identify_base, multiply_base_by_slanted_side, omit_perpendicularity_check, select_slanted_side_as_height (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

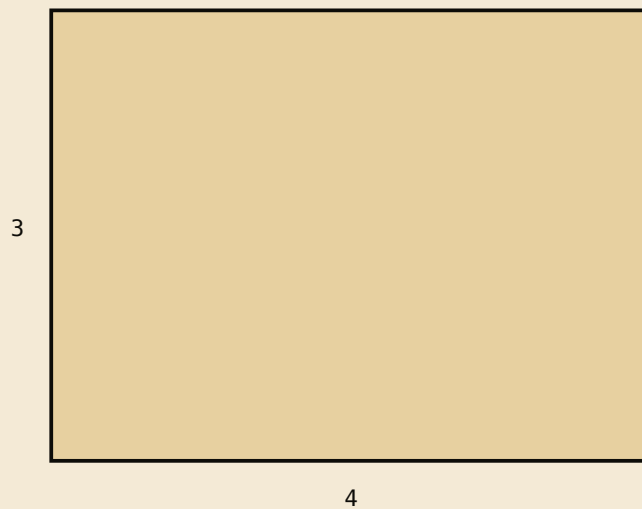
Input contract: {"base": "positive_integer", "height": "positive_integer", "kind": "parallelogram_with_unit", "offset": "integer", "slanted_side": "positive_integer", "unit": "atom"}; example {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_base	q_step_1	geometry_action_pairs.pl:897 observed on {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}
q_step_1	select_slanted_side_as_height	q_step_2	geometry_action_pairs.pl:897 observed on {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}
q_step_2	omit_perpendicularity_check	q_step_3	geometry_action_pairs.pl:897 observed on {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}
q_step_3	multiply_base_by_slanted_side	q_accept	geometry_action_pairs.pl:897 observed on {"base": 6, "height": 4, "kind": "parallelogram_with_unit", "offset": 3, "slanted_side": 5, "unit": "centimeter"}

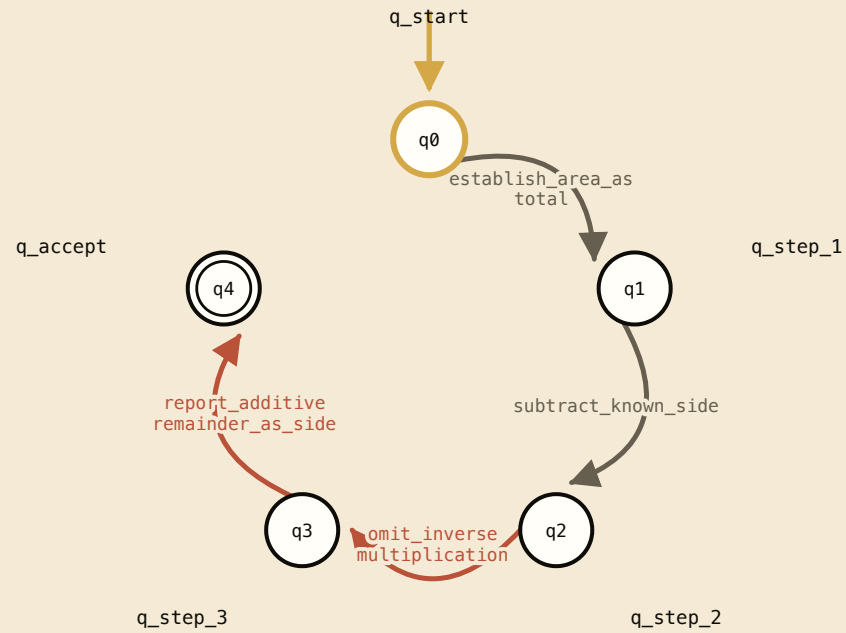
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

42. subtract_side_from_area



The run produced `side_length(24)`; the recorded expected result is `side_length(7)`.

The run produced `side_length(24)`; the recorded expected result is `side_length(7)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ establish_area_as_total, omit_inverse_multiplication,
report_additive_remainder_as_side, subtract_known_side (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"area": "positive_integer", "kind": "area_known_side", "known_side": "positive_integer"}; example {"area": 28, "kind": "area_known_side", "known_side": 4}; verified(strategy_trace_ok).

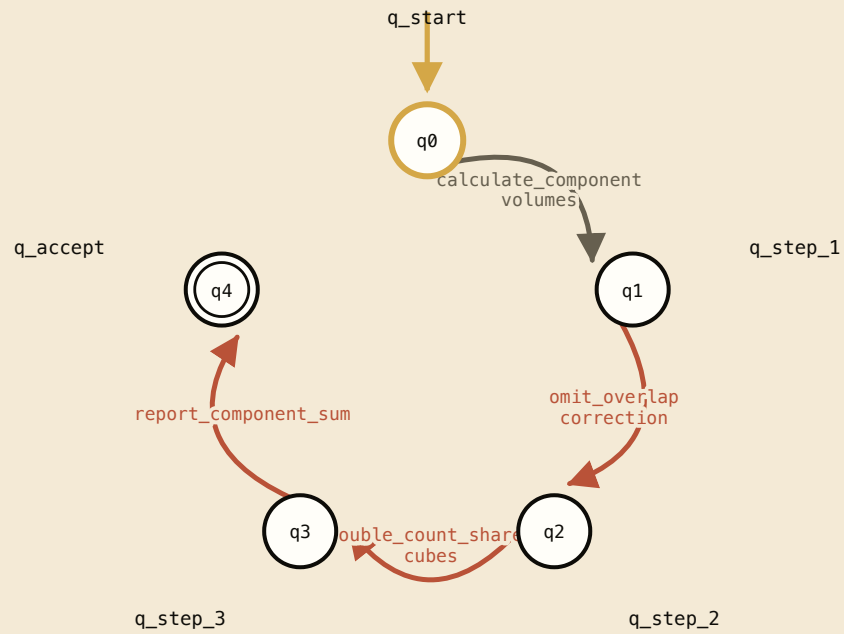
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_area_as_total	q_step_1	geometry_action_pairs.pl:565 observed on {"area": 28, "kind": "area_known_side", "known_side": 4}
q_step_1	subtract_known_side	q_step_2	geometry_action_pairs.pl:565 observed on {"area": 28, "kind": "area_known_side", "known_side": 4}
q_step_2	omit_inverse_multiplication	q_step_3	geometry_action_pairs.pl:565 observed on {"area": 28, "kind": "area_known_side", "known_side": 4}
q_step_3	report_additive_remainder_as_side	q_accept	geometry_action_pairs.pl:565 observed on {"area": 28, "kind": "area_known_side", "known_side": 4}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

43. sum_overlapping_prism_volumes

No domain scene: the static scene emitter has no geometry form for the overlapping_prisms contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{calculate_component_volumes, double_count_shared_cubes, omit_overlap_correction, report_component_sum (4 distinct)}$$

$$q_0 = q_start$$

$F \quad \{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"kind": "overlapping_prisms", "overlap_volume": "positive_integer", "prisms": [{"height": "positive_integer", "length": "positive_integer", "width": "positive_integer"}], "unit": "atom"}; example {"kind": "overlapping_prisms", "overlap_volume": 4, "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}; verified(strategy_trace_ok).

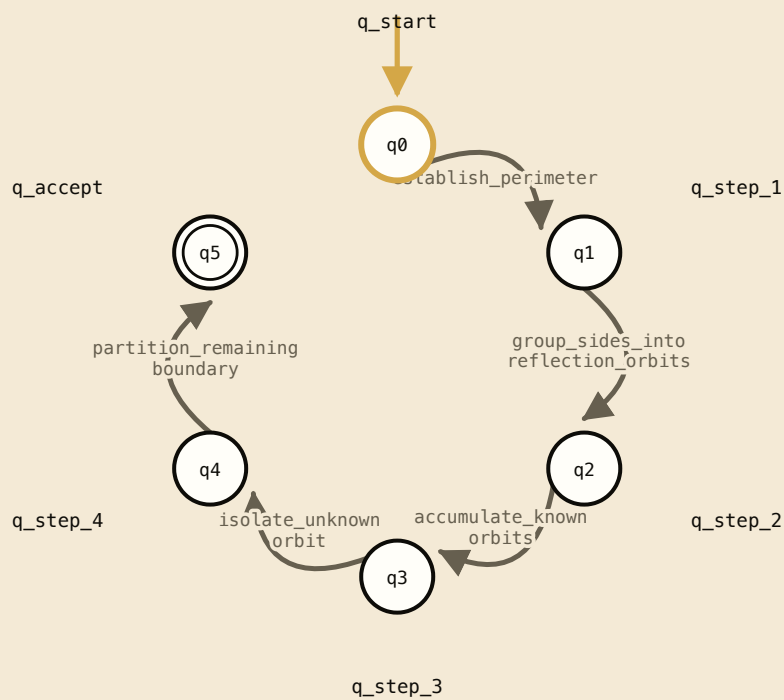
δ : recorded transitions

From	Action	To	Provenance
q_start	calculate_component_volumes	q_step_1	geometry_action_pairs.pl:706 observed on {"kind": "overlapping_prisms", "overlap_volume": 4, "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}
q_step_1	omit_overlap_correction	q_step_2	geometry_action_pairs.pl:706 observed on {"kind": "overlapping_prisms", "overlap_volume": 4, "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}
q_step_2	double_count_shared_cubes	q_step_3	geometry_action_pairs.pl:706 observed on {"kind": "overlapping_prisms", "overlap_volume": 4, "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}
q_step_3	report_component_sum	q_accept	geometry_action_pairs.pl:706 observed on {"kind": "overlapping_prisms", "overlap_volume": 4, "prisms": [{"height": 4, "length": 2, "width": 3}, {"height": 5, "length": 1, "width": 2}], "unit": "centimeter"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

44. symmetry_constrained_side_reconstruction

No domain scene: the static scene emitter has no geometry form for the symmetry_side_orbits contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$M = (Q, \Sigma, \delta, q_0, F)$

Q { $q_0, q_1, q_2, q_3, q_4, q_5$ }

Σ accumulate_known_orbits, establish_perimeter, group_sides_into_reflection_orbits, isolate_unknown_orbit, partition_remaining_boundary (5 distinct)

q_0 $q_0 = q_start$

F { q_5 }

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": "positive_integer", "length": "positive_integer"}], "perimeter": "positive_integer", "unit": "atom", "unknown_orbit": {"copies": "positive_integer", "name": "atom"}}; example {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}; verified(strategy_trace_ok).

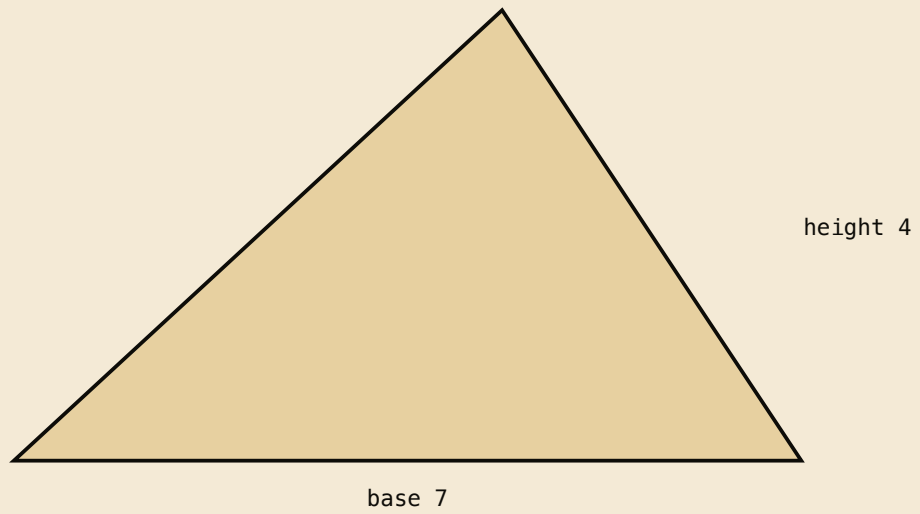
δ : recorded transitions

From	Action	To	Provenance
q_start	establish_perimeter	q_step_1	geometry_action_pairs.pl:333 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}

From	Action	To	Provenance
q_ste p_1	group_sides_in to_reflection_ orbits	q_st ep_2	geometry_action_pairs.pl:333 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}
q_ste p_2	accumulate_kno wn_orbits	q_st ep_3	geometry_action_pairs.pl:333 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}
q_ste p_3	isolate_unknow n_orbit	q_st ep_4	geometry_action_pairs.pl:333 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}
q_ste p_4	partition_rema ining_boundary	q_ac cept	geometry_action_pairs.pl:333 observed on {"kind": "symmetry_side_orbits", "known_orbits": [{"copies": 2, "length": 5}], "perimeter": 26, "unit": "centimeter", "unknown_orbit": {"copies": 2, "name": "slanted_side"}}

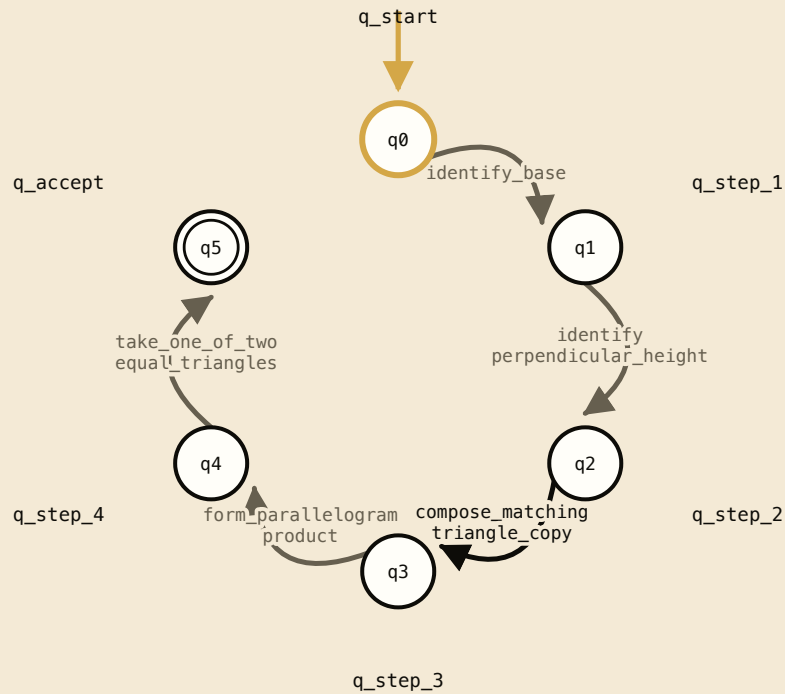
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

45. triangle_area_half_base_height



The run produced area(14,square(centimeter)).

The run produced area(14,square(centimeter)). The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ compose_matching_triangle_copy, form_parallelogram_product, identify_base, identify_perpendicular_height, take_one_of_two_equal_triangles (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

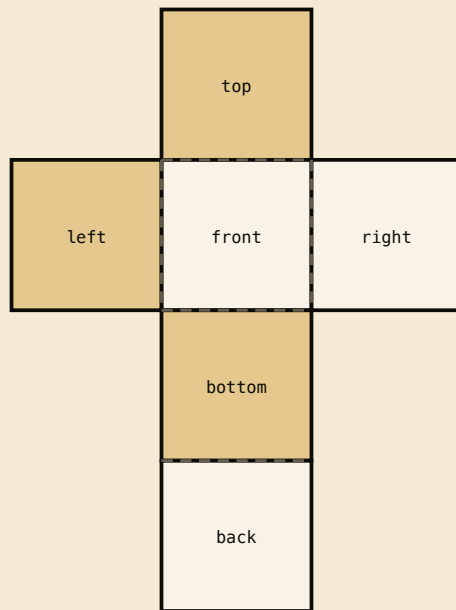
Input contract: {"base": "positive_integer", "height": "positive_integer", "kind": "triangle_with_unit", "unit": "atom"}; example {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_base	q_step_1	geometry_action_pairs.pl:923 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}
q_step_1	identify_perpendicular_height	q_step_2	geometry_action_pairs.pl:923 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}
q_step_2	compose_matching_triangle_copy	q_step_3	geometry_action_pairs.pl:923 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}
q_step_3	form_parallelogram_product	q_step_4	geometry_action_pairs.pl:923 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}
q_step_4	take_one_of_two_equal_triangles	q_accept	geometry_action_pairs.pl:923 observed on {"base": 7, "height": 4, "kind": "triangle_with_unit", "unit": "centimeter"}

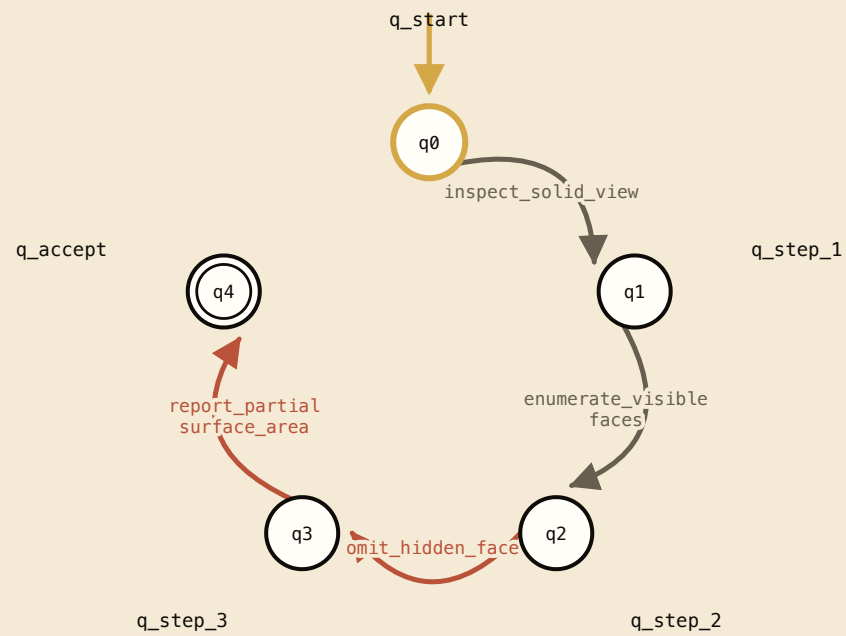
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

46. visible_faces_only_surface_area



The run produced `area(46,square(centimeter))`; the recorded expected result is `area(52,square(centimeter))`.

The run produced `area(46,square(centimeter))`; the recorded expected result is `area(52,square(centimeter))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{enumerate_visible_faces, inspect_solid_view, omit_hidden_face, report_partial_surface_area (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

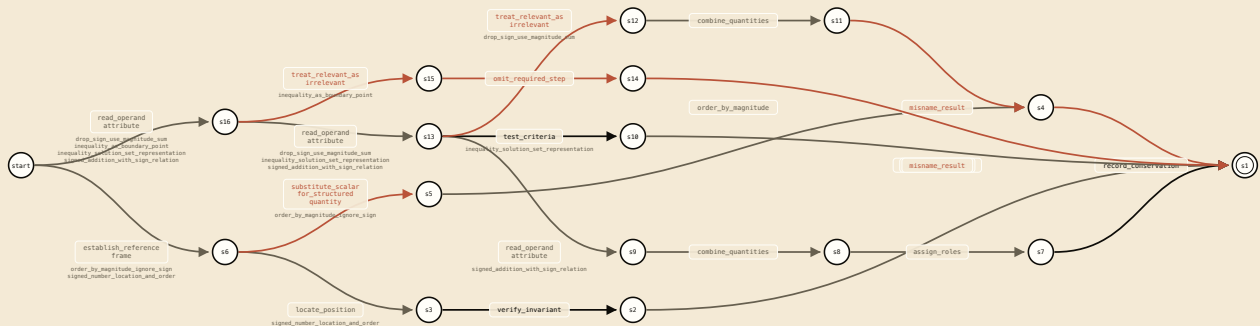
Input contract: {"face_areas": ["positive_integer"], "kind": "solid_net", "solid": "atom", "unit": "atom"}; example {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	inspect_solid_view	q_step_1	geometry_action_pairs.pl:995 observed on {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}
q_step_1	enumerate_visible_faces	q_step_2	geometry_action_pairs.pl:995 observed on {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}
q_step_2	omit_hidden_face	q_step_3	geometry_action_pairs.pl:995 observed on {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}
q_step_3	report_partial_surface_area	q_accept	geometry_action_pairs.pl:995 observed on {"face_areas": [12, 12, 8, 8, 6, 6], "kind": "solid_net", "solid": "cube", "unit": "centimeter"}

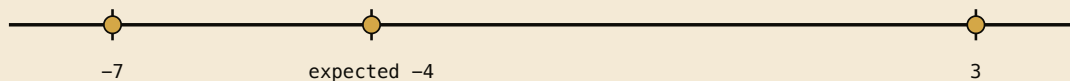
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

integer



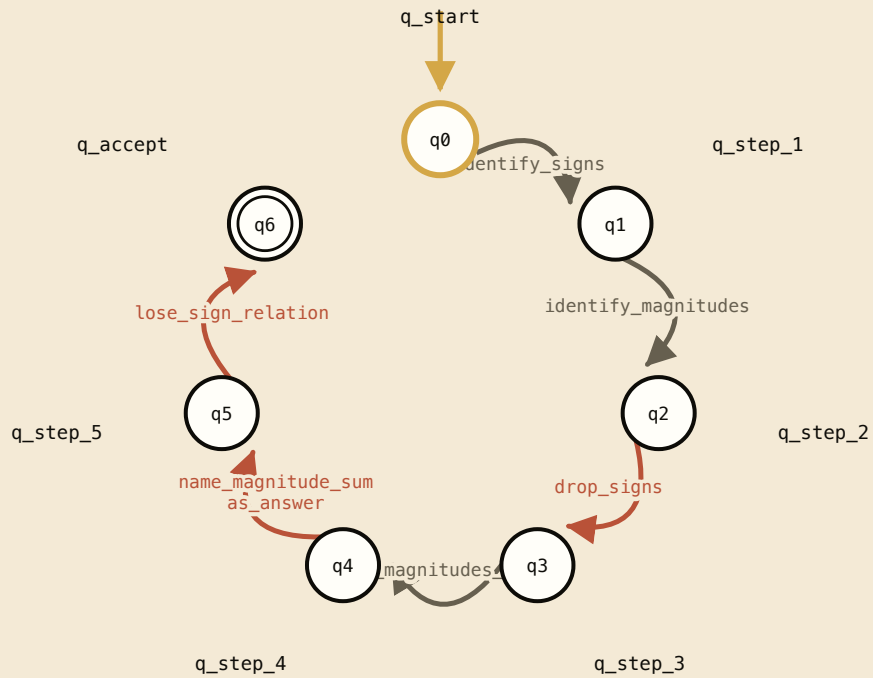
Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. drop_sign_use_magnitude_sum



The run produced 10; the recorded expected result is -4.

The run produced 10; the recorded expected result is -4. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \{\text{drop_signs, identify_magnitudes, identify_signs, lose_sign_relation, name_magnitude_sum_as_answer, sum_magnitudes_only}\} \text{ (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": -7, "b": 3};
verified(strategy_trace_ok).

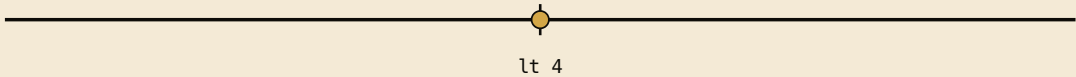
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_signs	q_step_1	integer_action_pairs.pl:53 observed on {"a": -7, "b": 3} observed on {"a": -6, "b": 3} (derived template)
q_step_1	identify_magnitudes	q_step_2	integer_action_pairs.pl:53 observed on {"a": -7, "b": 3} observed on {"a": -6, "b": 3} (derived template)
q_step_2	drop_signs	q_step_3	integer_action_pairs.pl:53 observed on {"a": -7, "b": 3} observed on {"a": -6, "b": 3} (derived template)
q_step_3	sum_magnitudes_only	q_step_4	integer_action_pairs.pl:53 observed on {"a": -7, "b": 3} observed on {"a": -6, "b": 3} (derived template)
q_step_4	name_magnitude_sum_as_answer	q_step_5	integer_action_pairs.pl:53 observed on {"a": -7, "b": 3} observed on {"a": -6, "b": 3} (derived template)

From	Action	To	Provenance
q_step_5	lose_sign_relation	q_accept	<code>integer_action_pairs.pl:53</code> observed on {"a": -7, "b": 3} observed on {"a": -6, "b": 3} (derived template)

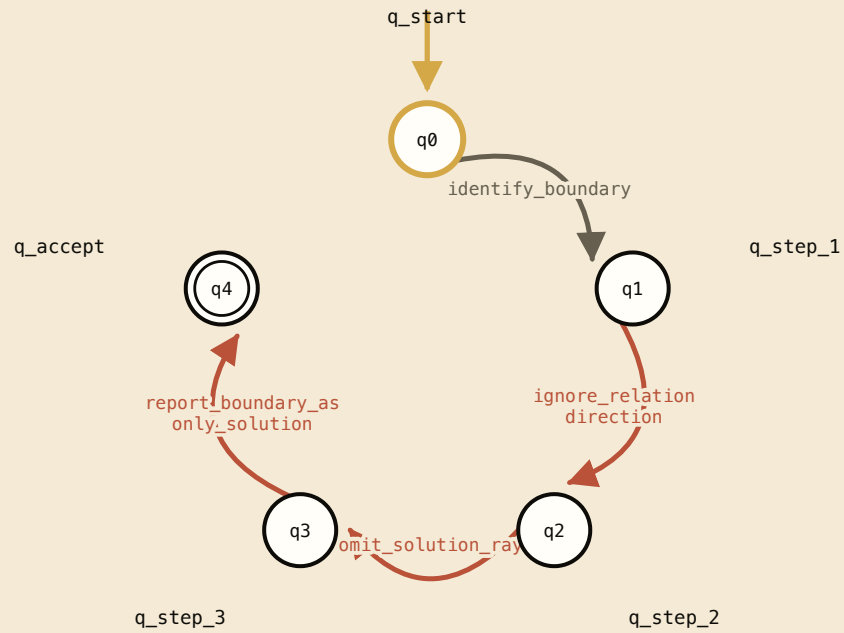
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

2. inequality_as_boundary_point



The run produced `single_value(4)`; the recorded expected result is `solution_set(ray(left,open,4))`.

The run produced `single_value(4)`; the recorded expected result is `solution_set(ray(left,open,4))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ identify_boundary, ignore_relation_direction, omit_solution_ray, report_boundary_as_only_solution (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

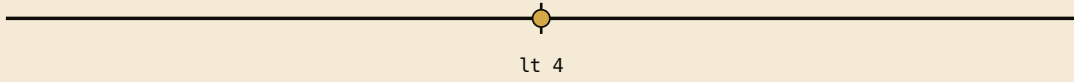
Input contract: {"bound": "integer", "kind": "inequality", "relation": "atom", "variable": "atom"}; example {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_boundary	q_step_1	integer_action_pairs.pl:160 observed on {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}
q_step_1	ignore_relation_direction	q_step_2	integer_action_pairs.pl:160 observed on {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}
q_step_2	omit_solution_ray	q_step_3	integer_action_pairs.pl:160 observed on {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}
q_step_3	report_boundary_as_onto_solution	q_accept	integer_action_pairs.pl:160 observed on {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}

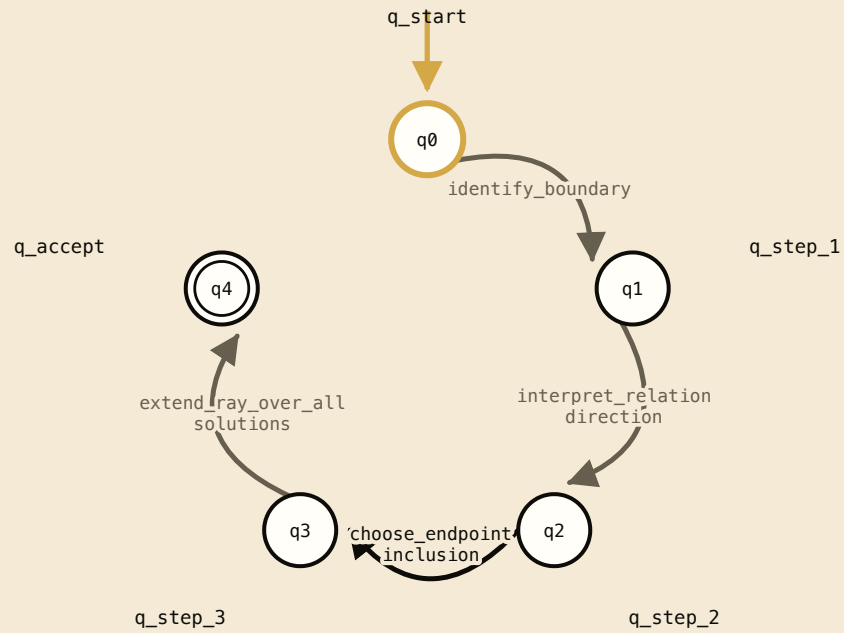
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

3. inequality_solution_set_representation



The run produced `solution_set(ray(left,open,4))`.

The run produced `solution_set(ray(left,open,4))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{choose_endpoint_inclusion, extend_ray_over_all_solutions, identify_boundary, interpret_relation_direction (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

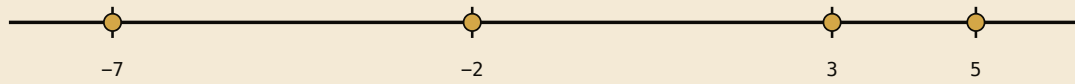
Input contract: {"bound": "integer", "kind": "inequality", "relation": "atom", "variable": "atom"}; example {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_boundary	q_step_1	integer_action_pairs.pl:133 observed on {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}
q_step_1	interpret_relation_direction	q_step_2	integer_action_pairs.pl:133 observed on {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}
q_step_2	choose_endpoint_inclusion	q_step_3	integer_action_pairs.pl:133 observed on {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}
q_step_3	extend_ray_over_all_solutions	q_accept	integer_action_pairs.pl:133 observed on {"bound": 4, "kind": "inequality", "relation": "lt", "variable": "x"}

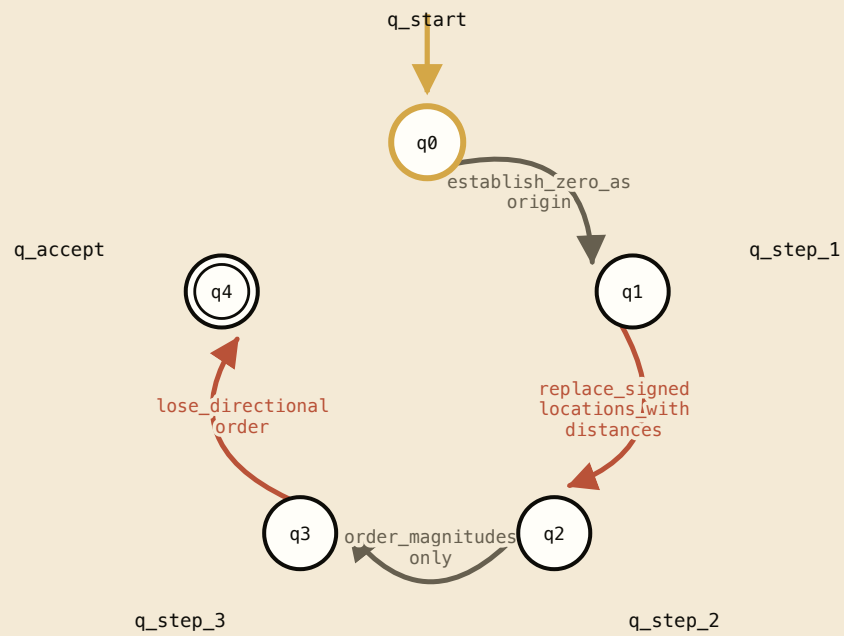
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

4. order_by_magnitude_ignore_sign



The run produced `ordered_values([-2,3,5,-7])`; the recorded expected result is `ordered_values([-7,-2,3,5])`.

The run produced `ordered_values([-2,3,5,-7])`; the recorded expected result is `ordered_values([-7,-2,3,5])`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ establish_zero_as_origin, lose_directional_order, order_magnitudes_only, replace_signed_locations_with_distances (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

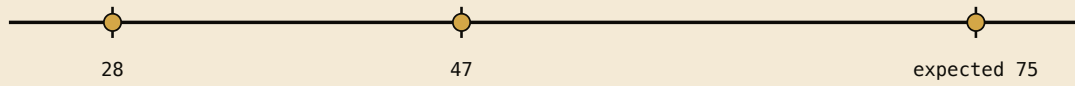
Input contract: {"kind": "signed_number_list", "values": ["integer"]}; example {"kind": "signed_number_list", "values": [-7, 3, -2, 5]};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_zero_as_origin	q_step_1	integer_action_pairs.pl:106 observed on {"kind": "signed_number_list", "values": [-7, 3, -2, 5]}
q_step_1	replace_signed_locations_with_distances	q_step_2	integer_action_pairs.pl:106 observed on {"kind": "signed_number_list", "values": [-7, 3, -2, 5]}
q_step_2	order_magnitudes_only	q_step_3	integer_action_pairs.pl:106 observed on {"kind": "signed_number_list", "values": [-7, 3, -2, 5]}
q_step_3	lose_directional_order	q_accept	integer_action_pairs.pl:106 observed on {"kind": "signed_number_list", "values": [-7, 3, -2, 5]}

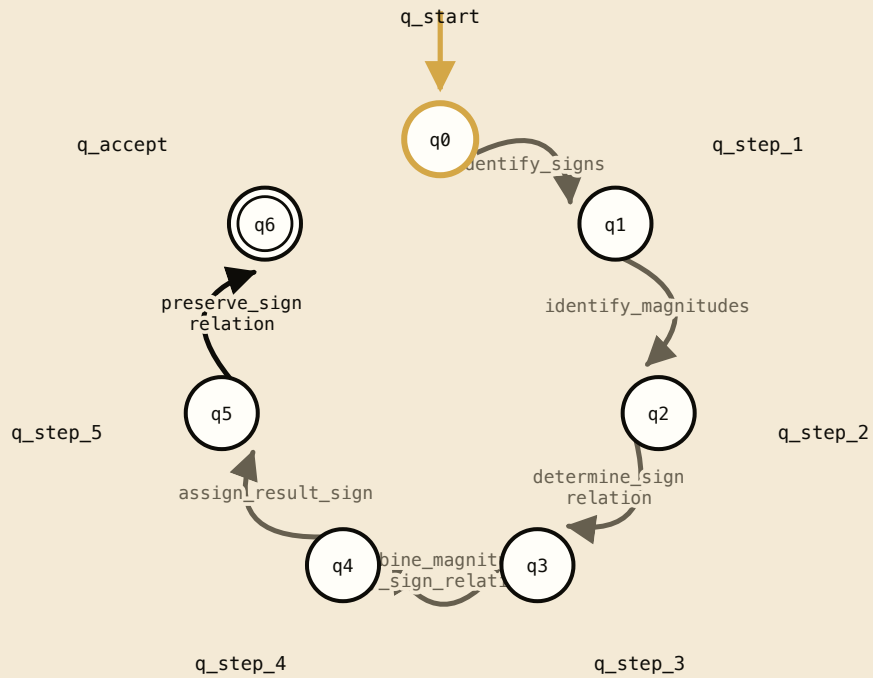
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

5. signed_addition_with_sign_relation



The run produced 75.

The run produced 75. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ assign_result_sign, combine_magnitudes_by_sign_relation, determine_sign_relation, identify_magnitudes, identify_signs, preserve_sign_relation (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

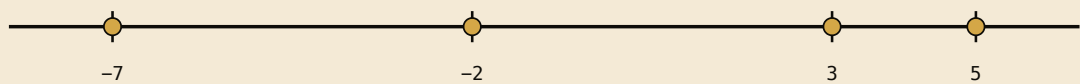
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_signs	q_step_1	integer_action_pairs.pl:27 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	identify_magnitudes	q_step_2	integer_action_pairs.pl:27 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	determine_sign_relation	q_step_3	integer_action_pairs.pl:27 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	combine_magnitudes_by_sign_relation	q_step_4	integer_action_pairs.pl:27 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	assign_result_sign	q_step_5	integer_action_pairs.pl:27 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_5	preserve_sign_relation	q_accept	<code>integer_action_pairs.pl:27</code> observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

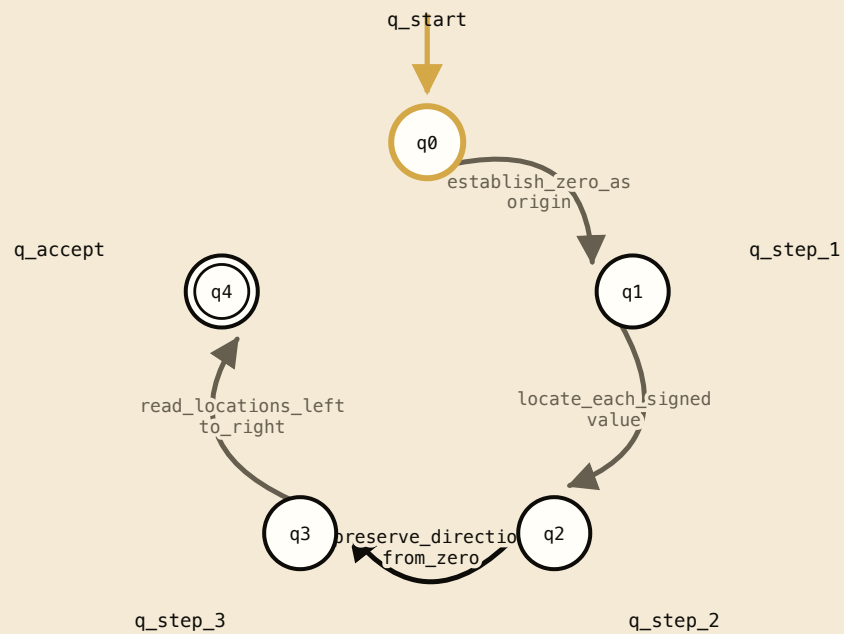
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

6. signed_number_location_and_order



The run produced `ordered_values([-7,-2,3,5])`.

The run produced `ordered_values([-7,-2,3,5])`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ $establish_zero_as_origin, locate_each_signed_value, preserve_direction_from_zero, read_locations_left_to_right$ (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

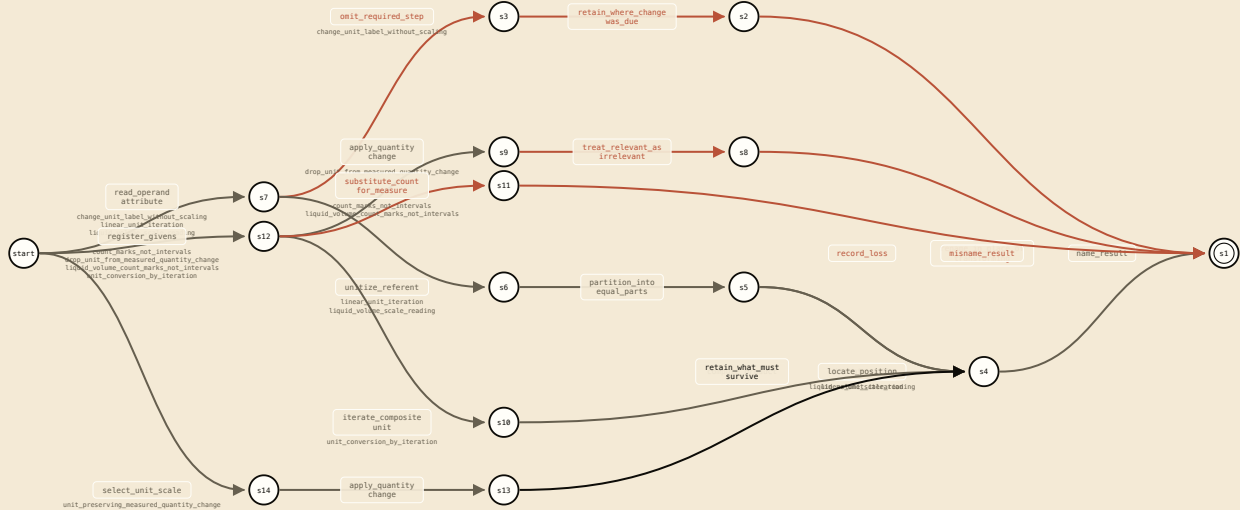
Input contract: {"kind": "signed_number_list", "values": ["integer"]}; example {"kind": "signed_number_list", "values": [-7, 3, -2, 5]};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_zero_as_origin	q_step_1	integer_action_pairs.pl:83 observed on {"kind": "signed_number_list", "values": [-7, 3, -2, 5]}
q_step_1	locate_each_signed_value	q_step_2	integer_action_pairs.pl:83 observed on {"kind": "signed_number_list", "values": [-7, 3, -2, 5]}
q_step_2	preserve_direction_from_zero	q_step_3	integer_action_pairs.pl:83 observed on {"kind": "signed_number_list", "values": [-7, 3, -2, 5]}
q_step_3	read_locations_left_to_right	q_accept	integer_action_pairs.pl:83 observed on {"kind": "signed_number_list", "values": [-7, 3, -2, 5]}

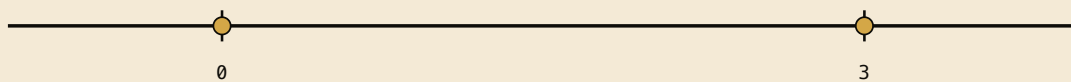
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

measurement



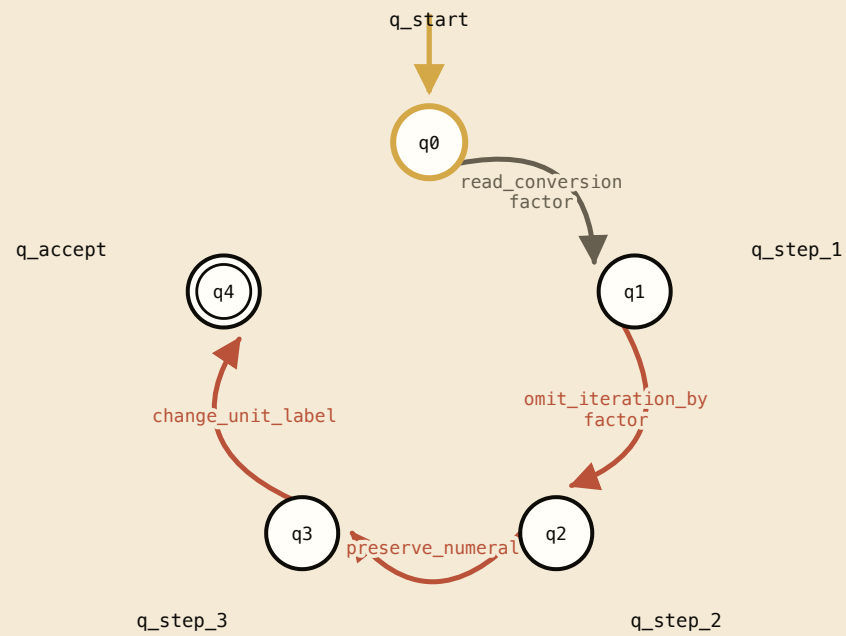
Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. change_unit_label_without_scaling



The run produced `quantity(3,foot)`; the recorded expected result is `quantity(9,foot)`.

The run produced `quantity(3,foot)`; the recorded expected result is `quantity(9,foot)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{change_unit_label, omit_iteration_by_factor, preserve_numeral, read_conversion_factor (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

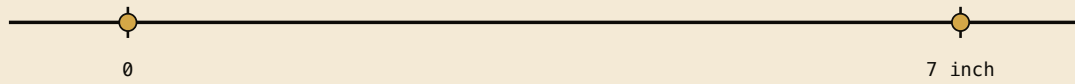
Input contract: {"count": "integer", "factor": "positive_integer", "from_unit": "atom", "kind": "quantity_conversion", "to_unit": "atom"}; example {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	read_conversion_factor	q_step_1	measurement_action_pairs.pl:162 observed on {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}
q_step_1	omit_iteration_by_factor	q_step_2	measurement_action_pairs.pl:162 observed on {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}
q_step_2	preserve_numeral	q_step_3	measurement_action_pairs.pl:162 observed on {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}
q_step_3	change_unit_label	q_accept	measurement_action_pairs.pl:162 observed on {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}

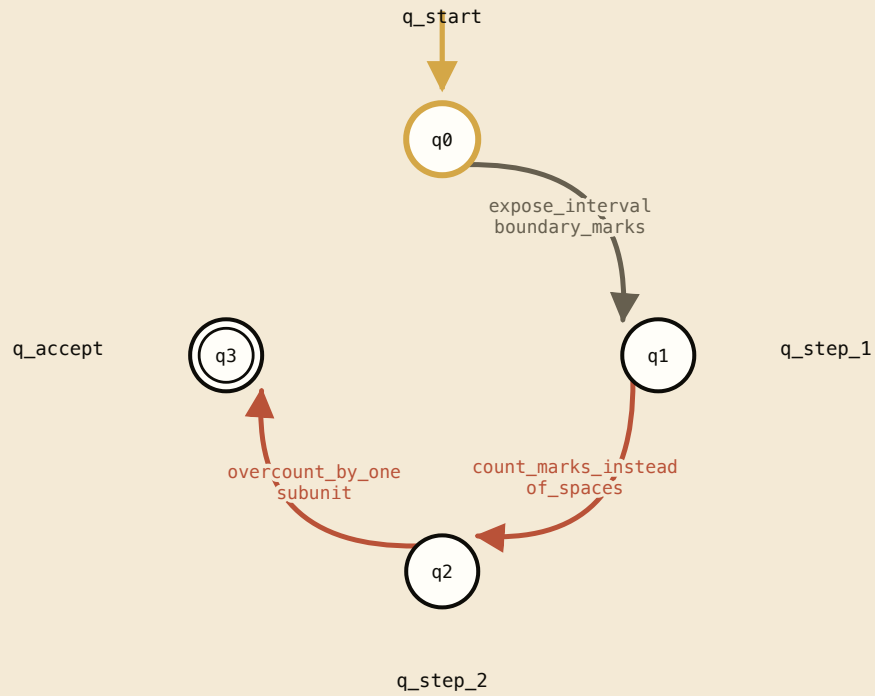
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

2. count_marks_not_intervals



The run produced `length(rational(2,1),inch)`; the recorded expected result is `length(rational(7,4),inch)`.

The run produced `length(rational(2,1),inch)`; the recorded expected result is `length(rational(7,4),inch)`.
The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3\}$

Σ count_marks_instead_of_spaces, expose_interval_boundary_marks,
overcount_by_one_subunit (3 distinct)

q_0 $q_0 = q_start$

F $\{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

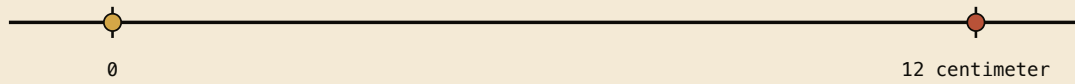
Input contract: {"interval_count": "positive_integer", "kind": "measure_with_unit", "subdivisions": "positive_integer", "unit": "atom"}; example {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	expose_interval_boundary_marks	q_step_1	measurement_action_pairs.pl:51 observed on {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}
q_step_1	count_marks_instead_of_spaces	q_step_2	measurement_action_pairs.pl:51 observed on {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}
q_step_2	overcount_by_one_subunit	q_accept	measurement_action_pairs.pl:51 observed on {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}

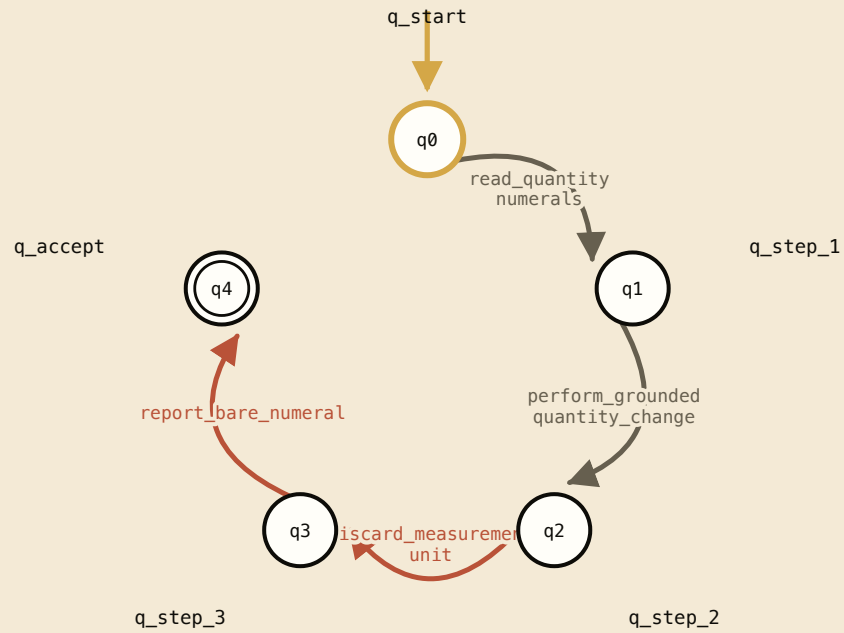
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

3. drop_unit_from_measured_quantity_change



The run produced 12; the recorded expected result is `quantity(12,centimeter)`.

The run produced 12; the recorded expected result is `quantity(12,centimeter)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{discard_measurement_unit, perform_grounded_quantity_change, read_quantity_numerals, report_bare_numeral (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

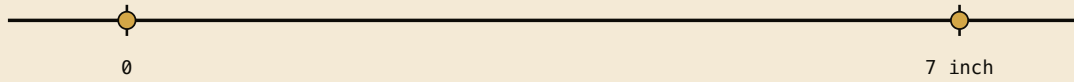
Input contract: {"a": "integer", "b": "integer", "kind": "measured_change", "operation": "atom", "unit": "atom"}; example {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	read_quantity_numerals	q_step_1	measurement_action_pairs.pl:216 observed on {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}
q_step_1	perform_grounded_quantity_change	q_step_2	measurement_action_pairs.pl:216 observed on {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}
q_step_2	discard_measurement_unit	q_step_3	measurement_action_pairs.pl:216 observed on {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}
q_step_3	report_bare_numeral	q_accept	measurement_action_pairs.pl:216 observed on {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}

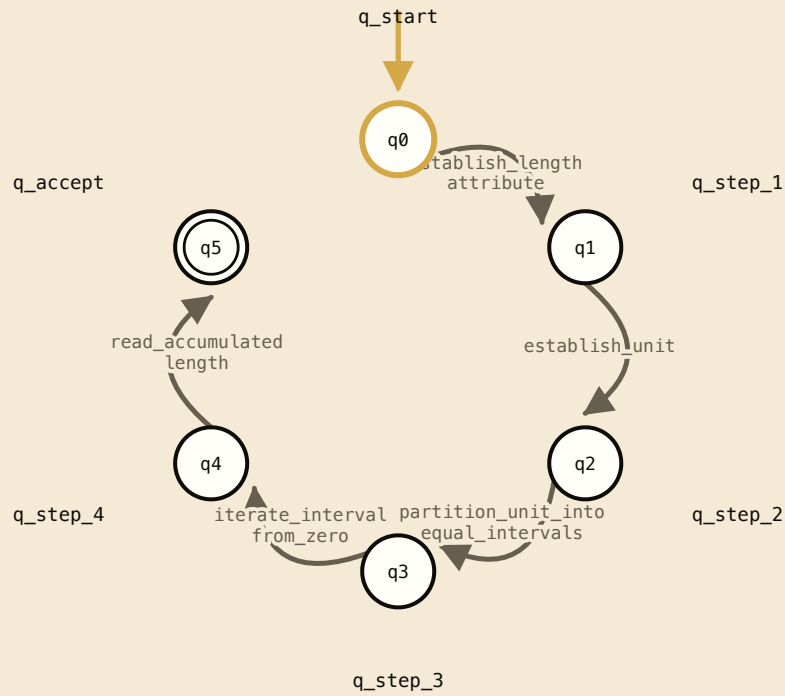
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

4. linear_unit_iteration



The run produced `length(rational(7,4),inch)`.

The run produced `length(rational(7,4),inch)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ establish_length_attribute, establish_unit, iterate_interval_from_zero, partition_unit_into_equal_intervals, read_accumulated_length (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

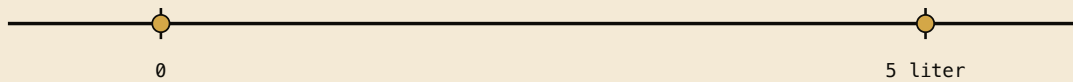
Input contract: {"interval_count": "positive_integer", "kind": "measure_with_unit", "subdivisions": "positive_integer", "unit": "atom"}; example {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_length_attribute	q_step_1	measurement_action_pairs.pl:20 observed on {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}
q_step_1	establish_unit	q_step_2	measurement_action_pairs.pl:20 observed on {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}
q_step_2	partition_unit_into_equal_intervals	q_step_3	measurement_action_pairs.pl:20 observed on {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}
q_step_3	iterate_interval_from_zero	q_step_4	measurement_action_pairs.pl:20 observed on {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}
q_step_4	read_accumulated_length	q_accept	measurement_action_pairs.pl:20 observed on {"interval_count": 7, "kind": "measure_with_unit", "subdivisions": 4, "unit": "inch"}

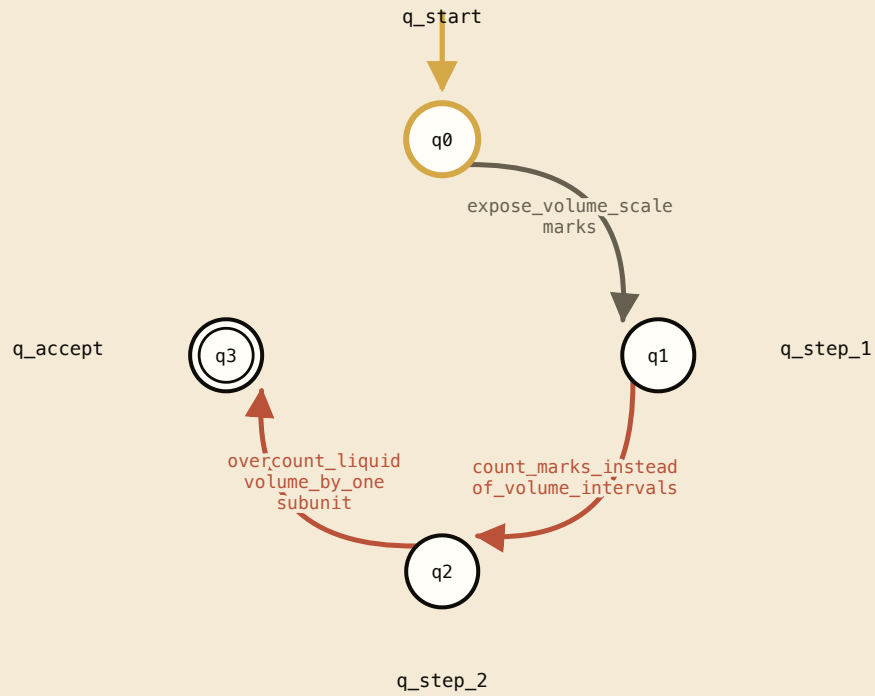
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

5. `liquid_volume_count_marks_not_intervals`



The run produced `volume(rational(3,2),liter)`; the recorded expected result is `volume(rational(5,4),l`.

The run produced `volume(rational(3,2),liter)`; the recorded expected result is `volume(rational(5,4),liter)`. The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad \text{count_marks_instead_of_volume_intervals, expose_volume_scale_marks, overcount_liquid_volume_by_one_subunit (3 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_3\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

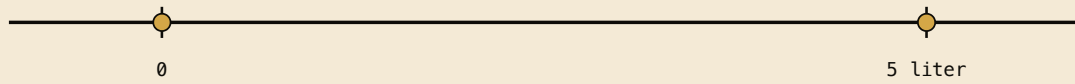
Input contract: {"interval_count": "positive_integer", "kind": "measure_with_unit", "subdivisions": "positive_integer", "unit": "atom"}; example {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	expose_volume_scale_marks	q_step_1	measurement_action_pairs.pl:104 observed on {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}
q_step_1	count_marks_instead_of_volume_intervals	q_step_2	measurement_action_pairs.pl:104 observed on {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}
q_step_2	overcount_liquid_volume_by_one_subunit	q_accept	measurement_action_pairs.pl:104 observed on {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}

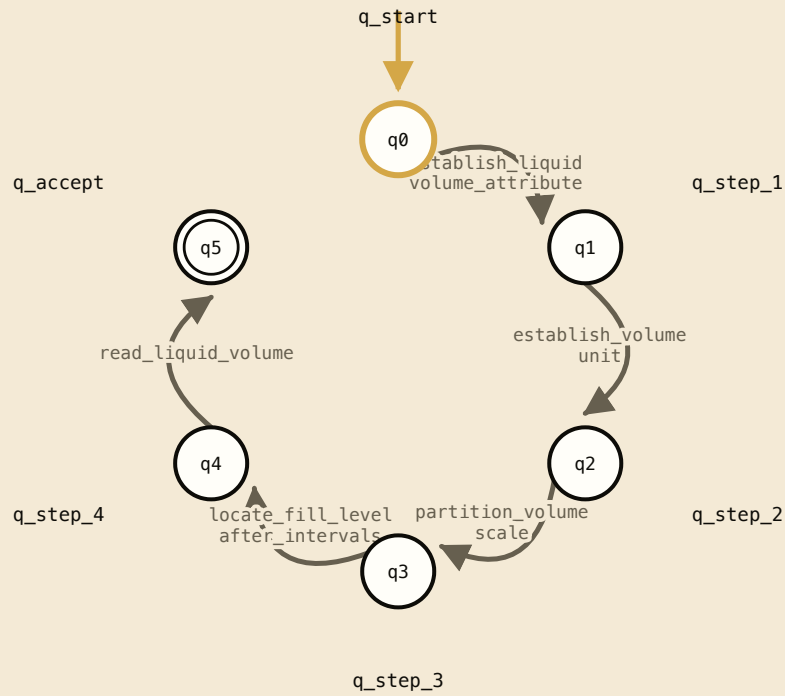
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

6. liquid_volume_scale_reading



The run produced `volume(rational(5,4),liter)`.

The run produced `volume(rational(5,4),liter)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \text{establish_liquid_volume_attribute, establish_volume_unit, locate_fill_level_after_intervals, partition_volume_scale, read_liquid_volume (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

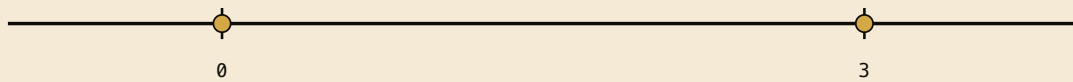
Input contract: {"interval_count": "positive_integer", "kind": "measure_with_unit", "subdivisions": "positive_integer", "unit": "atom"}; example {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_liquid_volume_attribute	q_step_1	measurement_action_pairs.pl:76 observed on {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}
q_step_1	establish_volume_unit	q_step_2	measurement_action_pairs.pl:76 observed on {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}
q_step_2	partition_volume_scale	q_step_3	measurement_action_pairs.pl:76 observed on {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}
q_step_3	locate_fill_level_after_intervals	q_step_4	measurement_action_pairs.pl:76 observed on {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}
q_step_4	read_liquid_volume	q_accept	measurement_action_pairs.pl:76 observed on {"interval_count": 5, "kind": "measure_with_unit", "subdivisions": 4, "unit": "liter"}

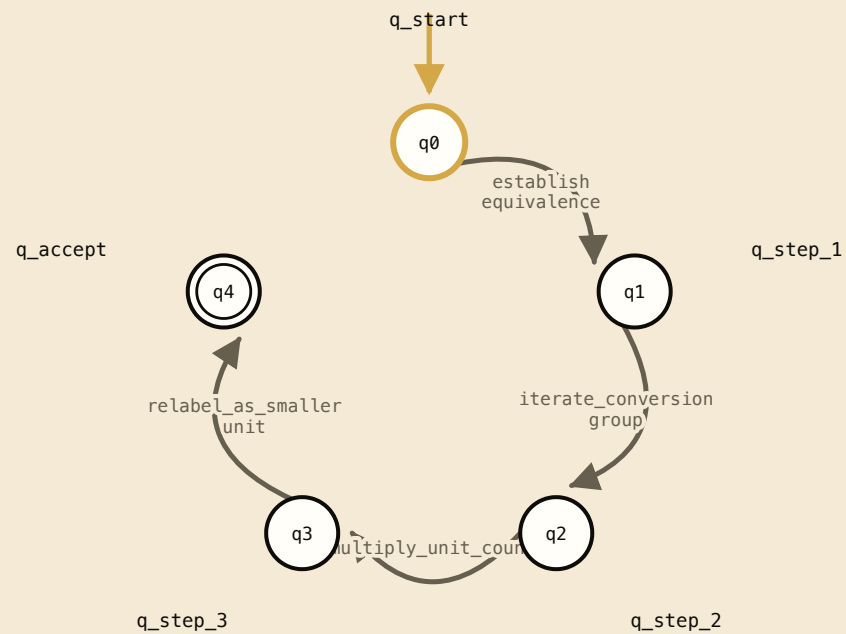
Computed structure: `linear_trace`. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

7. unit_conversion_by_iteration



The run produced `quantity(9,foot)`.

The run produced `quantity(9,foot)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ establish_equivalence, iterate_conversion_group, multiply_unit_count, relabel_as_smaller_unit (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

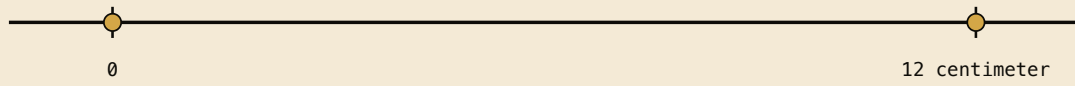
Input contract: {"count": "integer", "factor": "positive_integer", "from_unit": "atom", "kind": "quantity_conversion", "to_unit": "atom"}; example {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_equivalence	q_step_1	measurement_action_pairs.pl:133 observed on {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}
q_step_1	iterate_conversion_group	q_step_2	measurement_action_pairs.pl:133 observed on {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}
q_step_2	multiply_unit_count	q_step_3	measurement_action_pairs.pl:133 observed on {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}
q_step_3	relabel_as_smaller_unit	q_accept	measurement_action_pairs.pl:133 observed on {"count": 3, "factor": 3, "from_unit": "yard", "kind": "quantity_conversion", "to_unit": "foot"}

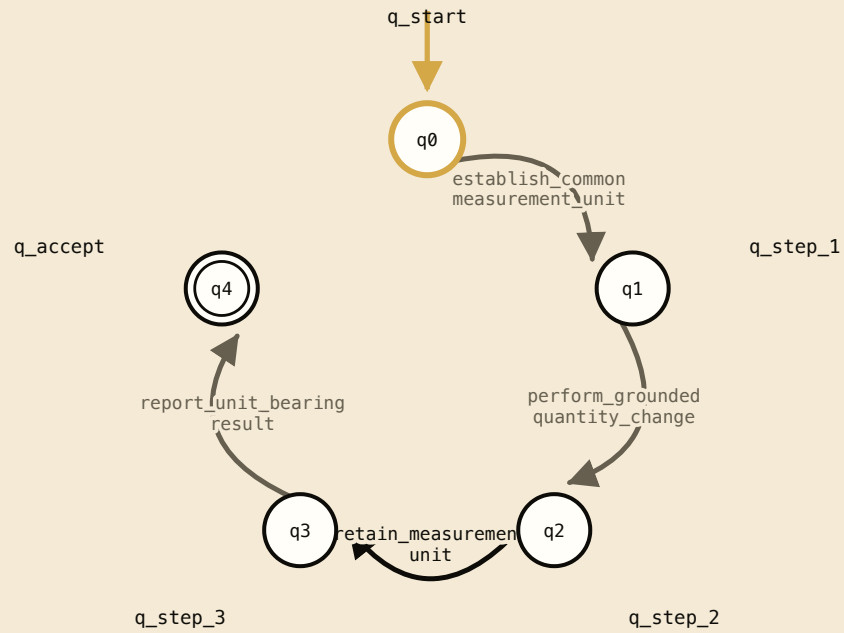
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

8. unit_preserving_measured_quantity_change



The run produced `quantity(12,centimeter)`.

The run produced `quantity(12,centimeter)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ establish_common_measurement_unit, perform_grounded_quantity_change, report_unit_bearing_result, retain_measurement_unit (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

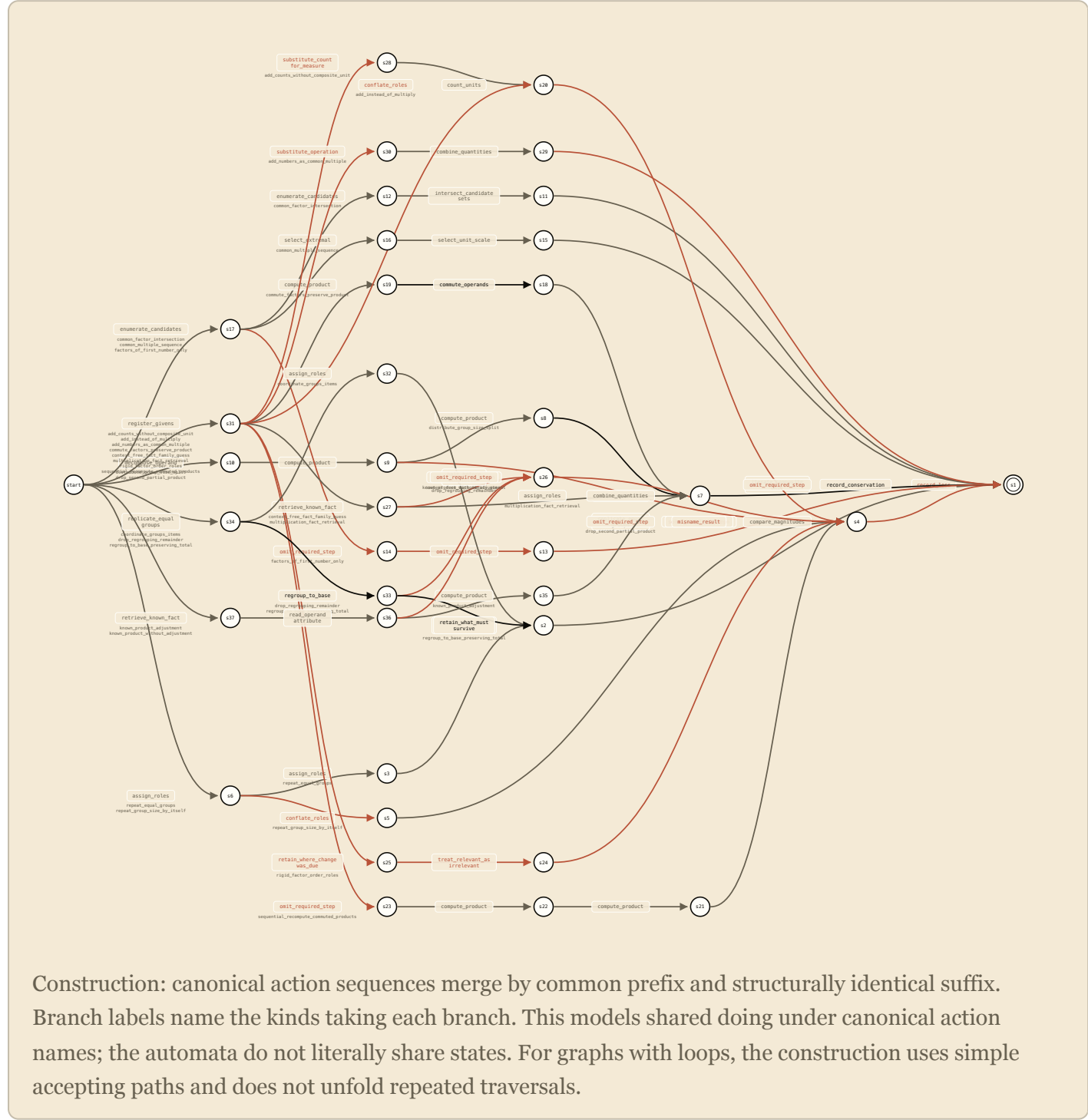
Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer", "kind": "measured_change", "operation": "atom", "unit": "atom"}; example {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_common_measurement_unit	q_step_1	measurement_action_pairs.pl:190 observed on {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}
q_step_1	perform_grounded_quantity_change	q_step_2	measurement_action_pairs.pl:190 observed on {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}
q_step_2	retain_measurement_unit	q_step_3	measurement_action_pairs.pl:190 observed on {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}
q_step_3	report_unit_bearing_result	q_accept	measurement_action_pairs.pl:190 observed on {"a": 7, "b": 5, "kind": "measured_change", "operation": "add", "unit": "centimeter"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

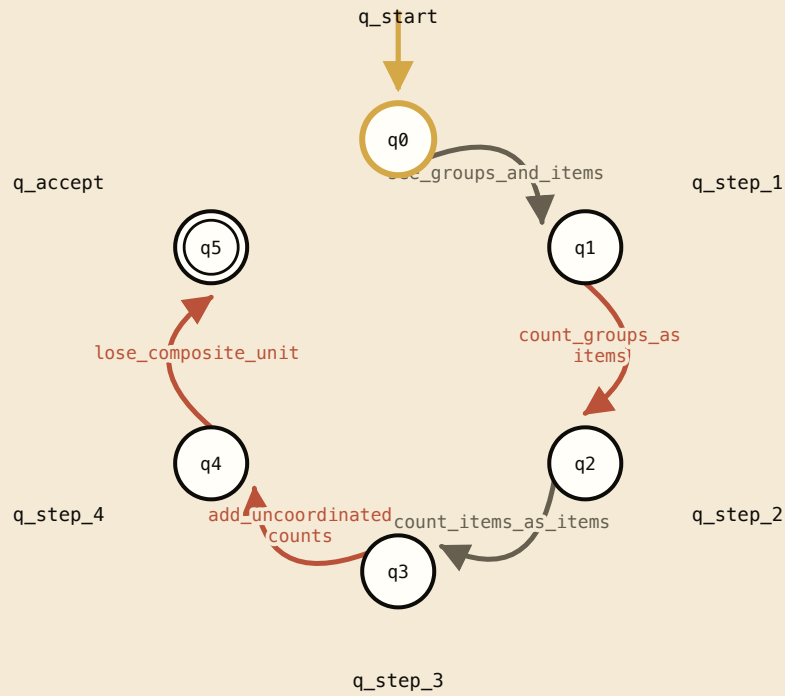


names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. add_counts_without_composite_unit

The run produced 15; the recorded expected result is 56.

The run produced 15; the recorded expected result is 56. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \text{add_uncoordinated_counts, count_groups_as_items, count_items_as_items, lose_composite_unit, see_groups_and_items (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

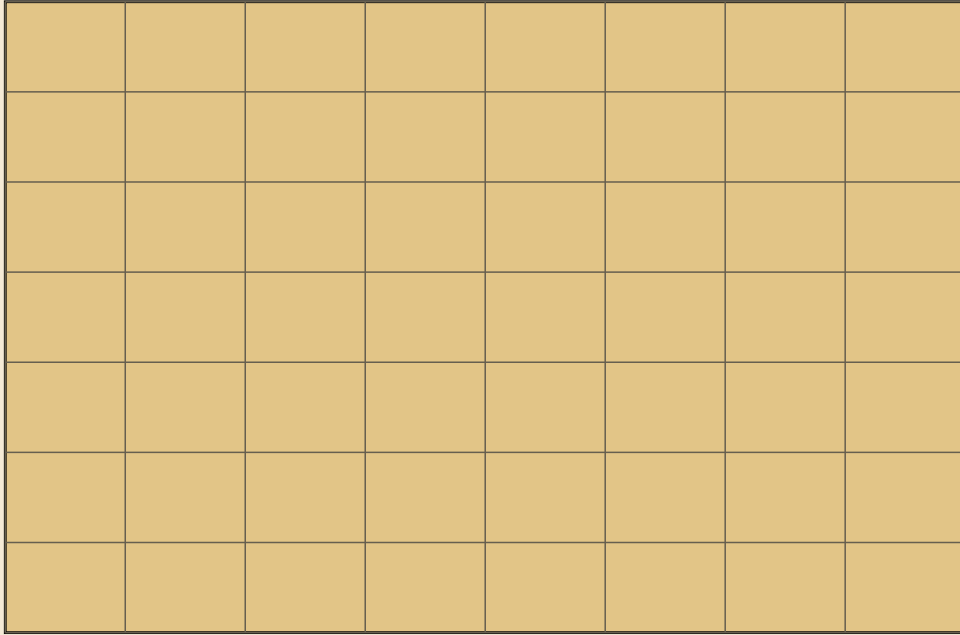
Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	see_groups_and_items	q_step_1	smr_mult_action_pairs.pl:83 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	count_groups_as_items	q_step_2	smr_mult_action_pairs.pl:83 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	count_items_as_items	q_step_3	smr_mult_action_pairs.pl:83 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	add_uncoordinated_counts	q_step_4	smr_mult_action_pairs.pl:83 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_4	lose_composite_unit	q_accept	smr_mult_action_pairs.pl:83 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

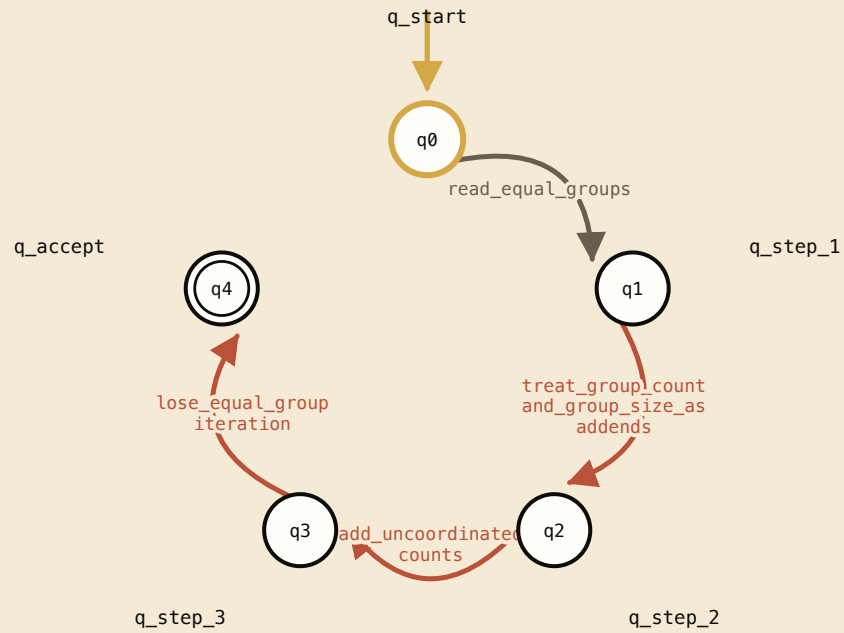
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

2. add_instead_of_multiply



The run produced 15; the recorded expected result is 56.

The run produced 15; the recorded expected result is 56. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{add_uncoordinated_counts, lose_equal_group_iteration, read_equal_groups, treat_group_count_and_group_size_as_addends (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

δ : recorded transitions

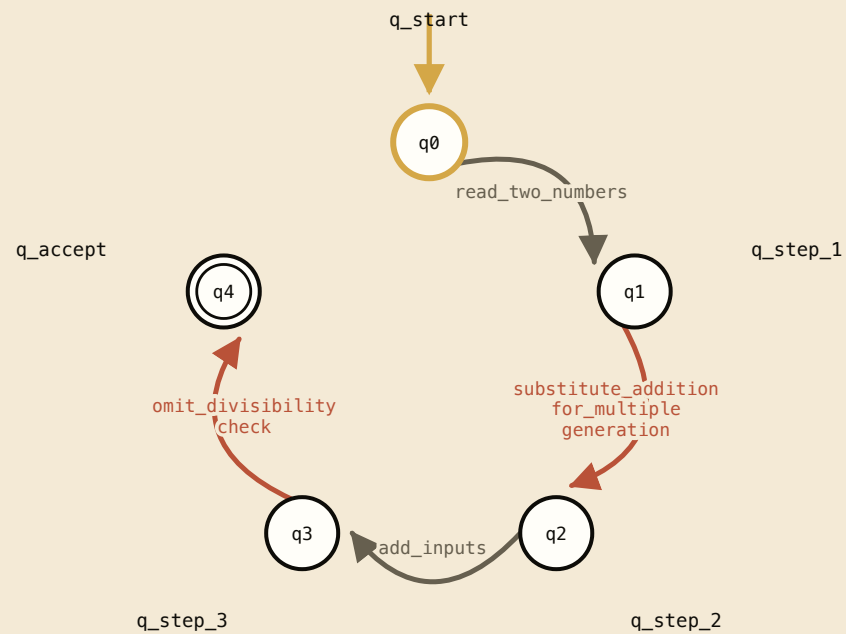
From	Action	To	Provenance
q_start	read_equal_groups	q_step_1	smr_mult_action_pairs.pl:108 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	treat_group_count_and_group_size_as_addends	q_step_2	smr_mult_action_pairs.pl:108 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	add_uncoordinated_counts	q_step_3	smr_mult_action_pairs.pl:108 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	lose_equal_group_iteration	q_accept	smr_mult_action_pairs.pl:108 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

3. add_numbers_as_common_multiple

The run produced `candidate_common_multiple(75)`; the recorded expected result is `common_multi`

The run produced `candidate_common_multiple(75)`; the recorded expected result is `common_multiples(generator(step(1316)),witnesses([1316,2632,3948,5264,6580]),least(1316))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ add_inputs, omit_divisibility_check, read_two_numbers, substitute_addition_for_multiple_generation (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

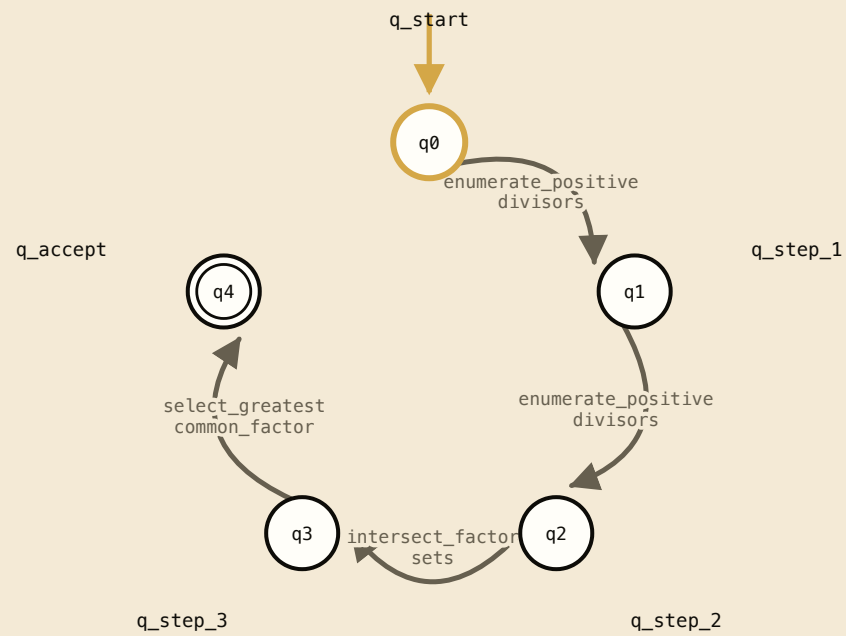
From	Action	To	Provenance
q_start	read_two_numbers	q_step_1	smr_mult_action_pairs.pl:270 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	substitute_addition_for_multiple_generation	q_step_2	smr_mult_action_pairs.pl:270 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	add_inputs	q_step_3	smr_mult_action_pairs.pl:270 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	omit_divisibility_check	q_accept	smr_mult_action_pairs.pl:270 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

4. common_factor_intersection

The run produced `common_factors([1],greatest(1)).`

The run produced `common_factors([1],greatest(1))`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{enumerate_positive_divisors, intersect_factor_sets, select_greatest_common_factor} \\ (3 \text{ distinct})$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

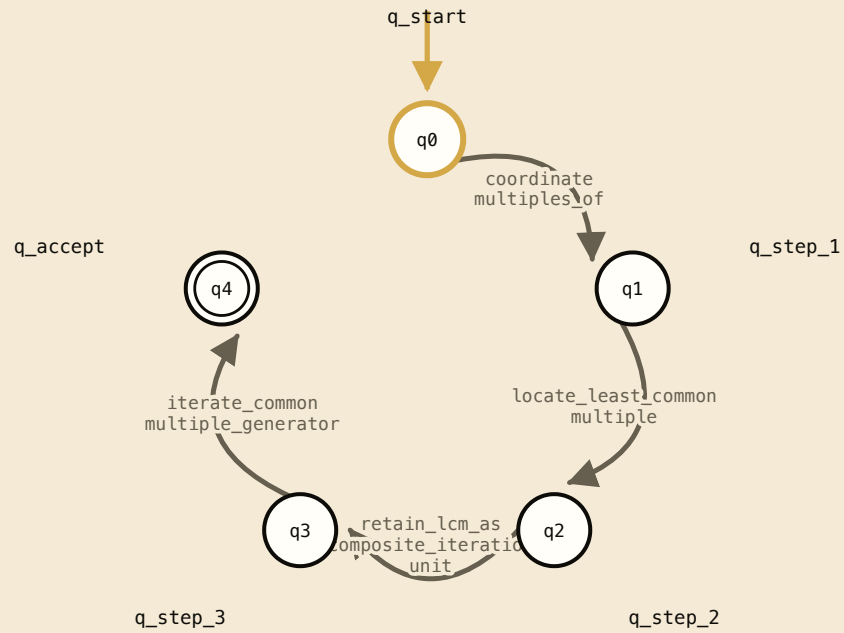
From	Action	To	Provenance
q_start	enumerate_positive_divisors	q_step_1	smr_mult_action_pairs.pl:188 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	enumerate_positive_divisors	q_step_2	smr_mult_action_pairs.pl:188 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	intersect_factor_sets	q_step_3	smr_mult_action_pairs.pl:188 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	select_greatest_common_factor	q_accept	smr_mult_action_pairs.pl:188 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 3 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

5. common_multiple_sequence

The run produced `common_multiples(generator(step(1316)),witnesses([1316,2632,3948,5264,6580])`

The run produced
`common_multiples(generator(step(1316)),witnesses([1316,2632,3948,5264,6580]),least(1316))`. The
 scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ coordinate_multiples_of, iterate_common_multiple_generator, locate_least_common_multiple, retain_lcm_as_composite_iteration_unit (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

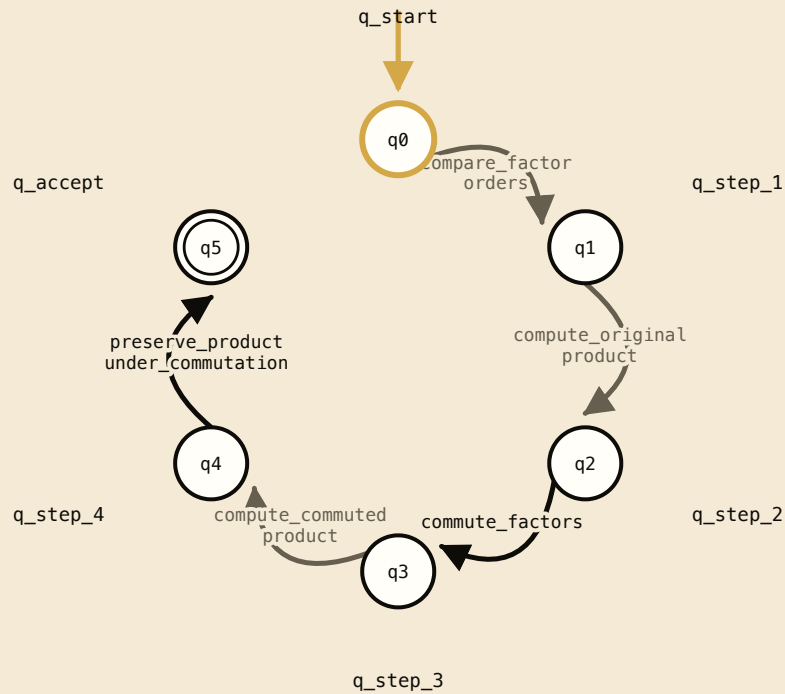
From	Action	To	Provenance
q_start	coordinate_multiples_of	q_step_1	smr_mult_action_pairs.pl:246 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	locate_least_common_multiple	q_step_2	smr_mult_action_pairs.pl:246 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	retain_lcm_as_composite_iteration_unit	q_step_3	smr_mult_action_pairs.pl:246 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	iterate_common_multiple_generator	q_accept	smr_mult_action_pairs.pl:246 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

6. commute_factors_preserve_product

The run produced 1316.

The run produced 1316. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma \quad \text{commute_factors, compare_factor_orders, compute_commuted_product, compute_original_product, preserve_product_under_commutation (5 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

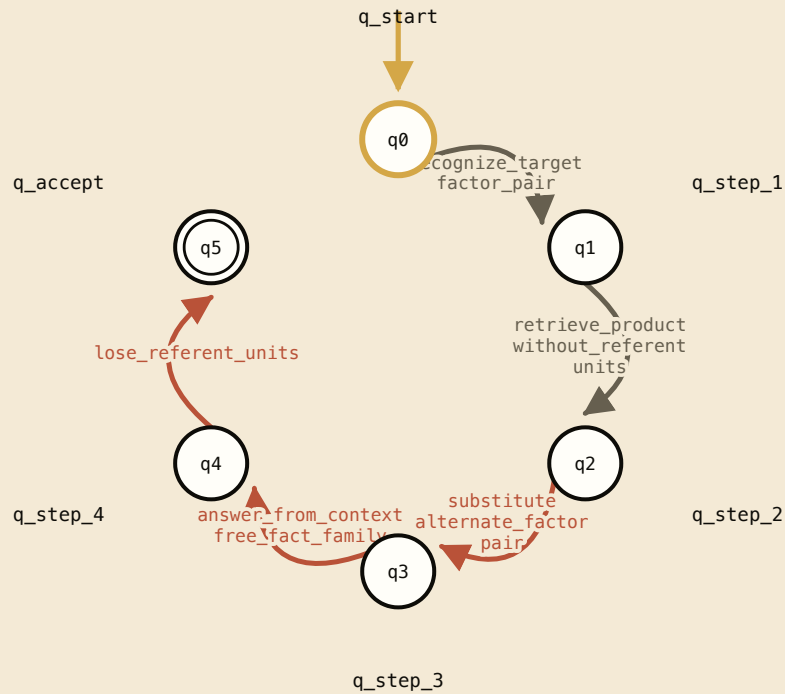
From	Action	To	Provenance
q_start	compare_factor_orders	q_step_1	smr_mult_action_pairs.pl:451 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	compute_original_product	q_step_2	smr_mult_action_pairs.pl:451 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	commute_factors	q_step_3	smr_mult_action_pairs.pl:451 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	compute_commutated_product	q_step_4	smr_mult_action_pairs.pl:451 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	preserve_product_under_commutation	q_accept	smr_mult_action_pairs.pl:451 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

7. context_free_fact_family_guess

The run produced `alternate_factor_pair(2,28,56)`; the recorded expected result is `factor_pair`

The run produced `alternate_factor_pair(2,28,56)`; the recorded expected result is `factor_pair(7,8,56)`.
The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ answer_from_context_free_fact_family, lose_referent_units, recognize_target_factor_pair, retrieve_product_without_referent_units, substitute_alternate_factor_pair (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

δ : recorded transitions

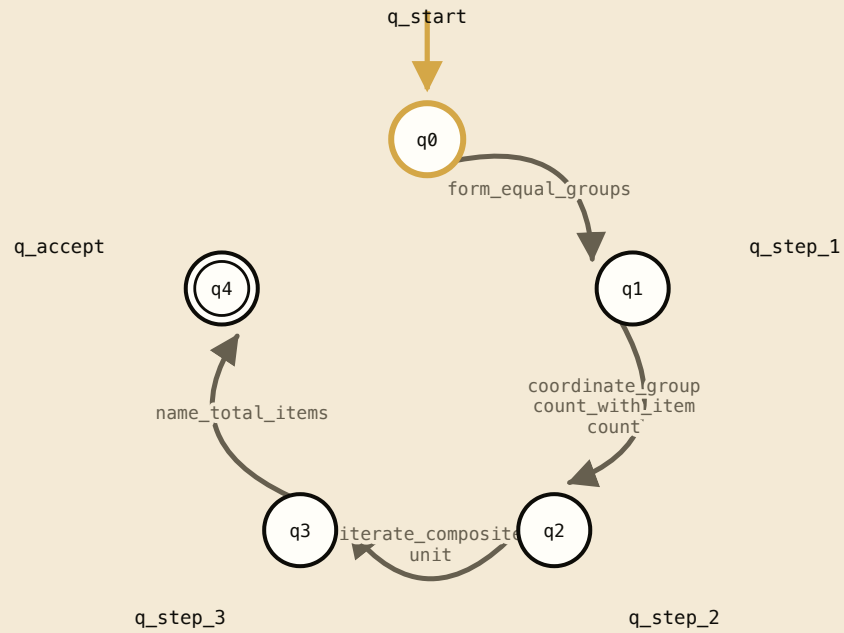
From	Action	To	Provenance
q_start	recognize_target_factor_pair	q_step_1	smr_mult_action_pairs.pl:545 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	retrieve_product_without_referent_units	q_step_2	smr_mult_action_pairs.pl:545 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	substitute_alternate_factor_pair	q_step_3	smr_mult_action_pairs.pl:545 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	answer_from_context_free_fact_fam_ily	q_step_4	smr_mult_action_pairs.pl:545 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_4	lose_referent_units	q_accept	smr_mult_action_pairs.pl:545 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

8. coordinate_groups_items

[illegible]

The run produced 1316. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ coordinate_group_count_with_item_count, form_equal_groups, iterate_composite_unit, name_total_items (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

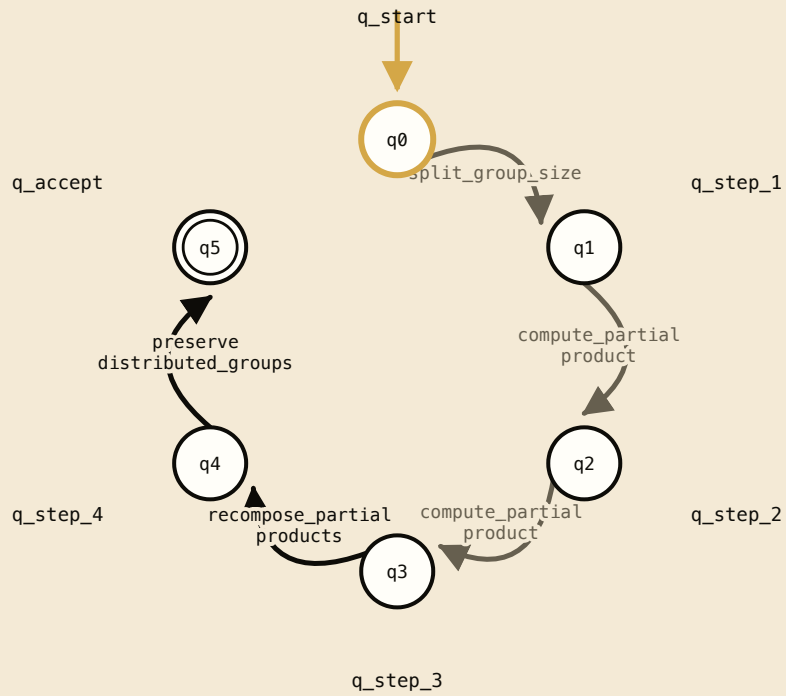
From	Action	To	Provenance
q_start	form_equal_groups	q_step_1	smr_mult_action_pairs.pl:61 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	coordinate_group_count_with_item_count	q_step_2	smr_mult_action_pairs.pl:61 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	iterate_composite_unit	q_step_3	smr_mult_action_pairs.pl:61 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	name_total_items	q_accept	smr_mult_action_pairs.pl:61 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

9. distribute_group_size_split

The run produced 1316.

The run produced 1316. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ compute_partial_product, preserve_distributed_groups, recompose_partial_products, split_group_size (4 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

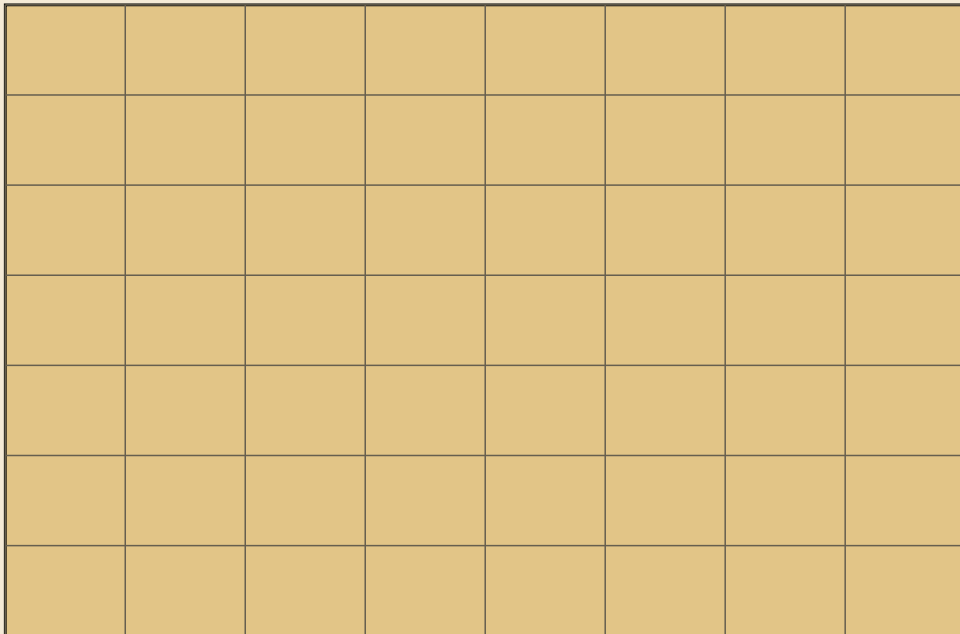
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	split_group_size	q_step_1	smr_mult_action_pairs.pl:299 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	compute_partial_product	q_step_2	smr_mult_action_pairs.pl:299 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	compute_partial_product	q_step_3	smr_mult_action_pairs.pl:299 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	recompose_partial_products	q_step_4	smr_mult_action_pairs.pl:299 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	preserve_distributed_groups	q_accept	smr_mult_action_pairs.pl:299 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

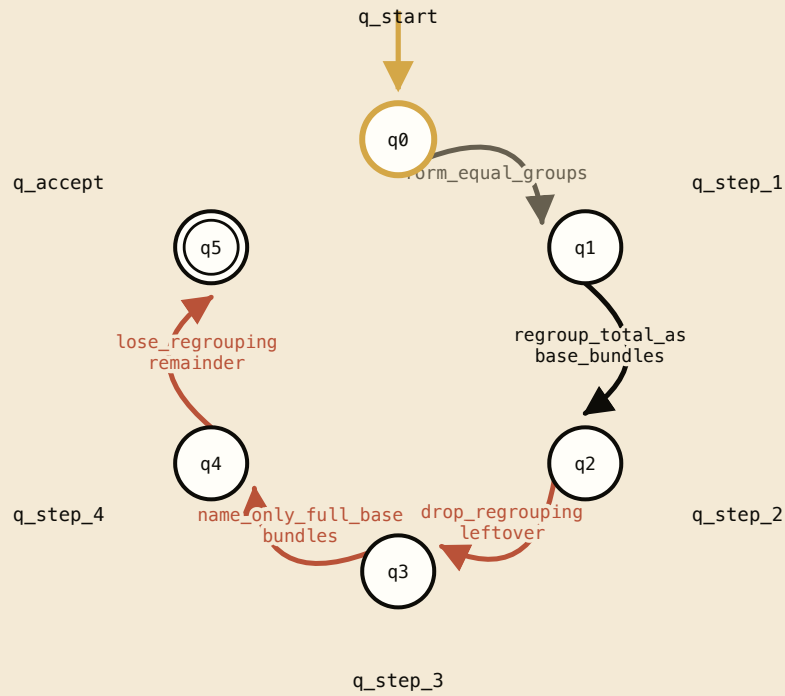
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 4 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

10. drop_regrouping_remainder



The run produced 50; the recorded expected result is 56.

The run produced 50; the recorded expected result is 56. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ drop_regrouping_leftover, form_equal_groups, lose_regrouping_remainder, name_only_full_base_bundles, regroup_total_as_base_bundles (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

δ : recorded transitions

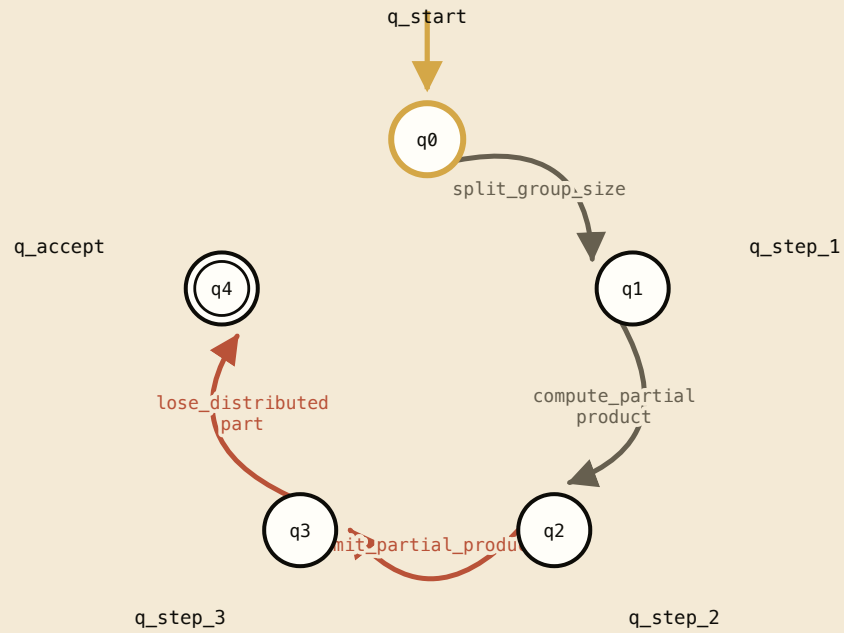
From	Action	To	Provenance
q_start	form_equal_groups	q_step_1	smr_mult_action_pairs.pl:375 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	regroup_total_as_base_bundles	q_step_2	smr_mult_action_pairs.pl:375 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	drop_regrouping_leftover	q_step_3	smr_mult_action_pairs.pl:375 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	name_only_full_base_bundles	q_step_4	smr_mult_action_pairs.pl:375 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_4	lose_regrouping_remainder	q_accept	smr_mult_action_pairs.pl:375 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

11. drop_second_partial_product

The run produced 35; the recorded expected result is 56.

The run produced 35; the recorded expected result is 56. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{compute_partial_product}, \text{lose_distributed_part}, \text{omit_partial_product}, \text{split_group_size}\} \text{ (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

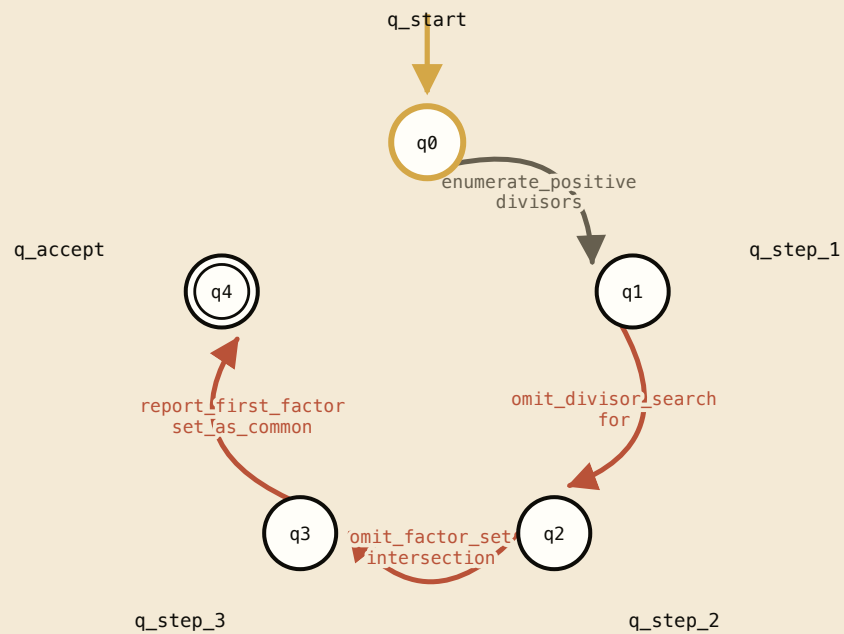
Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	split_group_size	q_step_1	smr_mult_action_pairs.pl:324 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	compute_partial_product	q_step_2	smr_mult_action_pairs.pl:324 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	omit_partial_product	q_step_3	smr_mult_action_pairs.pl:324 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	lose_distributed_part	q_accept	smr_mult_action_pairs.pl:324 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

12. factors_of_first_number_only



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ enumerate_positive_divisors, omit_divisor_search_for,
omit_factor_set_intersection, report_first_factor_set_as_common (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

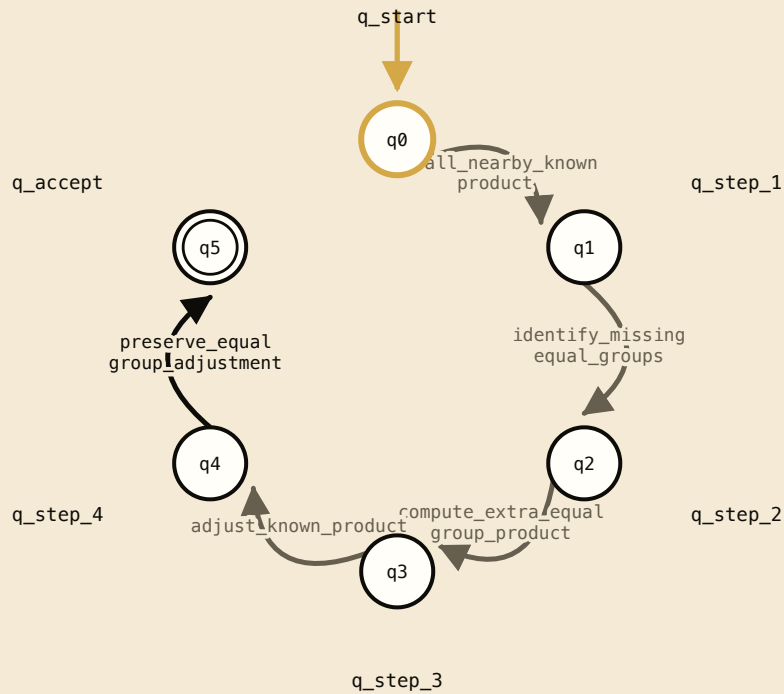
From	Action	To	Provenance
q_start	enumerate_positive_divisors	q_step_1	smr_mult_action_pairs.pl:217 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	omit_divisor_search_for	q_step_2	smr_mult_action_pairs.pl:217 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	omit_factor_set_intersection	q_step_3	smr_mult_action_pairs.pl:217 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	report_first_factor_set_as_common	q_accept	smr_mult_action_pairs.pl:217 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

13. known_product_adjustment

The run produced 56.

The run produced 56. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \text{adjust_known_product, compute_extra_equal_group_product, identify_missing_equal_groups, preserve_equal_group_adjustment, recall_nearby_known_product (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

δ : recorded transitions

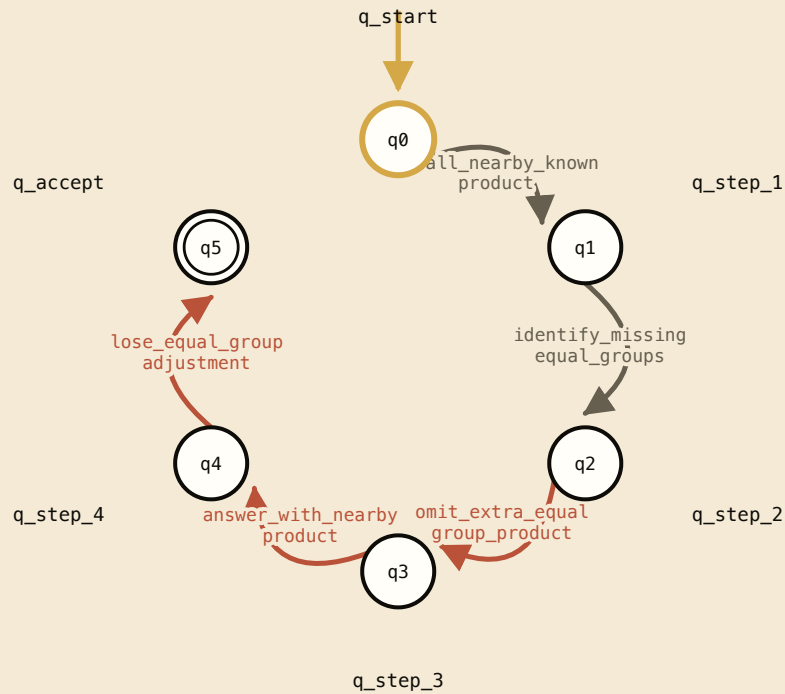
From	Action	To	Provenance
q_start	recall_nearby_known_product	q_step_1	smr_mult_action_pairs.pl:403 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	identify_missing_equal_groups	q_step_2	smr_mult_action_pairs.pl:403 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	compute_extra_equal_group_product	q_step_3	smr_mult_action_pairs.pl:403 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	adjust_known_product	q_step_4	smr_mult_action_pairs.pl:403 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_4	preserve_equal_group_adjustment	q_accept	smr_mult_action_pairs.pl:403 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

14. known_product_without_adjustment

The run produced 48; the recorded expected result is 56.

The run produced 48; the recorded expected result is 56. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ answer_with_nearby_product, identify_missing_equal_groups,
lose_equal_group_adjustment, omit_extra_equal_group_product,
recall_nearby_known_product (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

δ : recorded transitions

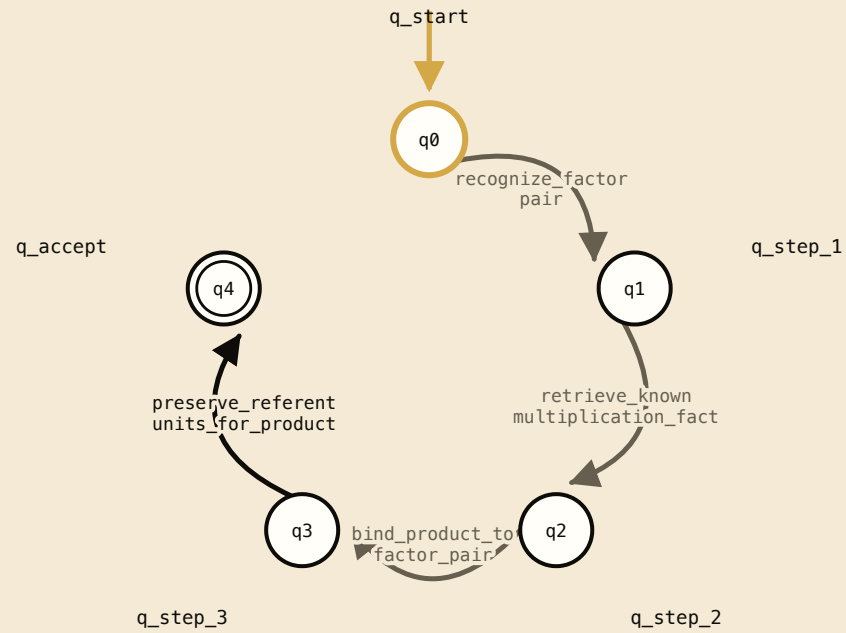
From	Action	To	Provenance
q_start	recall_nearby_known_product	q_step_1	smr_mult_action_pairs.pl:426 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	identify_missing_equal_group_s	q_step_2	smr_mult_action_pairs.pl:426 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	omit_extra_equal_group_product	q_step_3	smr_mult_action_pairs.pl:426 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	answer_with_nearby_product	q_step_4	smr_mult_action_pairs.pl:426 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_4	lose_equal_group_adjustment	q_accept	smr_mult_action_pairs.pl:426 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

15. multiplication_fact_retrieval

[illegible]

The run produced 1316. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{bind_product_to_factor_pair, preserve_referent_units_for_product, recognize_factor_pair, retrieve_known_multiplication_fact (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

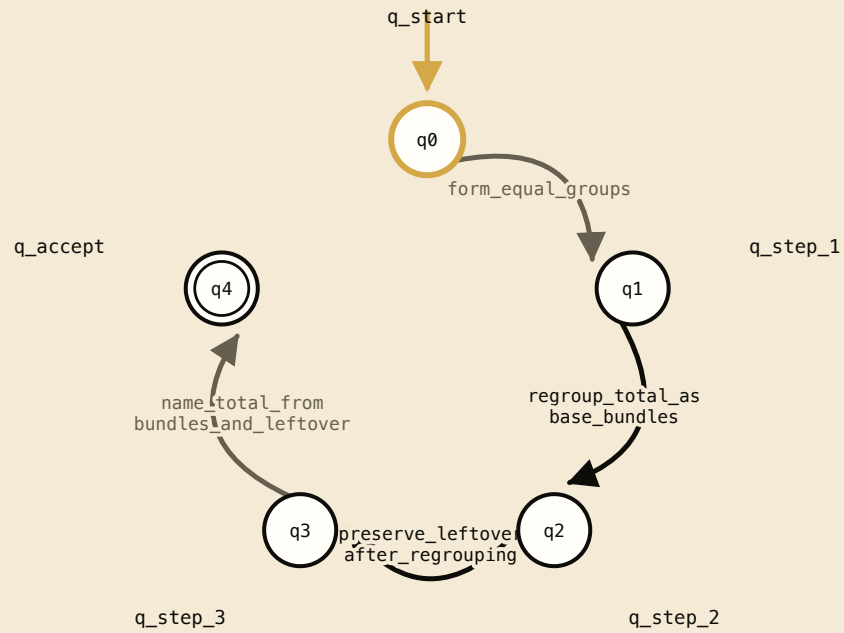
From	Action	To	Provenance
q_start	recognize_factor_pair	q_step_1	smr_mult_action_pairs.pl:524 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	retrieve_known_multiplication_fact	q_step_2	smr_mult_action_pairs.pl:524 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	bind_product_to_factor_pair	q_step_3	smr_mult_action_pairs.pl:524 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	preserve_referent_units_for_product	q_accept	smr_mult_action_pairs.pl:524 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

16. regroup_to_base_preserving_total

The run produced 36.

The run produced 36. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{form_equal_groups, name_total_from_bundles_and_leftover, preserve_leftover_after_regrouping, regroup_total_as_base_bundles (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 12, "b": 3};
verified(strategy_trace_ok).

δ : recorded transitions

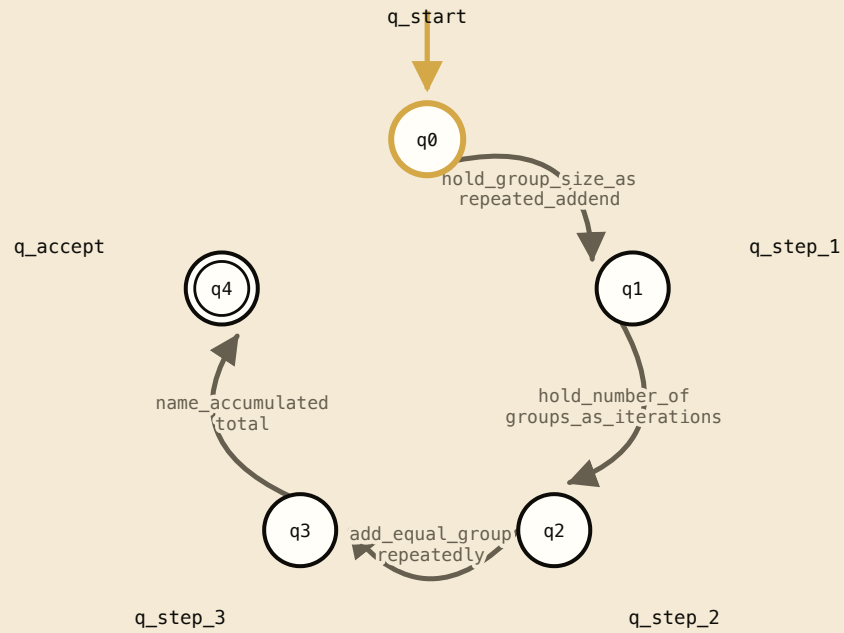
From	Action	To	Provenance
q_start	form_equal_groups	q_step_1	smr_mult_action_pairs.pl:349 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)
q_step_1	regroup_total_as_base_bundles	q_step_2	smr_mult_action_pairs.pl:349 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)
q_step_2	preserve_leftover_after_regrouping	q_step_3	smr_mult_action_pairs.pl:349 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)
q_step_3	name_total_from_bundles_and_leftover	q_accept	smr_mult_action_pairs.pl:349 observed on {"a": 12, "b": 3} observed on {"a": 13, "b": 3} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

17. repeat_equal_groups

The run produced 1316.

The run produced 1316. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad add_equal_group_repeatedly, hold_group_size_as_repeated_addend, \\ hold_number_of_groups_as_iterations, name_accumulated_total \text{ (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

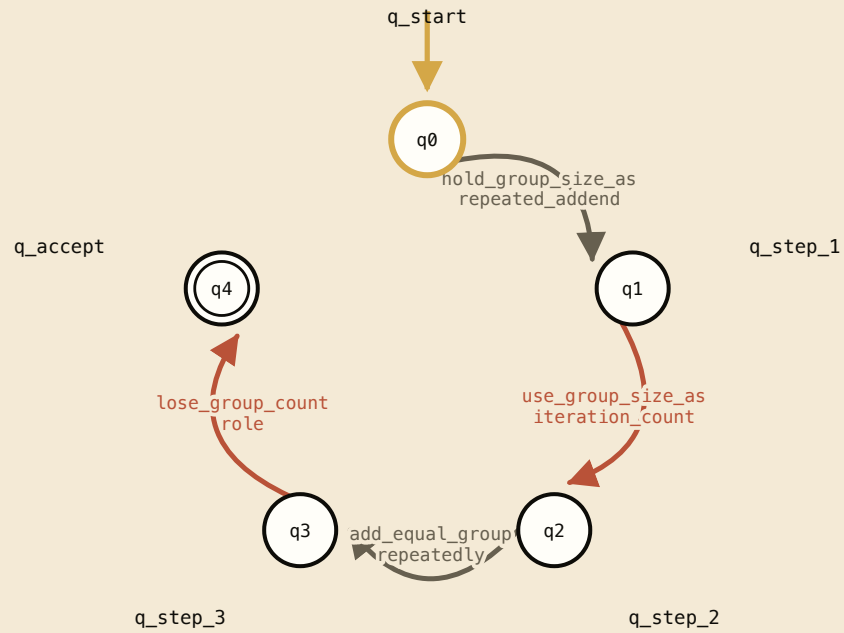
From	Action	To	Provenance
q_start	hold_group_size_as_repeated_ad dend	q_step_ 1	smr_mult_action_pairs.pl:134 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 1	hold_number_of_groups_as_itera tions	q_step_ 2	smr_mult_action_pairs.pl:134 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 2	add_equal_group_repeatedly	q_step_ 3	smr_mult_action_pairs.pl:134 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 3	name_accumulated_total	q_accep t	smr_mult_action_pairs.pl:134 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one
accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and
8 observed rows.

18. repeat_group_size_by_itself

The run produced 784; the recorded expected result is 1316.

The run produced 784; the recorded expected result is 1316. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad add_equal_group_repeatedly, hold_group_size_as_repeated_addend, \\ lose_group_count_role, use_group_size_as_iteration_count \text{ (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

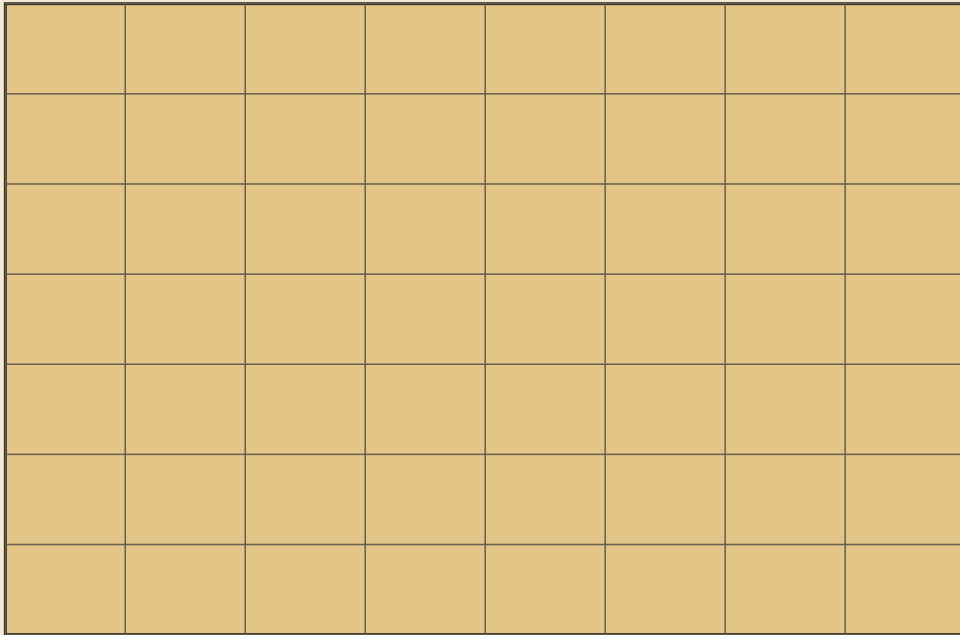
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	hold_group_size_as_repeated_ad dend	q_step_ 1	smr_mult_action_pairs.pl:158 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 1	use_group_size_as_iteration_co unt	q_step_ 2	smr_mult_action_pairs.pl:158 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 2	add_equal_group_repeatedly	q_step_ 3	smr_mult_action_pairs.pl:158 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_ 3	lose_group_count_role	q_accep t	smr_mult_action_pairs.pl:158 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

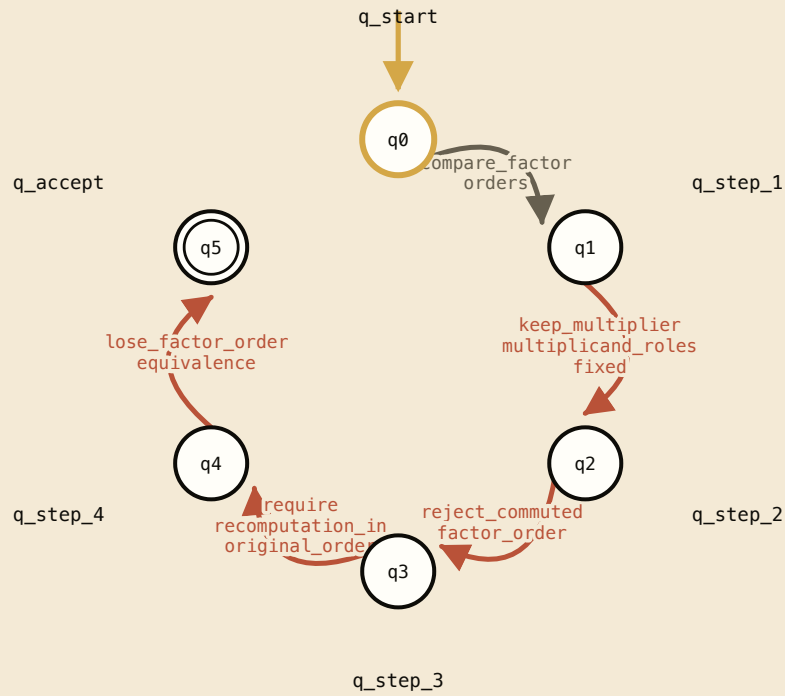
Computed structure: linear_trace. no recorded branch or loop; the table records one
accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and
8 observed rows.

19. rigid_factor_order_roles



The run produced `rejected_commutation(8,7)`; the recorded expected result is 56.

The run produced `rejected_commutation(8,7)`; the recorded expected result is 56. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ compare_factor_orders, keep_multiplier_multiplicand_roles_fixed, lose_factor_order_equivalence, reject_commuted_factor_order, require_recomputation_in_original_order (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

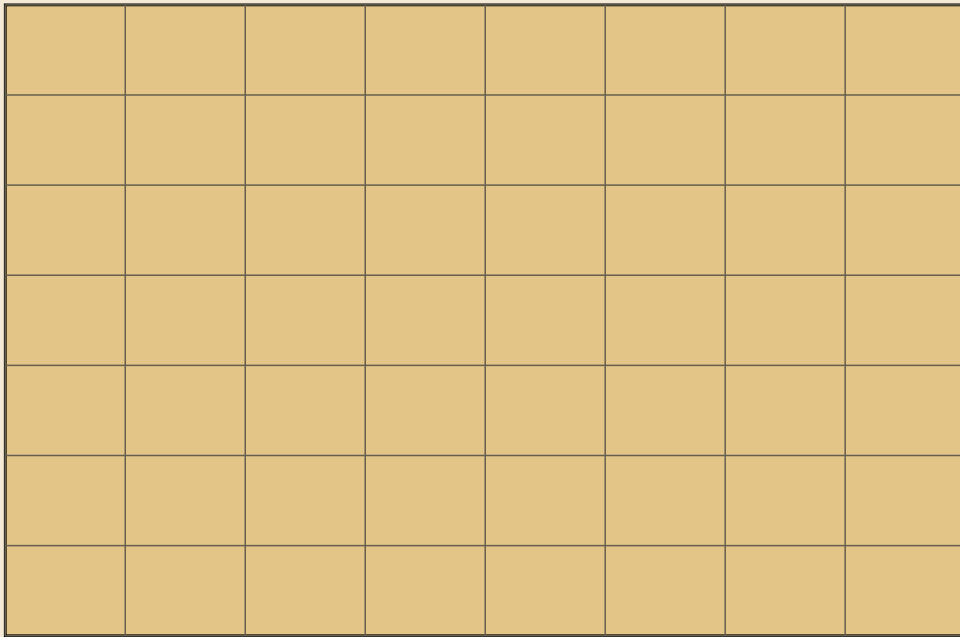
Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	compare_factor_orders	q_step_1	smr_mult_action_pairs.pl:474 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	keep_multiplier_multiplicand_roles_fixed	q_step_2	smr_mult_action_pairs.pl:474 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	reject_commuted_factor_order	q_step_3	smr_mult_action_pairs.pl:474 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	require_recomputation_in_original_order	q_step_4	smr_mult_action_pairs.pl:474 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_4	lose_factor_order_equivalence	q_accept	smr_mult_action_pairs.pl:474 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

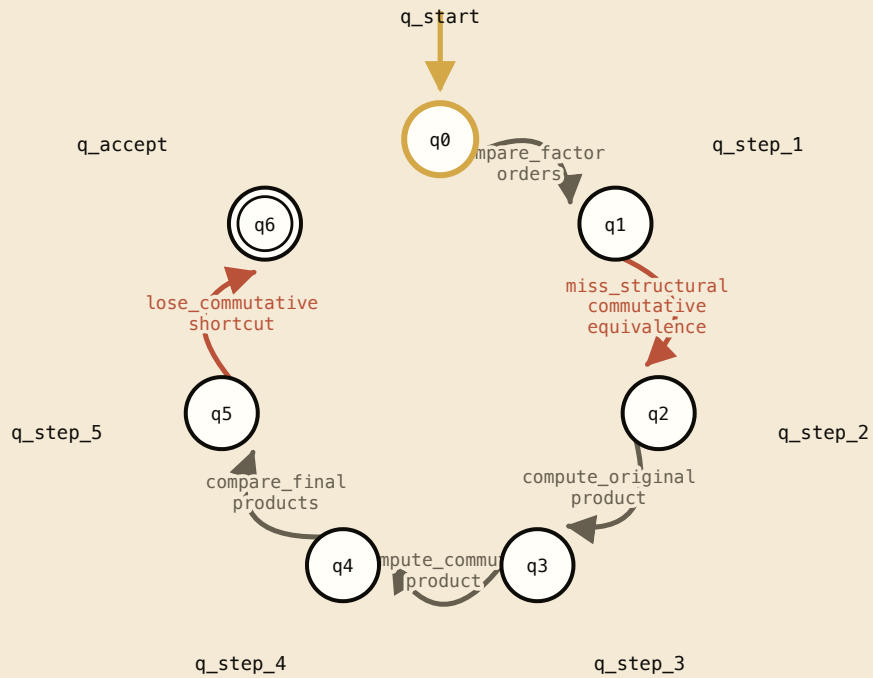
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

20. sequential_recompute_commuted_products



The run produced recomputed_both_products(56,56); the recorded expected result is structural

The run produced recomputed_both_products(56,56); the recorded expected result is structural_equivalence(56). The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \text{compare_factor_orders, compare_final_products, compute_commuted_product, compute_original_product, lose_commutative_shortcut, miss_structural_commutative_equivalence (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 7, "b": 8};
verified(strategy_trace_ok).

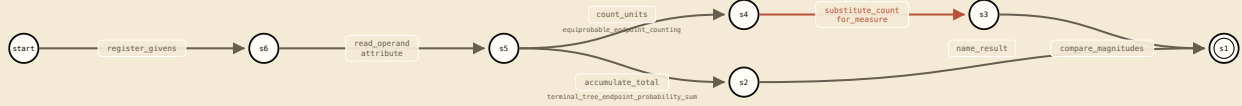
δ : recorded transitions

From	Action	To	Provenance
q_start	compare_factor_orders	q_step_1	smr_mult_action_pairs.pl:498 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_1	miss_structural_commutative_equivalence	q_step_2	smr_mult_action_pairs.pl:498 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_2	compute_original_product	q_step_3	smr_mult_action_pairs.pl:498 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_3	compute_commuted_product	q_step_4	smr_mult_action_pairs.pl:498 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)
q_step_4	compare_final_products	q_step_5	smr_mult_action_pairs.pl:498 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

From	Action	To	Provenance
q_step_5	lose_commutative_shortcut	q_accept	smr_mult_action_pairs.pl:498 observed on {"a": 7, "b": 8} observed on {"a": 8, "b": 8} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

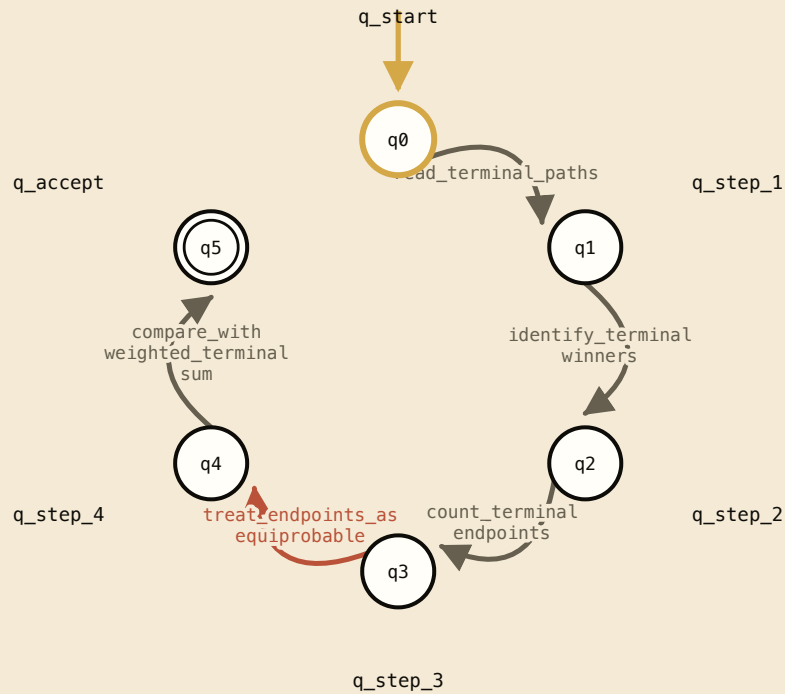
probability



Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. equiprobable_endpoint_counting

No domain scene: the static scene emitter has no probability form for the `terminal_path_tree` contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \text{compare_with_weighted_terminal_sum, count_terminal_endpoints, identify_terminal_winners, read_terminal_paths, treat_endpoints_as_equiprobable (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"kind": "terminal_path_tree", "paths": [{"events": ["atom"], "probability": {"d": "positive_integer", "n": "positive_integer"}, "winner": "atom"}], "stake": "positive_number"}; **example** {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60};
verified(strategy_trace_ok).

δ : recorded transitions

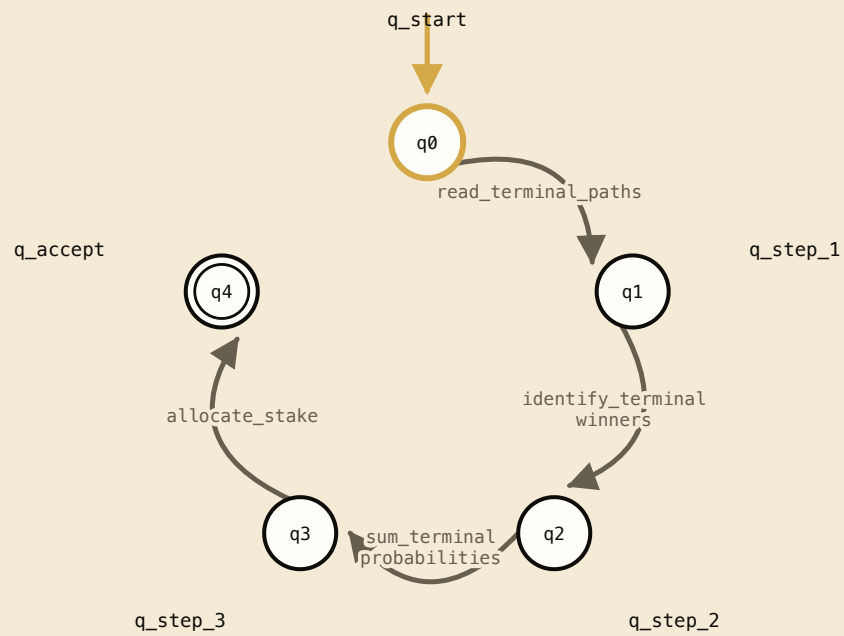
From	Action	To	Provenance
q_start	read_terminal_paths	q_step_1	probability_action_pairs.pl:59 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}
q_step_1	identify_terminal_winners	q_step_2	probability_action_pairs.pl:59 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}

From	Action	To	Provenance
q_ste p_2	count_term inal_endpo ints	q_s tep _3	probability_action_pairs.pl:59 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}
q_ste p_3	treat_endp oints_as_e quiprobabl e	q_s tep _4	probability_action_pairs.pl:59 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}
q_ste p_4	compare_wi th_weighte d_terminal _sum	q_a cce pt	probability_action_pairs.pl:59 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

2. terminal_tree_endpoint_probability_sum

No domain scene: the static scene emitter has no probability form for the terminal_path_tree contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{allocate_stake, identify_terminal_winners, read_terminal_paths, sum_terminal_probabilities (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"kind": "terminal_path_tree", "paths": [{"events": ["atom"], "probability": {"d": "positive_integer", "n": "positive_integer"}, "winner": "atom"}], "stake": "positive_number"}; example {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60};
verified(strategy_trace_ok).

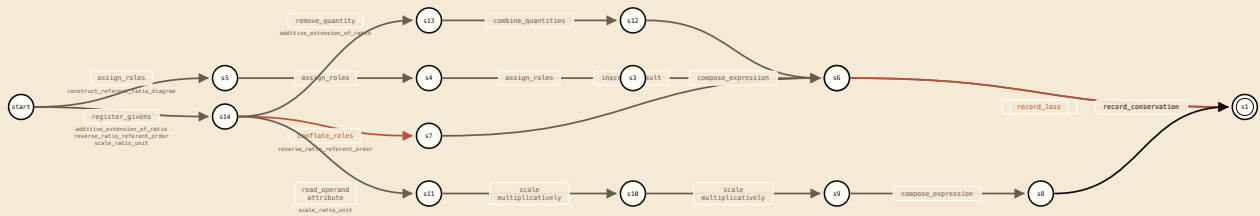
δ : recorded transitions

From	Action	To	Provenance
q_start	read_terminal_paths	q_step_1	probability_action_pairs.pl:27 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}
q_step_1	identify_terminal_winners	q_step_2	probability_action_pairs.pl:27 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}

From	Action	To	Provenance
q_ste p_2	sum_termi nal_proba bilities	q_s tep _3	probability_action_pairs.pl:27 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}
q_ste p_3	allocate_ stake	q_a cce pt	probability_action_pairs.pl:27 observed on {"kind": "terminal_path_tree", "paths": [{"events": ["alice_wins_first"], "probability": {"d": 2, "n": 1}, "winner": "alice"}, {"events": ["alice_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}, {"events": ["bob_wins_then_bob", "bob_wins"], "probability": {"d": 4, "n": 1}, "winner": "bob"}], "stake": 60}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

ratio



Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

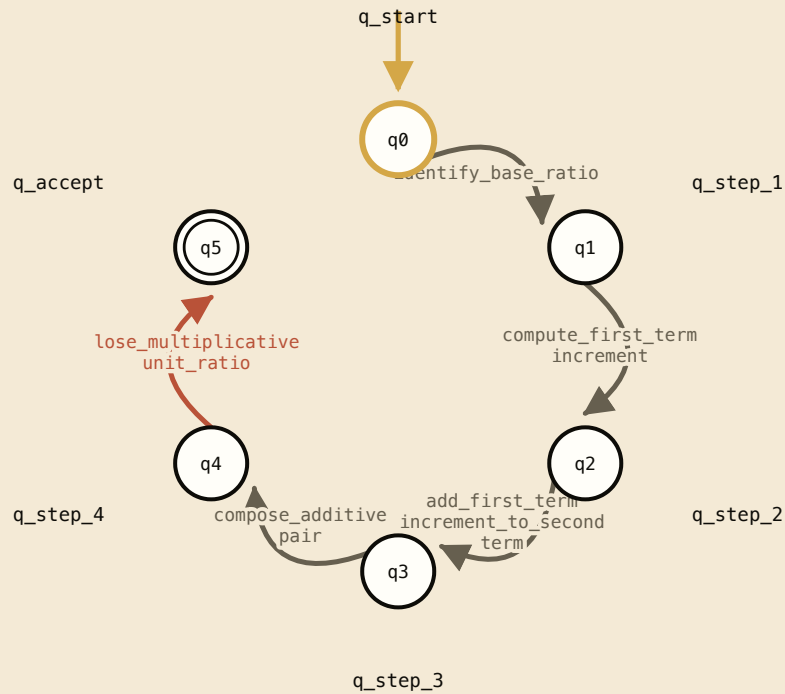
1. additive_extension_of_ratio

iterated unit: 47/28



The run produced `ratio_pair(94,75)`; the recorded expected result is `ratio_pair(94,56)`.

The run produced `ratio_pair(94,75)`; the recorded expected result is `ratio_pair(94,56)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma = \{\text{add_first_term_increment_to_second_term}, \text{compose_additive_pair}, \text{compute_first_term_increment}, \text{identify_base_ratio}, \text{lose_multiplicative_unit_ratio}\} \text{ (5 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_base_ratio	q_step_1	ratio_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	compute_first_term_increment	q_step_2	ratio_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	add_first_term_increment_to_second_term	q_step_3	ratio_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	compose_additive_pair	q_step_4	ratio_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	lose_multiplicative_unit_ratio	q_accept	ratio_action_pairs.pl:132 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

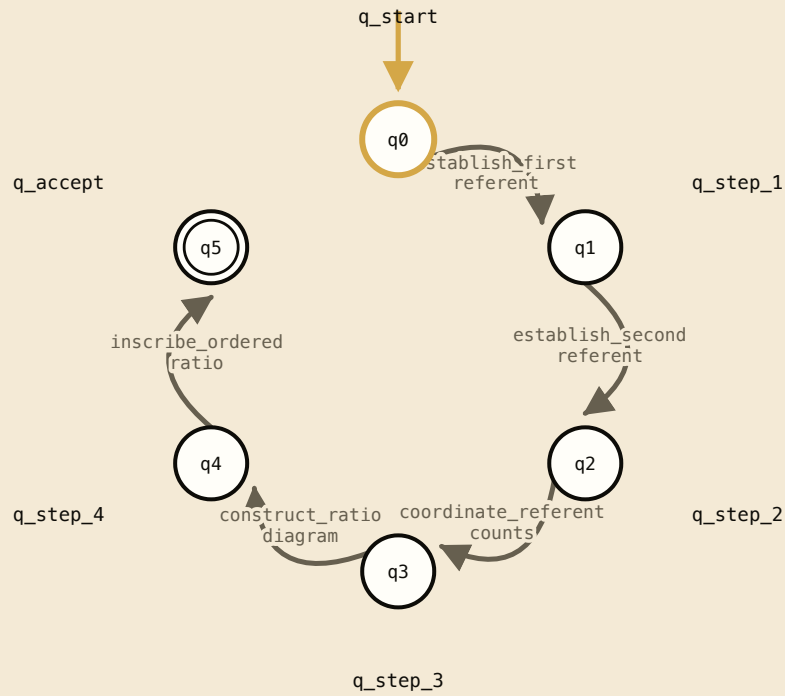
2. construct_referent_ratio_diagram

iterated unit: 1/1



The run produced `ratio_statement(red_tiles,3,blue_tiles,5)`.

The run produced `ratio_statement(red_tiles,3,blue_tiles,5)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ construct_ratio_diagram, coordinate_referent_counts, establish_first_referent, establish_second_referent, inscribe_ordered_ratio (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"first": {"count": "positive_integer", "label": "atom"}, "kind": "referent_pair", "second": {"count": "positive_integer", "label": "atom"}};
example {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_first_referent	q_step_1	ratio_action_pairs.pl:42 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}
q_step_1	establish_second_referent	q_step_2	ratio_action_pairs.pl:42 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}
q_step_2	coordinate_referent_counts	q_step_3	ratio_action_pairs.pl:42 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}
q_step_3	construct_ratio_diagram	q_step_4	ratio_action_pairs.pl:42 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}
q_step_4	inscribe_ordered_ratio	q_accept	ratio_action_pairs.pl:42 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

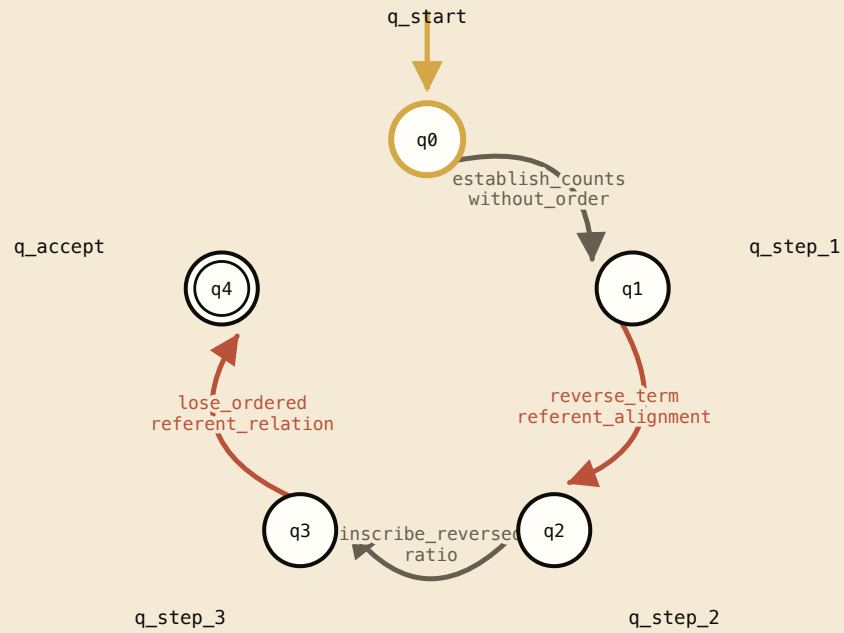
3. reverse_ratio_referent_order

iterated unit: 1/1



The run produced `ratio_statement(red_tiles,5,blue_tiles,3)`; the recorded expected result is `ratio_st`

The run produced `ratio_statement(red_tiles,5,blue_tiles,3)`; the recorded expected result is `ratio_statement(red_tiles,3,blue_tiles,5)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ establish_counts_without_order, inscribe_reversed_ratio,
lose_ordered_referent_relation, reverse_term_referent_alignment (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"first": {"count": "positive_integer", "label": "atom"}, "kind": "referent_pair", "second": {"count": "positive_integer", "label": "atom"}};
example {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair",
"second": {"count": 5, "label": "blue_tiles"}}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_counts_without_order	q_step_1	ratio_action_pairs.pl:70 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}
q_step_1	reverse_term_referent_alignment	q_step_2	ratio_action_pairs.pl:70 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}
q_step_2	inscribe_reverse_ratio	q_step_3	ratio_action_pairs.pl:70 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}
q_step_3	lose_ordered_referent_relation	q_accept	ratio_action_pairs.pl:70 observed on {"first": {"count": 3, "label": "red_tiles"}, "kind": "referent_pair", "second": {"count": 5, "label": "blue_tiles"}}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

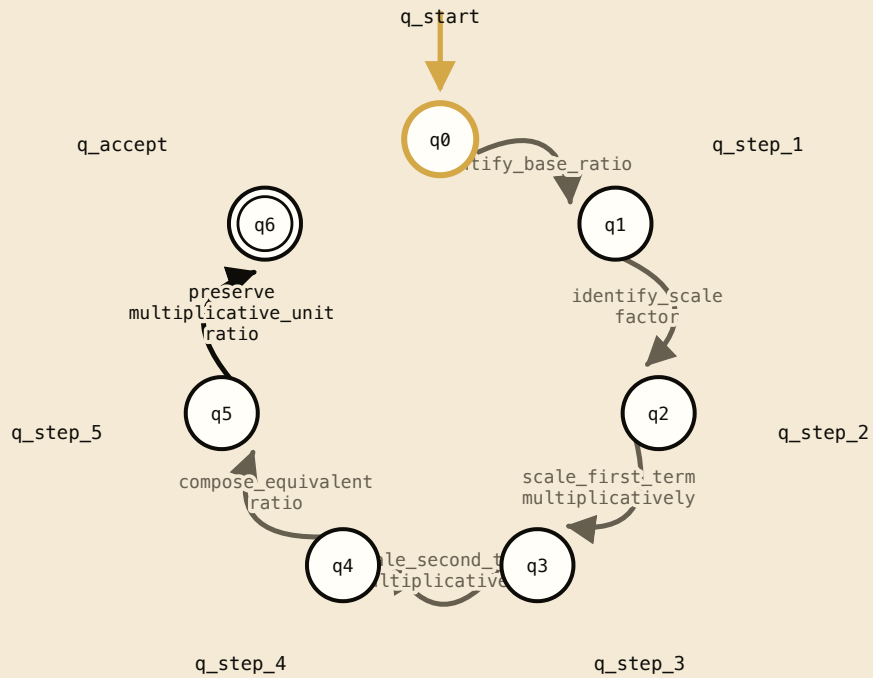
4. scale_ratio_unit

iterated unit: 47/28



The run produced `ratio_pair(94,56)`.

The run produced `ratio_pair(94,56)`. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \{\text{compose_equivalent_ratio}, \text{identify_base_ratio}, \text{identify_scale_factor}, \text{preserve_multiplicative_unit_ratio}, \text{scale_first_term_multiplicatively}, \text{scale_second_term_multiplicatively}\} \text{ (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

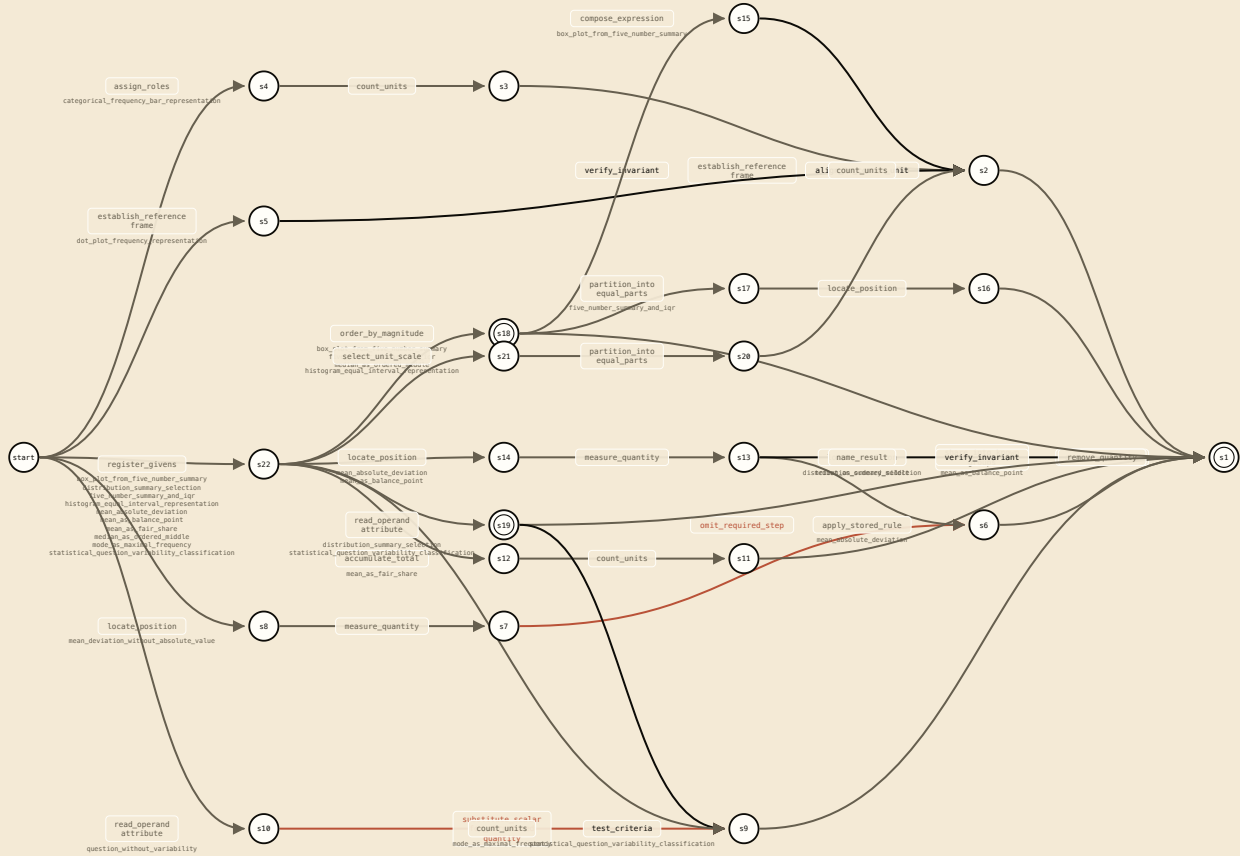
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_base_ratio	q_step_1	ratio_action_pairs.pl:107 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	identify_scale_factor	q_step_2	ratio_action_pairs.pl:107 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	scale_first_term_multiplicatively	q_step_3	ratio_action_pairs.pl:107 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	scale_second_term_multiplicatively	q_step_4	ratio_action_pairs.pl:107 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	compose_equivalent_ratio	q_step_5	ratio_action_pairs.pl:107 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

From	Action	To	Provenance
q_step_5	preserve_multiplicative_unit_ratio	q_accept	ratio_action_pairs.pl:107 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

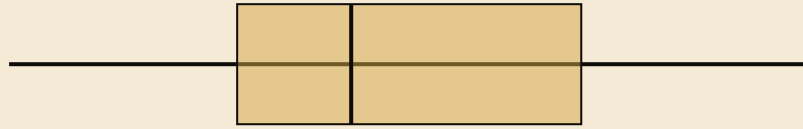
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

statistics



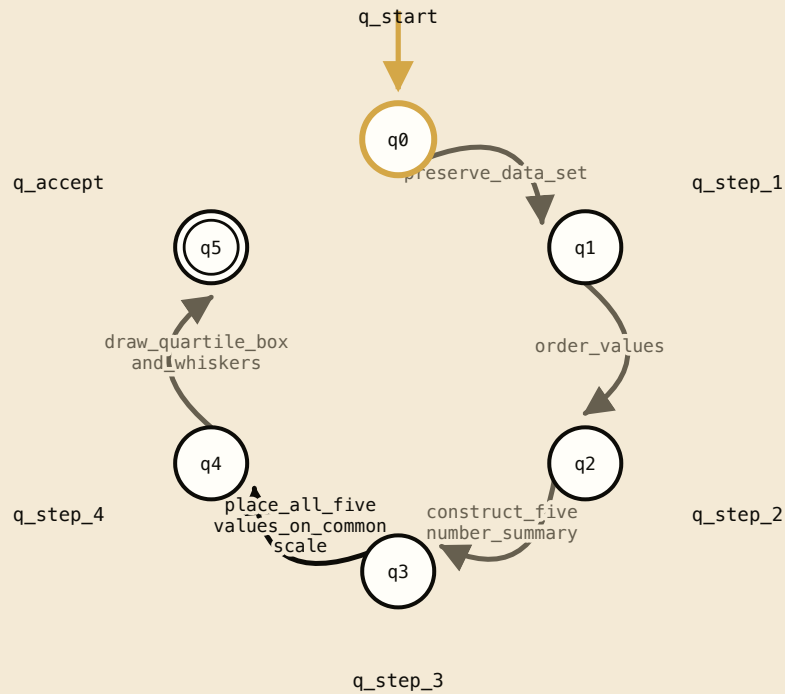
Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. box_plot_from_five_number_summary



The run produced `box_plot(five_number(2,rational(3,1),rational(5,1),rational(8,1),9)`

The run produced `box_plot(five_number(2,rational(3,1),rational(5,1),rational(8,1),9))`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ construct_five_number_summary, draw_quartile_box_and_whiskers, order_values, place_all_five_values_on_common_scale, preserve_data_set (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

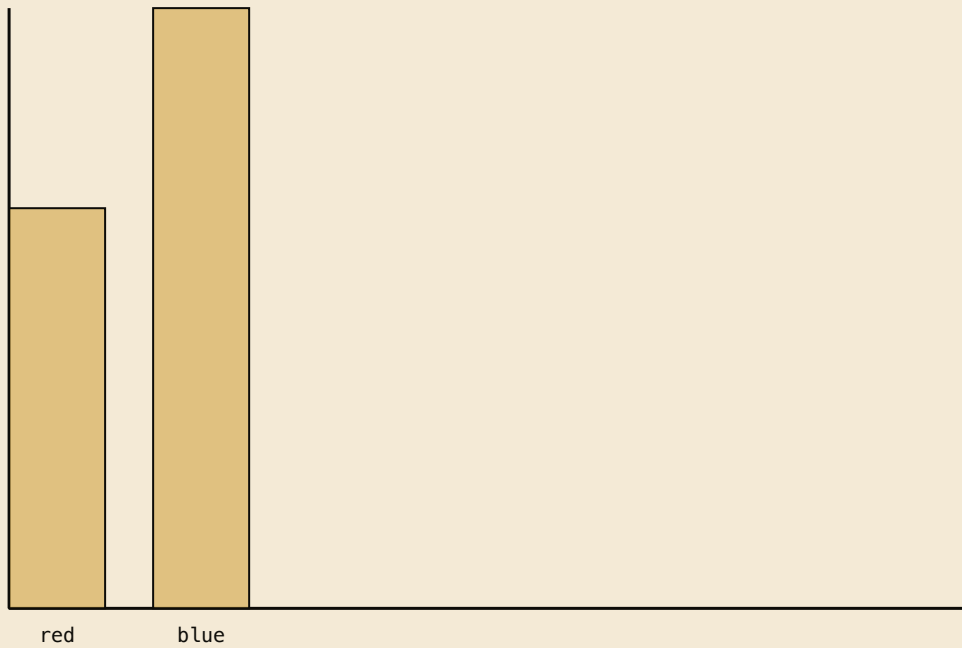
Input contract: {"display": "box_plot", "kind": "box_plot_data", "values": ["integer"]}; example {"display": "box_plot", "kind": "box_plot_data", "values": [2, 4, 5, 7, 9]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:299 observed on {"display": "box_plot", "kind": "box_plot_data", "values": [2, 4, 5, 7, 9]}
q_step_1	order_values	q_step_2	statistics_action_pairs.pl:299 observed on {"display": "box_plot", "kind": "box_plot_data", "values": [2, 4, 5, 7, 9]}
q_step_2	construct_five_number_summary	q_step_3	statistics_action_pairs.pl:299 observed on {"display": "box_plot", "kind": "box_plot_data", "values": [2, 4, 5, 7, 9]}
q_step_3	place_all_five_values_on_common_scale	q_step_4	statistics_action_pairs.pl:299 observed on {"display": "box_plot", "kind": "box_plot_data", "values": [2, 4, 5, 7, 9]}
q_step_4	draw_quartile_box_and_whiskers	q_accept	statistics_action_pairs.pl:299 observed on {"display": "box_plot", "kind": "box_plot_data", "values": [2, 4, 5, 7, 9]}

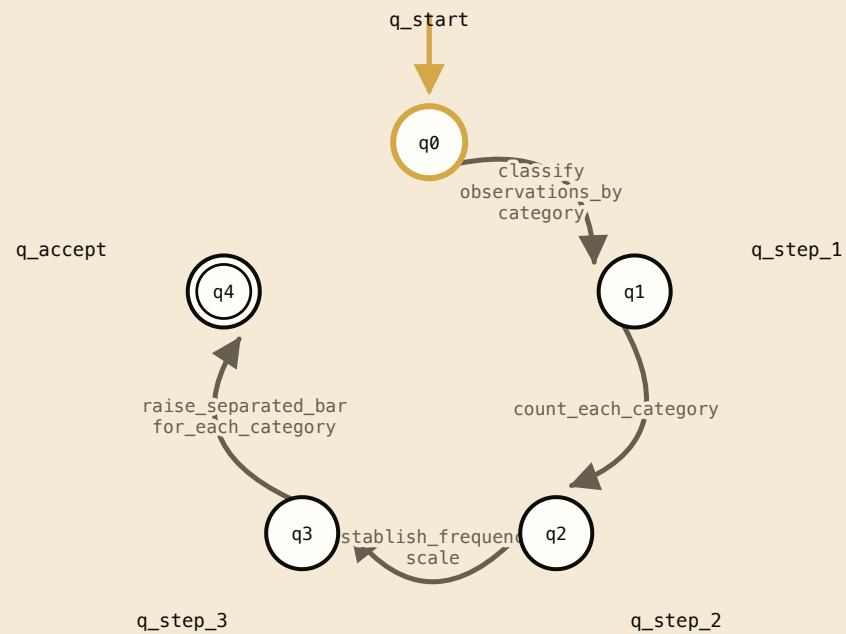
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

2. categorical_frequency_bar_representation



The run produced `frequency_representation([red-4,blue-6])`.

The run produced `frequency_representation([red-4,blue-6])`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ classify_observations_by_category, count_each_category, establish_frequency_scale, raise_separated_bar_for_each_category (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

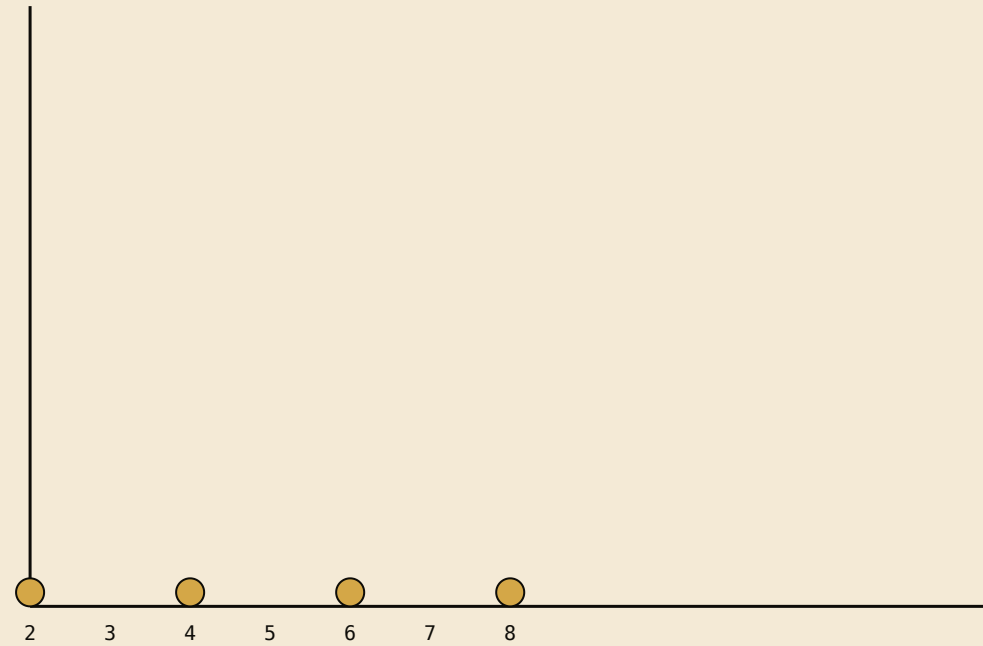
Input contract: {"display": "bar_chart", "kind": "categorical_frequencies", "pairs": [{"category": "atom", "count": "integer"}]}; example {"display": "bar_chart", "kind": "categorical_frequencies", "pairs": [{"category": "red", "count": 4}, {"category": "blue", "count": 6}]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	classify_observations_by_category	q_step_1	statistics_action_pairs.pl:19 observed on {"display": "bar_chart", "kind": "categorical_frequencies", "pairs": [{"category": "red", "count": 4}, {"category": "blue", "count": 6}]}
q_step_1	count_each_category	q_step_2	statistics_action_pairs.pl:19 observed on {"display": "bar_chart", "kind": "categorical_frequencies", "pairs": [{"category": "red", "count": 4}, {"category": "blue", "count": 6}]}
q_step_2	establish_frequency_scale	q_step_3	statistics_action_pairs.pl:19 observed on {"display": "bar_chart", "kind": "categorical_frequencies", "pairs": [{"category": "red", "count": 4}, {"category": "blue", "count": 6}]}
q_step_3	raise_separated_bar_for_each_category	q_accept	statistics_action_pairs.pl:19 observed on {"display": "bar_chart", "kind": "categorical_frequencies", "pairs": [{"category": "red", "count": 4}, {"category": "blue", "count": 6}]}

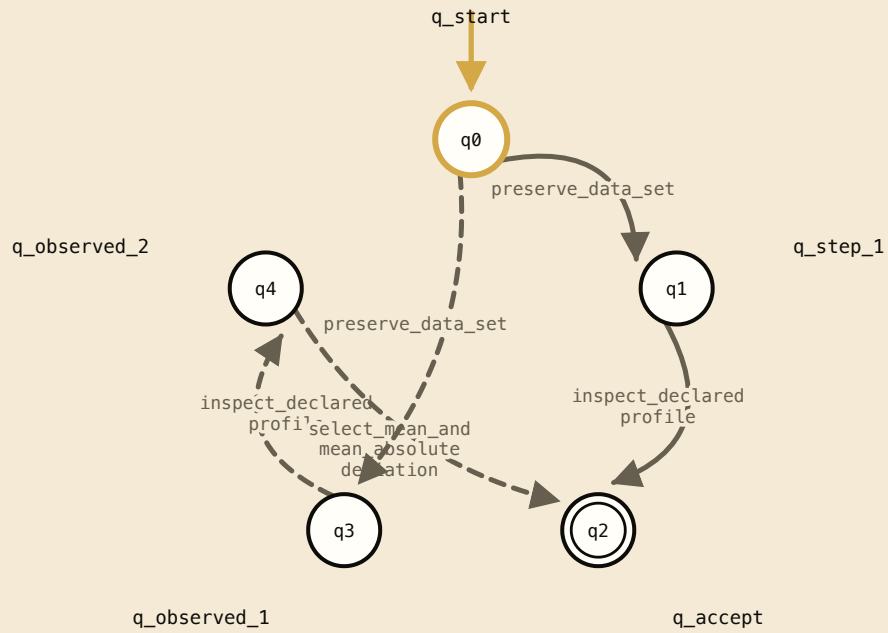
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

3. distribution_summary_selection



The run produced `summary_choice(mean_and_mad,measures(mean(rational(5,1)),mad(ratio`

The run produced
`summary_choice(mean_and_mad,measures(mean(rational(5,1)),mad(rational(2,1))))`. The scene is
derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{inspect_declared_profile, preserve_data_set, select_mean_and_mean_absolute_deviation (3 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_2\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_accept	none recorded
q_3	q_observed_1	none recorded
q_4	q_observed_2	none recorded

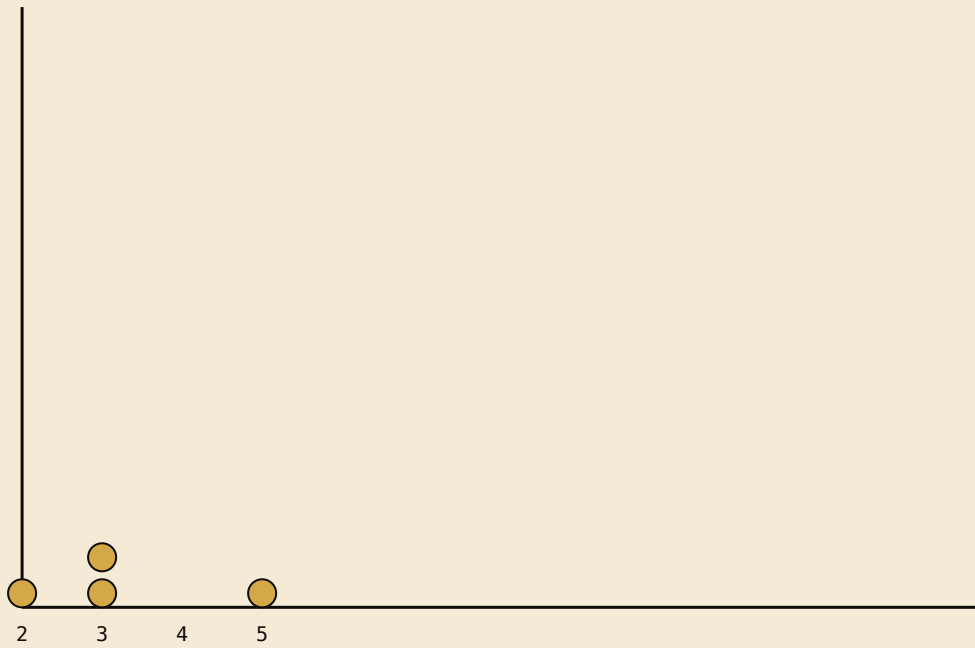
Input contract: {"kind": "distribution_data", "profile": "atom", "values": ["integer"]}; example {"kind": "distribution_data", "profile": "symmetric_without_outliers", "values": [2, 4, 6, 8]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:323
q_step_1	inspect_declared_profile	q_accept	statistics_action_pairs.pl:323
q_start	preserve_data_set	q_observed_1	observed on {"kind": "distribution_data", "profile": "symmetric_without_outliers", "values": [2, 4, 6, 8]}
q_observed_1	inspect_declared_profile	q_observed_2	observed on {"kind": "distribution_data", "profile": "symmetric_without_outliers", "values": [2, 4, 6, 8]}
q_observed_2	select_mean_and_mean_absolute_deviation	q_accept	observed on {"kind": "distribution_data", "profile": "symmetric_without_outliers", "values": [2, 4, 6, 8]}

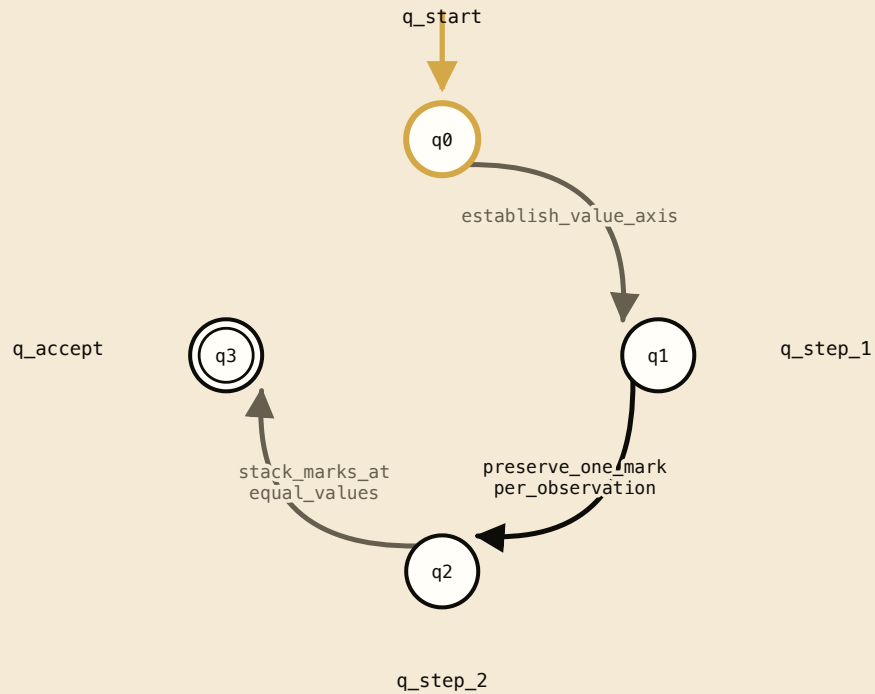
Computed structure: branching. one or more recorded states have multiple outgoing transitions; no loop edge is recorded. 5 states, 3 distinct actions, 1 branching states, 0 loop edges; 2 static rows and 3 observed rows.

4. dot_plot_frequency_representation



The run produced `dot_plot([2,3,3,5])`.

The run produced `dot_plot([2,3,3,5])`. The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3\}$

Σ establish_value_axis, preserve_one_mark_per_observation,
stack_marks_at_equal_values (3 distinct)

q_0 $q_0 = q_start$

F $\{q_3\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

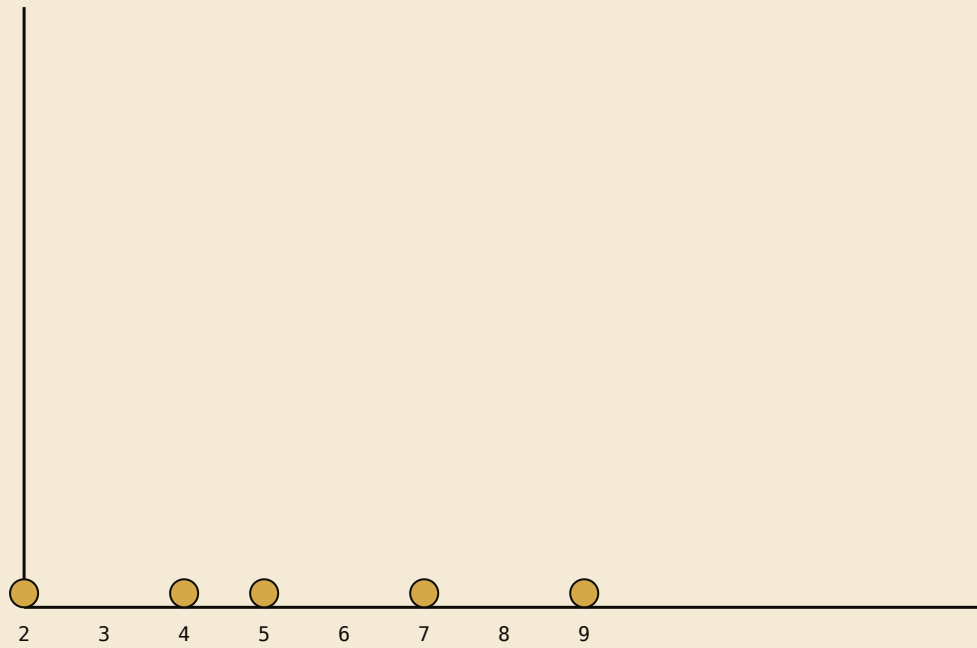
Input contract: {"display": "dot_plot", "kind": "numeric_data_display", "values": ["integer"]}; example {"display": "dot_plot", "kind": "numeric_data_display", "values": [2, 3, 3, 5]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	establish_value_axis	q_step_1	statistics_action_pairs.pl:44 observed on {"display": "dot_plot", "kind": "numeric_data_display", "values": [2, 3, 3, 5]}
q_step_1	preserve_one_mark_per_observation	q_step_2	statistics_action_pairs.pl:44 observed on {"display": "dot_plot", "kind": "numeric_data_display", "values": [2, 3, 3, 5]}
q_step_2	stack_marks_at_equal_values	q_accept	statistics_action_pairs.pl:44 observed on {"display": "dot_plot", "kind": "numeric_data_display", "values": [2, 3, 3, 5]}

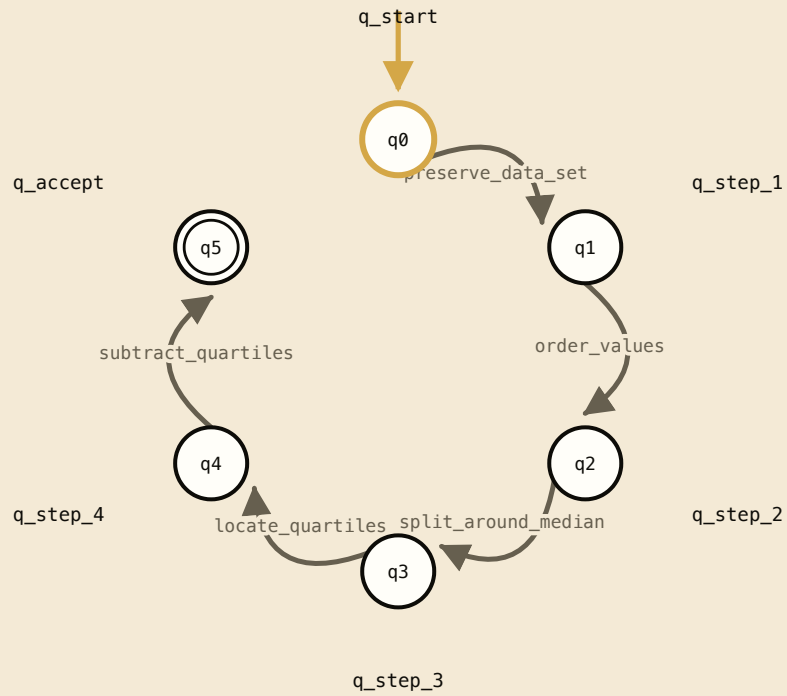
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

5. five_number_summary_and_iqr



The run produced `summary(five_number(2,rational(3,1),rational(5,1),rational(8,1),9),`

The run produced
`summary(five_number(2,rational(3,1),rational(5,1),rational(8,1),9),iqr(rational(5,1)))`. The scene is
derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ locate_quartiles, order_values, preserve_data_set, split_around_median, subtract_quartiles (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

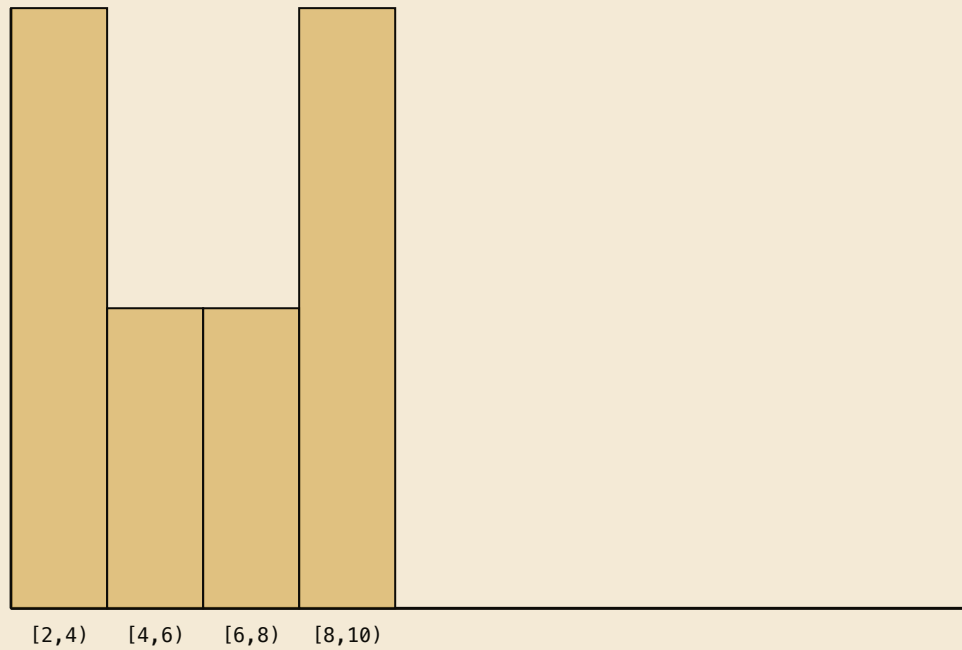
Input contract: {"kind": "numeric_data_with_unit", "unit": "atom", "values": ["integer"]}; example {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:278 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}
q_step_1	order_values	q_step_2	statistics_action_pairs.pl:278 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}
q_step_2	split_around_median	q_step_3	statistics_action_pairs.pl:278 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}
q_step_3	locate_quartiles	q_step_4	statistics_action_pairs.pl:278 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}
q_step_4	subtract_quartiles	q_accept	statistics_action_pairs.pl:278 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}

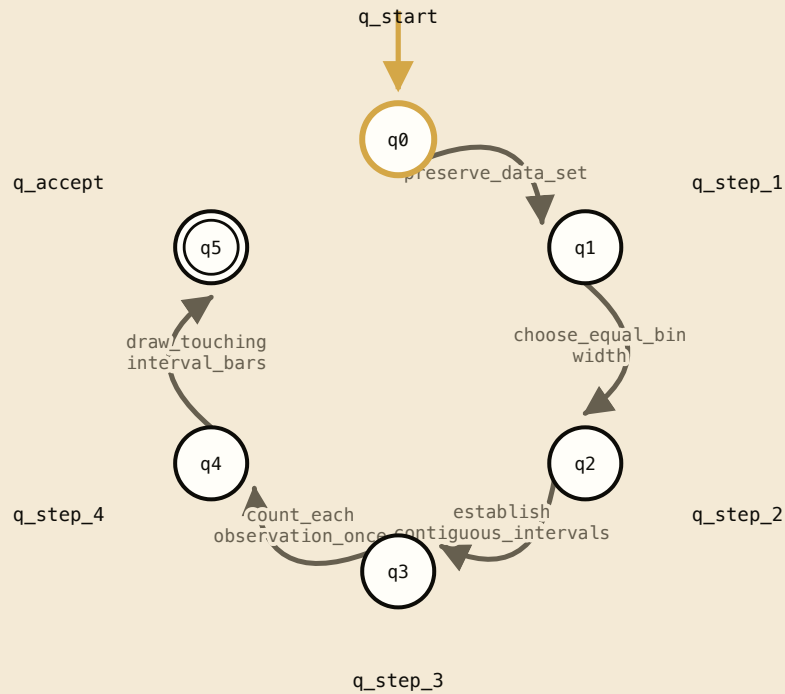
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

6. histogram_equal_interval_representation



The run produced `histogram([bin(2,4)-2,bin(4,6)-1,bin(6,8)-1,bin(8,10)-2])`.

The run produced `histogram([bin(2,4)-2,bin(4,6)-1,bin(6,8)-1,bin(8,10)-2])`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ choose_equal_bin_width, count_each_observation_once, draw_touching_intervalBars, establish_contiguous_intervals, preserve_data_set (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

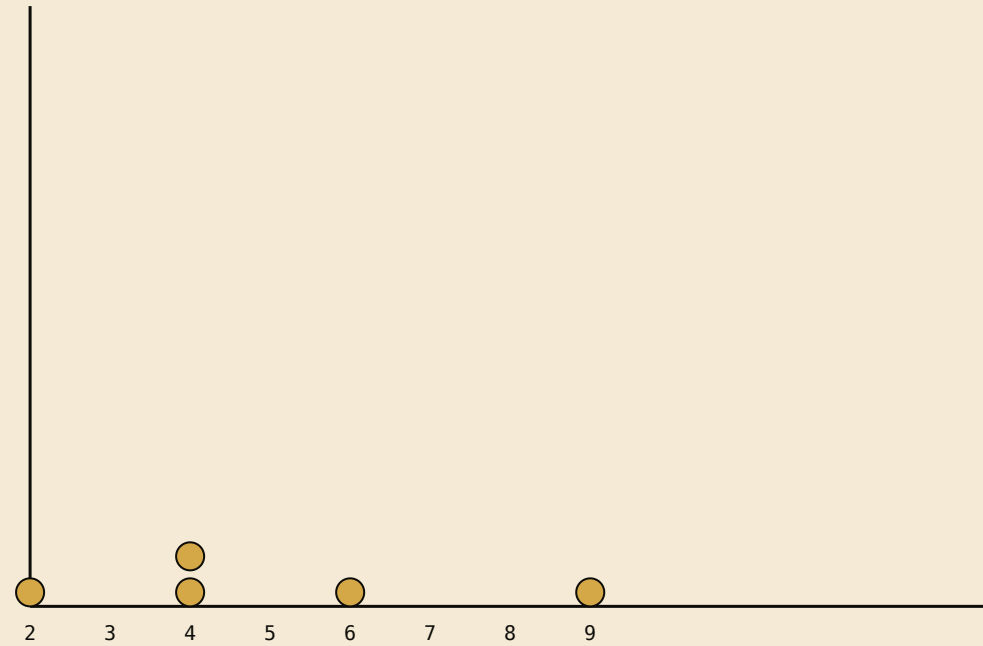
Input contract: {"bin_width": "positive_integer", "kind": "histogram_data", "values": ["integer"]}; example {"bin_width": 2, "kind": "histogram_data", "values": [2, 3, 5, 7, 8, 9]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:115 observed on {"bin_width": 2, "kind": "histogram_data", "values": [2, 3, 5, 7, 8, 9]}
q_step_1	choose_equal_bin_width	q_step_2	statistics_action_pairs.pl:115 observed on {"bin_width": 2, "kind": "histogram_data", "values": [2, 3, 5, 7, 8, 9]}
q_step_2	establish_contiguous_intervals	q_step_3	statistics_action_pairs.pl:115 observed on {"bin_width": 2, "kind": "histogram_data", "values": [2, 3, 5, 7, 8, 9]}
q_step_3	count_each_observation_once	q_step_4	statistics_action_pairs.pl:115 observed on {"bin_width": 2, "kind": "histogram_data", "values": [2, 3, 5, 7, 8, 9]}
q_step_4	draw_touching_interval_bars	q_accept	statistics_action_pairs.pl:115 observed on {"bin_width": 2, "kind": "histogram_data", "values": [2, 3, 5, 7, 8, 9]}

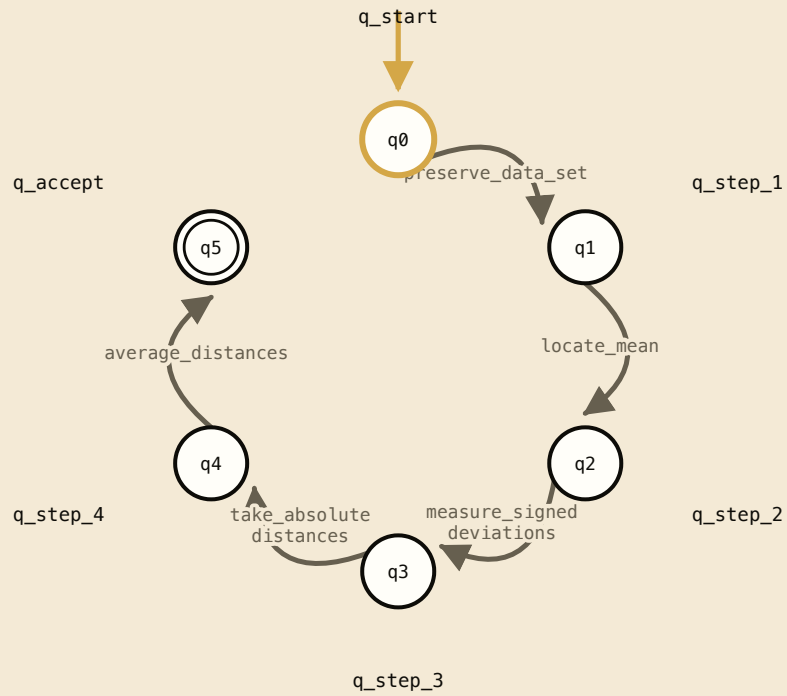
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

7. mean_absolute_deviation



The run produced `rational(2,1)`.

The run produced `rational(2,1)`. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ average_distances, locate_mean, measure_signed_deviations, preserve_data_set, take_absolute_distances (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

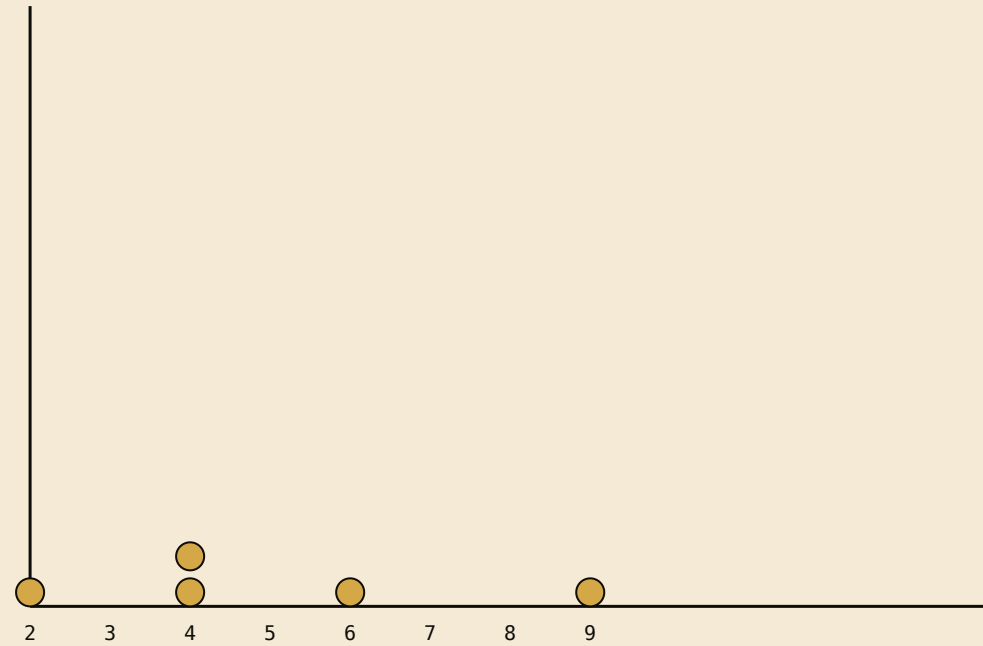
Input contract: {"kind": "numeric_data_with_unit", "unit": "atom", "values": ["integer"]}; example {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:189 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_1	locate_mean	q_step_2	statistics_action_pairs.pl:189 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_2	measure_signed_deviations	q_step_3	statistics_action_pairs.pl:189 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_3	take_absolute_distances	q_step_4	statistics_action_pairs.pl:189 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_4	average_distances	q_accept	statistics_action_pairs.pl:189 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}

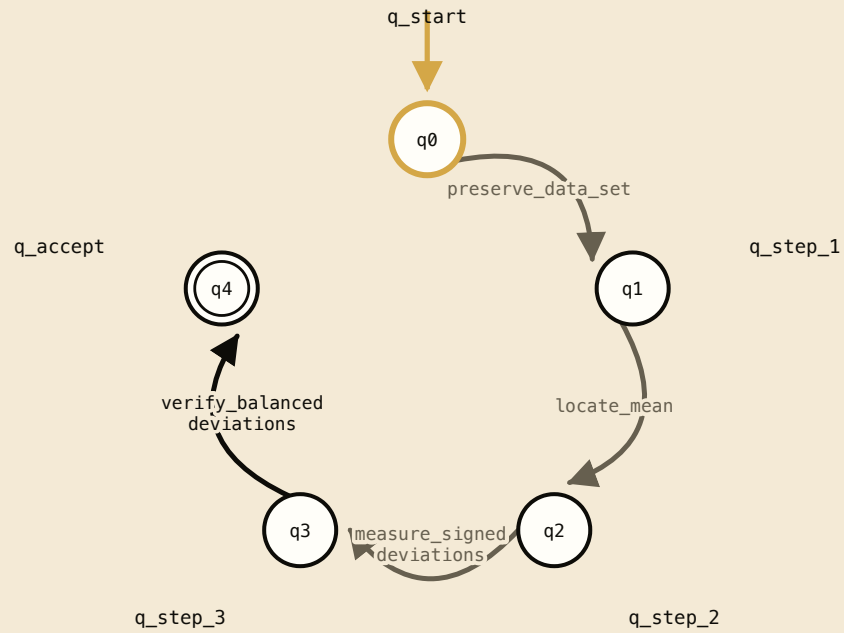
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

8. mean_as_balance_point



The run produced `rational(5,1)`.

The run produced `rational(5,1)`. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{locate_mean, measure_signed_deviations, preserve_data_set, verify_balanced_deviations (4 distinct)}$$

$$q_0 \quad q_0 = \text{q_start}$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

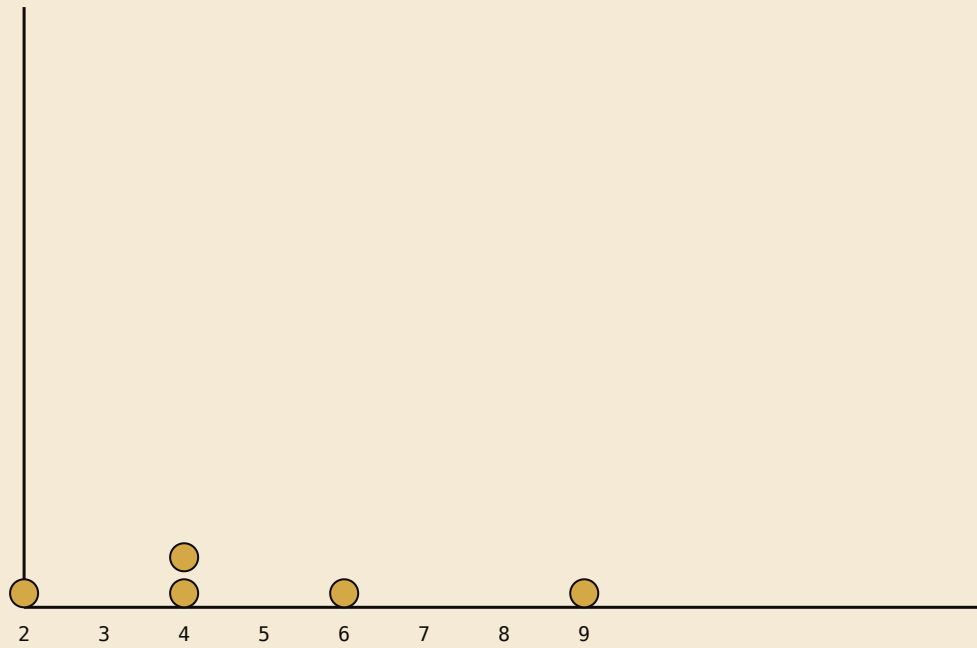
Input contract: {"kind": "numeric_data_with_unit", "unit": "atom", "values": ["integer"]}; example {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:164 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_1	locate_mean	q_step_2	statistics_action_pairs.pl:164 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_2	measure_signed_deviations	q_step_3	statistics_action_pairs.pl:164 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_3	verify_balanced_deviations	q_accept	statistics_action_pairs.pl:164 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}

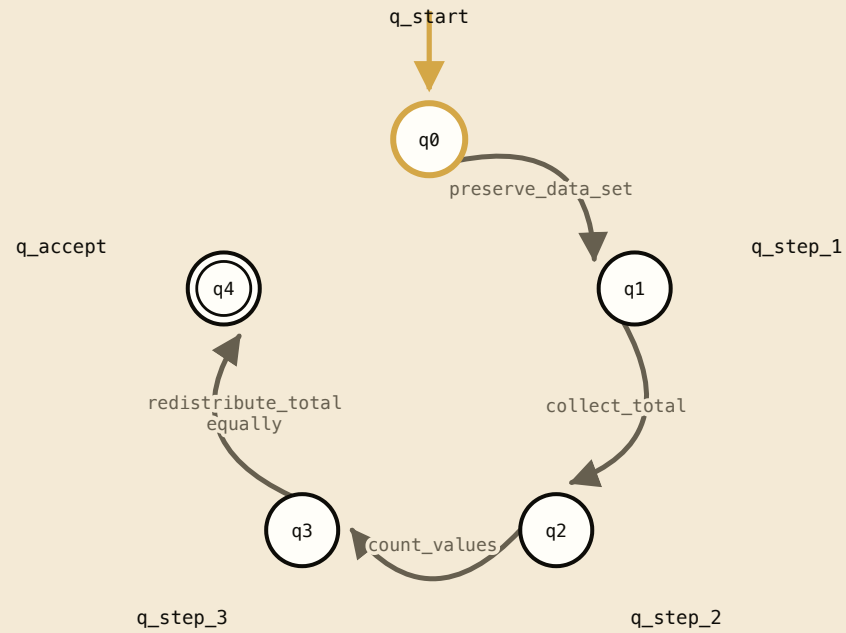
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

9. mean_as_fair_share



The run produced $\text{rational}(5,1)$.

The run produced $\text{rational}(5,1)$. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ collect_total, count_values, preserve_data_set, redistribute_total_equally (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

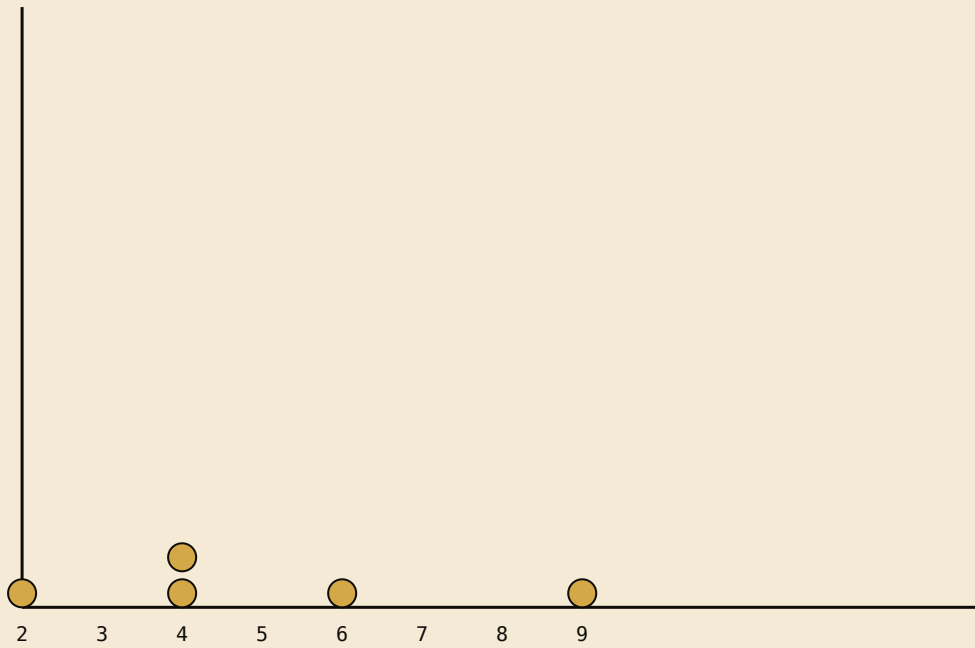
Input contract: {"kind": "numeric_data_with_unit", "unit": "atom", "values": ["integer"]}; example {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:141 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_1	collect_total	q_step_2	statistics_action_pairs.pl:141 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_2	count_values	q_step_3	statistics_action_pairs.pl:141 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_3	redistribute_total_equally	q_accept	statistics_action_pairs.pl:141 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}

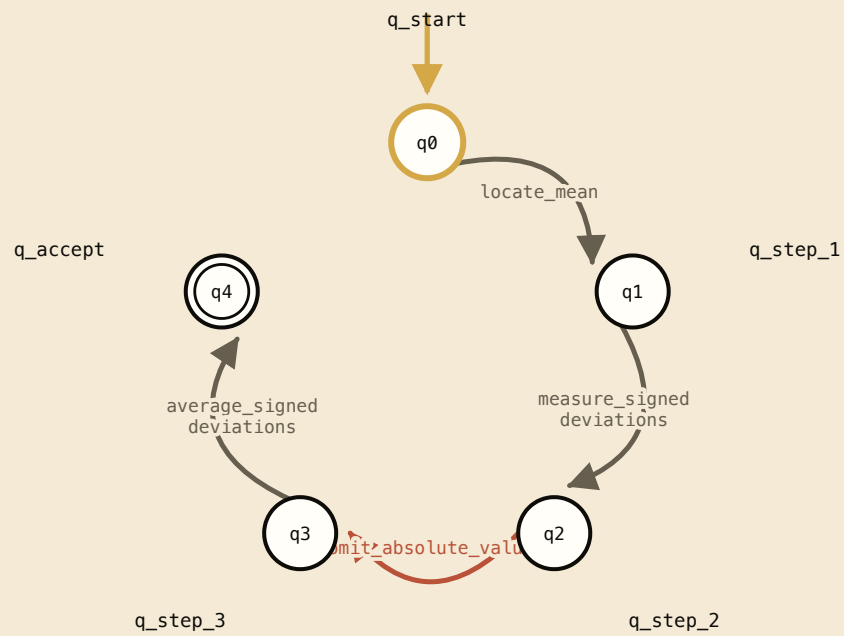
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

10. mean_deviation_without_absolute_value



The run produced $\text{rational}(0,1)$; the recorded expected result is $\text{rational}(2,1)$.

The run produced $\text{rational}(0,1)$; the recorded expected result is $\text{rational}(2,1)$. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \text{average_signed_deviations, locate_mean, measure_signed_deviations, omit_absolute_value (4 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

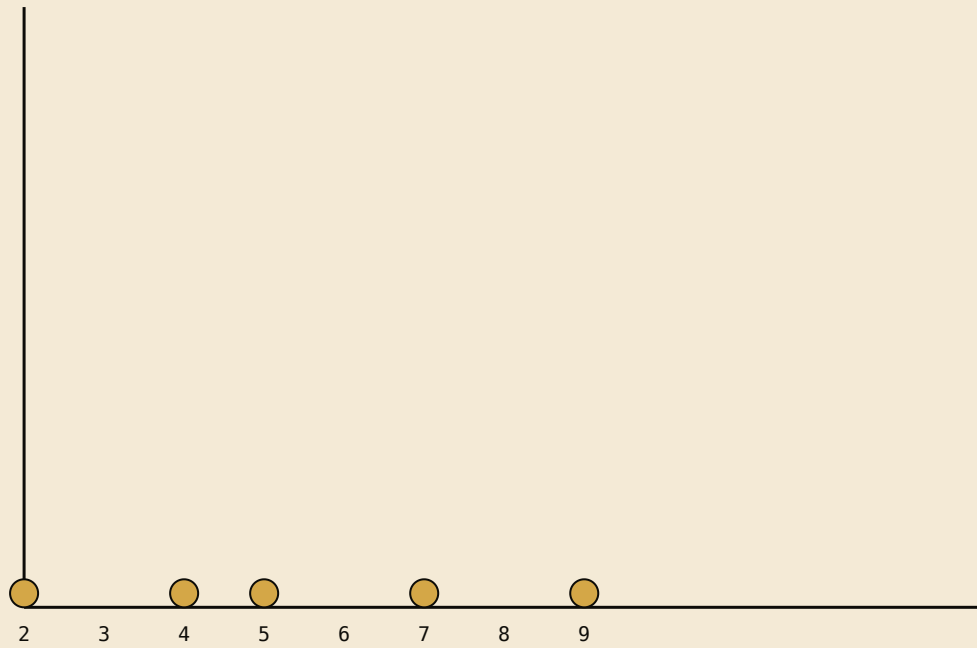
Input contract: {"kind": "numeric_data_with_unit", "unit": "atom", "values": ["integer"]}; example {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	locate_mean	q_step_1	statistics_action_pairs.pl:213 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_1	measure_signed_deviations	q_step_2	statistics_action_pairs.pl:213 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_2	omit_absolute_value	q_step_3	statistics_action_pairs.pl:213 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}
q_step_3	average_signed_deviations	q_accept	statistics_action_pairs.pl:213 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 4, 6, 9]}

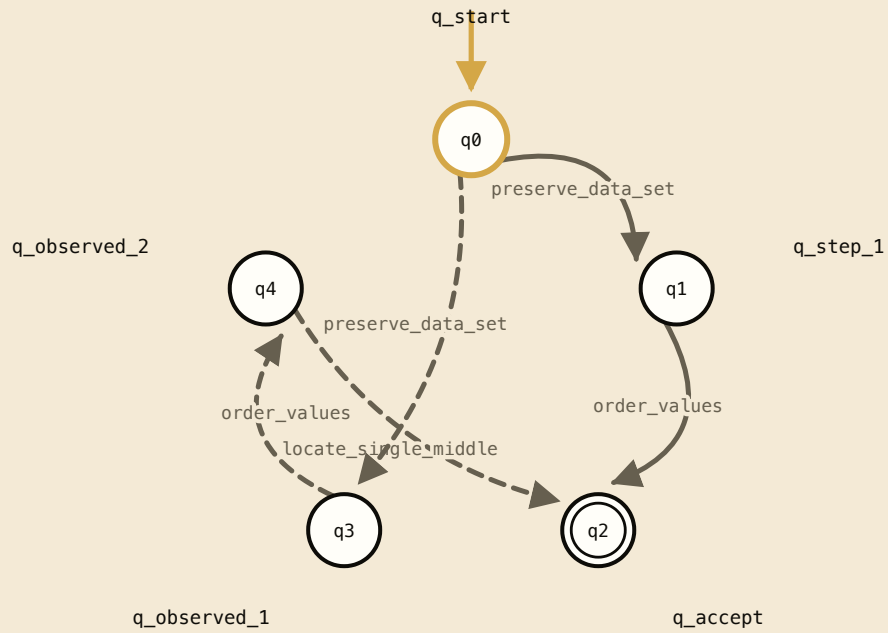
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

11. median_as_ordered_middle



The run produced 5.

The run produced 5. The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{\text{locate_single_middle}, \text{order_values}, \text{preserve_data_set}\} \text{ (3 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_2\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_accept	none recorded
q_3	q_observed_1	none recorded
q_4	q_observed_2	none recorded

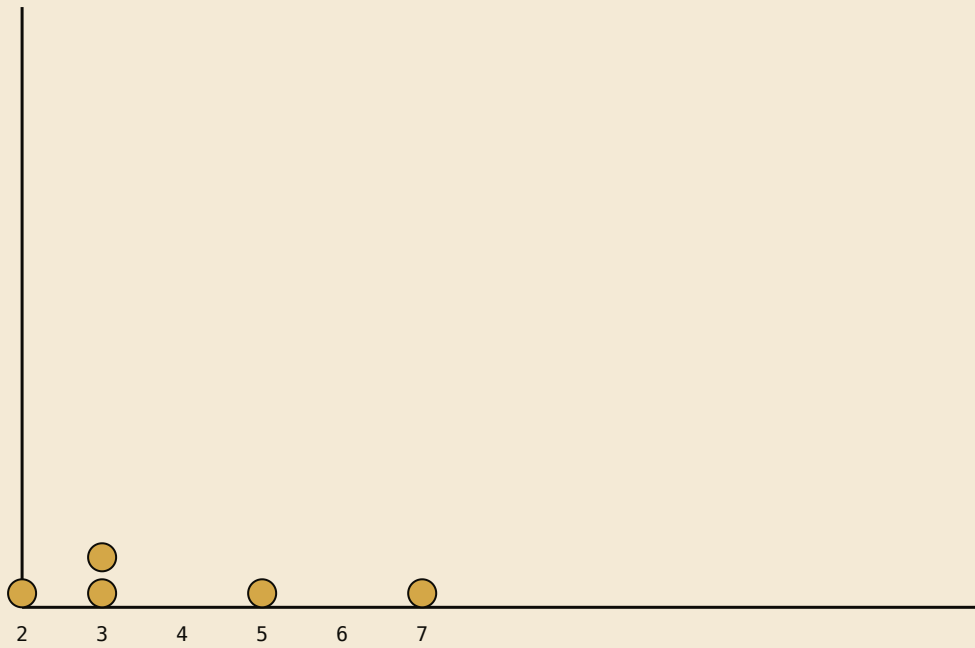
Input contract: {"kind": "numeric_data_with_unit", "unit": "atom", "values": ["integer"]}; example {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:234
q_step_1	order_values	q_accept	statistics_action_pairs.pl:234
q_start	preserve_data_set	q_observed_1	observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}
q_observed_1	order_values	q_observed_2	observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}
q_observed_2	locate_single_middle	q_accept	observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 4, 5, 7, 9]}

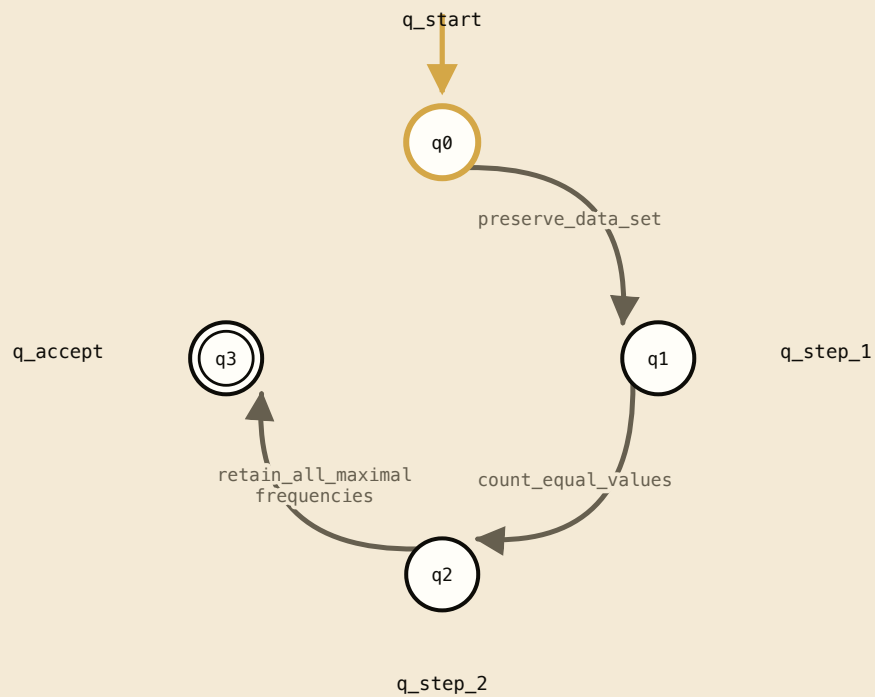
Computed structure: branching. one or more recorded states have multiple outgoing transitions; no loop edge is recorded. 5 states, 3 distinct actions, 1 branching states, 0 loop edges; 2 static rows and 3 observed rows.

12. mode_as_maximal_frequency



The run produced `modes([3])`.

The run produced `modes([3])`. The scene is derived from the contract run (3 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad \text{count_equal_values, preserve_data_set, retain_all_maximal_frequencies (3 distinct)}$$

$$q_0 \quad q_0 = \text{q_start}$$

$$F \quad \{q_3\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

Input contract: {"kind": "numeric_data_with_unit", "unit": "atom", "values": ["integer"]}; example {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 3, 3, 5, 7]}; verified(strategy_trace_ok).

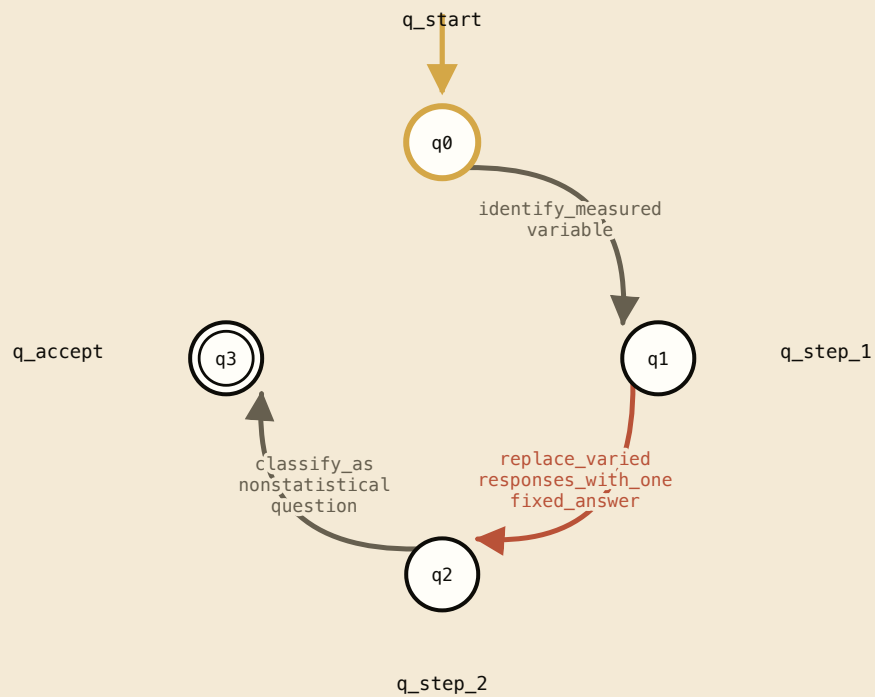
δ : recorded transitions

From	Action	To	Provenance
q_start	preserve_data_set	q_step_1	statistics_action_pairs.pl:254 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 3, 3, 5, 7]}
q_step_1	count_equal_values	q_step_2	statistics_action_pairs.pl:254 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 3, 3, 5, 7]}
q_step_2	retain_all_maximal_frequencies	q_accept	statistics_action_pairs.pl:254 observed on {"kind": "numeric_data_with_unit", "unit": "minutes", "values": [2, 3, 3, 5, 7]}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

13. question_without_variability

No domain scene: the static scene emitter has no statistics form for the statistical_question contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3\}$$

$$\Sigma \quad \text{classify_as_nonstatistical_question, identify_measured_variable,} \\ \text{replace_varied_responses_with_one_fixed_answer (3 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_3\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_accept	none recorded

Input contract: {"kind": "statistical_question", "population": "atom", "variable": "atom"}; example {"kind": "statistical_question", "population": "students", "variable": "commute_time"}; verified(strategy_trace_ok).

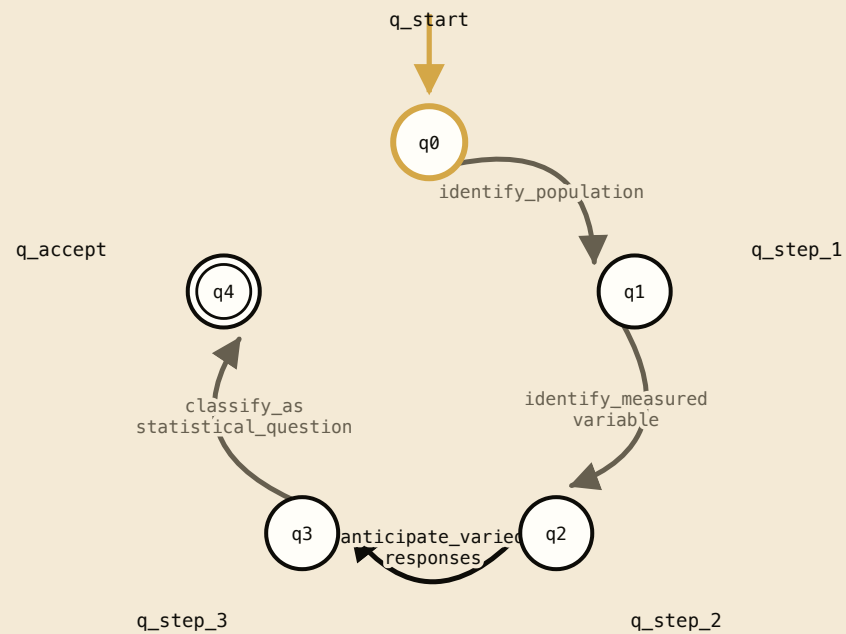
δ : recorded transitions

From	Action	To	Provenance
q_start	identify_measured_variable	q_step_1	statistics_action_pairs.pl:93 observed on {"kind": "statistical_question", "population": "students", "variable": "commute_time"}
q_step_1	replace_varied_responses_with_one_fixed_answer	q_step_2	statistics_action_pairs.pl:93 observed on {"kind": "statistical_question", "population": "students", "variable": "commute_time"}
q_step_2	classify_as_nonstatistical_question	q_accept	statistics_action_pairs.pl:93 observed on {"kind": "statistical_question", "population": "students", "variable": "commute_time"}

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 4 states, 3 distinct actions, 0 branching states, 0 loop edges; 3 static rows and 3 observed rows.

14. statistical_question_variability_classification

No domain scene: the static scene emitter has no statistics form for the statistical_question contract used by this machine.



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4\}$

Σ anticipate_varied_responses, classify_as_statistical_question,
identify_measured_variable, identify_population (4 distinct)

q_0 $q_0 = q_start$

F $\{q_4\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

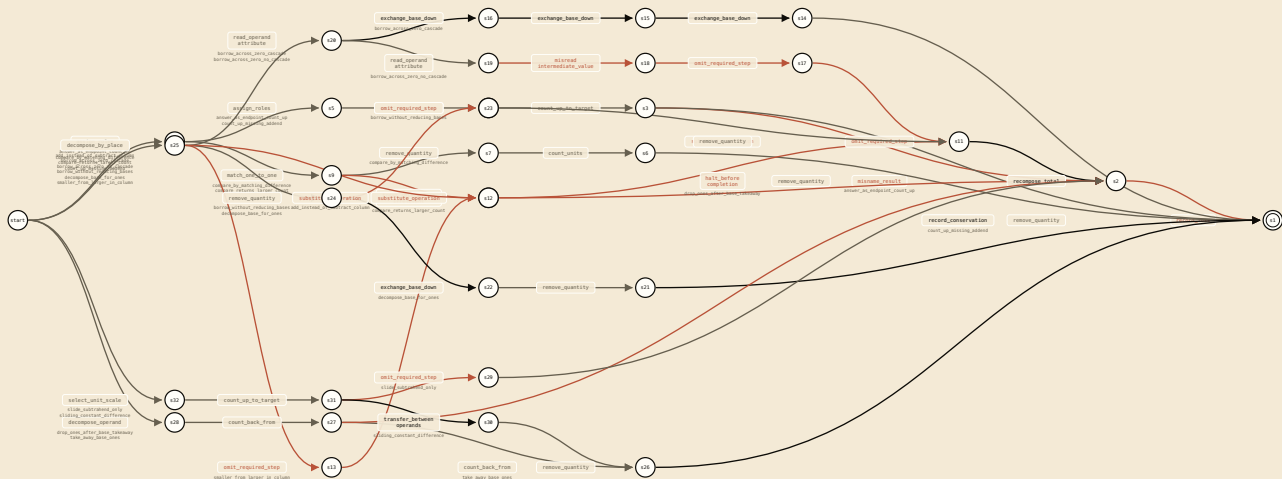
Input contract: {"kind": "statistical_question", "population": "atom", "variable": "atom"}; example {"kind": "statistical_question", "population": "students", "variable": "commute_time"}; verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_population	q_step_1	statistics_action_pairs.pl:70 observed on {"kind": "statistical_question", "population": "students", "variable": "commute_time"}
q_step_1	identify_measured_variable	q_step_2	statistics_action_pairs.pl:70 observed on {"kind": "statistical_question", "population": "students", "variable": "commute_time"}
q_step_2	anticipate_varied_responses	q_step_3	statistics_action_pairs.pl:70 observed on {"kind": "statistical_question", "population": "students", "variable": "commute_time"}
q_step_3	classify_as_statistical_question	q_accept	statistics_action_pairs.pl:70 observed on {"kind": "statistical_question", "population": "students", "variable": "commute_time"}

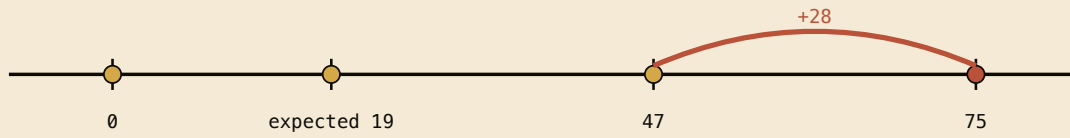
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 4 observed rows.

subtraction



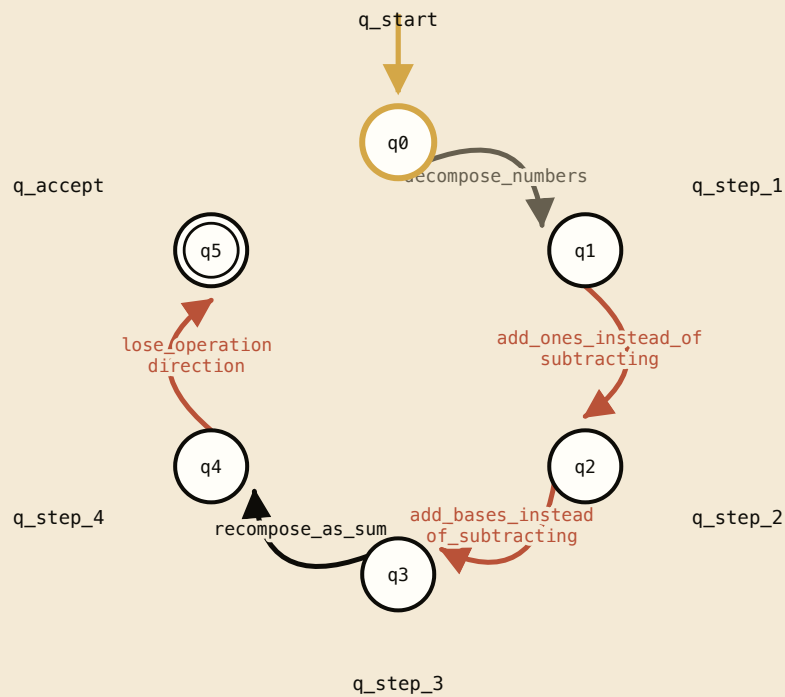
Construction: canonical action sequences merge by common prefix and structurally identical suffix. Branch labels name the kinds taking each branch. This models shared doing under canonical action names; the automata do not literally share states. For graphs with loops, the construction uses simple accepting paths and does not unfold repeated traversals.

1. add_instead_of_subtract_column



The run produced 75; the recorded expected result is 19.

The run produced 75; the recorded expected result is 19. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ add_bases_instead_of_subtracting, add_ones_instead_of_subtracting,
decompose_numbers, lose_operation_direction, recompose_as_sum (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

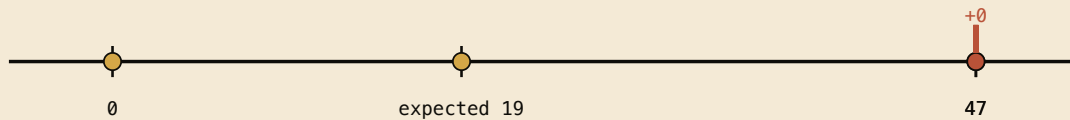
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_numbers	q_step_1	sar_sub_action_pairs.pl:365 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	add_ones_instead_of_subtracting	q_step_2	sar_sub_action_pairs.pl:365 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	add_bases_instead_of_subtracting	q_step_3	sar_sub_action_pairs.pl:365 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	recompose_as_sum	q_step_4	sar_sub_action_pairs.pl:365 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	lose_operation_direction	q_accept	sar_sub_action_pairs.pl:365 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

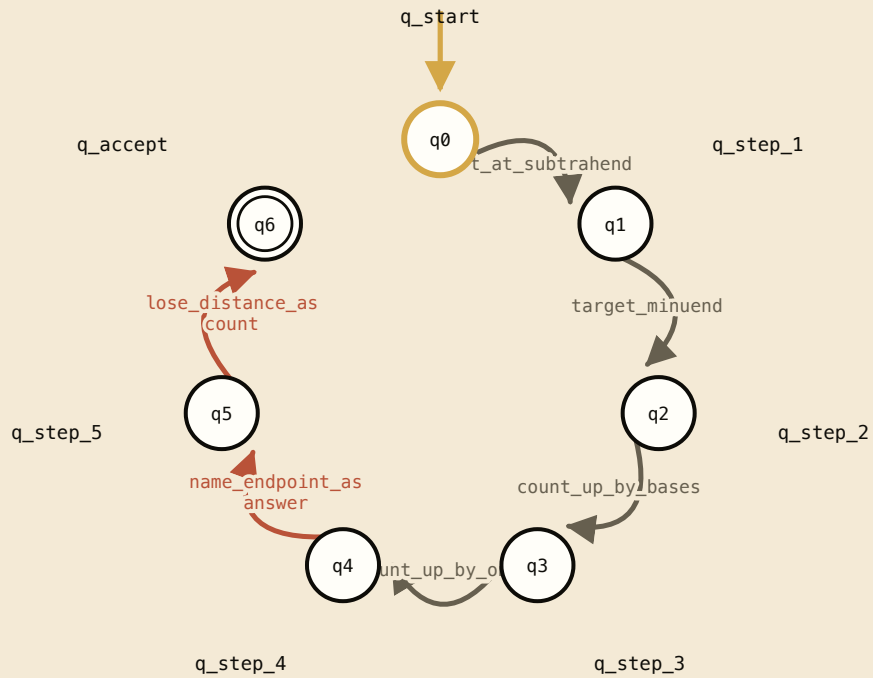
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

2. answer_as_endpoint_count_up



The run produced 47; the recorded expected result is 19.

The run produced 47; the recorded expected result is 19. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \text{count_up_by_bases, count_up_by_ones, lose_distance_as_count, name_endpoint_as_answer, start_at_subtrahend, target_minuend (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

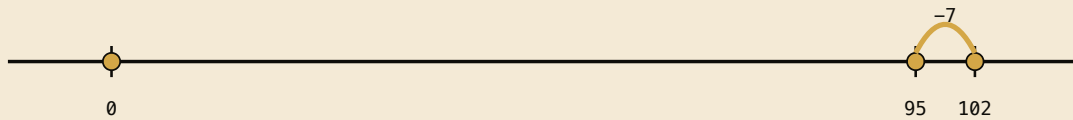
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	start_at_subtrahend	q_step_1	sar_sub_action_pairs.pl:147 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	target_minuend	q_step_2	sar_sub_action_pairs.pl:147 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	count_up_by_bases	q_step_3	sar_sub_action_pairs.pl:147 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	count_up_by_ones	q_step_4	sar_sub_action_pairs.pl:147 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	name_endpoint_as_answer	q_step_5	sar_sub_action_pairs.pl:147 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_5	lose_distance_as_count	q_accept	sar_sub_action_pairs.pl:147 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

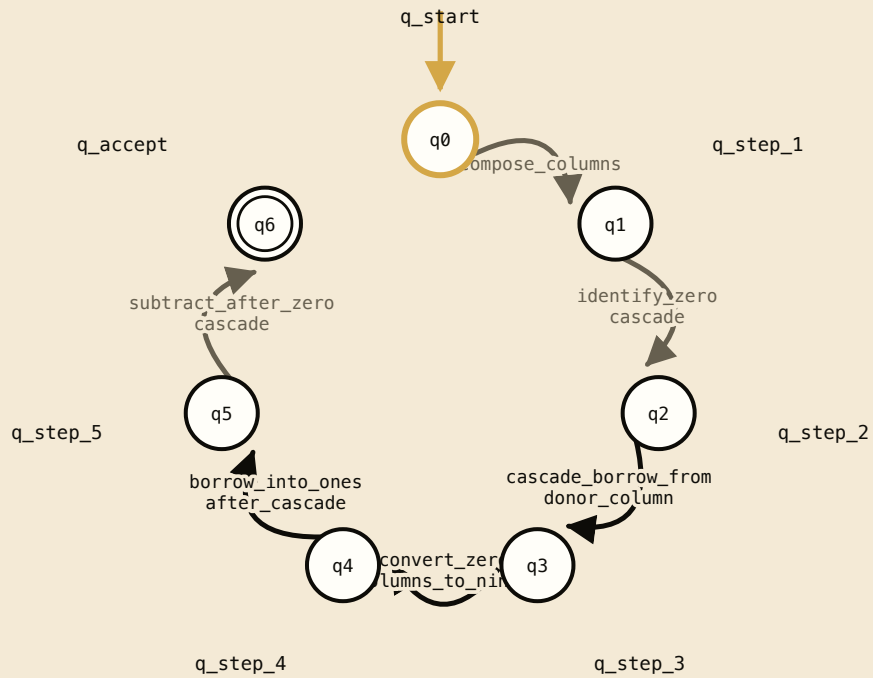
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

3. borrow_across_zero_cascade



The run produced 95.

The run produced 95. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Σ borrow_into_ones_after_cascade, cascade_borrow_from_donor_column, convert_zero_columns_to_nines, decompose_columns, identify_zero_cascade, subtract_after_zero_cascade (6 distinct)

q_0 $q_0 = q_start$

F $\{q_6\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 102, "b": 7};
verified(strategy_trace_ok).

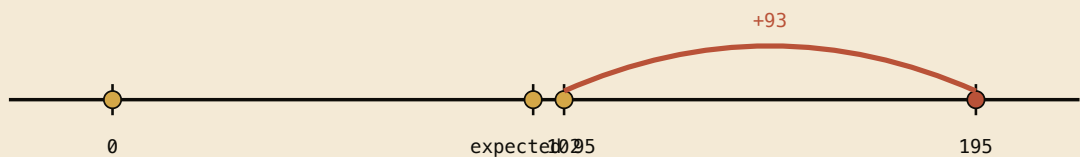
δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_columns	q_step_1	sar_sub_action_pairs.pl:394 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_1	identify_zero_cascade	q_step_2	sar_sub_action_pairs.pl:394 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_2	cascade_borrow_from_donor_column	q_step_3	sar_sub_action_pairs.pl:394 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_3	convert_zero_columns_to_nines	q_step_4	sar_sub_action_pairs.pl:394 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_4	borrow_into_ones_after_cascade	q_step_5	sar_sub_action_pairs.pl:394 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)

From	Action	To	Provenance
q_step_5	subtract_after_zero_cascade	q_accept	sar_sub_action_pairs.pl:394 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)

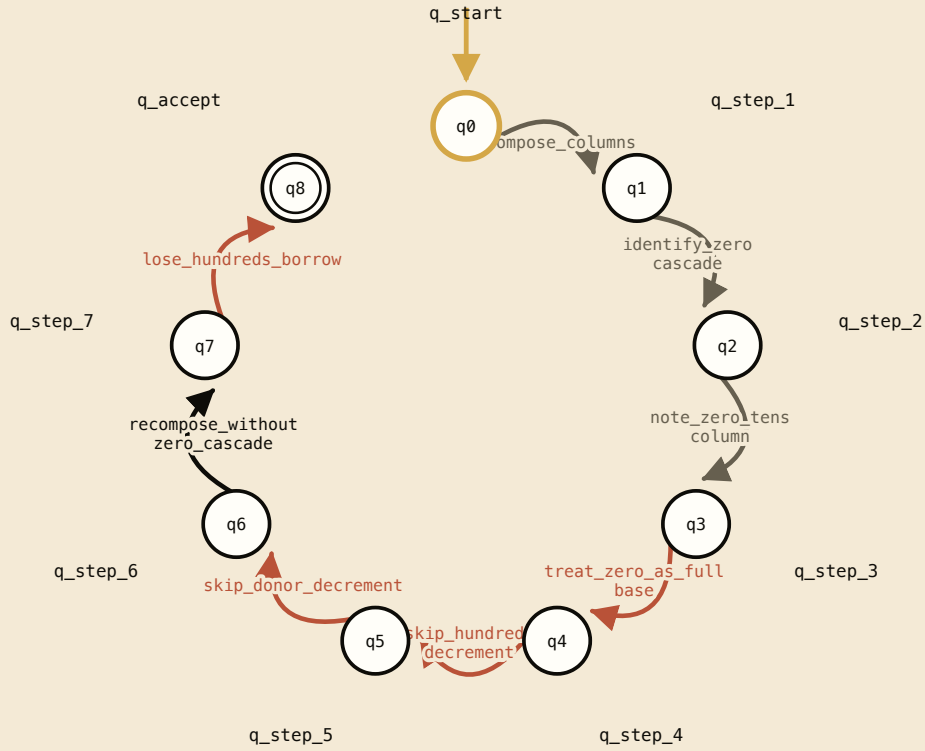
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

4. borrow_across_zero_no_cascade



The run produced 195; the recorded expected result is 95.

The run produced 195; the recorded expected result is 95. The scene is derived from the contract run (8 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

Σ decompose_columns, identify_zero_cascade, lose_hundreds_borrow, note_zero_tens_column, recompose_without_zero_cascade, skip_donor_decrement, skip_hundreds_decrement, treat_zero_as_full_base (8 distinct)

q_0 $q_0 = q_start$

F $\{q_8\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_step_6	none recorded
q_7	q_step_7	none recorded
q_8	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 102, "b": 7};
verified(strategy_trace_ok).

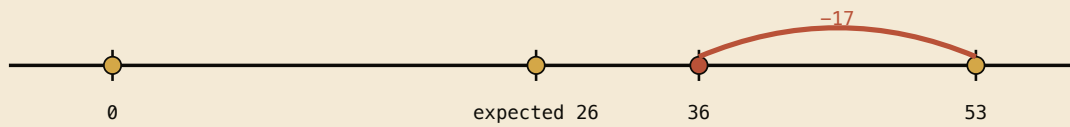
δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_columns	q_step_1	sar_sub_action_pairs.pl:420 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_1	identify_zero_cascade	q_step_2	sar_sub_action_pairs.pl:420 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_2	note_zero_tens_column	q_step_3	sar_sub_action_pairs.pl:420 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_3	treat_zero_as_full_base	q_step_4	sar_sub_action_pairs.pl:420 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)

From	Action	To	Provenance
q_step_4	skip_hundreds_decrement	q_step_5	sar_sub_action_pairs.pl:420 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_5	skip_donor_decrement	q_step_6	sar_sub_action_pairs.pl:420 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_6	recompose_without_zero_cascade	q_step_7	sar_sub_action_pairs.pl:420 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)
q_step_7	lose_hundreds_borrow	q_accept	sar_sub_action_pairs.pl:420 observed on {"a": 102, "b": 7} observed on {"a": 103, "b": 7} (derived template)

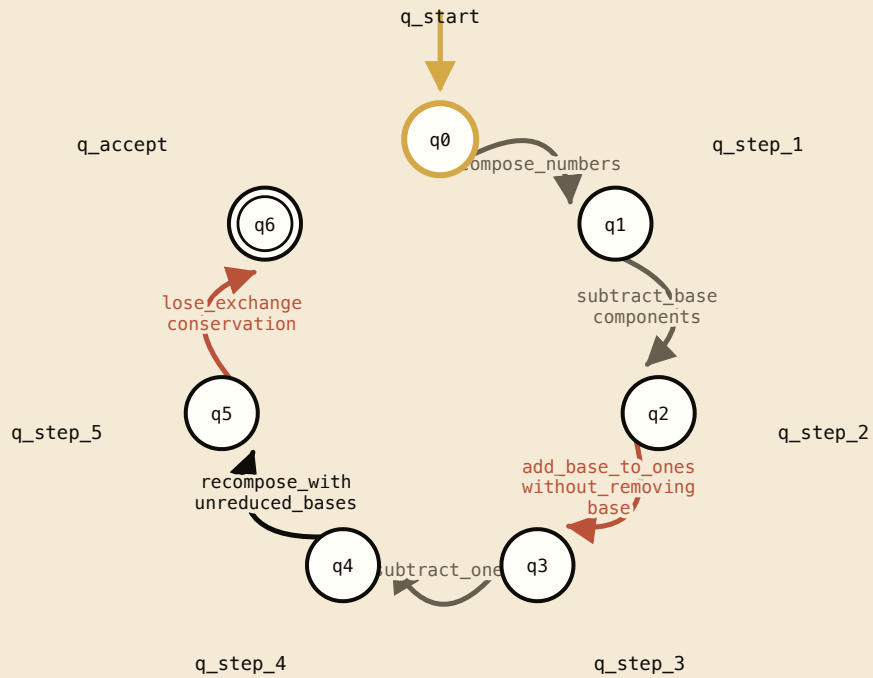
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 9 states, 8 distinct actions, 0 branching states, 0 loop edges; 8 static rows and 16 observed rows.

5. borrow_without_reducing_bases



The run produced 36; the recorded expected result is 26.

The run produced 36; the recorded expected result is 26. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \{\text{add_base_to_ones_without_removing_base, decompose_numbers, lose_exchange_conservation, recompose_with_unreduced_bases, subtract_base_components, subtract_ones (6 distinct)}\}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 53, "b": 27};
verified(strategy_trace_ok).

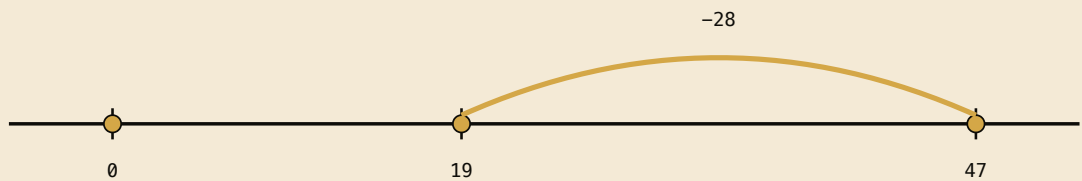
δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_numbers	q_step_1	sar_sub_action_pairs.pl:301 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_1	subtract_base_components	q_step_2	sar_sub_action_pairs.pl:301 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_2	add_base_to_ones_without_removal_base	q_step_3	sar_sub_action_pairs.pl:301 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_3	subtract_ones	q_step_4	sar_sub_action_pairs.pl:301 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_4	recompose_with_unreduced_bases	q_step_5	sar_sub_action_pairs.pl:301 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)

From	Action	To	Provenance
q_step_5	lose_exchange_conservation	q_accept	sar_sub_action_pairs.pl:301 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)

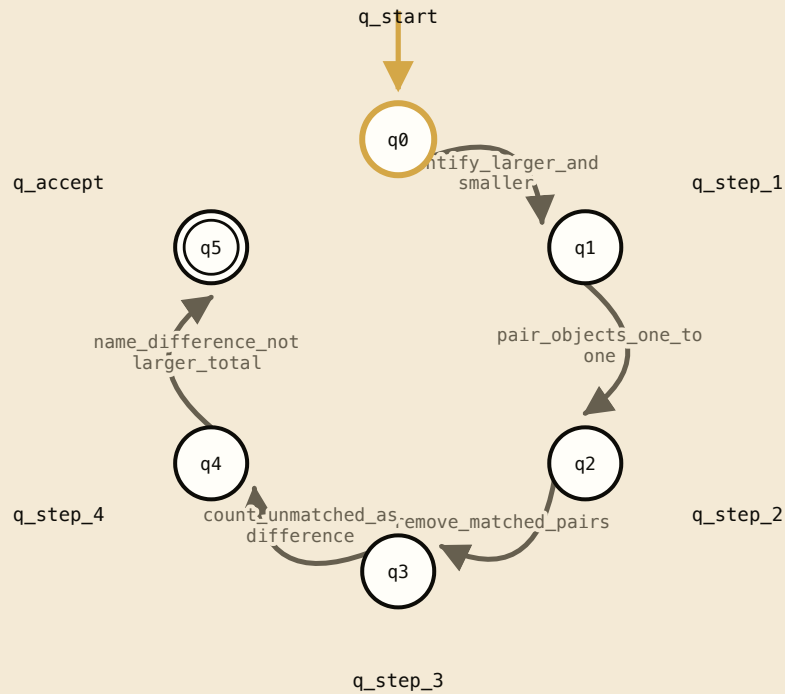
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 12 observed rows.

6. compare_by_matching_difference



The run produced 19.

The run produced 19. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ count_unmatched_as_difference, identify_larger_and_smaller, name_difference_not_larger_total, pair_objects_one_to_one, remove_matched_pairs (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_larger_and_smaller	q_step_1	sar_sub_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	pair_objects_one_to_one	q_step_2	sar_sub_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	remove_matched_pairs	q_step_3	sar_sub_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	count_unmatched_as_difference	q_step_4	sar_sub_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	name_difference_not_larger_to tal	q_accept	sar_sub_action_pairs.pl:171 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

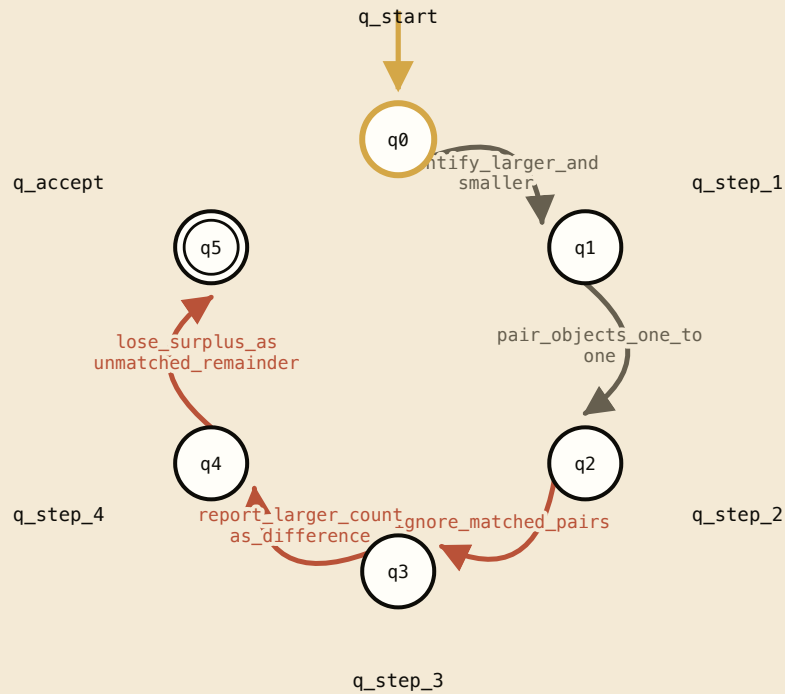
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

7. compare_returns_larger_count



The run produced 9; the recorded expected result is 4.

The run produced 9; the recorded expected result is 4. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ identify_larger_and_smaller, ignore_matched_pairs,
lose_surplus_as_unmatched_remainder, pair_objects_one_to_one,
report_larger_count_as_difference (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

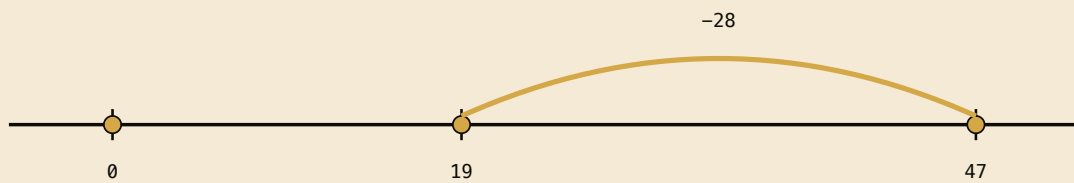
Input contract: {"a": "integer", "b": "integer"}; example {"a": 9, "b": 5};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_larger_and_smaller	q_step_1	sar_sub_action_pairs.pl:193 observed on {"a": 9, "b": 5} observed on {"a": 10, "b": 5} (derived template)
q_step_1	pair_objects_one_to_one	q_step_2	sar_sub_action_pairs.pl:193 observed on {"a": 9, "b": 5} observed on {"a": 10, "b": 5} (derived template)
q_step_2	ignore_matched_pairs	q_step_3	sar_sub_action_pairs.pl:193 observed on {"a": 9, "b": 5} observed on {"a": 10, "b": 5} (derived template)
q_step_3	report_larger_count_as_difference	q_step_4	sar_sub_action_pairs.pl:193 observed on {"a": 9, "b": 5} observed on {"a": 10, "b": 5} (derived template)
q_step_4	lose_surplus_as_unmatched_remainder	q_accept	sar_sub_action_pairs.pl:193 observed on {"a": 9, "b": 5} observed on {"a": 10, "b": 5} (derived template)

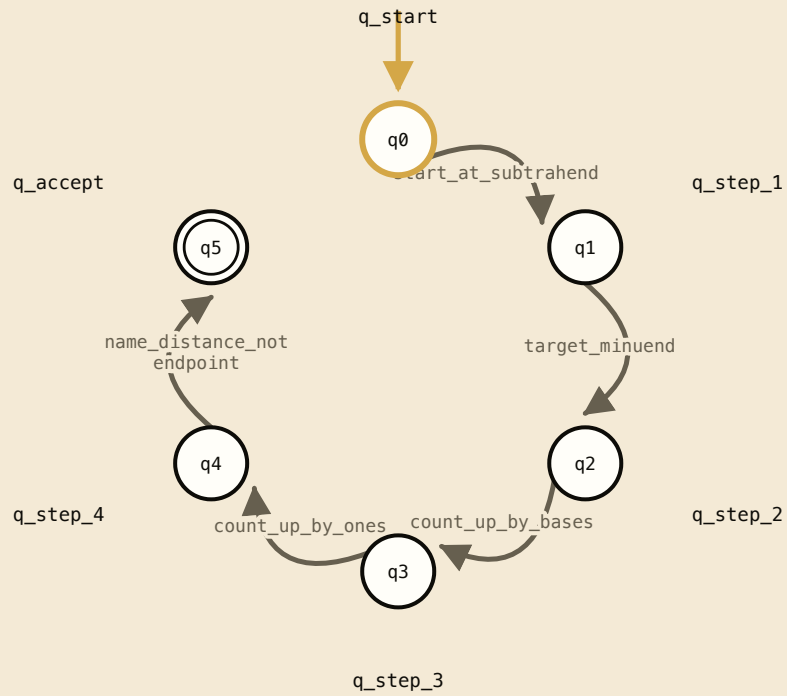
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

8. count_up_missing_addend



The run produced 19.

The run produced 19. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ $count_up_by_bases, count_up_by_ones, name_distance_not_endpoint, start_at_subtrahend, target_minuend$ (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

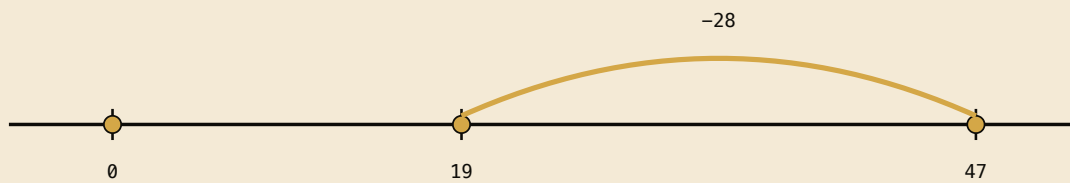
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	start_at_subtrahend	q_step_1	sar_sub_action_pairs.pl:122 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	target_minuend	q_step_2	sar_sub_action_pairs.pl:122 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	count_up_by_bases	q_step_3	sar_sub_action_pairs.pl:122 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	count_up_by_ones	q_step_4	sar_sub_action_pairs.pl:122 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_4	name_distance_not_endpoint	q_accept	sar_sub_action_pairs.pl:122 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

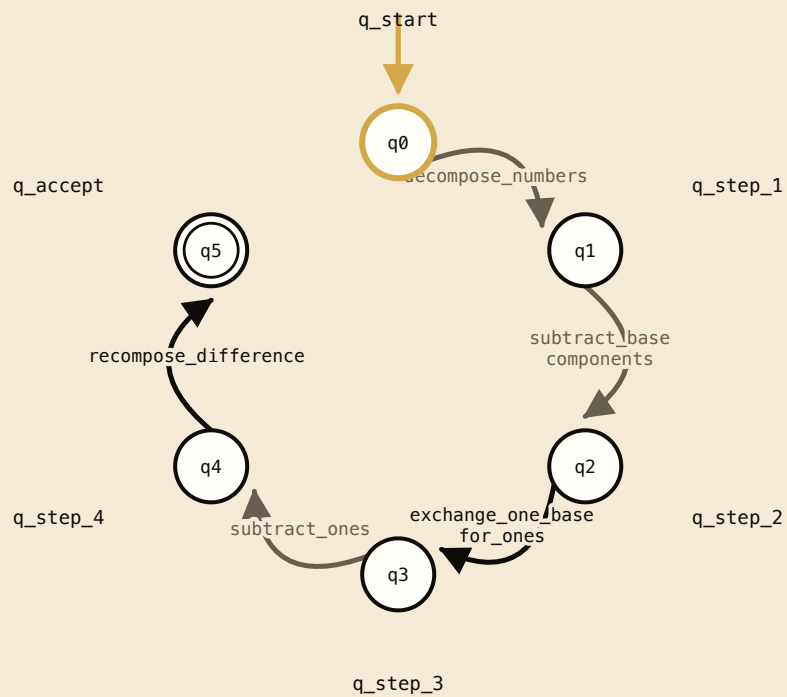
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

9. decompose_base_for_ones



The run produced 19.

The run produced 19. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ decompose_numbers, exchange_one_base_for_ones, recompose_difference, subtract_base_components, subtract_ones (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

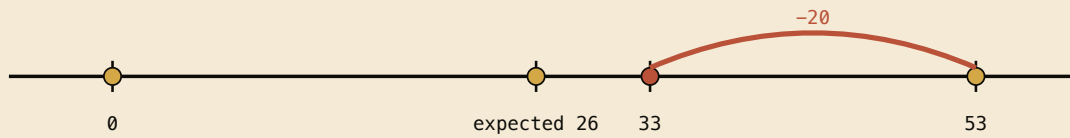
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_numbers	q_step_1	sar_sub_action_pairs.pl:270 observed on {"a": 47, "b": 28}
q_step_1	subtract_base_components	q_step_2	sar_sub_action_pairs.pl:270 observed on {"a": 47, "b": 28}
q_step_2	exchange_one_base_for_ones	q_step_3	sar_sub_action_pairs.pl:270 observed on {"a": 47, "b": 28}
q_step_3	subtract_ones	q_step_4	sar_sub_action_pairs.pl:270 observed on {"a": 47, "b": 28}
q_step_4	recompose_difference	q_accept	sar_sub_action_pairs.pl:270 observed on {"a": 47, "b": 28}

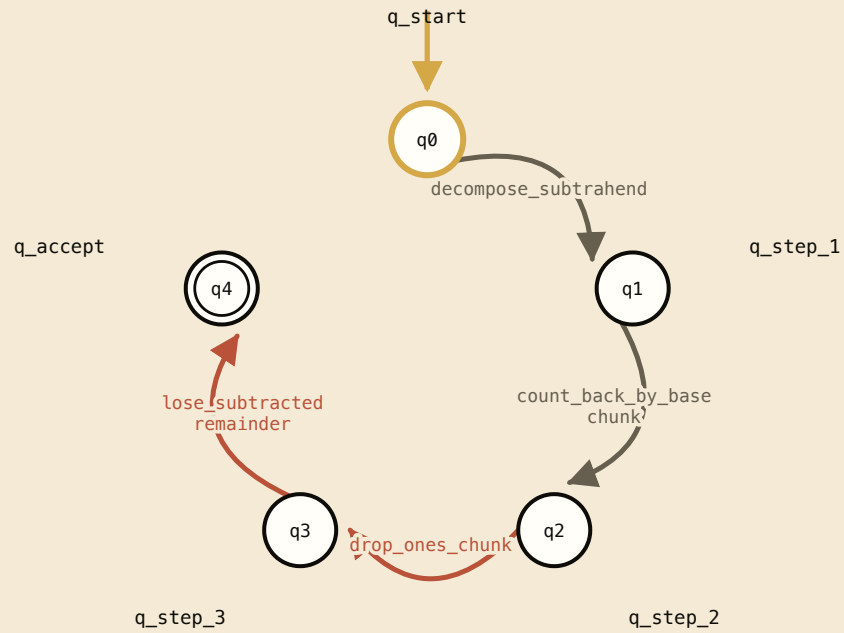
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 5 observed rows.

10. drop_ones_after_base_takeaway



The run produced 33; the recorded expected result is 26.

The run produced 33; the recorded expected result is 26. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{count_back_by_base_chunk, decompose_subtrahend, drop_ones_chunk, lose_subtracted_remainder (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

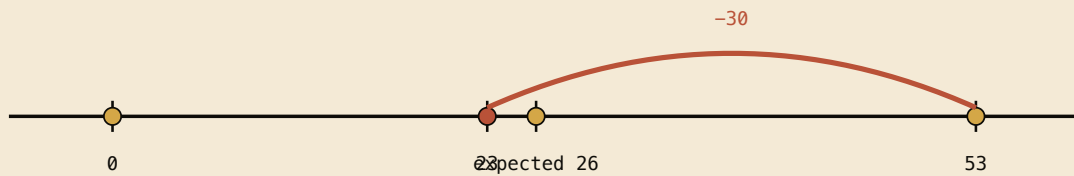
Input contract: {"a": "integer", "b": "integer"}; example {"a": 53, "b": 27};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_subtrahend	q_step_1	sar_sub_action_pairs.pl:97 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_1	count_back_by_base_chunk	q_step_2	sar_sub_action_pairs.pl:97 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_2	drop_ones_chunk	q_step_3	sar_sub_action_pairs.pl:97 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_3	lose_subtracted_remainder	q_accept	sar_sub_action_pairs.pl:97 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)

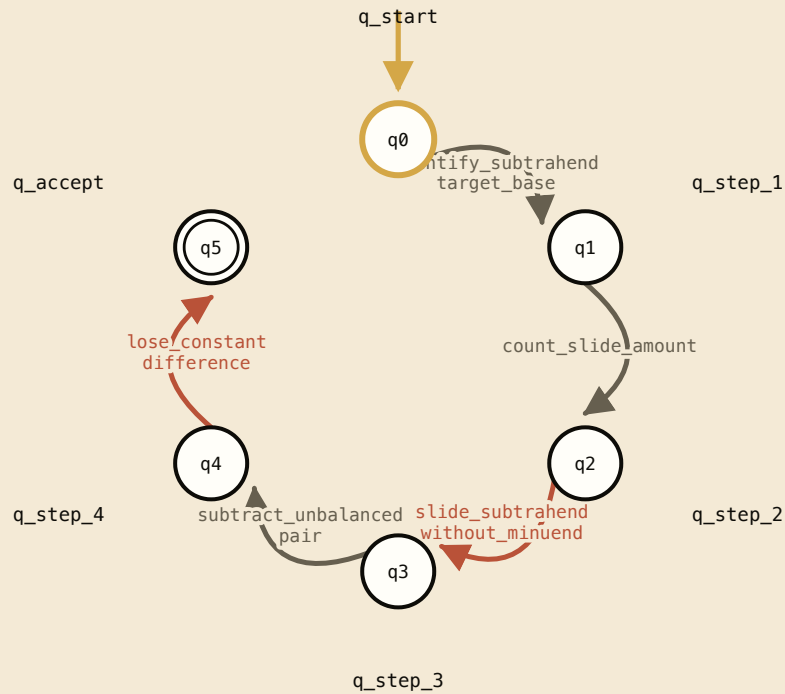
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

11. slide_subtrahend_only



The run produced 23; the recorded expected result is 26.

The run produced 23; the recorded expected result is 26. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

Σ $count_slide_amount, identify_subtrahend_target_base, lose_constant_difference, slide_subtrahend_without_minuend, subtract_unbalanced_pair$ (5 distinct)

q_0 $q_0 = q_start$

F $\{q_5\}$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

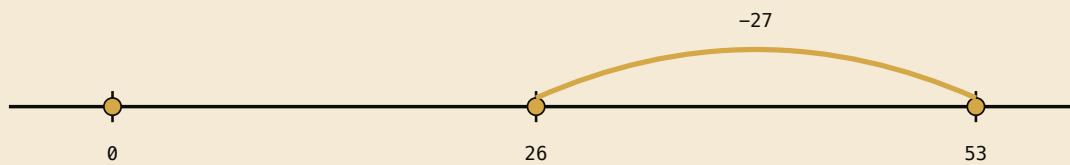
Input contract: {"a": "integer", "b": "integer"}; example {"a": 53, "b": 27};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_subtrahend_target_base	q_step_1	sar_sub_action_pairs.pl:245 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_1	count_slide_amount	q_step_2	sar_sub_action_pairs.pl:245 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_2	slide_subtrahend_without_minuend	q_step_3	sar_sub_action_pairs.pl:245 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_3	subtract_unbalanced_pair	q_step_4	sar_sub_action_pairs.pl:245 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_4	lose_constant_difference	q_accept	sar_sub_action_pairs.pl:245 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)

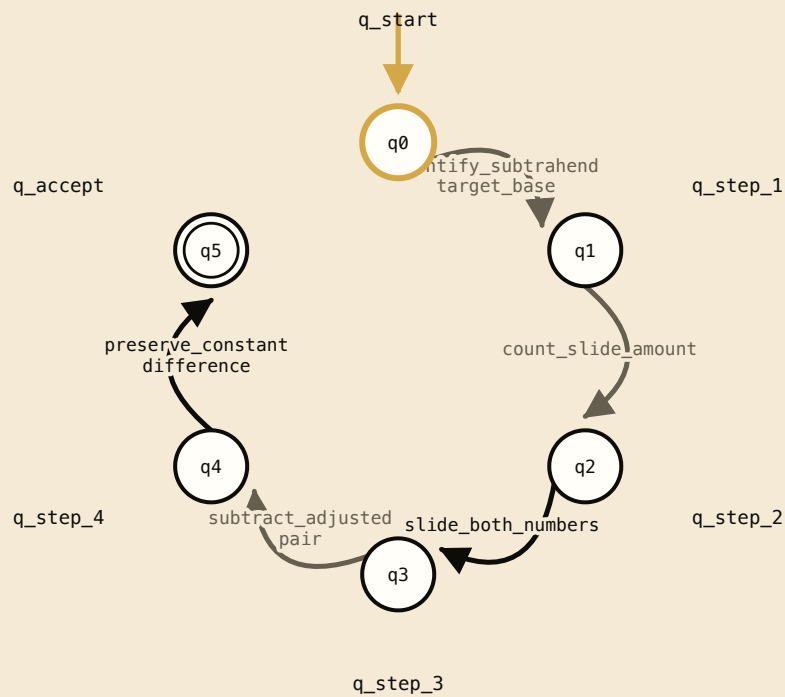
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

12. sliding_constant_difference



The run produced 26.

The run produced 26. The scene is derived from the contract run (5 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

$$\Sigma \quad \text{count_slide_amount, identify_subtrahend_target_base, preserve_constant_difference, slide_both_numbers, subtract_adjusted_pair (5 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_5\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_accept	none recorded

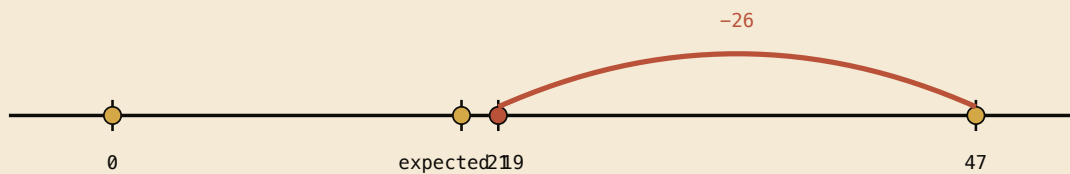
Input contract: {"a": "integer", "b": "integer"}; example {"a": 53, "b": 27};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	identify_subtrahend_target_base	q_step_1	sar_sub_action_pairs.pl:219 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_1	count_slide_amount	q_step_2	sar_sub_action_pairs.pl:219 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_2	slide_both_numbers	q_step_3	sar_sub_action_pairs.pl:219 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_3	subtract_adjusted_pair	q_step_4	sar_sub_action_pairs.pl:219 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)
q_step_4	preserve_constant_difference	q_accept	sar_sub_action_pairs.pl:219 observed on {"a": 53, "b": 27} observed on {"a": 54, "b": 27} (derived template)

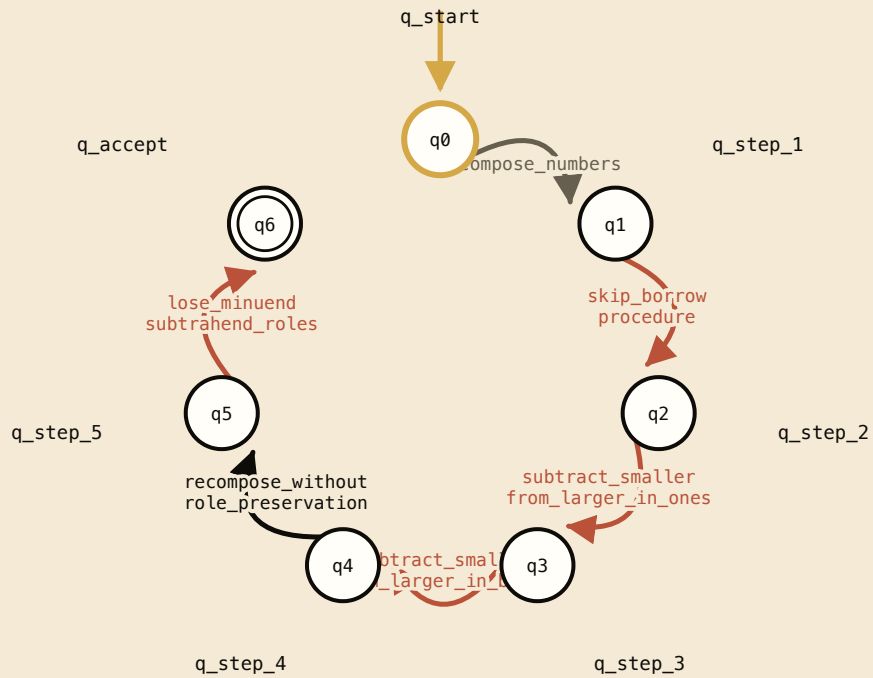
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 6 states, 5 distinct actions, 0 branching states, 0 loop edges; 5 static rows and 10 observed rows.

13. smaller_from_larger_in_column



The run produced 21; the recorded expected result is 19.

The run produced 21; the recorded expected result is 19. The scene is derived from the contract run (6 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \{decompose_numbers, lose_minuend_subtrahend_roles, \\ recompose_without_role_preservation, skip_borrow_procedure, \\ subtract_smaller_from_larger_in_bases, subtract_smaller_from_larger_in_ones\} \text{ (6 distinct)}$$

$$q_0 = q_start$$

$$F = \{q_6\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_step_4	none recorded
q_5	q_step_5	none recorded
q_6	q_accept	none recorded

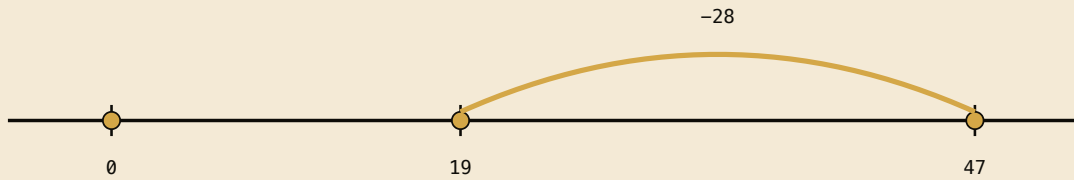
Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_numbers	q_step_1	sar_sub_action_pairs.pl:332 observed on {"a": 47, "b": 28}
q_step_1	skip_borrow_procedure	q_step_2	sar_sub_action_pairs.pl:332 observed on {"a": 47, "b": 28}
q_step_2	subtract_smaller_from_larger_in_ones	q_step_3	sar_sub_action_pairs.pl:332 observed on {"a": 47, "b": 28}
q_step_3	subtract_smaller_from_larger_in_bases	q_step_4	sar_sub_action_pairs.pl:332 observed on {"a": 47, "b": 28}
q_step_4	recompose_without_role_preservation	q_step_5	sar_sub_action_pairs.pl:332 observed on {"a": 47, "b": 28}
q_step_5	lose_minuend_subtrahend_roles	q_accept	sar_sub_action_pairs.pl:332 observed on {"a": 47, "b": 28}

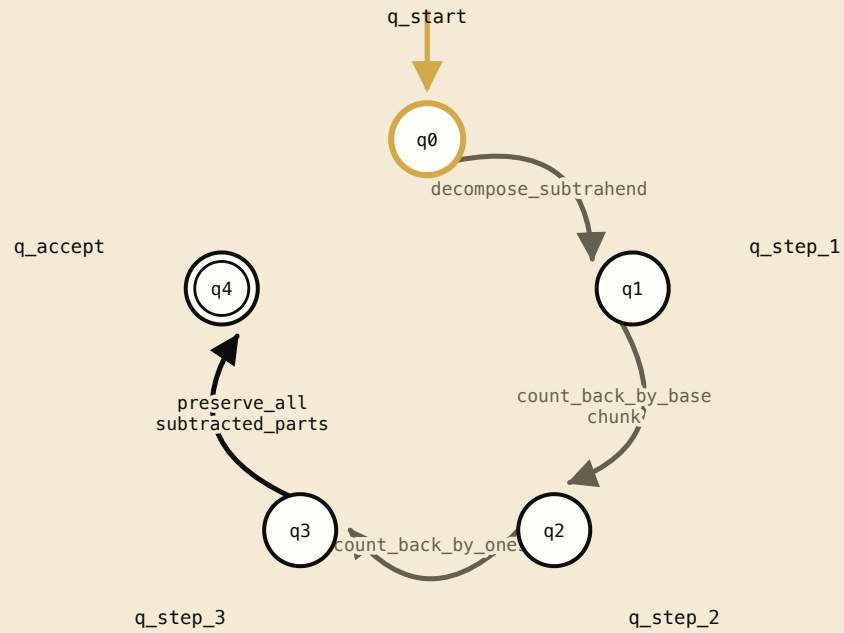
Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 7 states, 6 distinct actions, 0 branching states, 0 loop edges; 6 static rows and 6 observed rows.

14. take_away_base_ones



The run produced 19.

The run produced 19. The scene is derived from the contract run (4 trace steps).



Recorded transition graph; local action names remain on the edges.

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q \quad \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma \quad \text{count_back_by_base_chunk, count_back_by_ones, decompose_subtrahend, preserve_all_subtracted_parts (4 distinct)}$$

$$q_0 \quad q_0 = q_start$$

$$F \quad \{q_4\}$$

Formal state	ASCII atom	Literature label and citation
q_0	q_start	none recorded
q_1	q_step_1	none recorded
q_2	q_step_2	none recorded
q_3	q_step_3	none recorded
q_4	q_accept	none recorded

Input contract: {"a": "integer", "b": "integer"}; example {"a": 47, "b": 28};
verified(strategy_trace_ok).

δ : recorded transitions

From	Action	To	Provenance
q_start	decompose_subtrahend	q_step_1	sar_sub_action_pairs.pl:73 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_1	count_back_by_base_chunk	q_step_2	sar_sub_action_pairs.pl:73 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_2	count_back_by_ones	q_step_3	sar_sub_action_pairs.pl:73 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)
q_step_3	preserve_all_subtracted_parts	q_accept	sar_sub_action_pairs.pl:73 observed on {"a": 47, "b": 28} observed on {"a": 48, "b": 28} (derived template)

Computed structure: linear_trace. no recorded branch or loop; the table records one accepting run. 5 states, 4 distinct actions, 0 branching states, 0 loop edges; 4 static rows and 8 observed rows.

Artifacts and regeneration: knowledge/strategies/transition_tables/*.pl, python3
scripts/research/build_transition_tables.py; knowledge/strategies/machine_typology.pl, python3
scripts/research/build_machine_typology.py; radial assets
docs/research/assets/automata/<family>/<kind>.svg, python3
scripts/research/render_automaton_svg.py; scene and composite assets, python3
scripts/research/render_automaton_context_svg.py; the hub docs/research/2026-08-03-automata-
compendium.html and family pages under docs/research/automata-compendium/, python3
scripts/research/build_automata_compendium.py. Authored input:
knowledge/strategies/machine_class_attestations.pl. Parsed inputs:
knowledge/strategies/automaton_input_contracts.pl,
knowledge/strategies/math/state_vocabulary.pl, and
knowledge/strategies/deformation_coincidence.pl; the coincidence file records its sweep generator and
finite grids in its header.